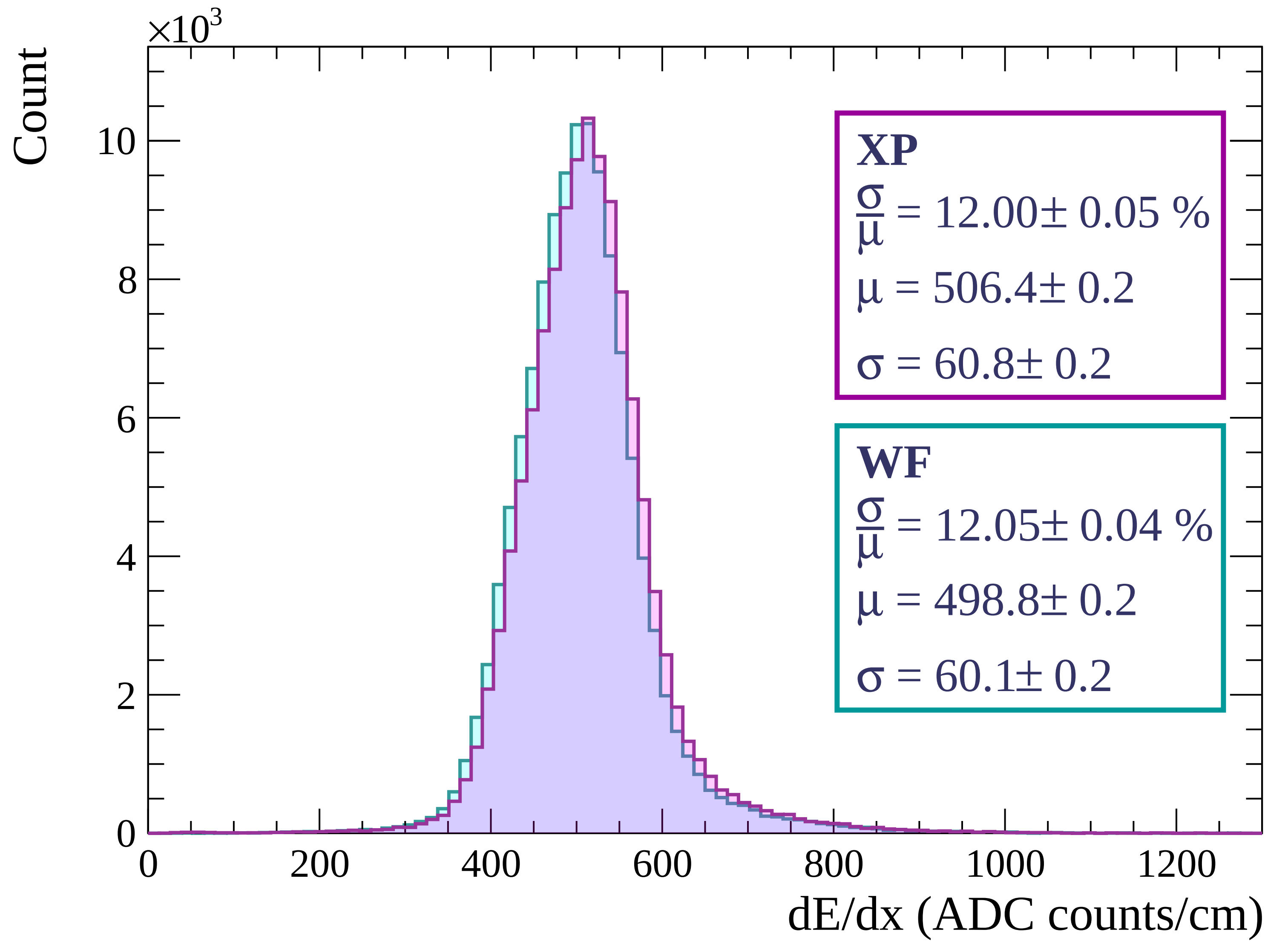
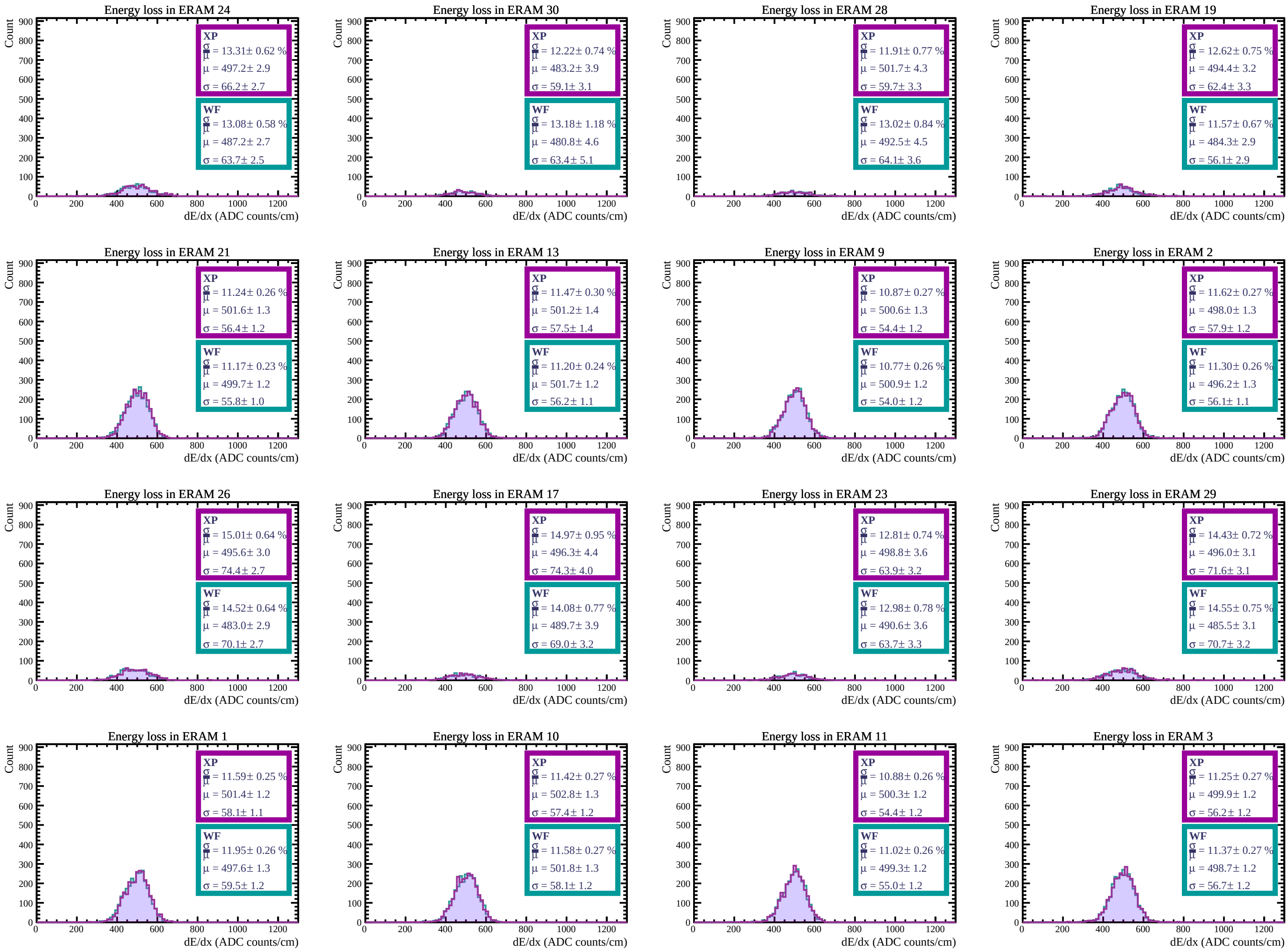
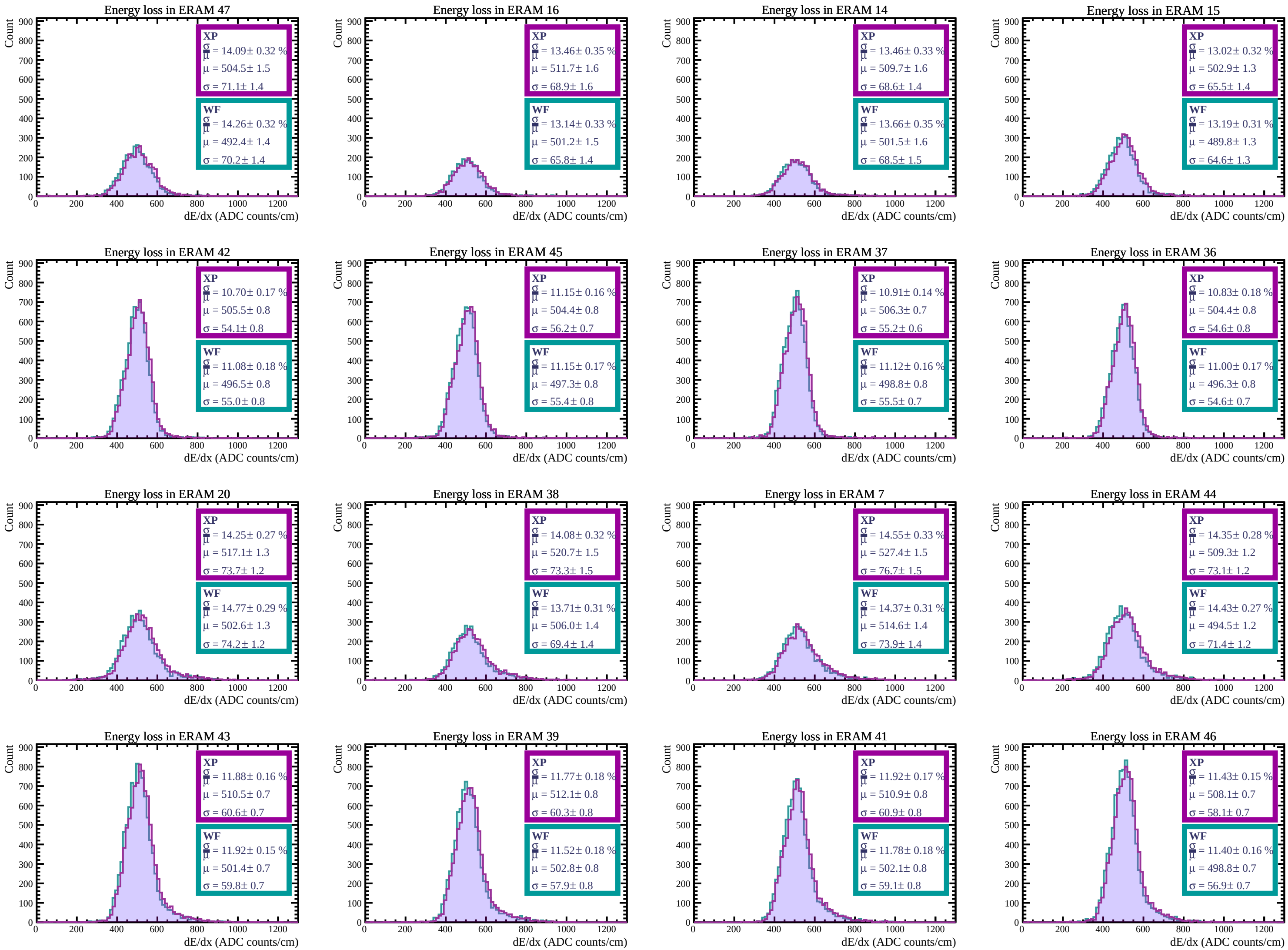
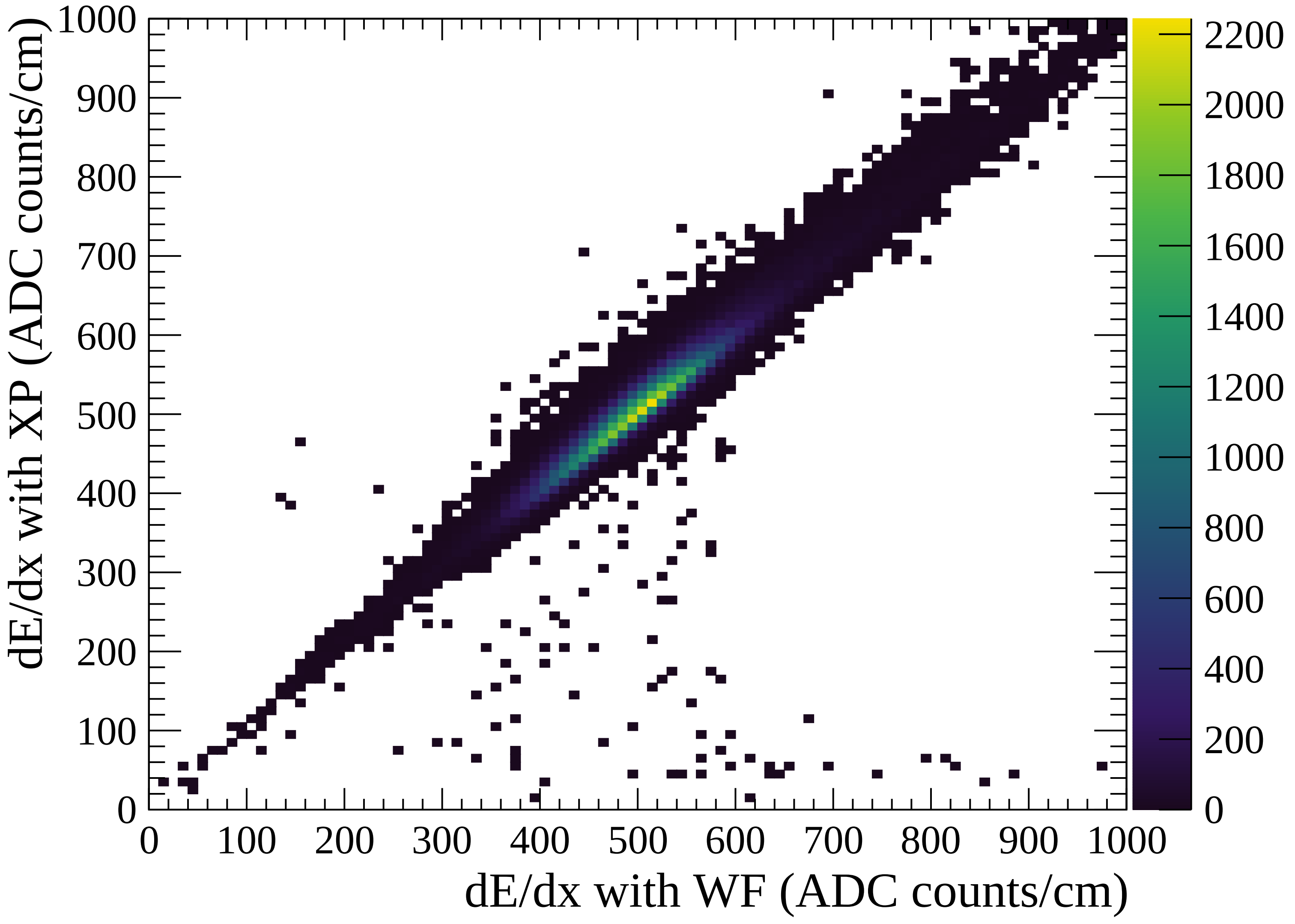


×10^3
Count
0
2
4
6
8
10
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 12.00± 0.05 %
μ = 506.4± 0.2
σ = 60.8± 0.2
WF
σ/μ = 12.05± 0.04 %
μ = 498.8± 0.2
σ = 60.1± 0.2

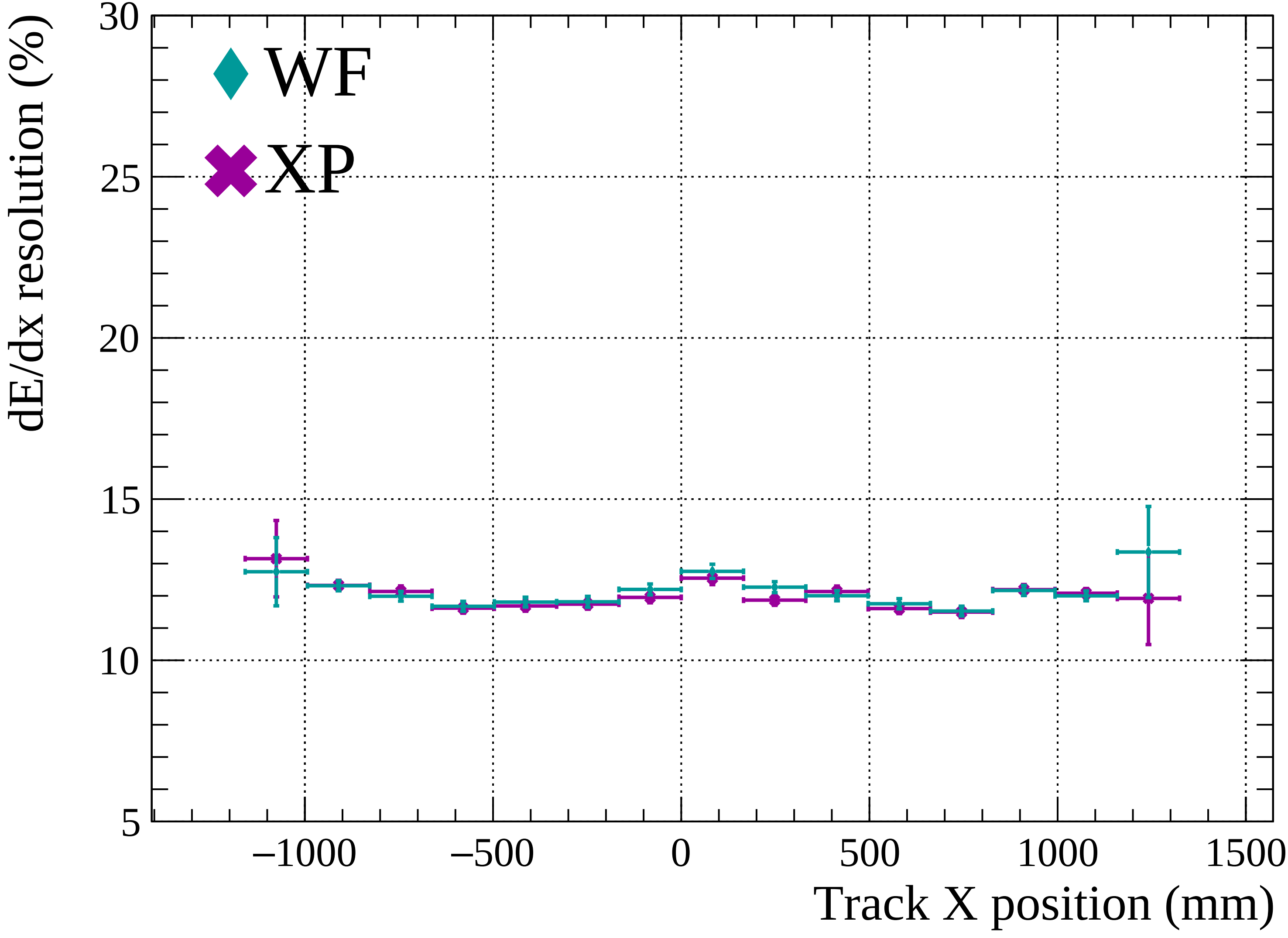

Energy loss in ERAM 24
XP
σ/μ = 13.31± 0.62 %
μ = 497.2± 2.9
σ = 66.2± 2.7
WF
σ/μ = 13.08± 0.58 %
μ = 487.2± 2.7
σ = 63.7± 2.5
Energy loss in ERAM 30
XP
σ/μ = 12.22± 0.74 %
μ = 483.2± 3.9
σ = 59.1± 3.1
WF
σ/μ = 13.18± 1.18 %
μ = 480.8± 4.6
σ = 63.4± 5.1
Energy loss in ERAM 28
XP
σ/μ = 11.91± 0.77 %
μ = 501.7± 4.3
σ = 59.7± 3.3
WF
σ/μ = 13.02± 0.84 %
μ = 492.5± 4.5
σ = 64.1± 3.6
Energy loss in ERAM 19
XP
σ/μ = 12.62± 0.75 %
μ = 494.4± 3.2
σ = 62.4± 3.3
WF
σ/μ = 11.57± 0.67 %
μ = 484.3± 2.9
σ = 56.1± 2.9
Energy loss in ERAM 21
XP
σ/μ = 11.24± 0.26 %
μ = 501.6± 1.3
σ = 56.4± 1.2
WF
σ/μ = 11.17± 0.23 %
μ = 499.7± 1.2
σ = 55.8± 1.0
Energy loss in ERAM 13
XP
σ/μ = 11.47± 0.30 %
μ = 501.2± 1.4
σ = 57.5± 1.4
WF
σ/μ = 11.20± 0.24 %
μ = 501.7± 1.2
σ = 56.2± 1.1
Energy loss in ERAM 9
XP
σ/μ = 10.87± 0.27 %
μ = 500.6± 1.3
σ = 54.4± 1.2
WF
σ/μ = 10.77± 0.26 %
μ = 500.9± 1.2
σ = 54.0± 1.2
Energy loss in ERAM 2
XP
σ/μ = 11.62± 0.27 %
μ = 498.0± 1.3
σ = 57.9± 1.2
WF
σ/μ = 11.30± 0.26 %
μ = 496.2± 1.3
σ = 56.1± 1.1
Energy loss in ERAM 26
XP
σ/μ = 15.01± 0.64 %
μ = 495.6± 3.0
σ = 74.4± 2.7
WF
σ/μ = 14.52± 0.64 %
μ = 483.0± 2.9
σ = 70.1± 2.7
Energy loss in ERAM 17
XP
σ/μ = 14.97± 0.95 %
μ = 496.3± 4.4
σ = 74.3± 4.0
WF
σ/μ = 14.08± 0.77 %
μ = 489.7± 3.9
σ = 69.0± 3.2
Energy loss in ERAM 23
XP
σ/μ = 12.81± 0.74 %
μ = 498.8± 3.6
σ = 63.9± 3.2
WF
σ/μ = 12.98± 0.78 %
μ = 490.6± 3.6
σ = 63.7± 3.3
Energy loss in ERAM 29
XP
σ/μ = 14.43± 0.72 %
μ = 496.0± 3.1
σ = 71.6± 3.1
WF
σ/μ = 14.55± 0.75 %
μ = 485.5± 3.1
σ = 70.7± 3.2
Energy loss in ERAM 1
XP
σ/μ = 11.59± 0.25 %
μ = 501.4± 1.2
σ = 58.1± 1.1
WF
σ/μ = 11.95± 0.26 %
μ = 497.6± 1.3
σ = 59.5± 1.2
Energy loss in ERAM 10
XP
σ/μ = 11.42± 0.27 %
μ = 502.8± 1.3
σ = 57.4± 1.2
WF
σ/μ = 11.58± 0.27 %
μ = 501.8± 1.3
σ = 58.1± 1.2
Energy loss in ERAM 11
XP
σ/μ = 10.88± 0.26 %
μ = 500.3± 1.2
σ = 54.4± 1.2
WF
σ/μ = 11.02± 0.26 %
μ = 499.3± 1.2
σ = 55.0± 1.2
Energy loss in ERAM 3
XP
σ/μ = 11.25± 0.27 %
μ = 499.9± 1.2
σ = 56.2± 1.2
WF
σ/μ = 11.37± 0.27 %
μ = 498.7± 1.2
σ = 56.7± 1.2
Count
dE/dx (ADC counts/cm)



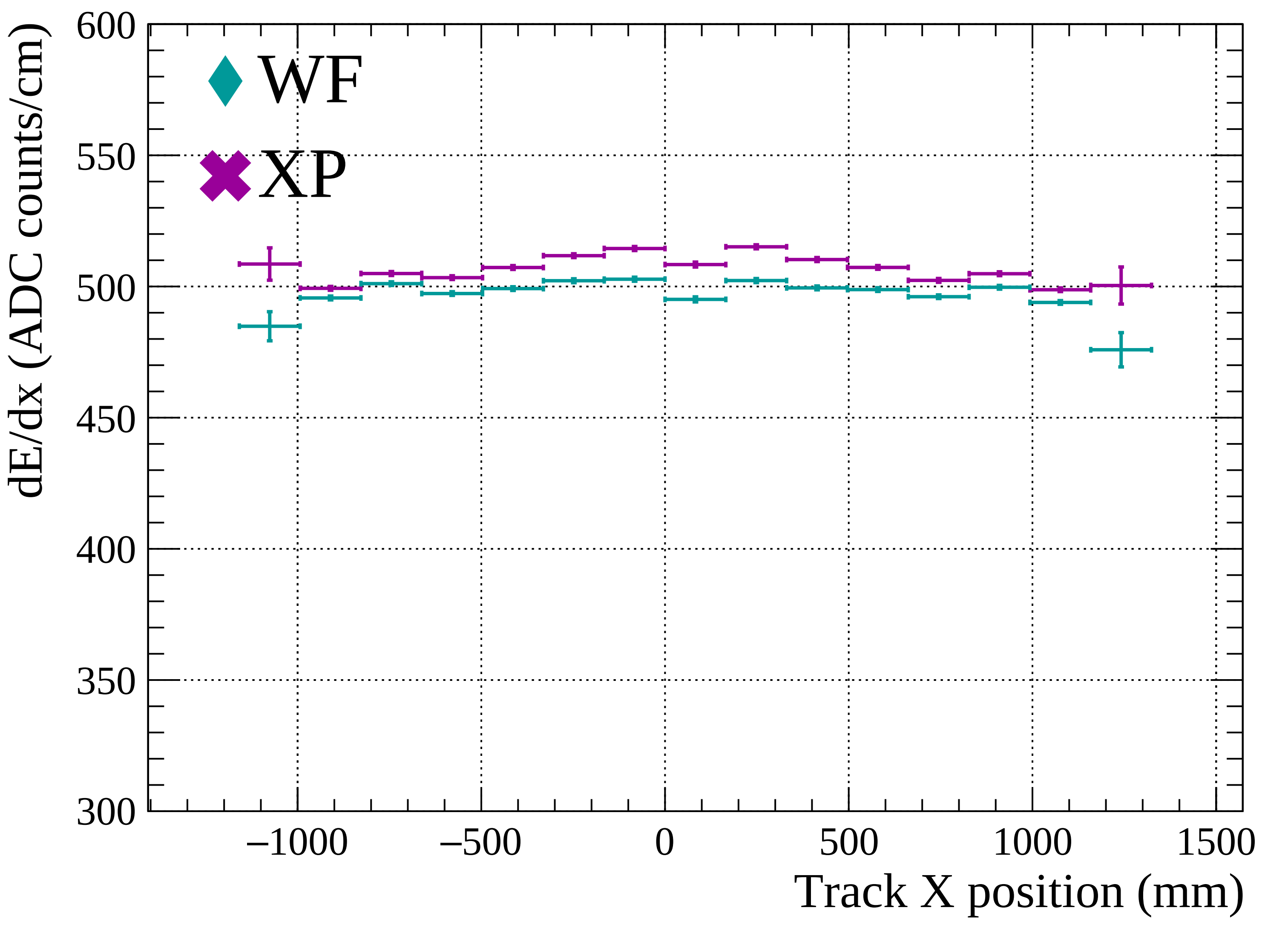
Energy loss in ERAM 47
XP
σ/μ = 14.09± 0.32 %
μ = 504.5± 1.5
σ = 71.1± 1.4
WF
σ/μ = 14.26± 0.32 %
μ = 492.4± 1.4
σ = 70.2± 1.4
Energy loss in ERAM 16
XP
σ/μ = 13.46± 0.35 %
μ = 511.7± 1.6
σ = 68.9± 1.6
WF
σ/μ = 13.14± 0.33 %
μ = 501.2± 1.5
σ = 65.8± 1.4
Energy loss in ERAM 14
XP
σ/μ = 13.46± 0.33 %
μ = 509.7± 1.6
σ = 68.6± 1.4
WF
σ/μ = 13.66± 0.35 %
μ = 501.5± 1.6
σ = 68.5± 1.5
Energy loss in ERAM 15
XP
σ/μ = 13.02± 0.32 %
μ = 502.9± 1.3
σ = 65.5± 1.4
WF
σ/μ = 13.19± 0.31 %
μ = 489.8± 1.3
σ = 64.6± 1.3
Energy loss in ERAM 42
XP
σ/μ = 10.70± 0.17 %
μ = 505.5± 0.8
σ = 54.1± 0.8
WF
σ/μ = 11.08± 0.18 %
μ = 496.5± 0.8
σ = 55.0± 0.8
Energy loss in ERAM 45
XP
σ/μ = 11.15± 0.16 %
μ = 504.4± 0.8
σ = 56.2± 0.7
WF
σ/μ = 11.15± 0.17 %
μ = 497.3± 0.8
σ = 55.4± 0.8
Energy loss in ERAM 37
XP
σ/μ = 10.91± 0.14 %
μ = 506.3± 0.7
σ = 55.2± 0.6
WF
σ/μ = 11.12± 0.16 %
μ = 498.8± 0.8
σ = 55.5± 0.7
Energy loss in ERAM 36
XP
σ/μ = 10.83± 0.18 %
μ = 504.4± 0.8
σ = 54.6± 0.8
WF
σ/μ = 11.00± 0.17 %
μ = 496.3± 0.8
σ = 54.6± 0.7
Energy loss in ERAM 20
XP
σ/μ = 14.25± 0.27 %
μ = 517.1± 1.3
σ = 73.7± 1.2
WF
σ/μ = 14.77± 0.29 %
μ = 502.6± 1.3
σ = 74.2± 1.2
Energy loss in ERAM 38
XP
σ/μ = 14.08± 0.32 %
μ = 520.7± 1.5
σ = 73.3± 1.5
WF
σ/μ = 13.71± 0.31 %
μ = 506.0± 1.4
σ = 69.4± 1.4
Energy loss in ERAM 7
XP
σ/μ = 14.55± 0.33 %
μ = 527.4± 1.5
σ = 76.7± 1.5
WF
σ/μ = 14.37± 0.31 %
μ = 514.6± 1.4
σ = 73.9± 1.4
Energy loss in ERAM 44
XP
σ/μ = 14.35± 0.28 %
μ = 509.3± 1.2
σ = 73.1± 1.2
WF
σ/μ = 14.43± 0.27 %
μ = 494.5± 1.2
σ = 71.4± 1.2
Energy loss in ERAM 43
XP
σ/μ = 11.88± 0.16 %
μ = 510.5± 0.7
σ = 60.6± 0.7
WF
σ/μ = 11.92± 0.15 %
μ = 501.4± 0.7
σ = 59.8± 0.7
Energy loss in ERAM 39
XP
σ/μ = 11.77± 0.18 %
μ = 512.1± 0.8
σ = 60.3± 0.8
WF
σ/μ = 11.52± 0.18 %
μ = 502.8± 0.8
σ = 57.9± 0.8
Energy loss in ERAM 41
XP
σ/μ = 11.92± 0.17 %
μ = 510.9± 0.8
σ = 60.9± 0.8
WF
σ/μ = 11.78± 0.18 %
μ = 502.1± 0.8
σ = 59.1± 0.8
Energy loss in ERAM 46
XP
σ/μ = 11.43± 0.15 %
μ = 508.1± 0.7
σ = 58.1± 0.7
WF
σ/μ = 11.40± 0.16 %
μ = 498.8± 0.7
σ = 56.9± 0.7
Count
dE/dx (ADC counts/cm)

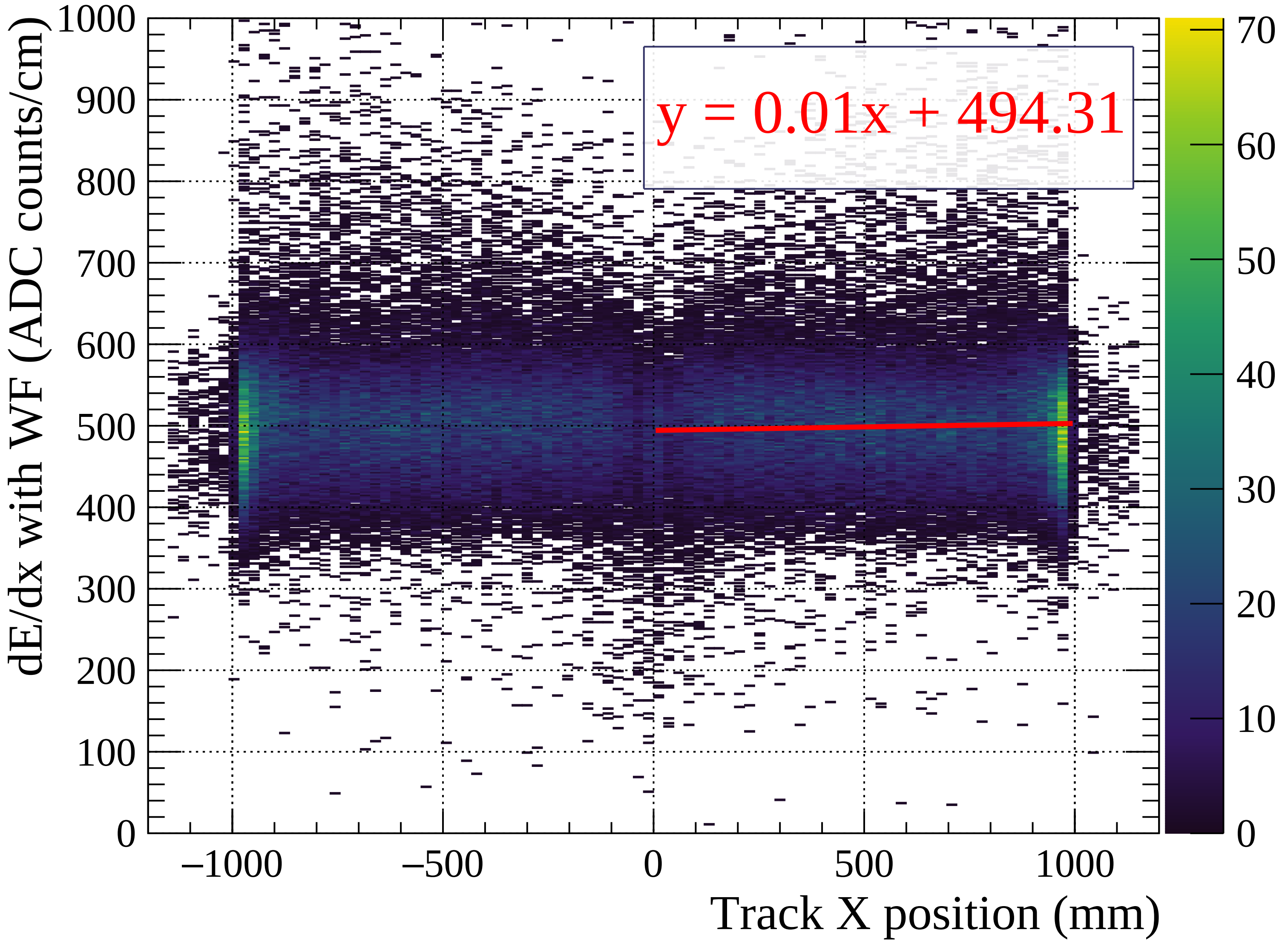


dE/dx with XP (ADC counts/cm)
dE/dx with WF (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
1000
2200
2000
1800
1600
1400
1200
1000
800
600
400
200
0

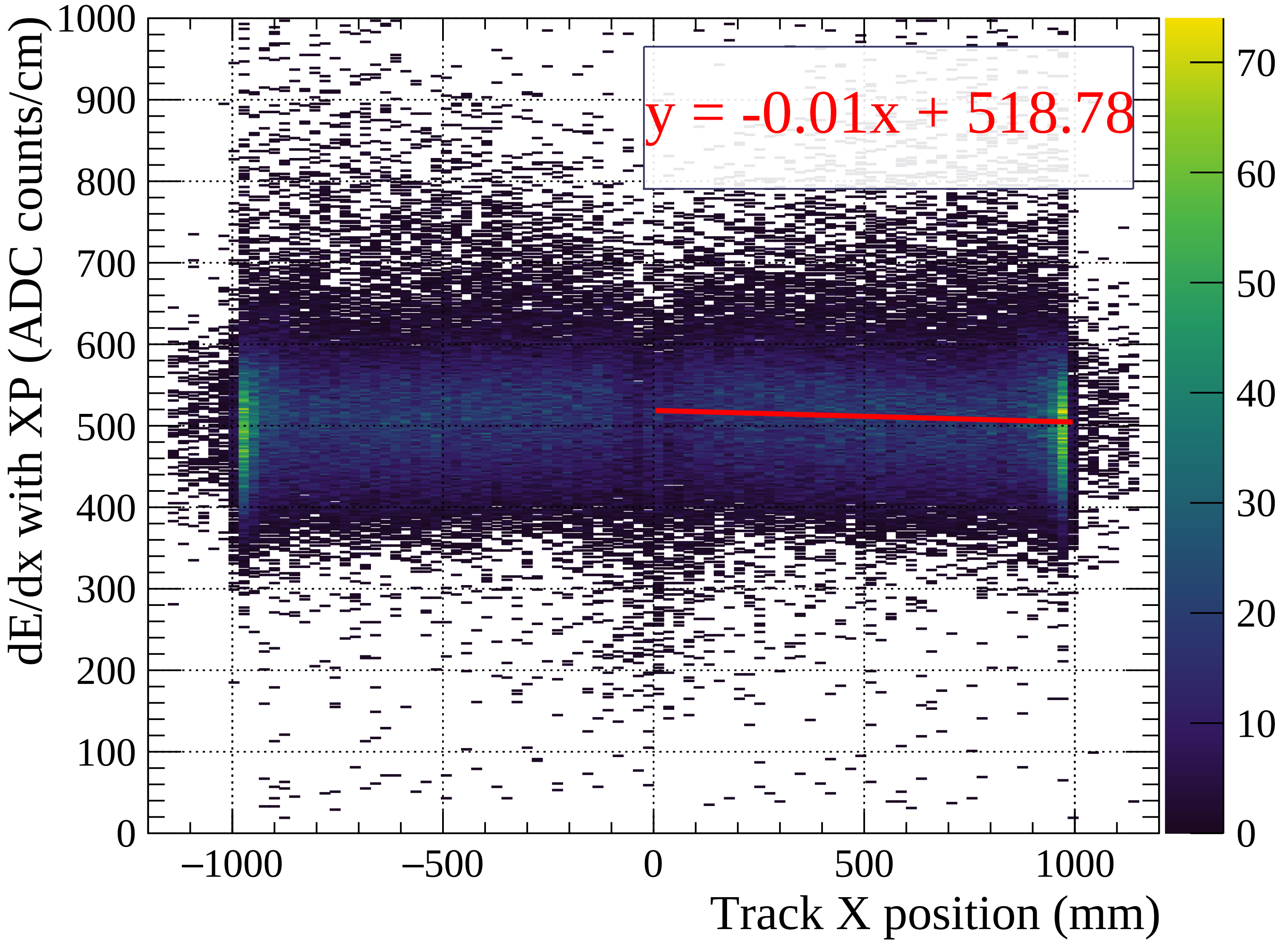


dE/dx resolution (%)
Track X position (mm)
WF
XP
30
25
20
15
10
5
−1000
−500
0
500
1000
1500

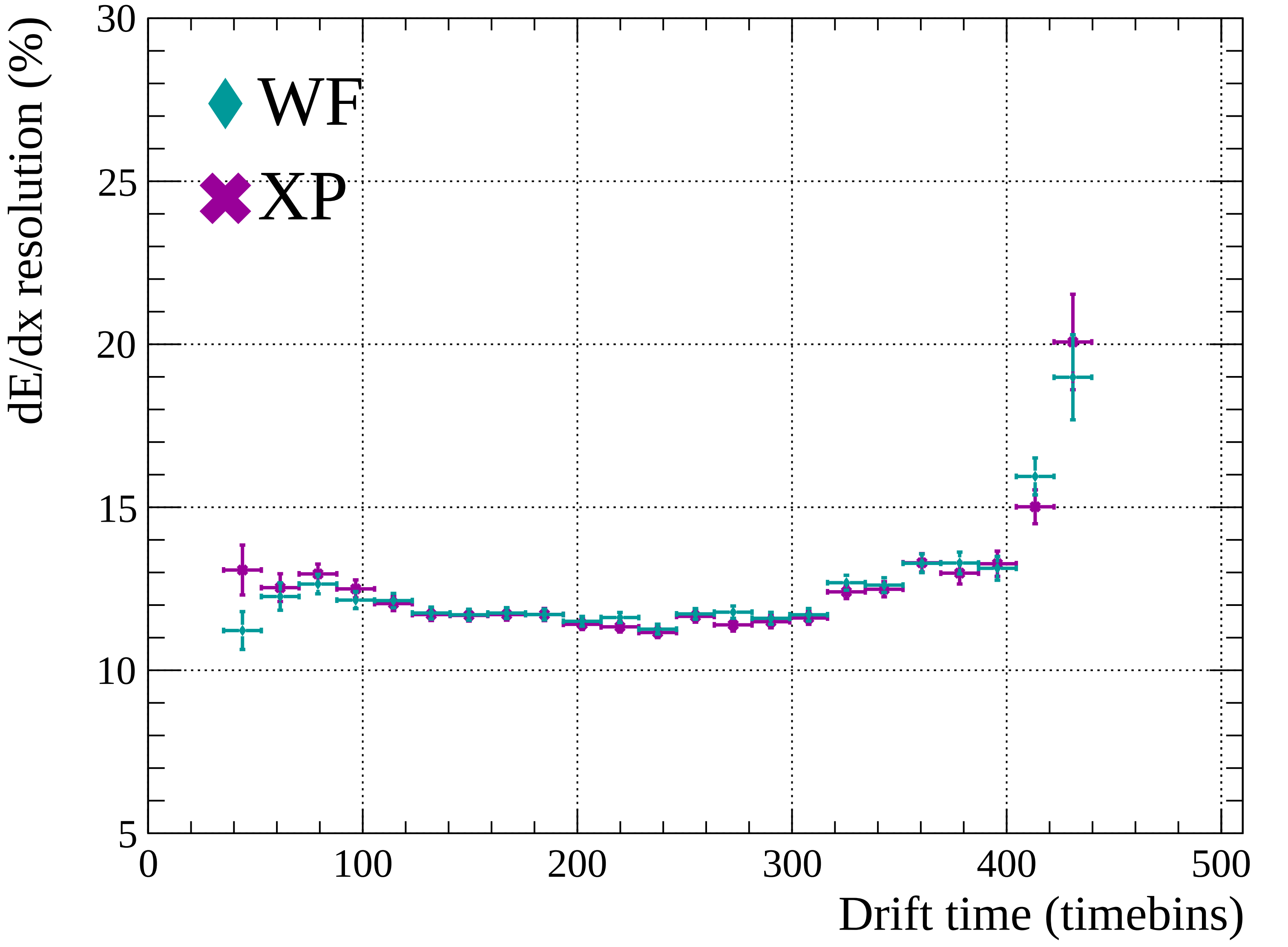

WF
XP
dE/dx (ADC counts/cm)
Track X position (mm)
600
550
500
450
400
350
300
−1000
−500
0
500
1000
1500

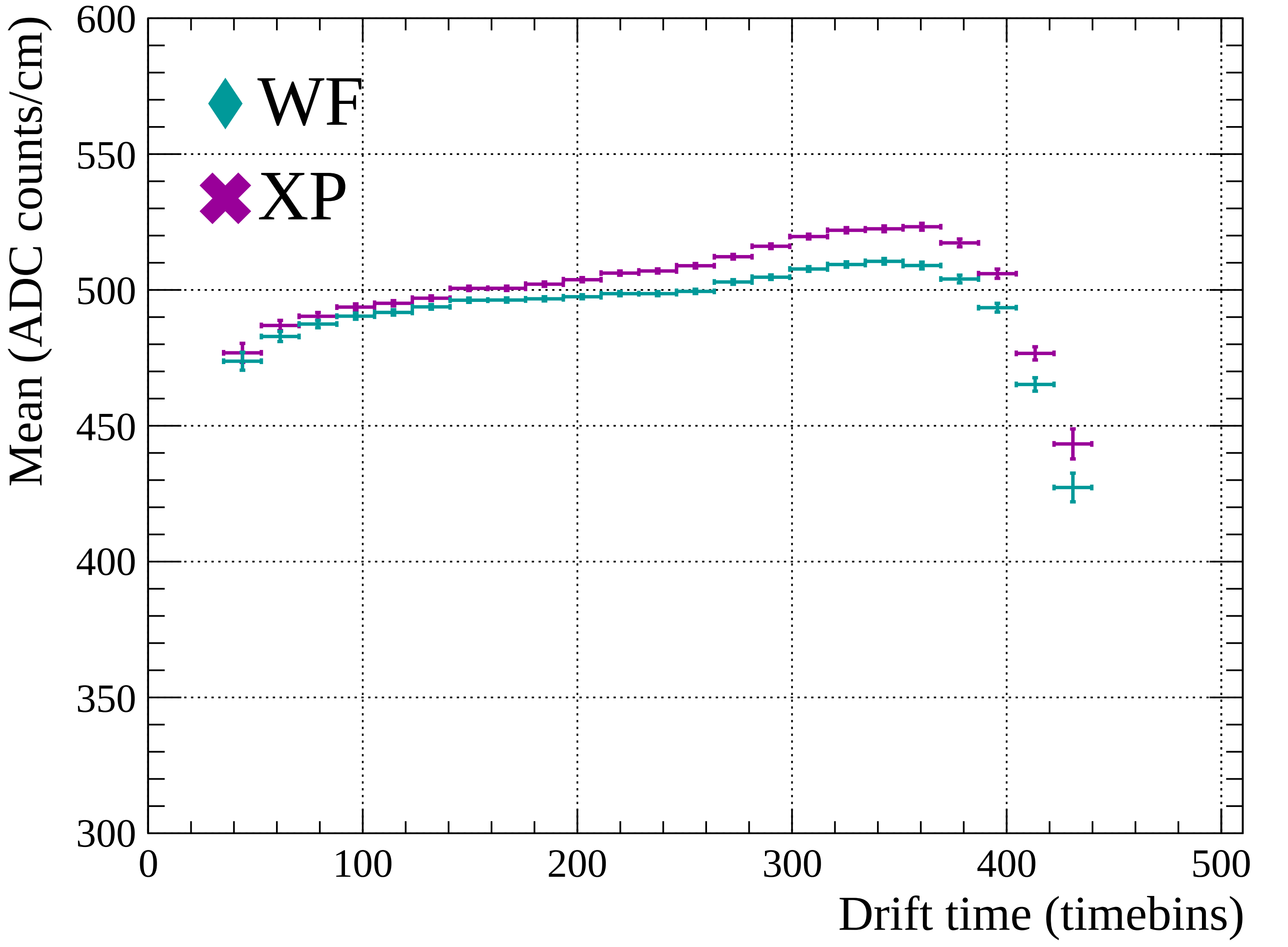


y = 0.01x + 494.31
dE/dx with WF (ADC counts/cm)
Track X position (mm)
0
100
200
300
400
500
600
700
800
900
1000
−1000
−500
0
500
1000
0
10
20
30
40
50
60
70

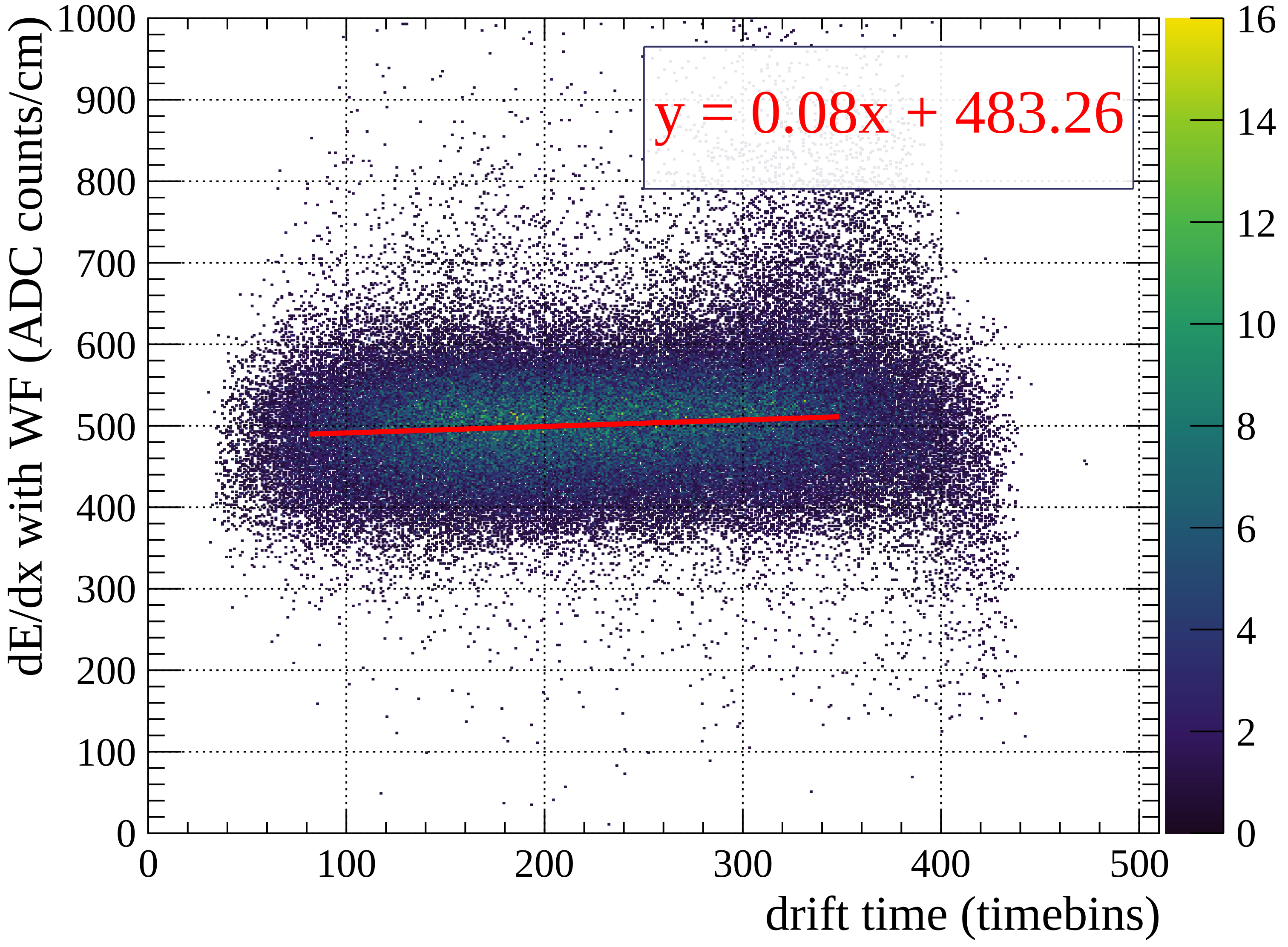


y = -0.01x + 518.78
dE/dx with XP (ADC counts/cm)
Track X position (mm)
0
100
200
300
400
500
600
700
800
900
1000
−1000
−500
0
500
1000
0
10
20
30
40
50
60
70

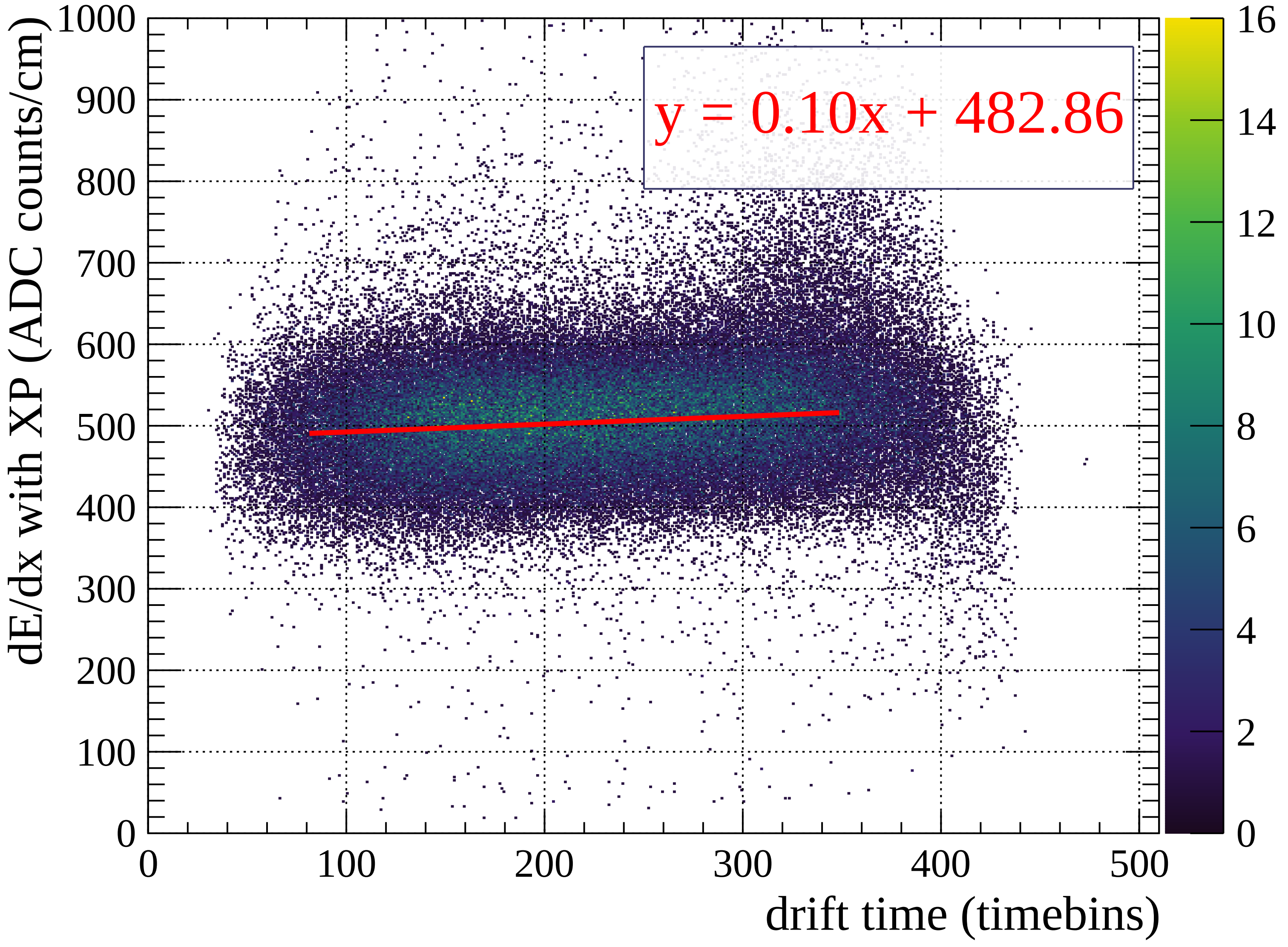

dE/dx resolution (%)
30
25
20
15
10
5
WF
XP
0
100
200
300
400
500
Drift time (timebins)



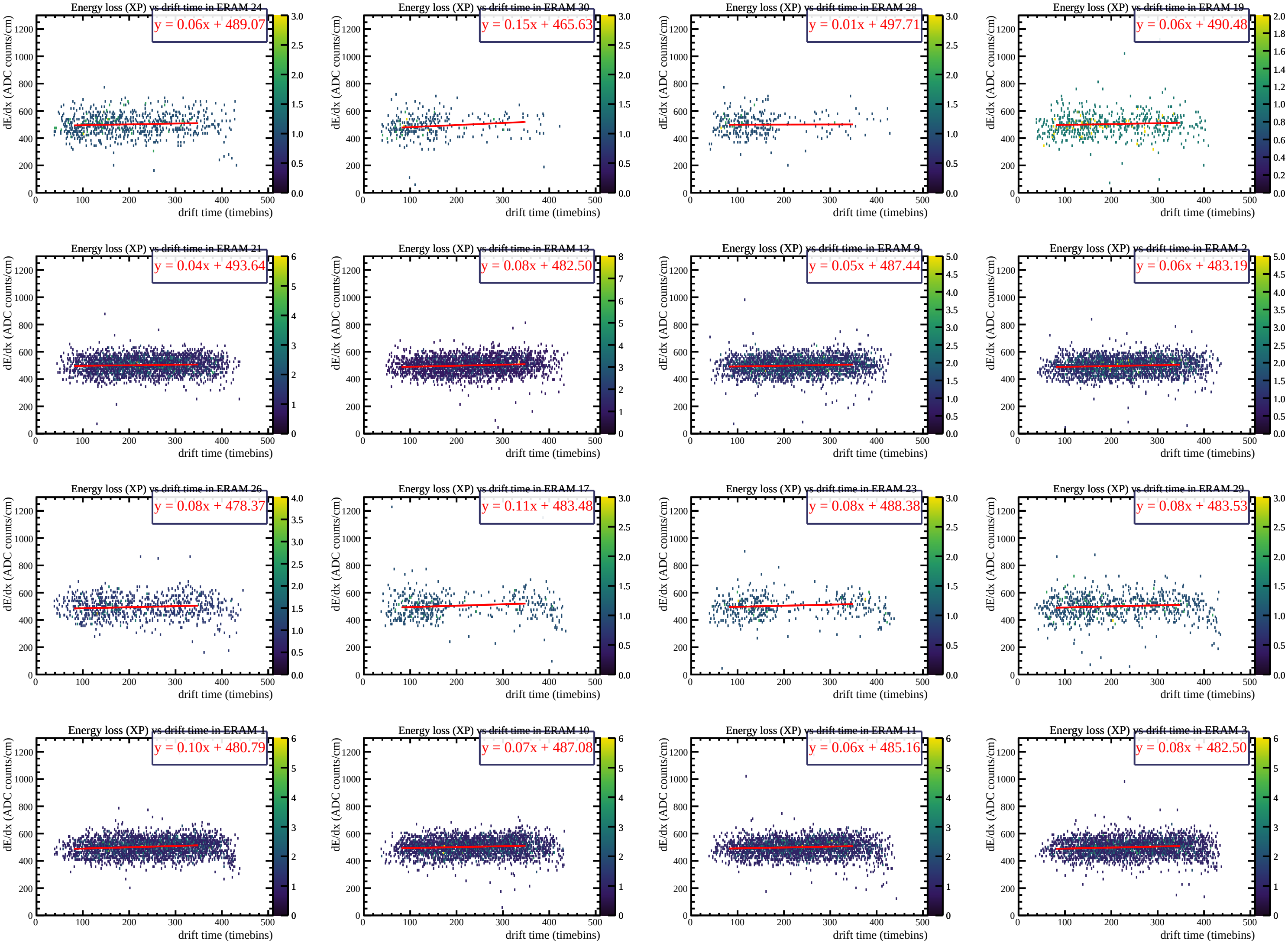
Mean (ADC counts/cm)
600
550
500
450
400
350
300
WF
XP
0
100
200
300
400
500
Drift time (timebins)



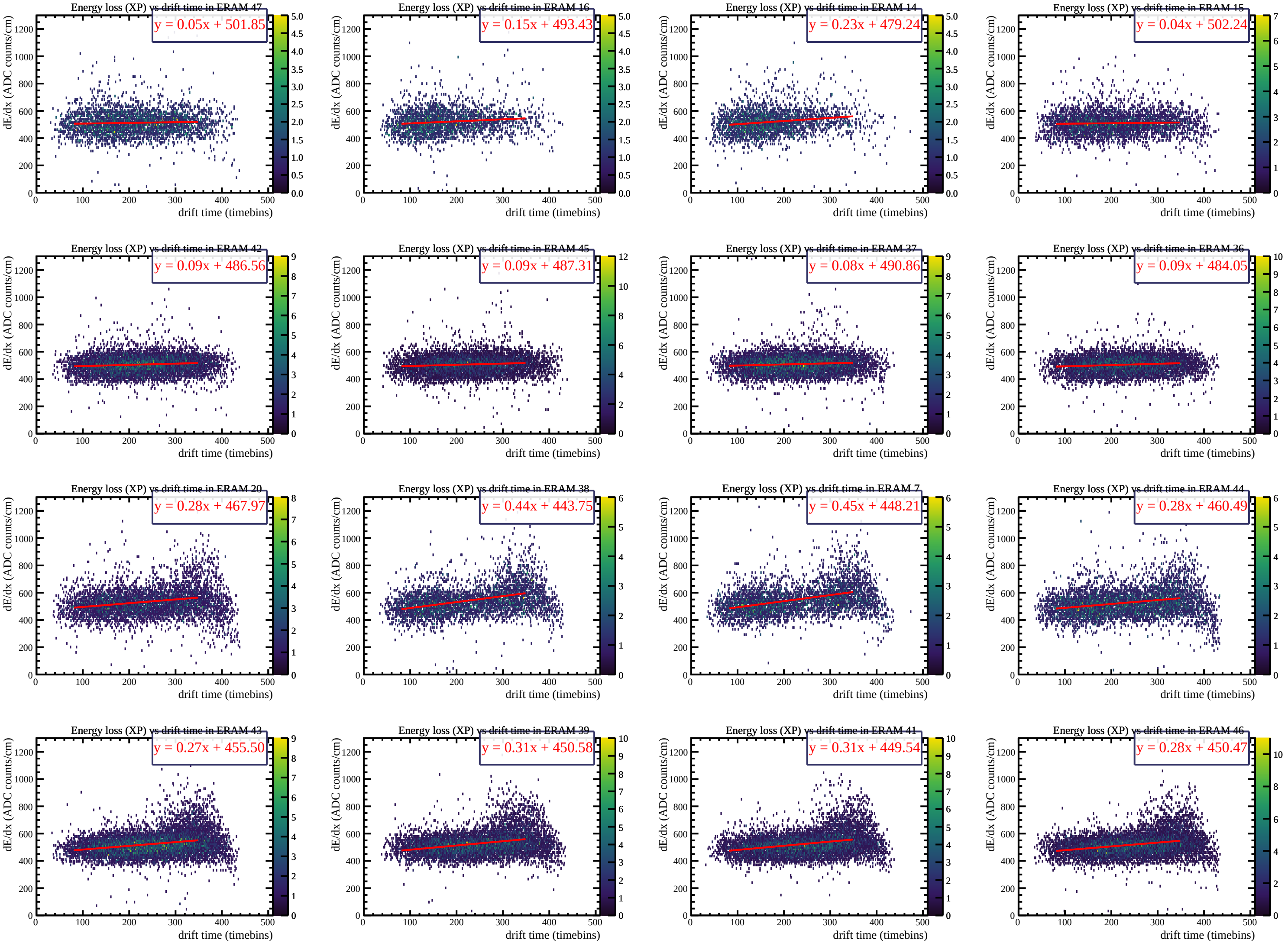
y = 0.08x + 483.26
dE/dx with WF (ADC counts/cm)
drift time (timebins)
0
100
200
300
400
500
600
700
800
900
1000
0
100
200
300
400
500
0
2
4
6
8
10
12
14
16



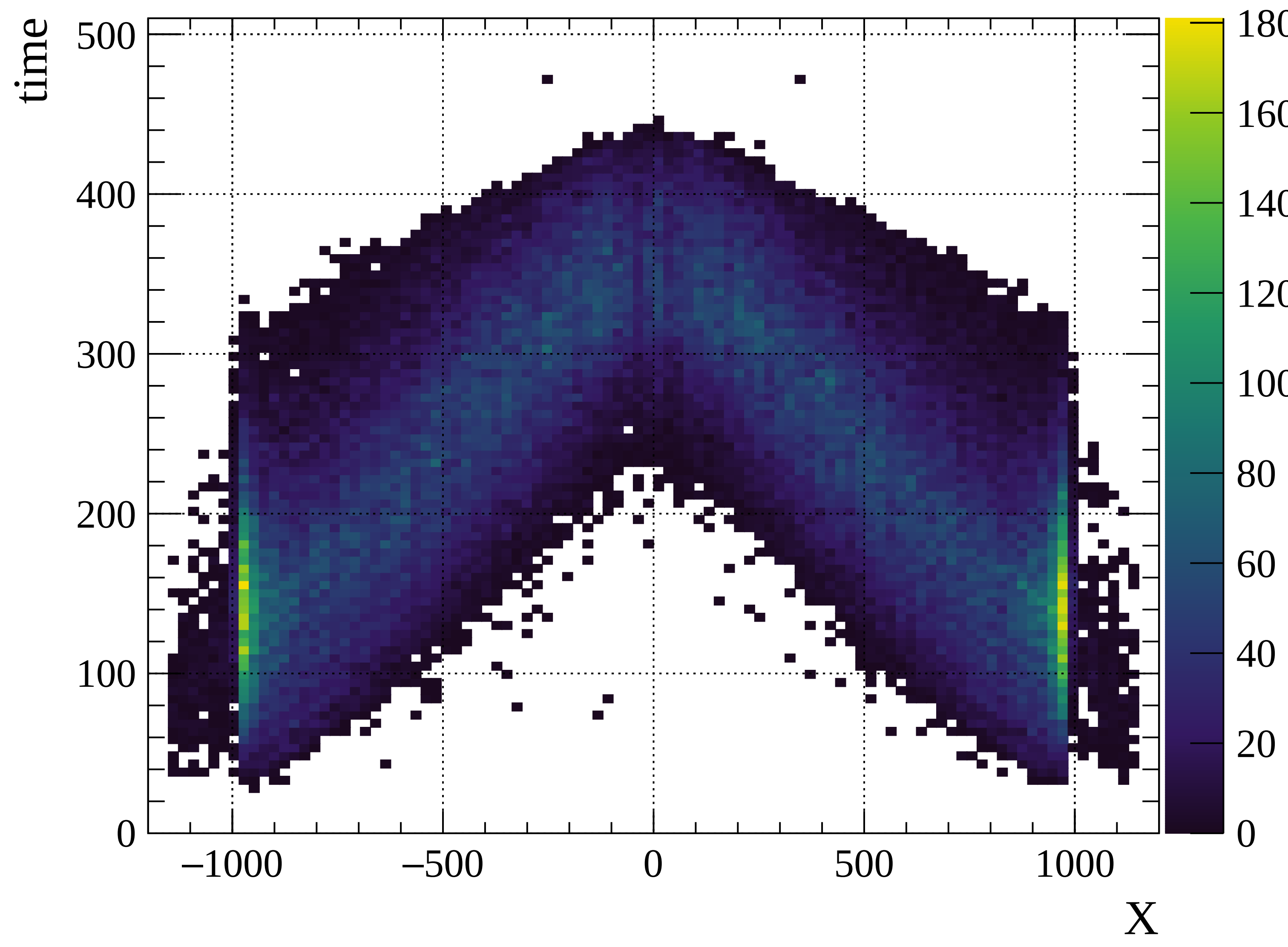
y = 0.10x + 482.86
dE/dx with XP (ADC counts/cm)
drift time (timebins)
0
100
200
300
400
500
600
700
800
900
1000
0
100
200
300
400
500
0
2
4
6
8
10
12
14
16



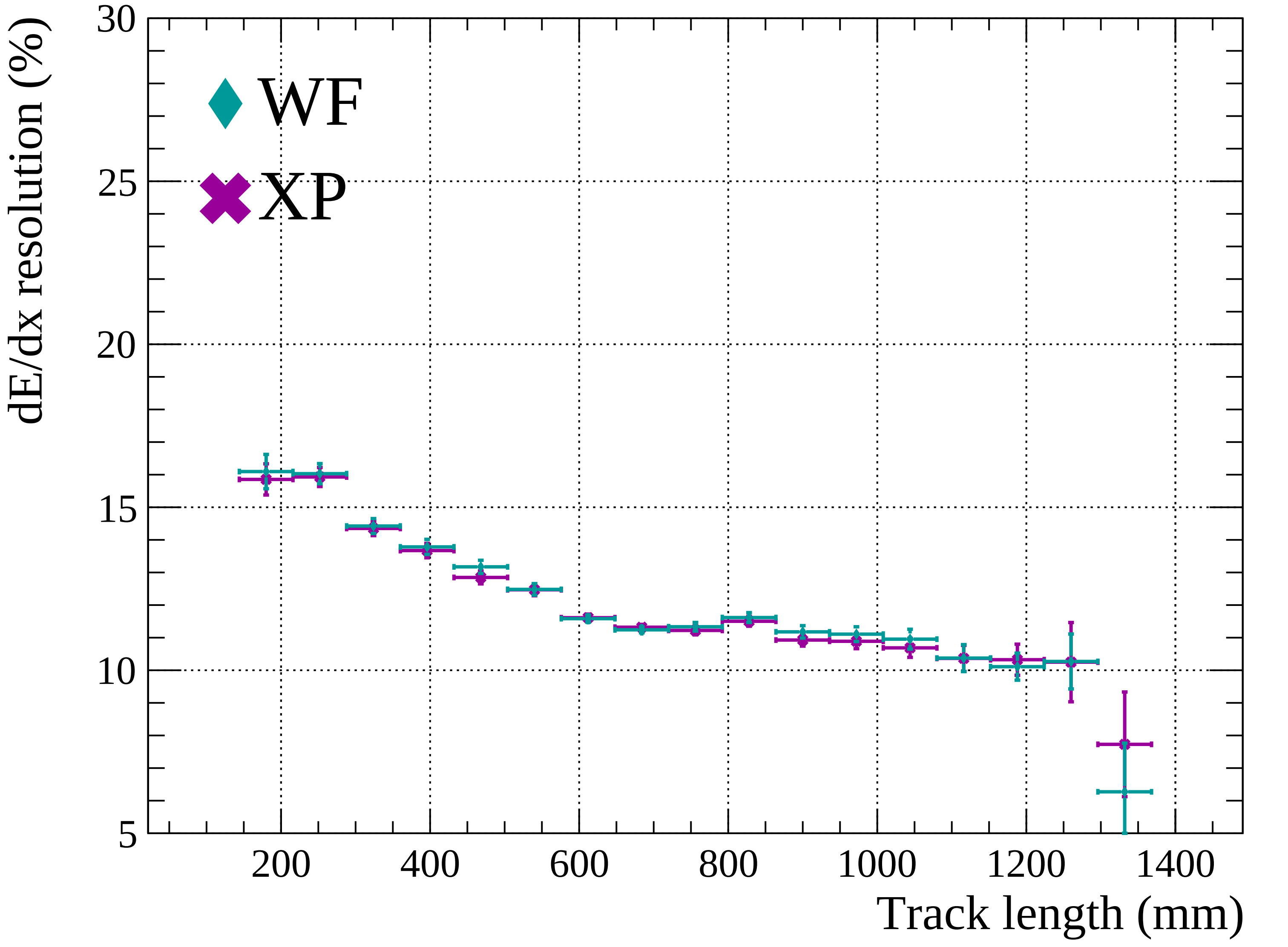
Energy loss (XP) vs drift time in ERAM 24
y = 0.06x + 489.07
Energy loss (XP) vs drift time in ERAM 30
y = 0.15x + 465.63
Energy loss (XP) vs drift time in ERAM 28
y = 0.01x + 497.71
Energy loss (XP) vs drift time in ERAM 19
y = 0.06x + 490.48
Energy loss (XP) vs drift time in ERAM 21
y = 0.04x + 493.64
Energy loss (XP) vs drift time in ERAM 13
y = 0.08x + 482.50
Energy loss (XP) vs drift time in ERAM 9
y = 0.05x + 487.44
Energy loss (XP) vs drift time in ERAM 2
y = 0.06x + 483.19
Energy loss (XP) vs drift time in ERAM 26
y = 0.08x + 478.37
Energy loss (XP) vs drift time in ERAM 17
y = 0.11x + 483.48
Energy loss (XP) vs drift time in ERAM 23
y = 0.08x + 488.38
Energy loss (XP) vs drift time in ERAM 29
y = 0.08x + 483.53
Energy loss (XP) vs drift time in ERAM 1
y = 0.10x + 480.79
Energy loss (XP) vs drift time in ERAM 10
y = 0.07x + 487.08
Energy loss (XP) vs drift time in ERAM 11
y = 0.06x + 485.16
Energy loss (XP) vs drift time in ERAM 3
y = 0.08x + 482.50
dE/dx (ADC counts/cm)
drift time (timebins)



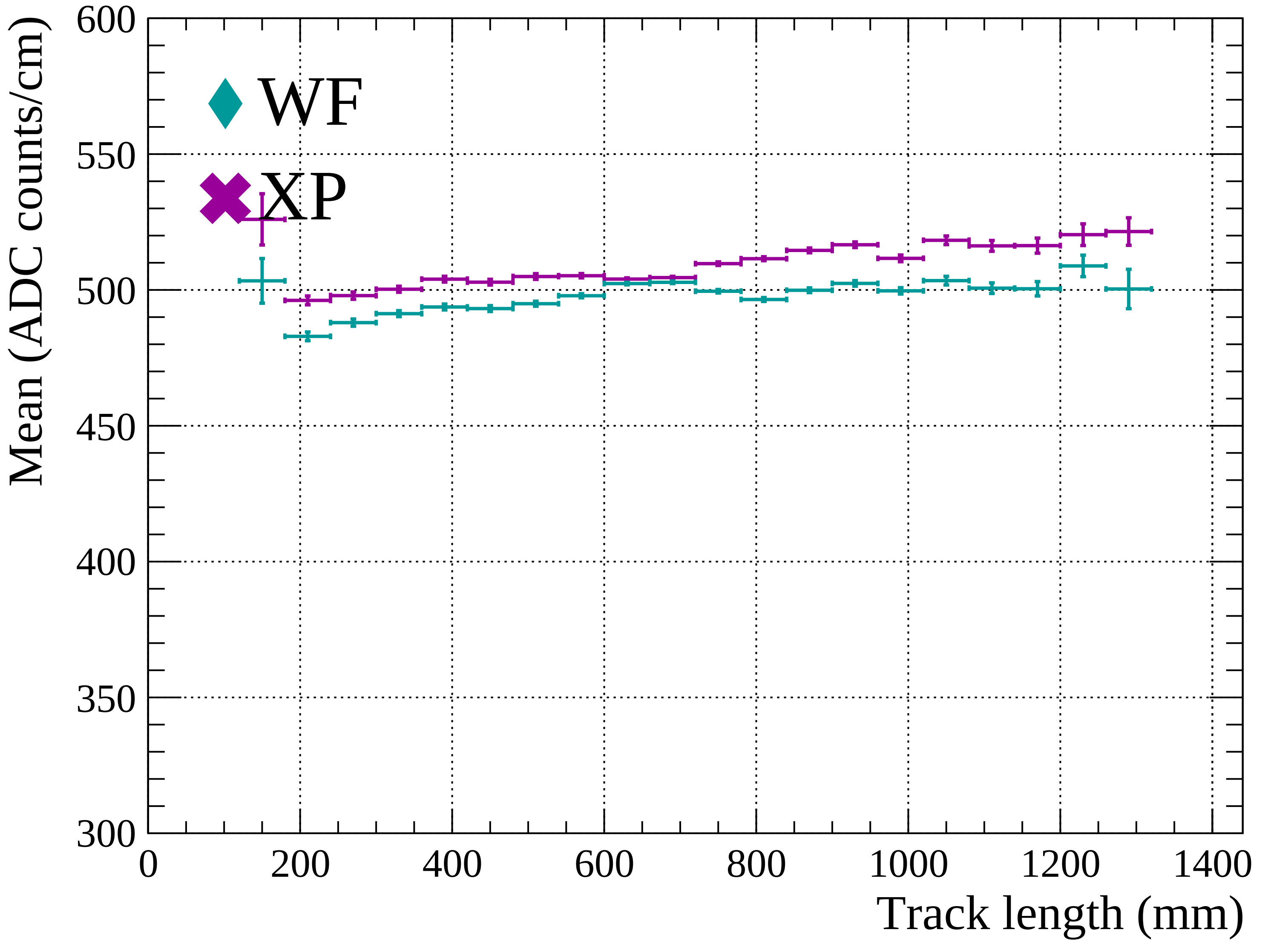
Energy loss (XP) vs drift time in ERAM 47
y = 0.05x + 501.85
Energy loss (XP) vs drift time in ERAM 16
y = 0.15x + 493.43
Energy loss (XP) vs drift time in ERAM 14
y = 0.23x + 479.24
Energy loss (XP) vs drift time in ERAM 15
y = 0.04x + 502.24
Energy loss (XP) vs drift time in ERAM 42
y = 0.09x + 486.56
Energy loss (XP) vs drift time in ERAM 45
y = 0.09x + 487.31
Energy loss (XP) vs drift time in ERAM 37
y = 0.08x + 490.86
Energy loss (XP) vs drift time in ERAM 36
y = 0.09x + 484.05
Energy loss (XP) vs drift time in ERAM 20
y = 0.28x + 467.97
Energy loss (XP) vs drift time in ERAM 38
y = 0.44x + 443.75
Energy loss (XP) vs drift time in ERAM 7
y = 0.45x + 448.21
Energy loss (XP) vs drift time in ERAM 44
y = 0.28x + 460.49
Energy loss (XP) vs drift time in ERAM 43
y = 0.27x + 455.50
Energy loss (XP) vs drift time in ERAM 39
y = 0.31x + 450.58
Energy loss (XP) vs drift time in ERAM 41
y = 0.31x + 449.54
Energy loss (XP) vs drift time in ERAM 46
y = 0.28x + 450.47
dE/dx (ADC counts/cm)
drift time (timebins)



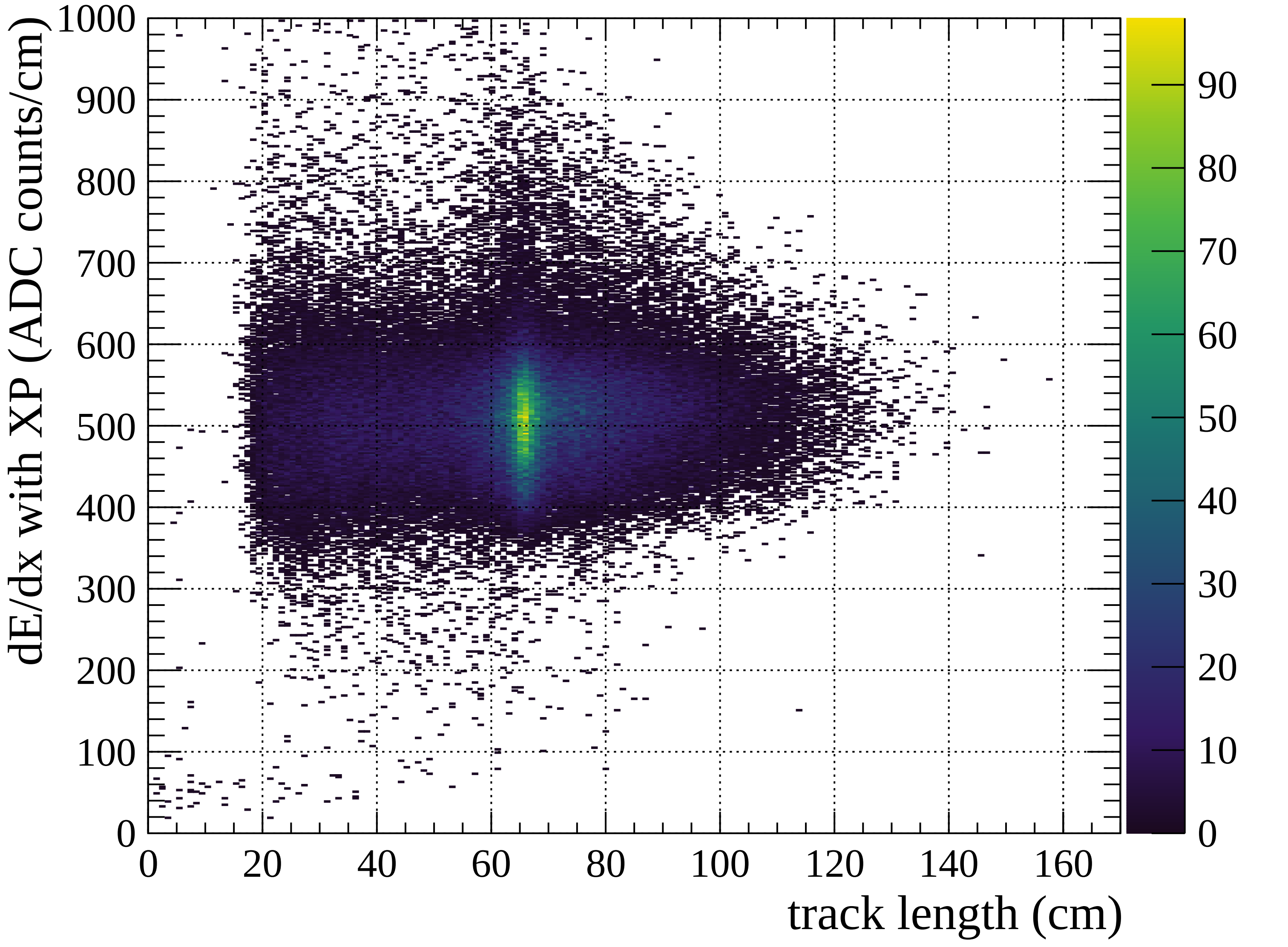
time
500
400
300
200
100
0
−1000
−500
0
500
1000
X
180
160
140
120
100
80
60
40
20
0



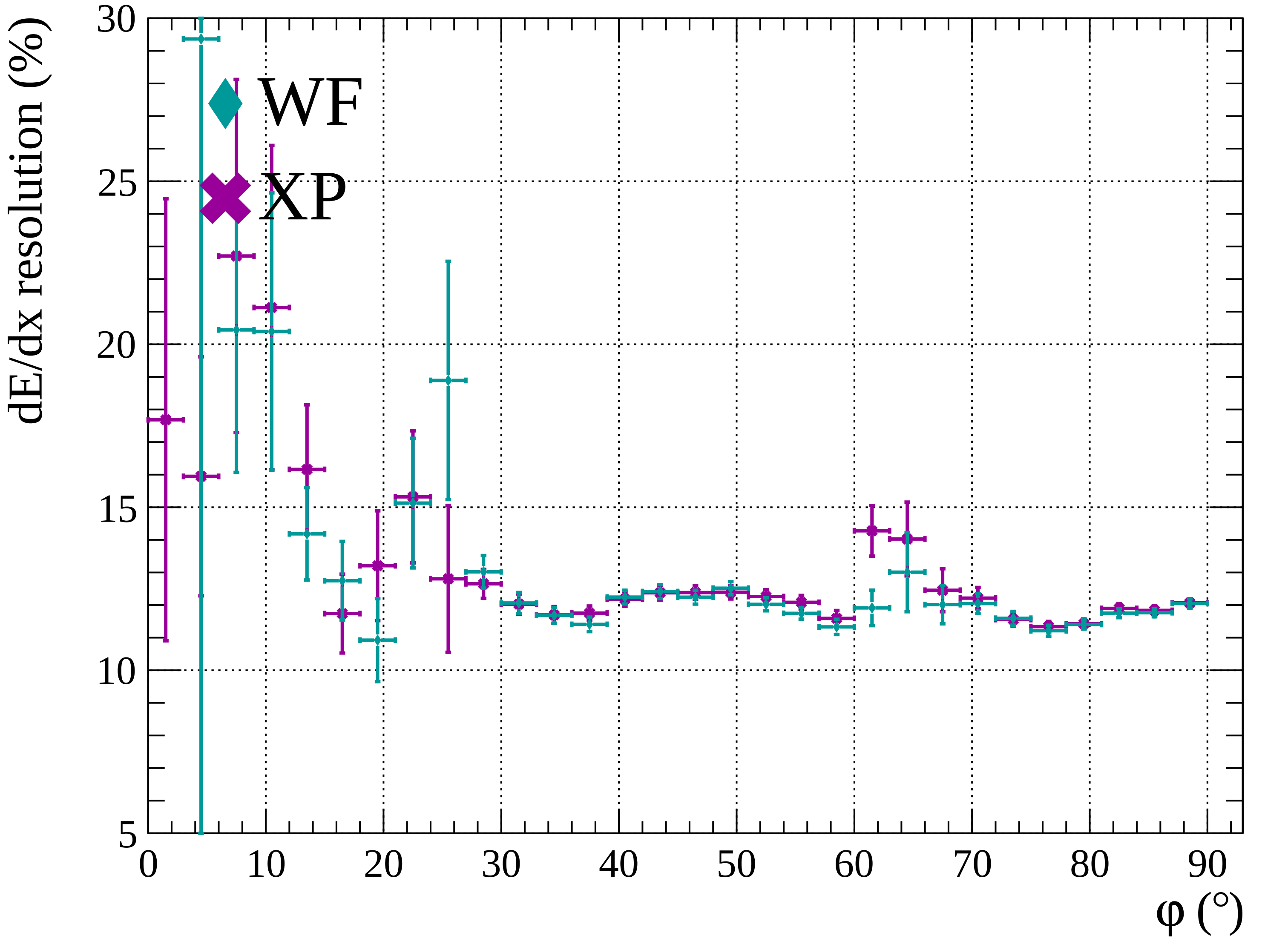
dE/dx resolution (%)
30
25
20
15
10
5
WF
XP
200
400
600
800
1000
1200
1400
Track length (mm)



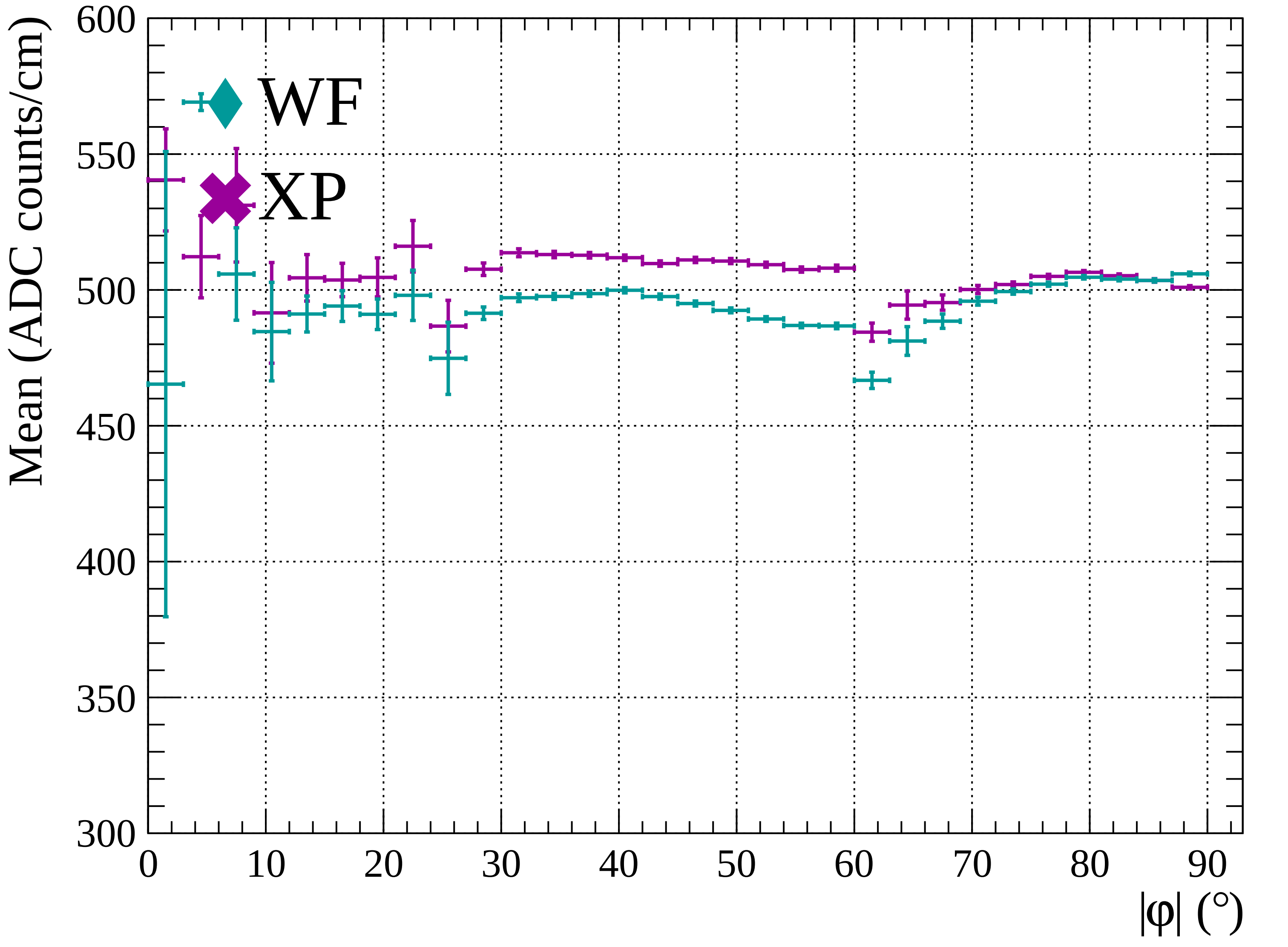
WF
XP
Mean (ADC counts/cm)
300
350
400
450
500
550
600
0
200
400
600
800
1000
1200
1400
Track length (mm)



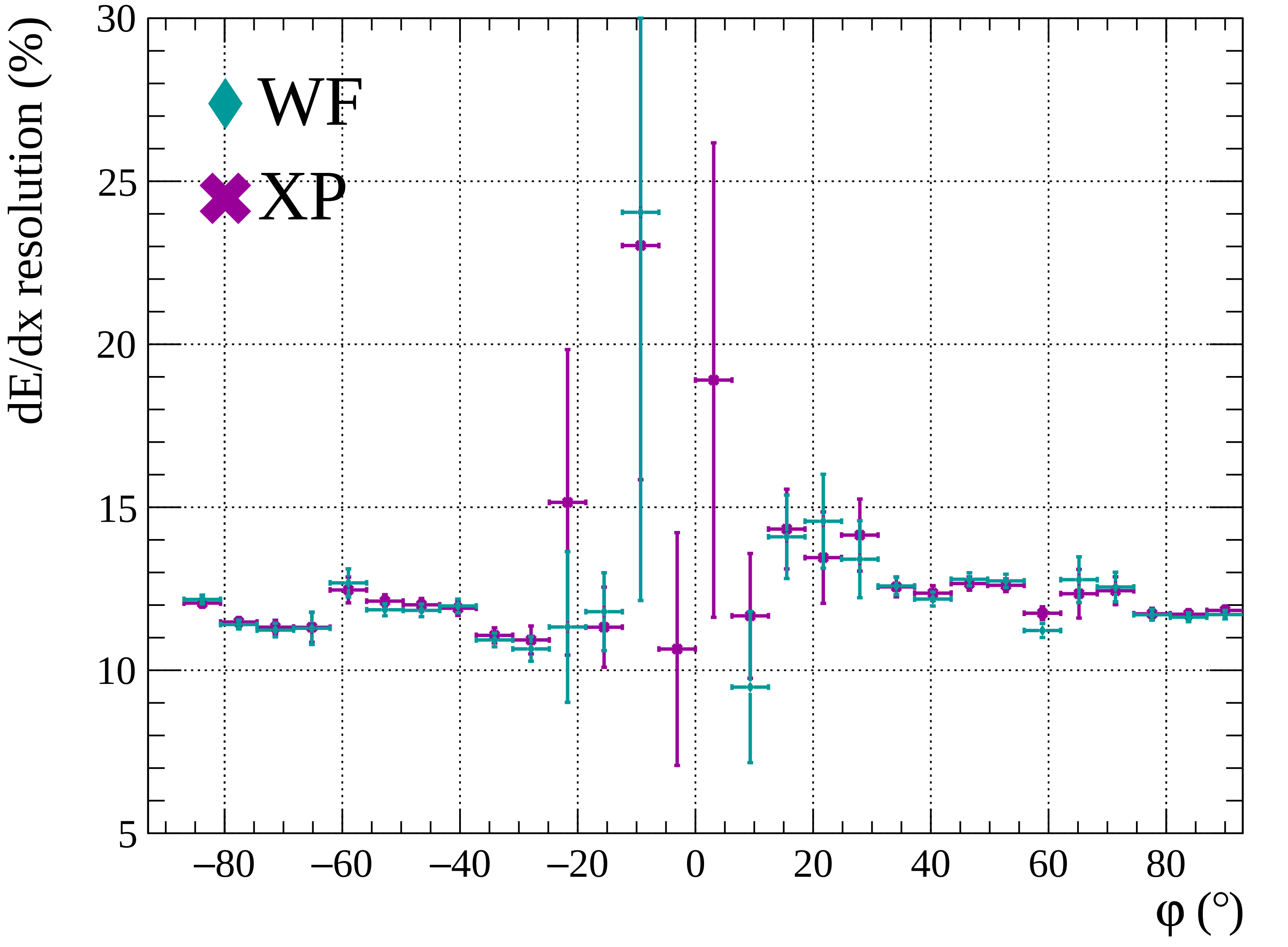
dE/dx with XP (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
0
20
40
60
80
100
120
140
160
track length (cm)
90
80
70
60
50
40
30
20
10
0



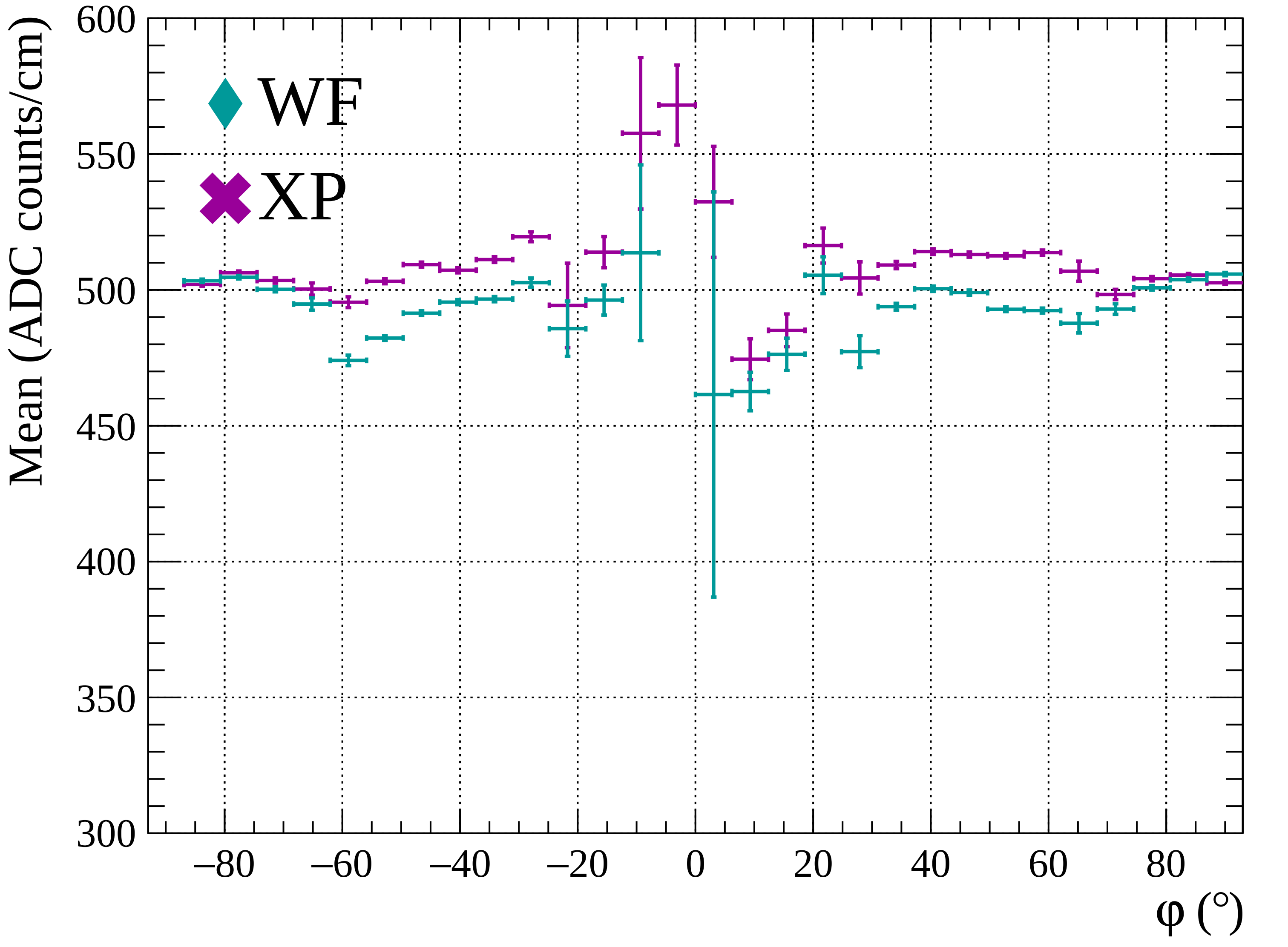
WF
XP
dE/dx resolution (%)
φ (°)
5
10
15
20
25
30
0
10
20
30
40
50
60
70
80
90



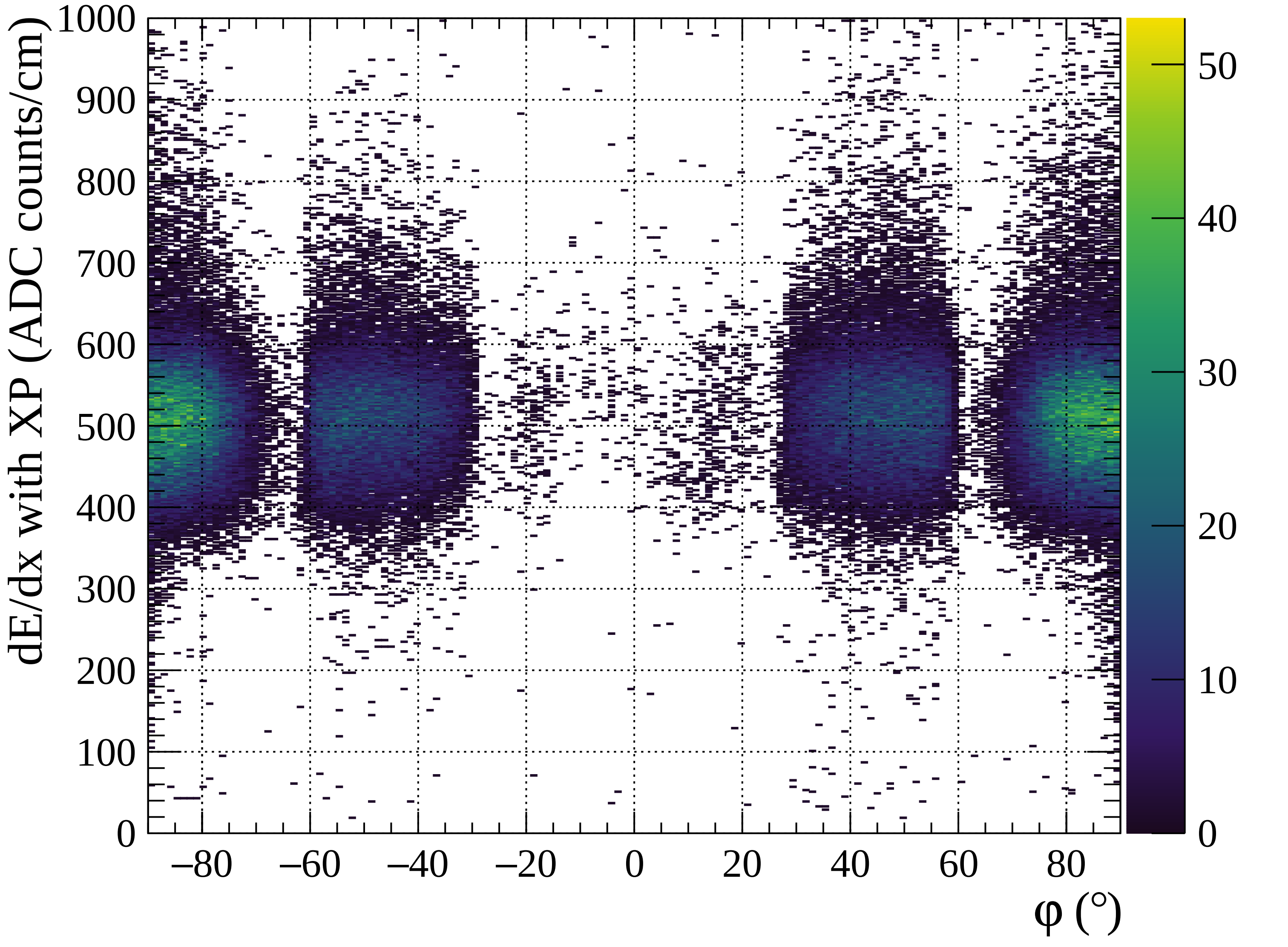
WF
XP
Mean (ADC counts/cm)
600
550
500
450
400
350
300
0
10
20
30
40
50
60
70
80
90
|φ| (°)



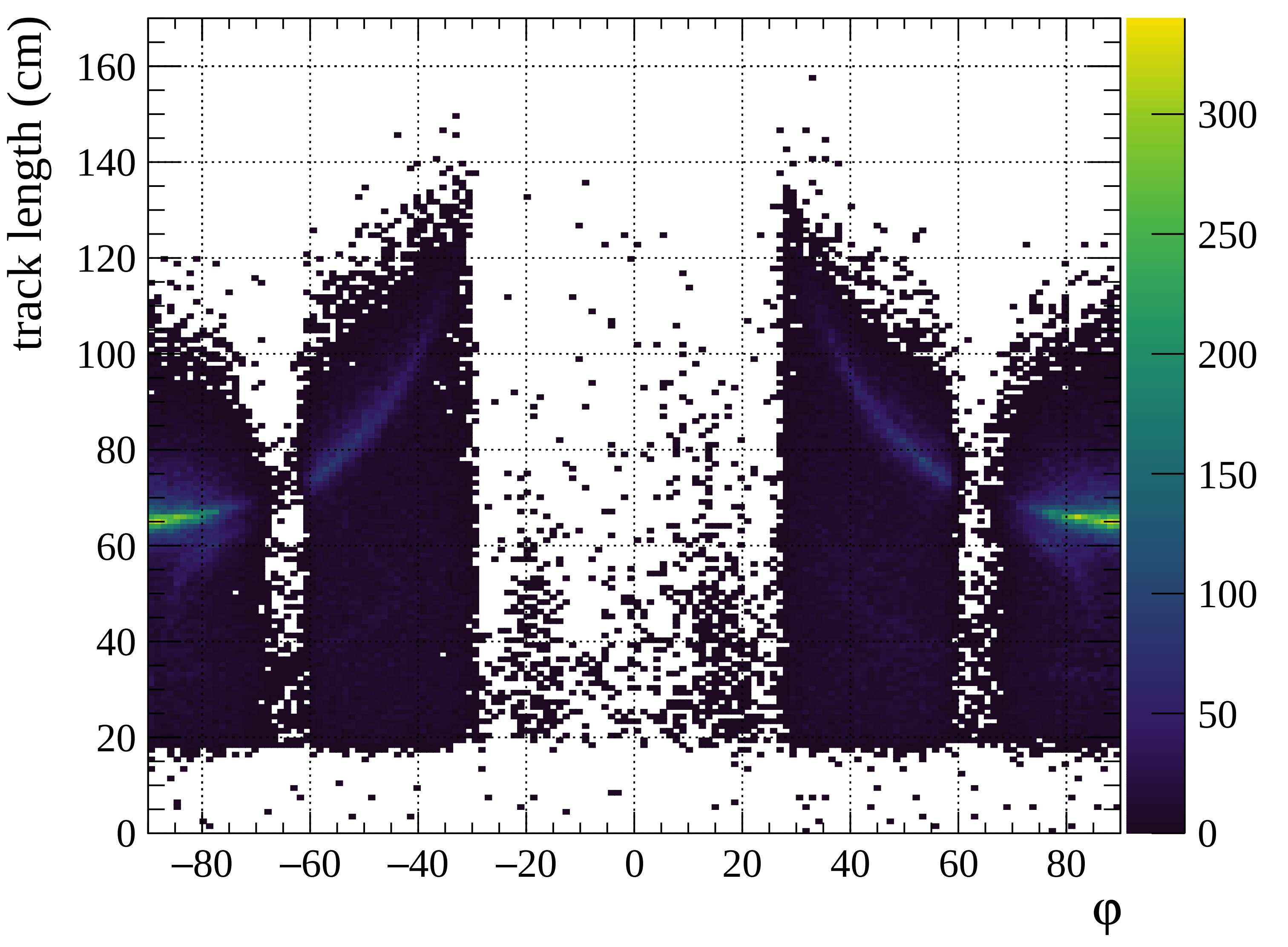
WF
XP
dE/dx resolution (%)
φ (°)
5
10
15
20
25
30
−80
−60
−40
−20
0
20
40
60
80

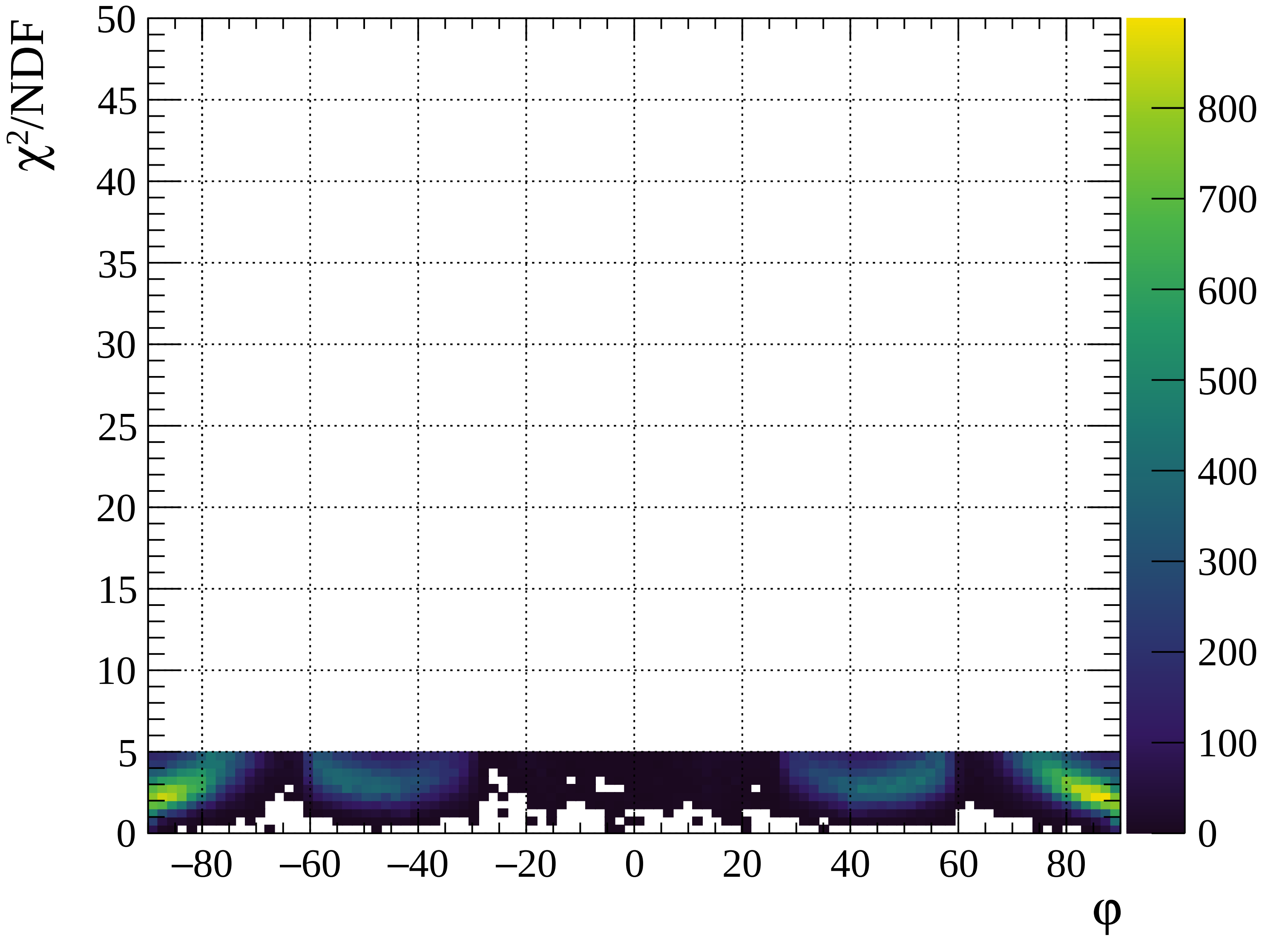


Mean (ADC counts/cm)
600
550
500
450
400
350
300
−80
−60
−40
−20
0
20
40
60
80
φ (°)
WF
XP

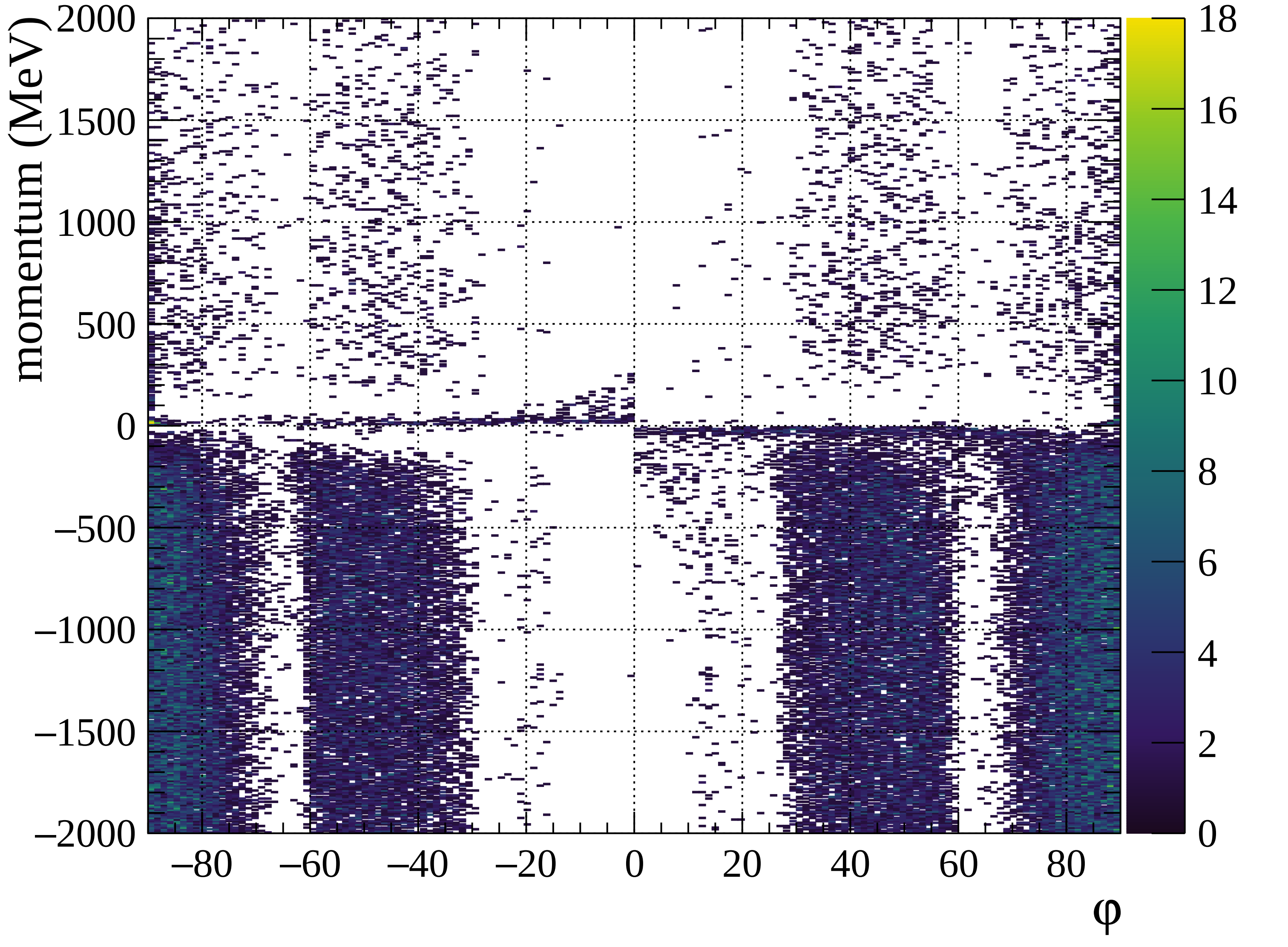


dE/dx with XP (ADC counts/cm)
φ (°)
1000
900
800
700
600
500
400
300
200
100
0
−80
−60
−40
−20
0
20
40
60
80
50
40
30
20
10
0

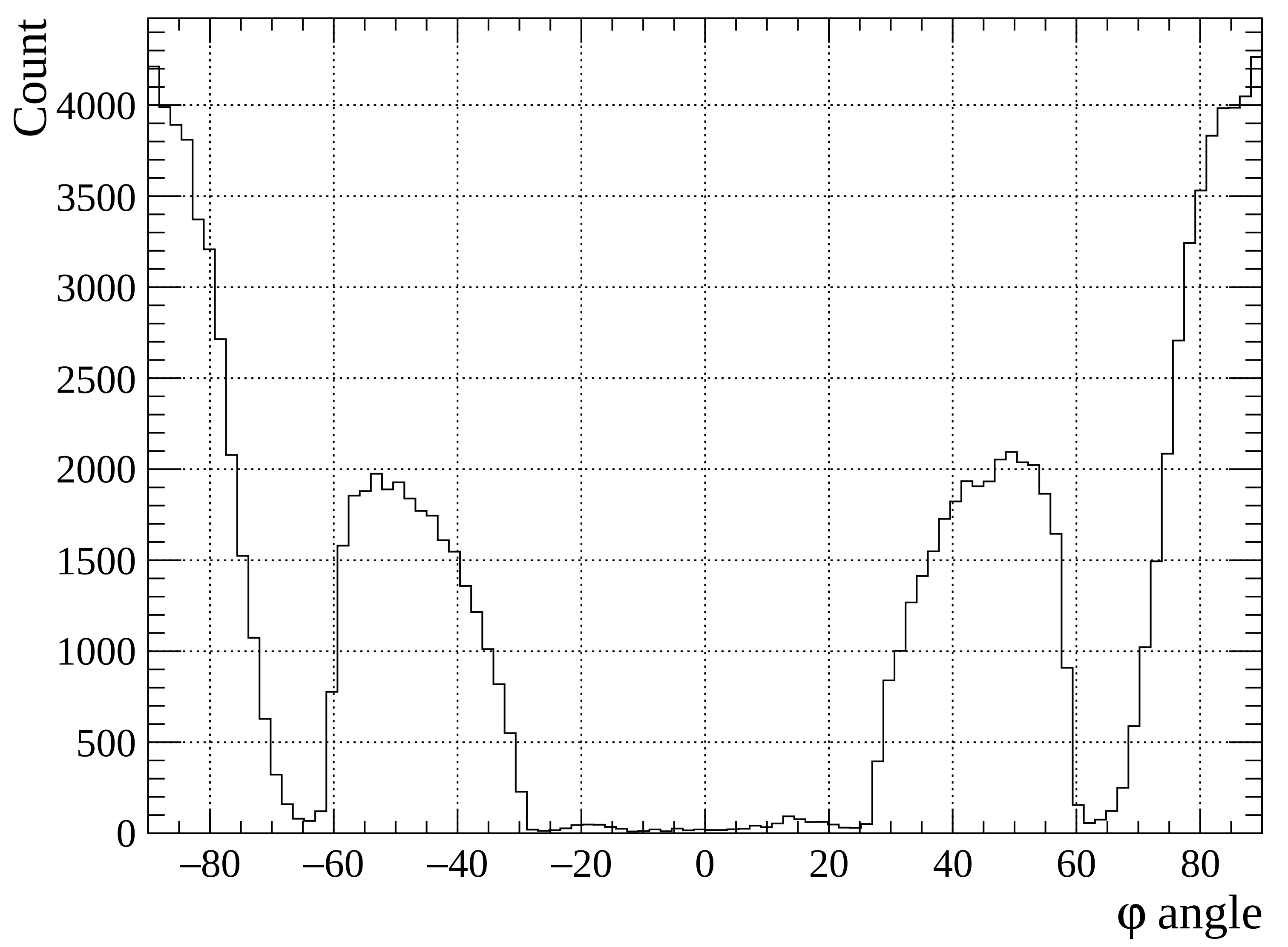

track length (cm)
160
140
120
100
80
60
40
20
0
−80
−60
−40
−20
0
20
40
60
80
φ
300
250
200
150
100
50
0



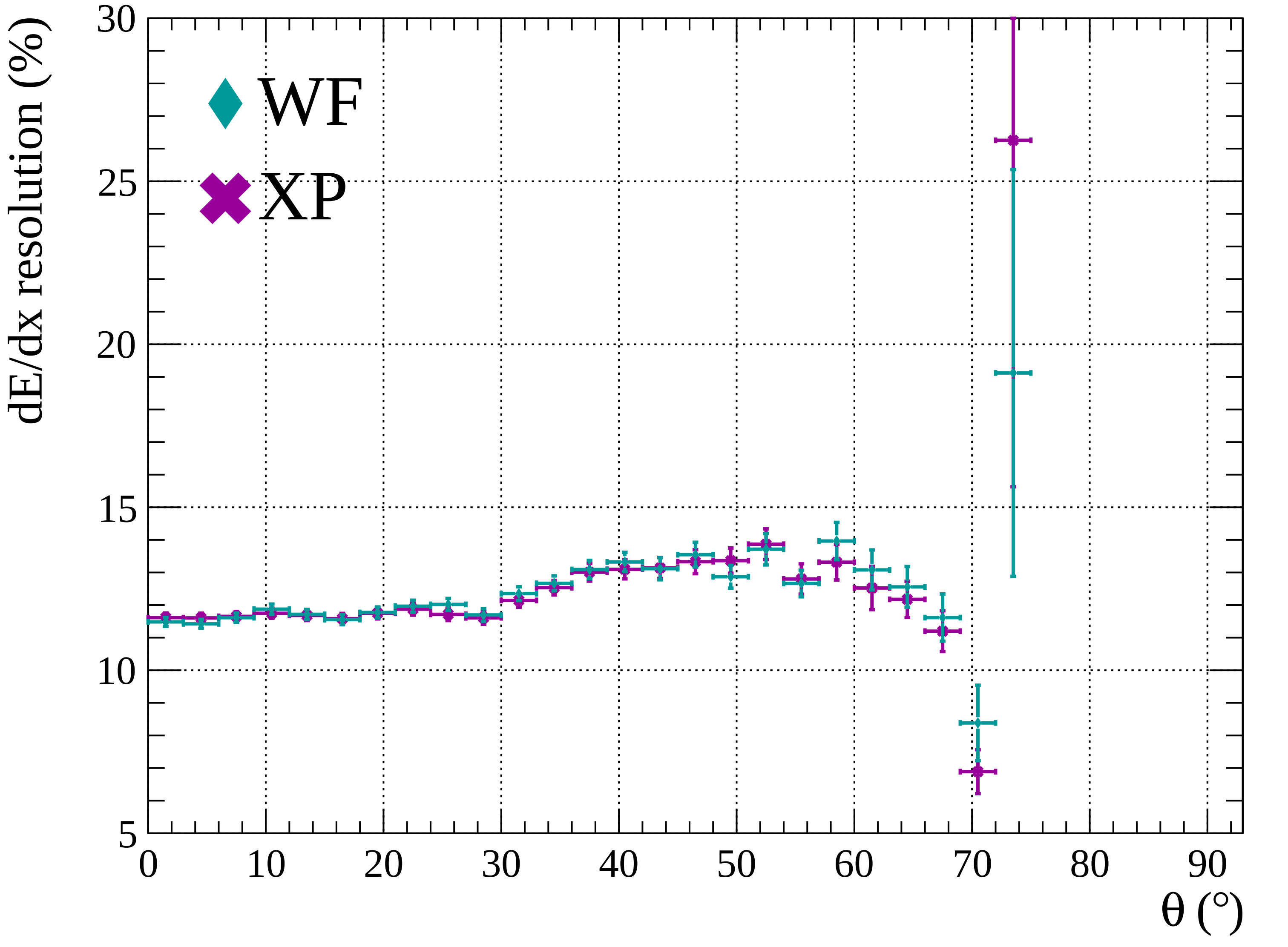
χ^2/NDF
50
45
40
35
30
25
20
15
10
5
0
−80
−60
−40
−20
0
20
40
60
80
φ
800
700
600
500
400
300
200
100
0



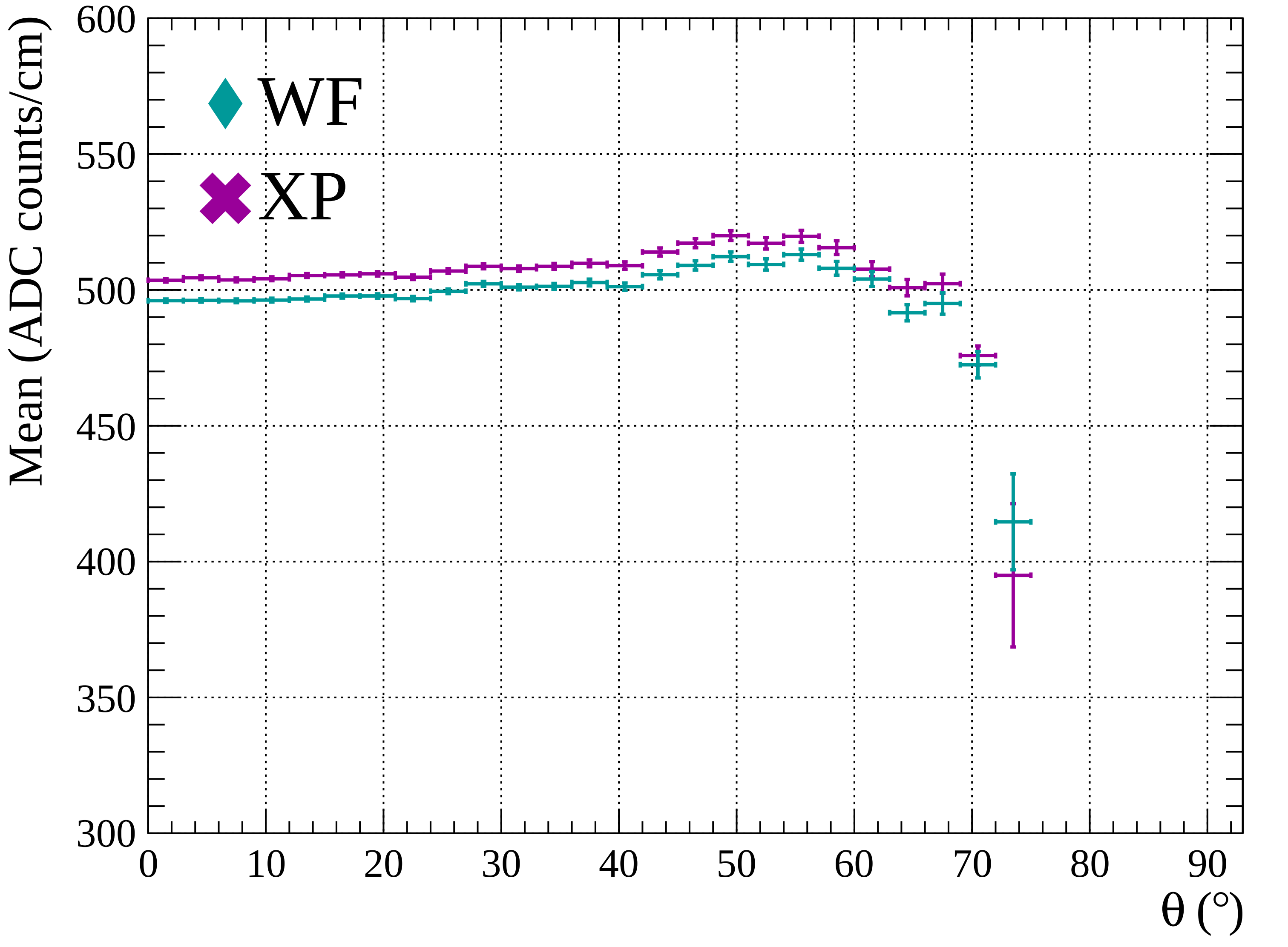
momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−80
−60
−40
−20
0
20
40
60
80
φ
18
16
14
12
10
8
6
4
2
0



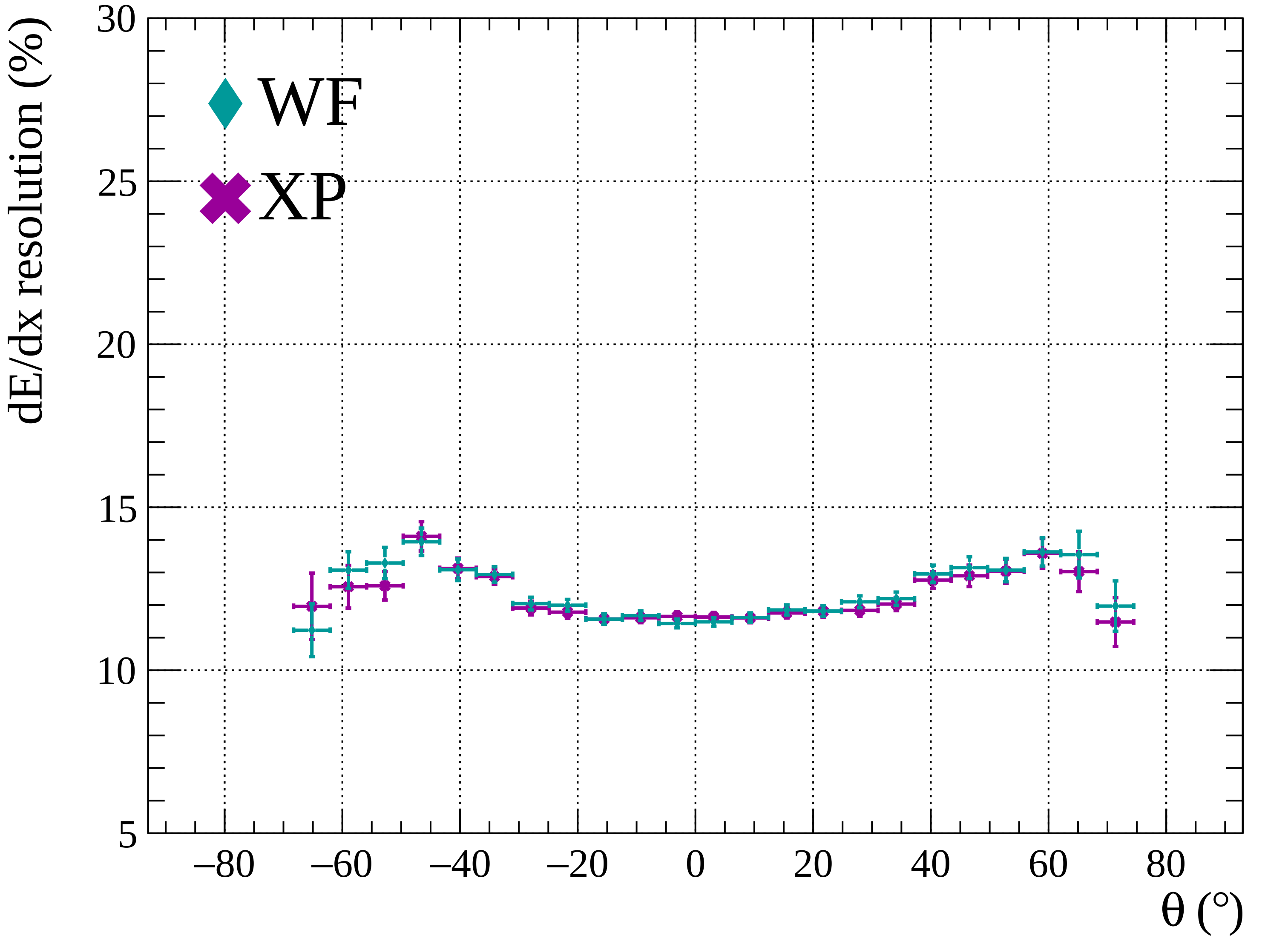
Count
4000
3500
3000
2500
2000
1500
1000
500
0
−80
−60
−40
−20
0
20
40
60
80
φ angle



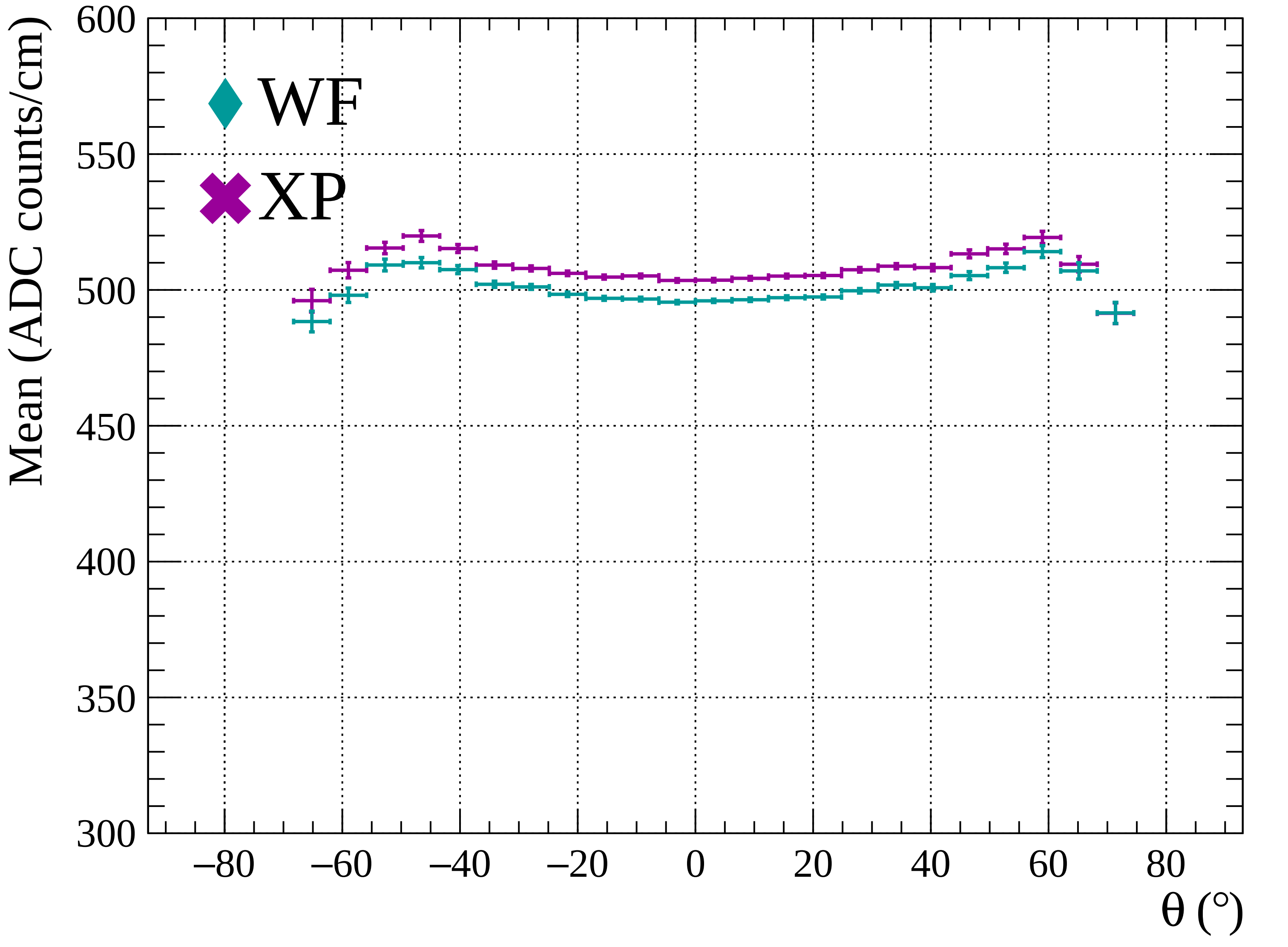
dE/dx resolution (%)
WF
XP
θ (°)
5
10
15
20
25
30
0
10
20
30
40
50
60
70
80
90



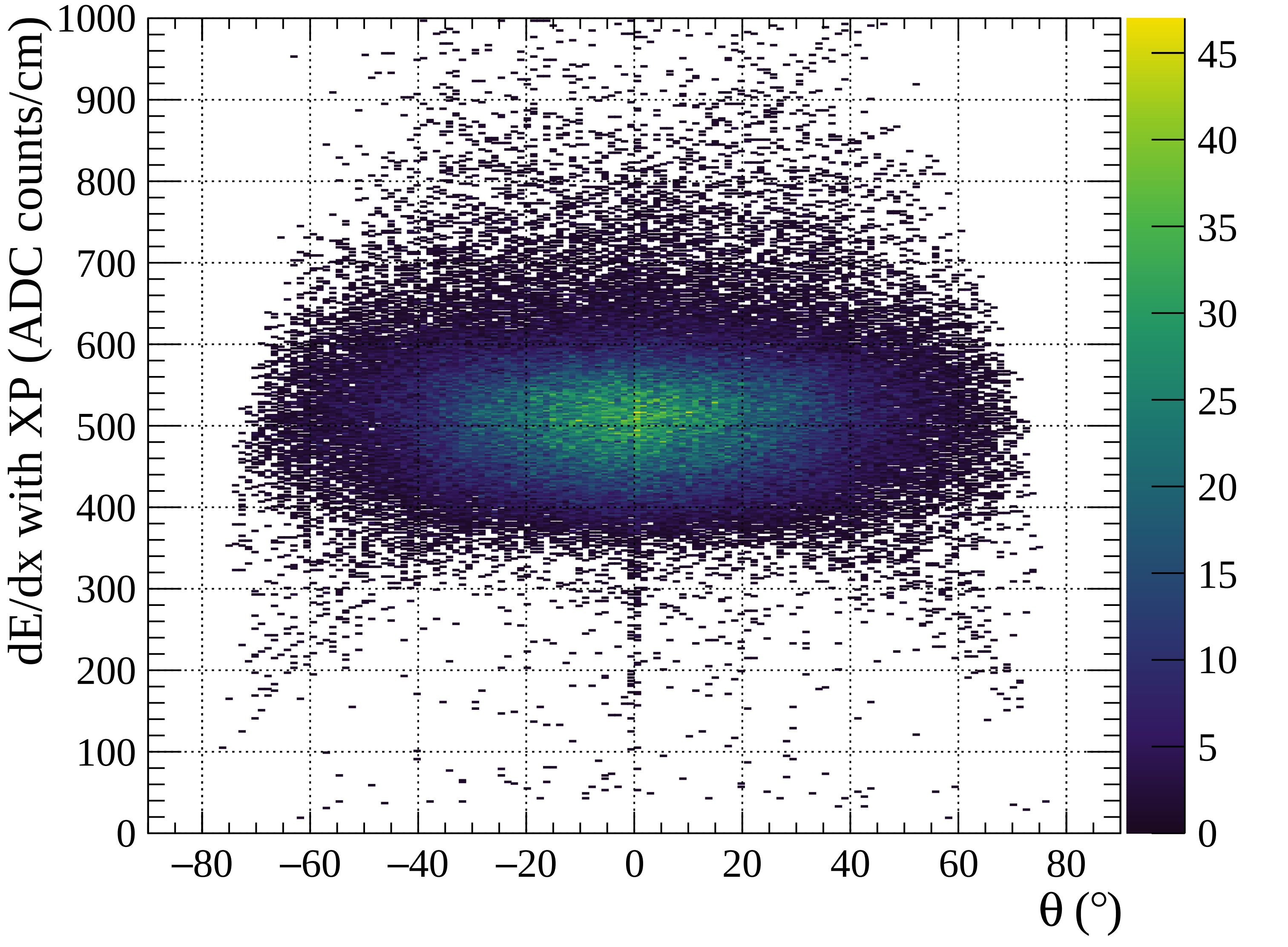
WF
XP
Mean (ADC counts/cm)
600
550
500
450
400
350
300
0
10
20
30
40
50
60
70
80
90
θ (°)



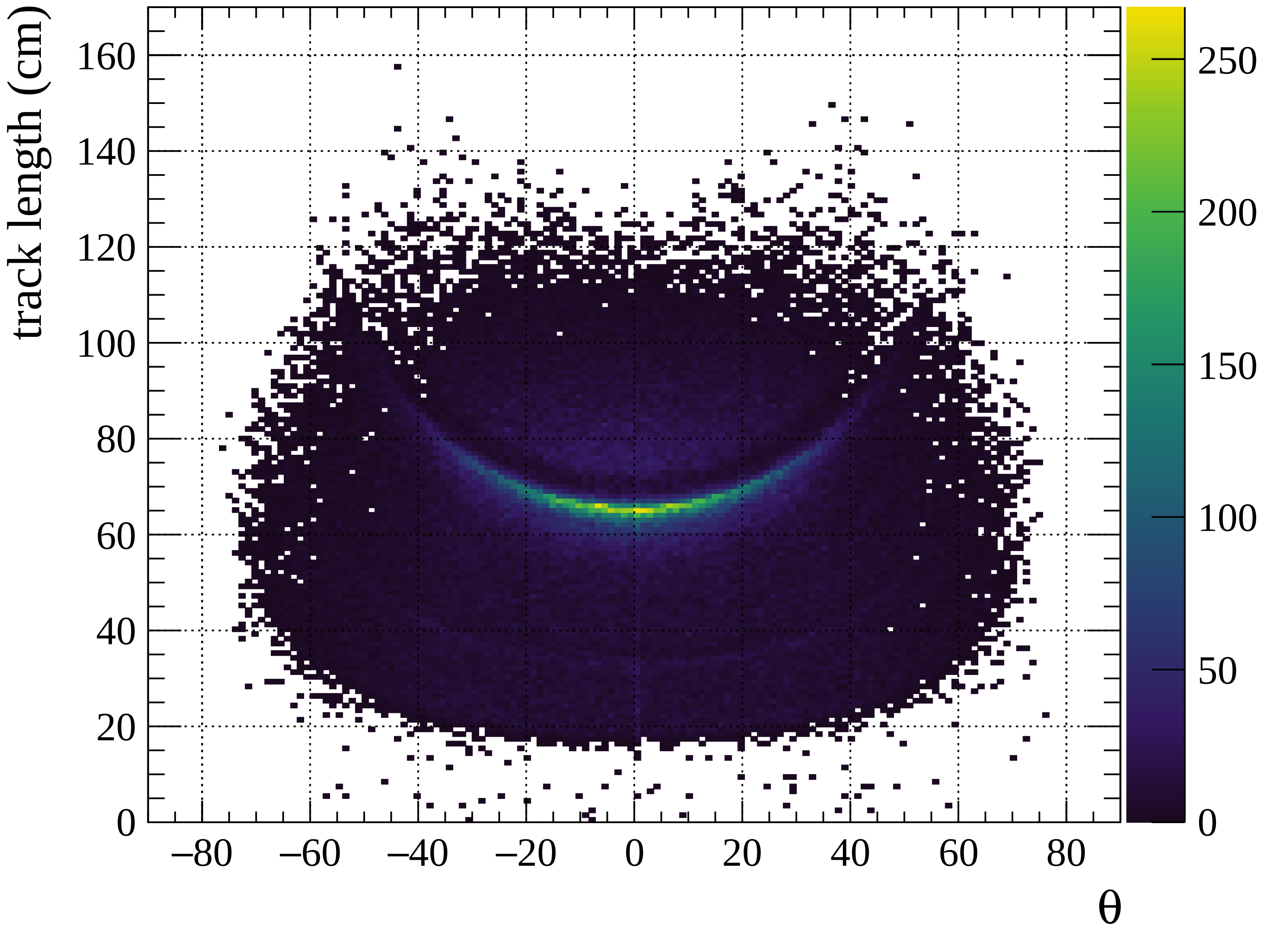
dE/dx resolution (%)
30
25
20
15
10
5
WF
XP
−80
−60
−40
−20
0
20
40
60
80
θ (°)



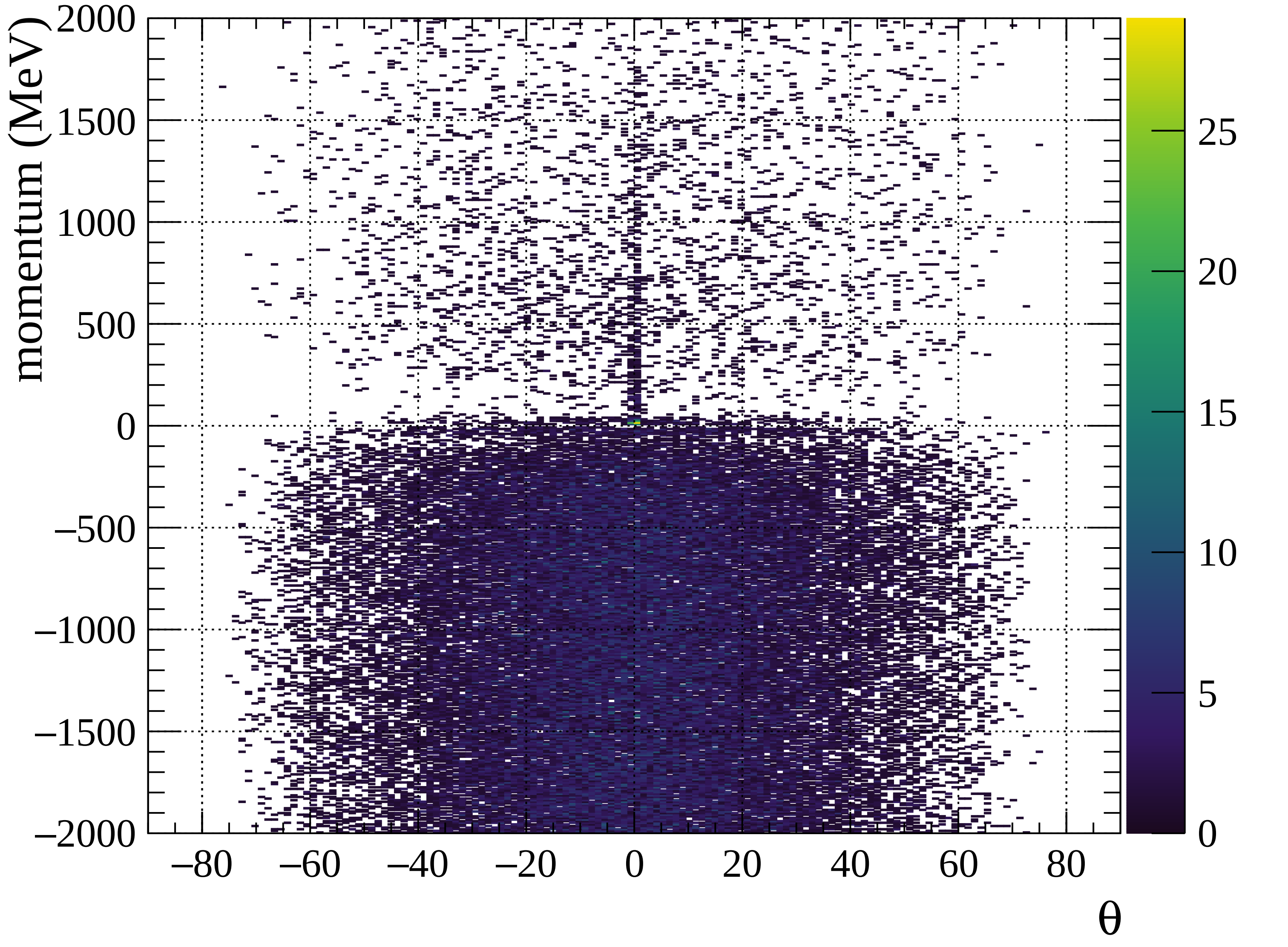
Mean (ADC counts/cm)
θ (°)
WF
XP
600
550
500
450
400
350
300
−80
−60
−40
−20
0
20
40
60
80



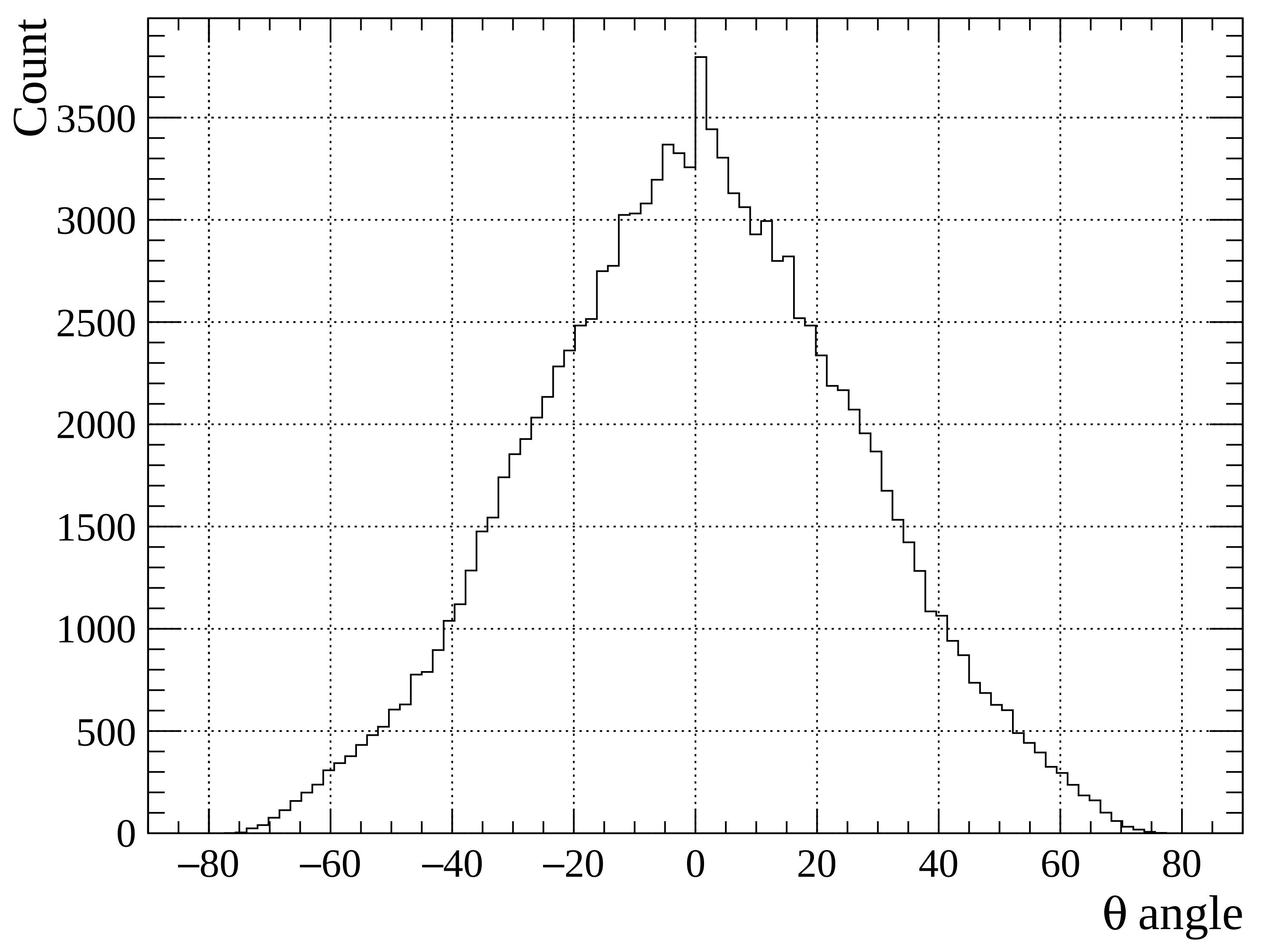
dE/dx with XP (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
−80
−60
−40
−20
0
20
40
60
80
θ (°)
45
40
35
30
25
20
15
10
5
0



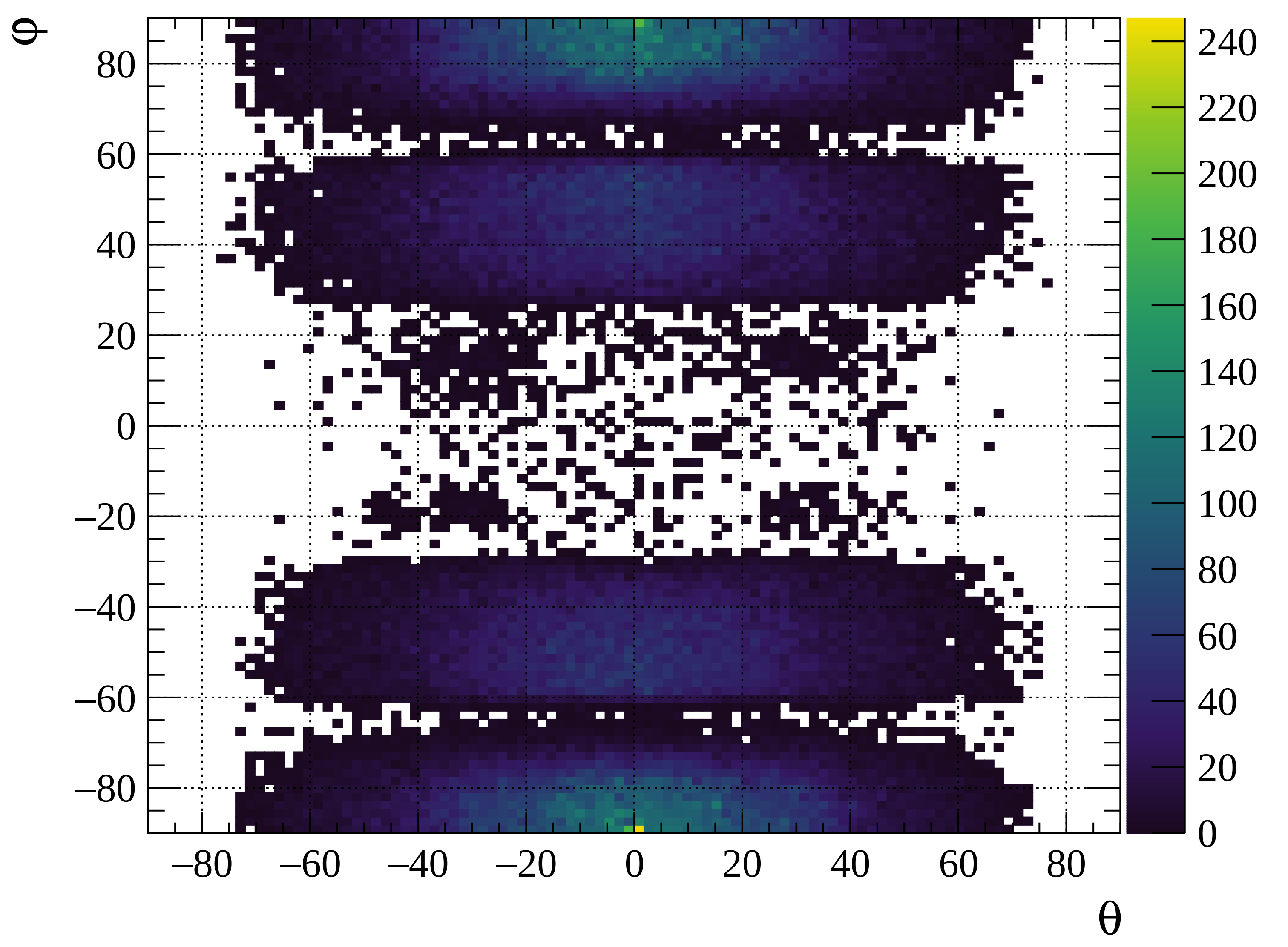
track length (cm)
160
140
120
100
80
60
40
20
0
−80
−60
−40
−20
0
20
40
60
80
θ
250
200
150
100
50
0



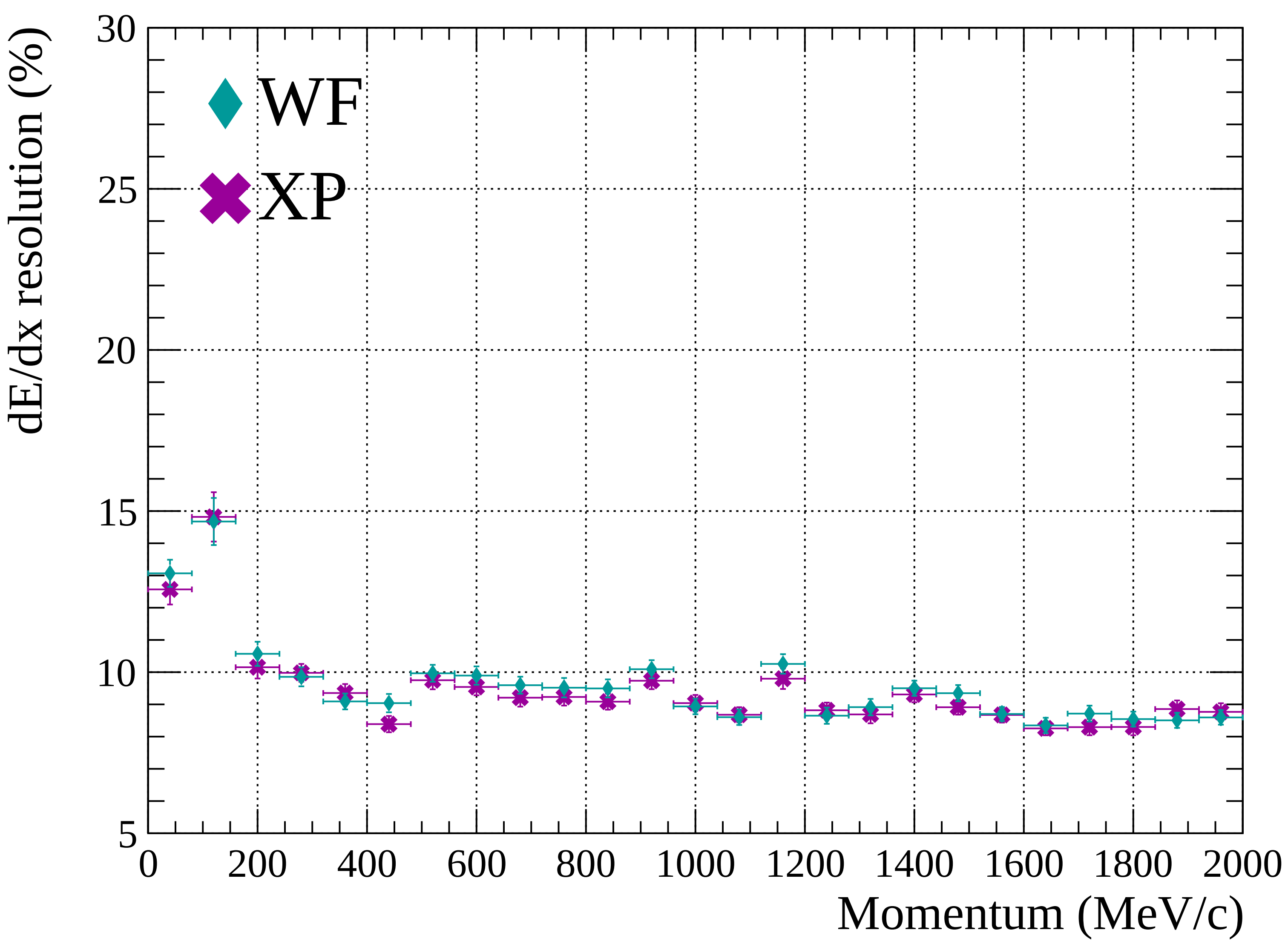
momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−80
−60
−40
−20
0
20
40
60
80
θ
25
20
15
10
5
0



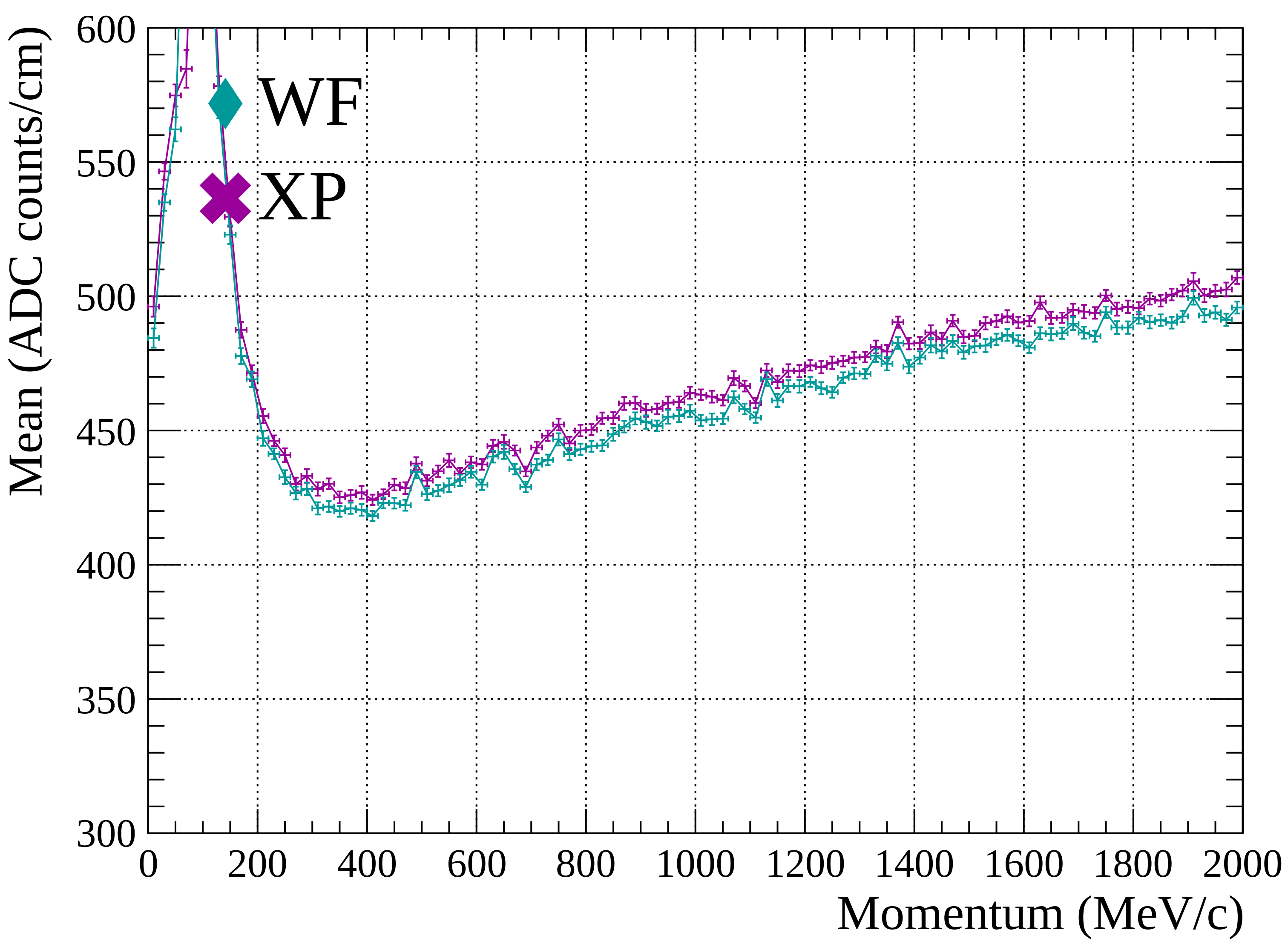
Count
3500
3000
2500
2000
1500
1000
500
0
−80
−60
−40
−20
0
20
40
60
80
θ angle



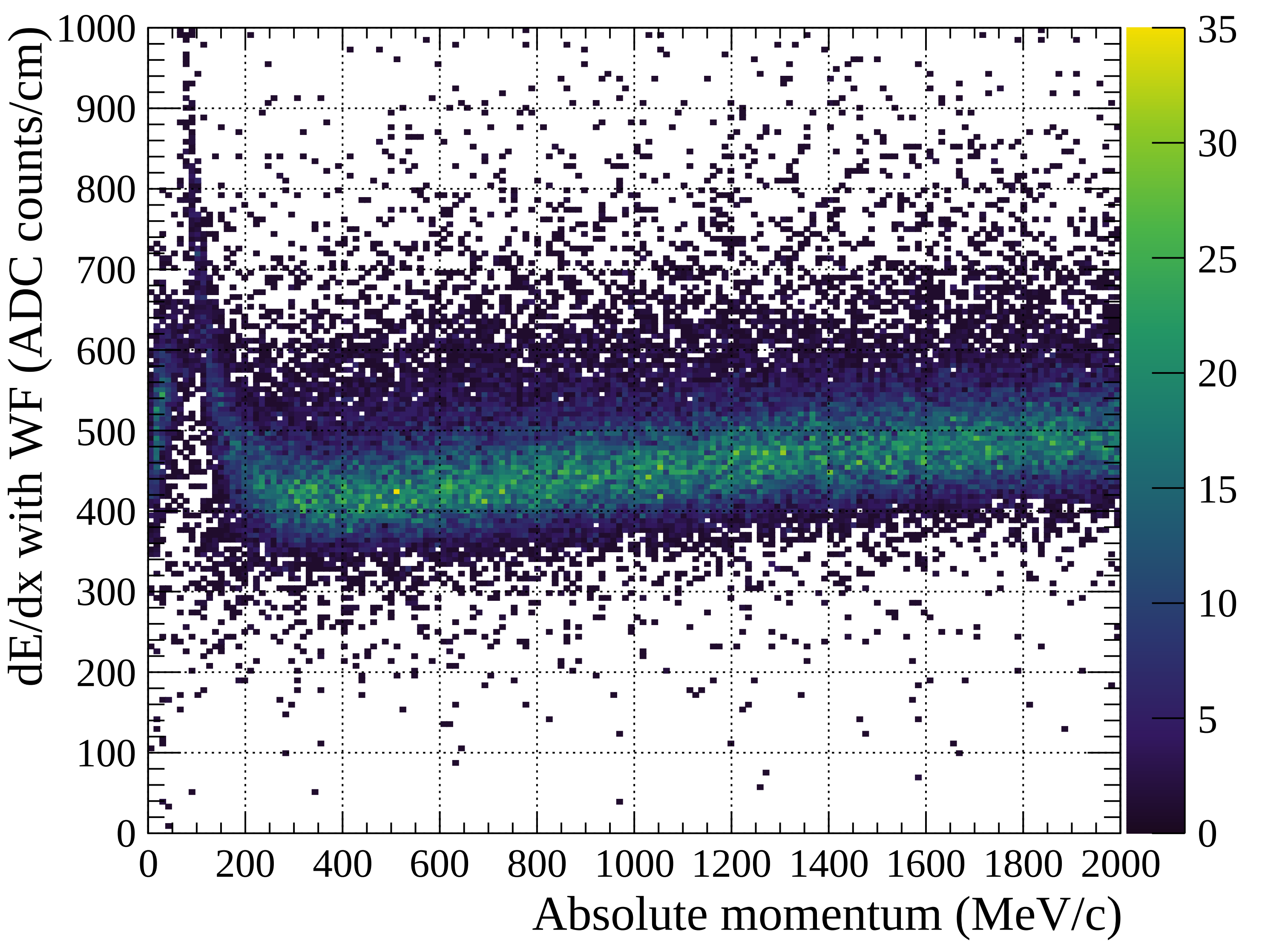
φ
80
60
40
20
0
−20
−40
−60
−80
−80
−60
−40
−20
0
20
40
60
80
θ
240
220
200
180
160
140
120
100
80
60
40
20
0

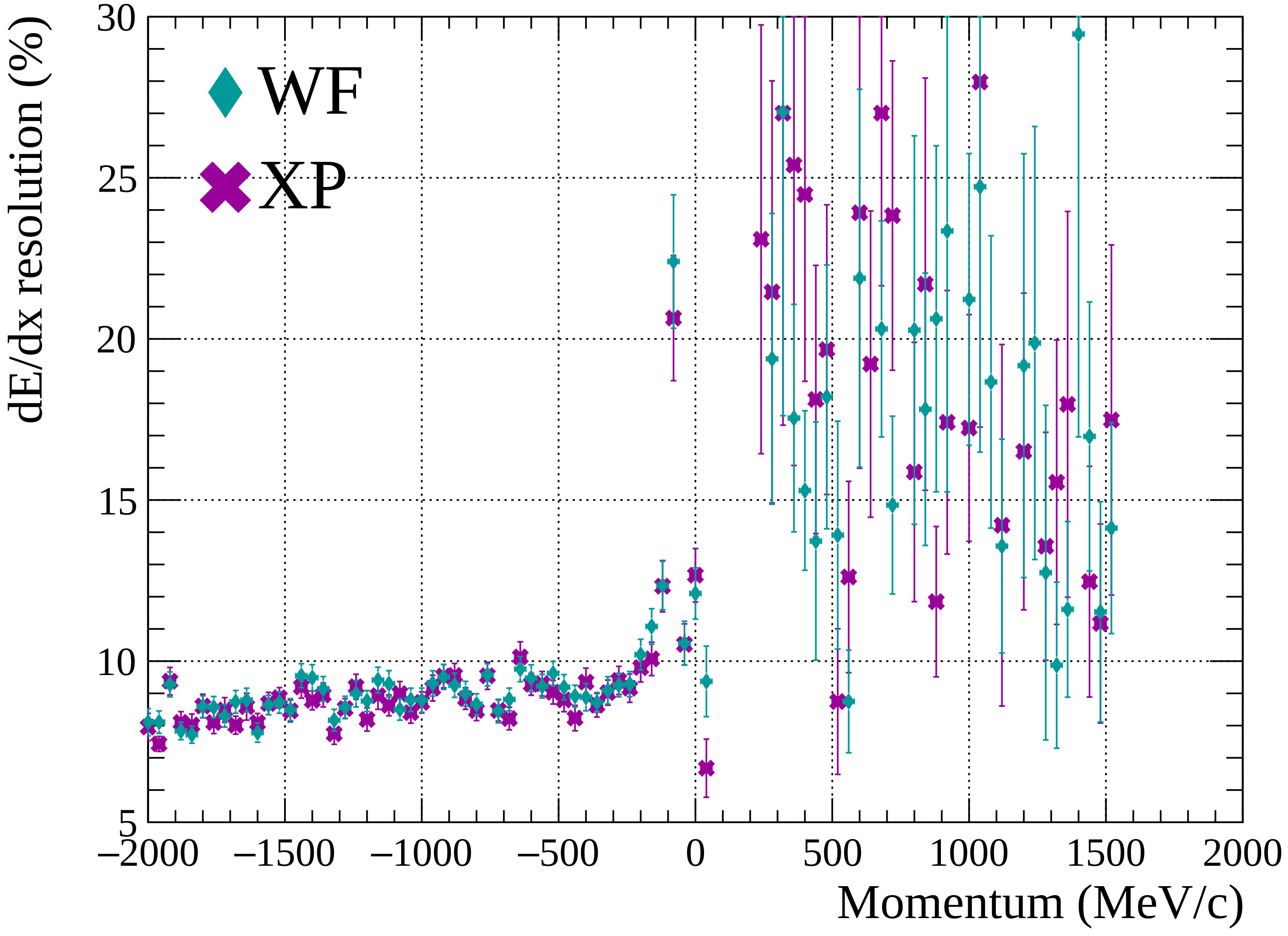


dE/dx resolution (%)
30
25
20
15
10
5
WF
XP
0
200
400
600
800
1000
1200
1400
1600
1800
2000
Momentum (MeV/c)

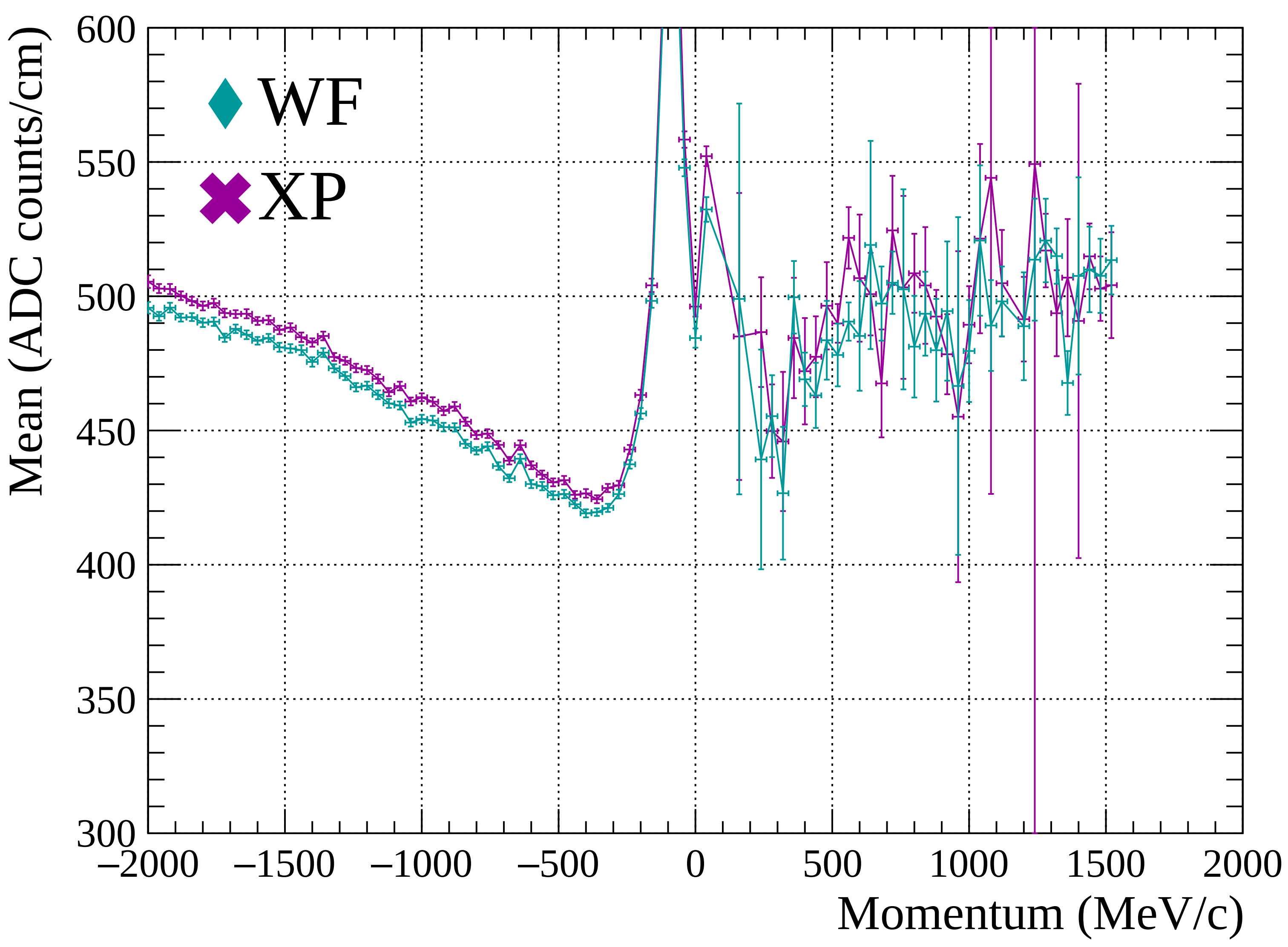


Mean (ADC counts/cm)
600
550
500
450
400
350
300
WF
XP
0
200
400
600
800
1000
1200
1400
1600
1800
2000
Momentum (MeV/c)

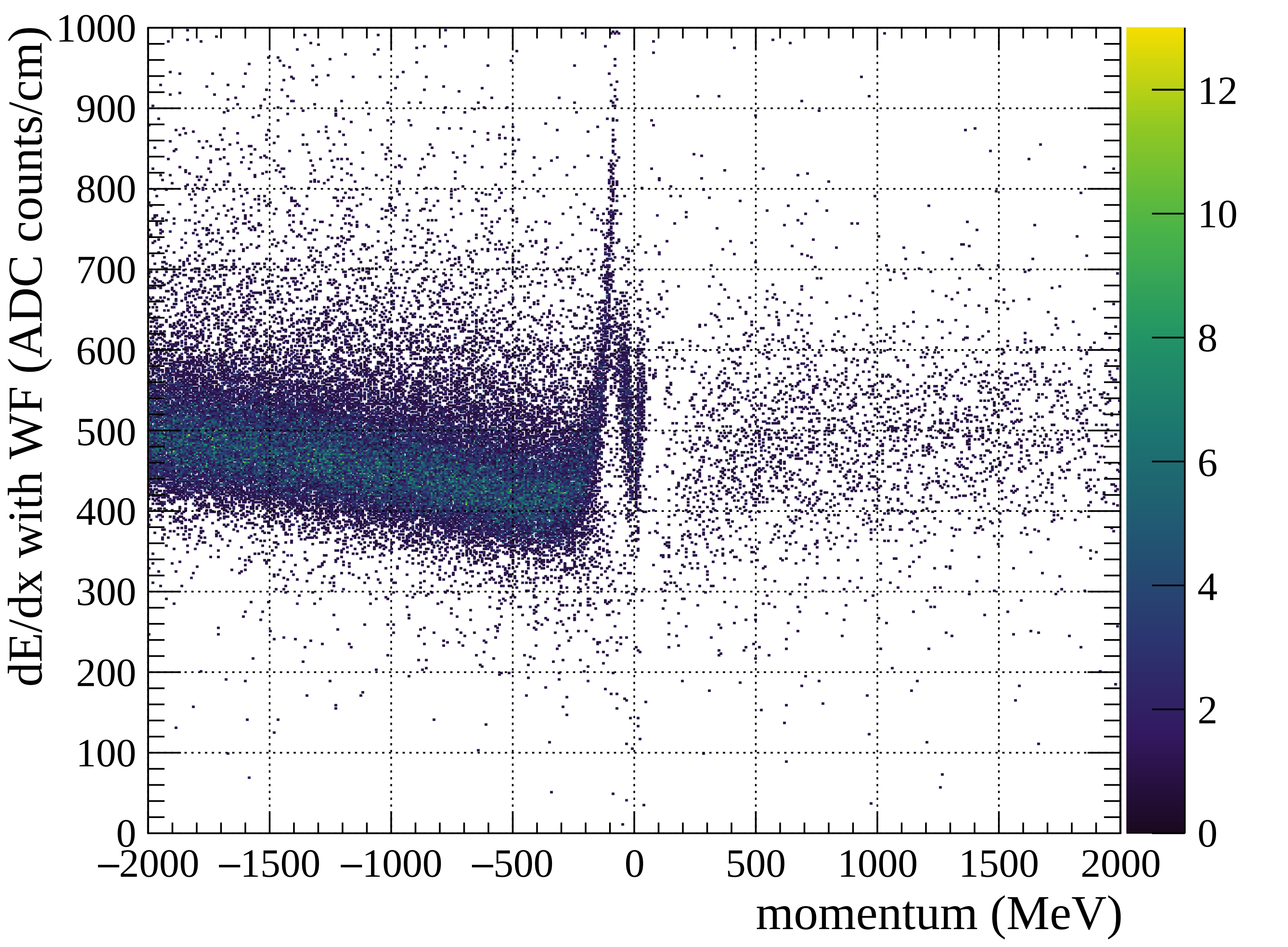

dE/dx with WF (ADC counts/cm)
Absolute momentum (MeV/c)
0
100
200
300
400
500
600
700
800
900
1000
0
200
400
600
800
1000
1200
1400
1600
1800
2000
0
5
10
15
20
25
30
35



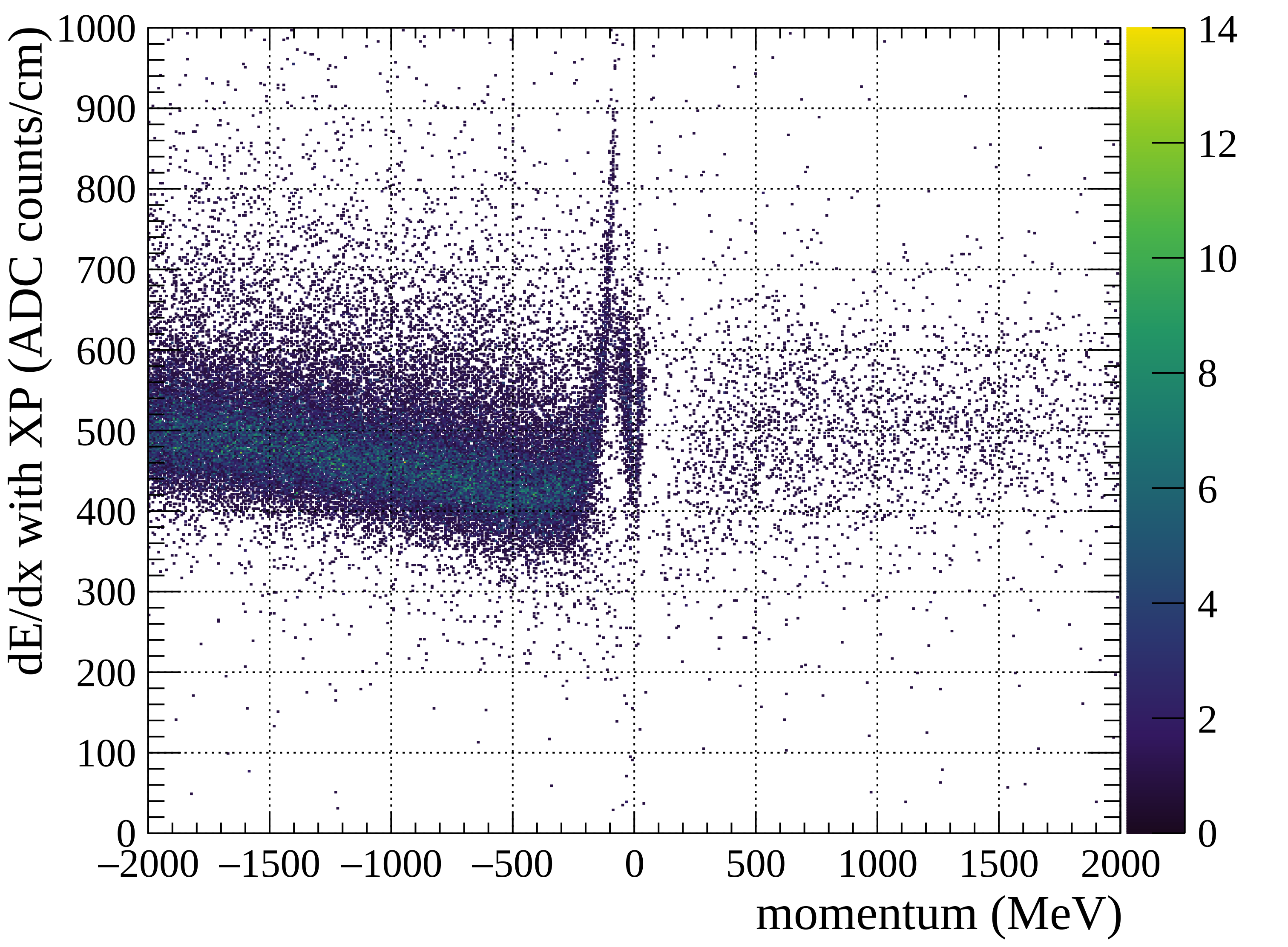
WF
XP
dE/dx resolution (%)
Momentum (MeV/c)
30
25
20
15
10
5
−2000
−1500
−1000
−500
0
500
1000
1500
2000



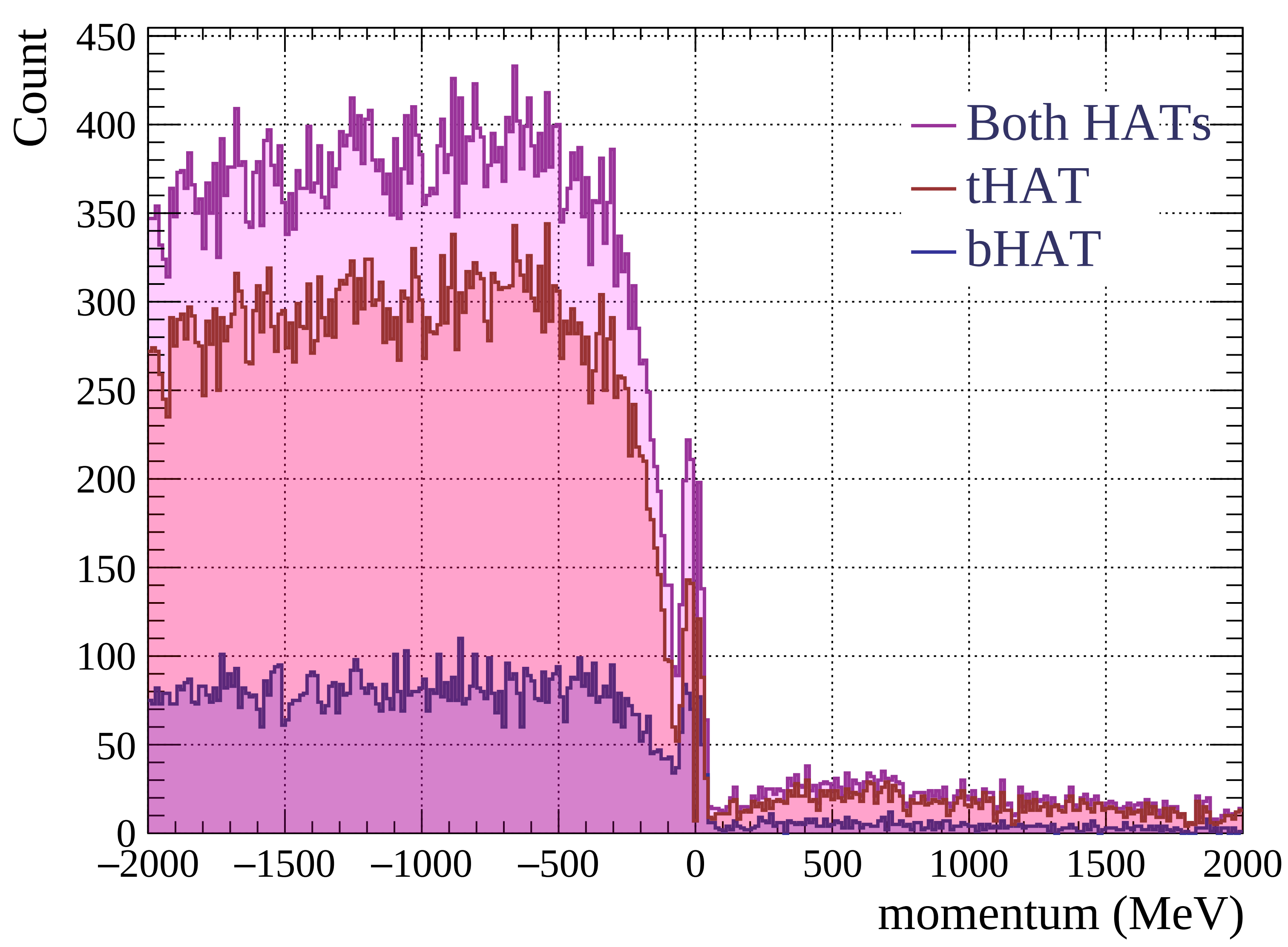
WF
XP
Mean (ADC counts/cm)
Momentum (MeV/c)
600
550
500
450
400
350
300
−2000
−1500
−1000
−500
0
500
1000
1500
2000

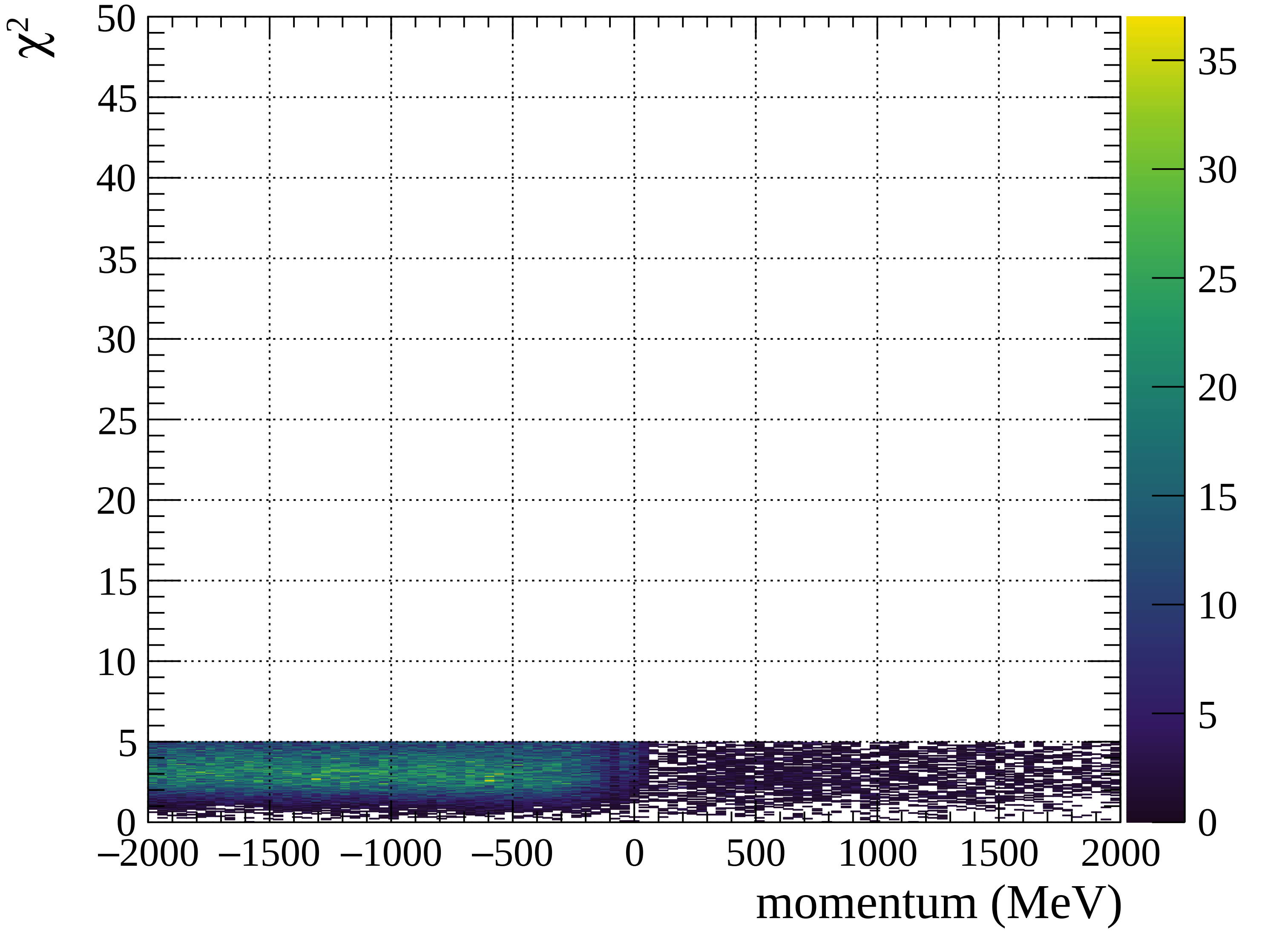


dE/dx with WF (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
momentum (MeV)
12
10
8
6
4
2
0

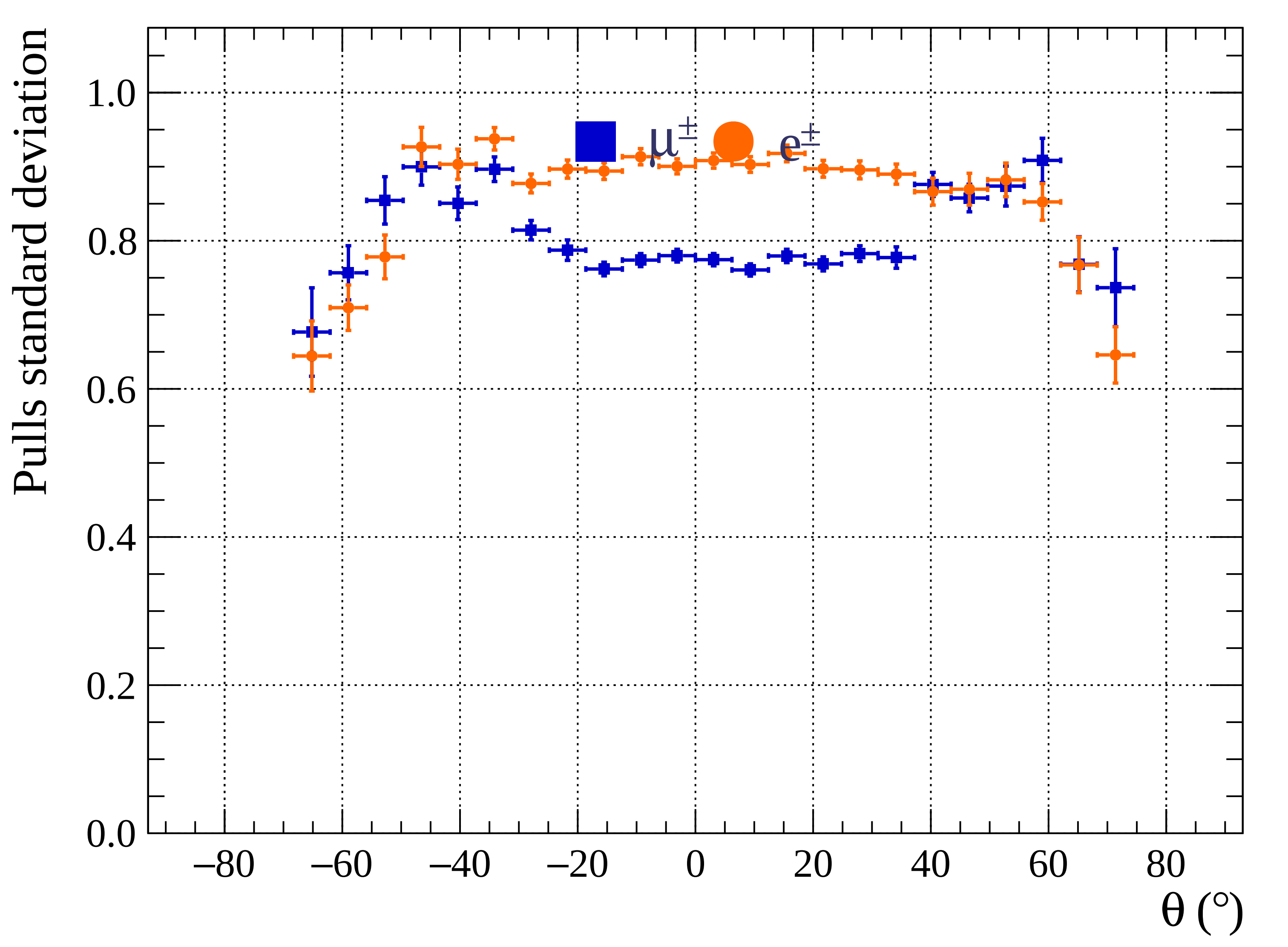


dE/dx with XP (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
1000
−2000
−1500
−1000
−500
0
500
1000
1500
2000
momentum (MeV)
0
2
4
6
8
10
12
14

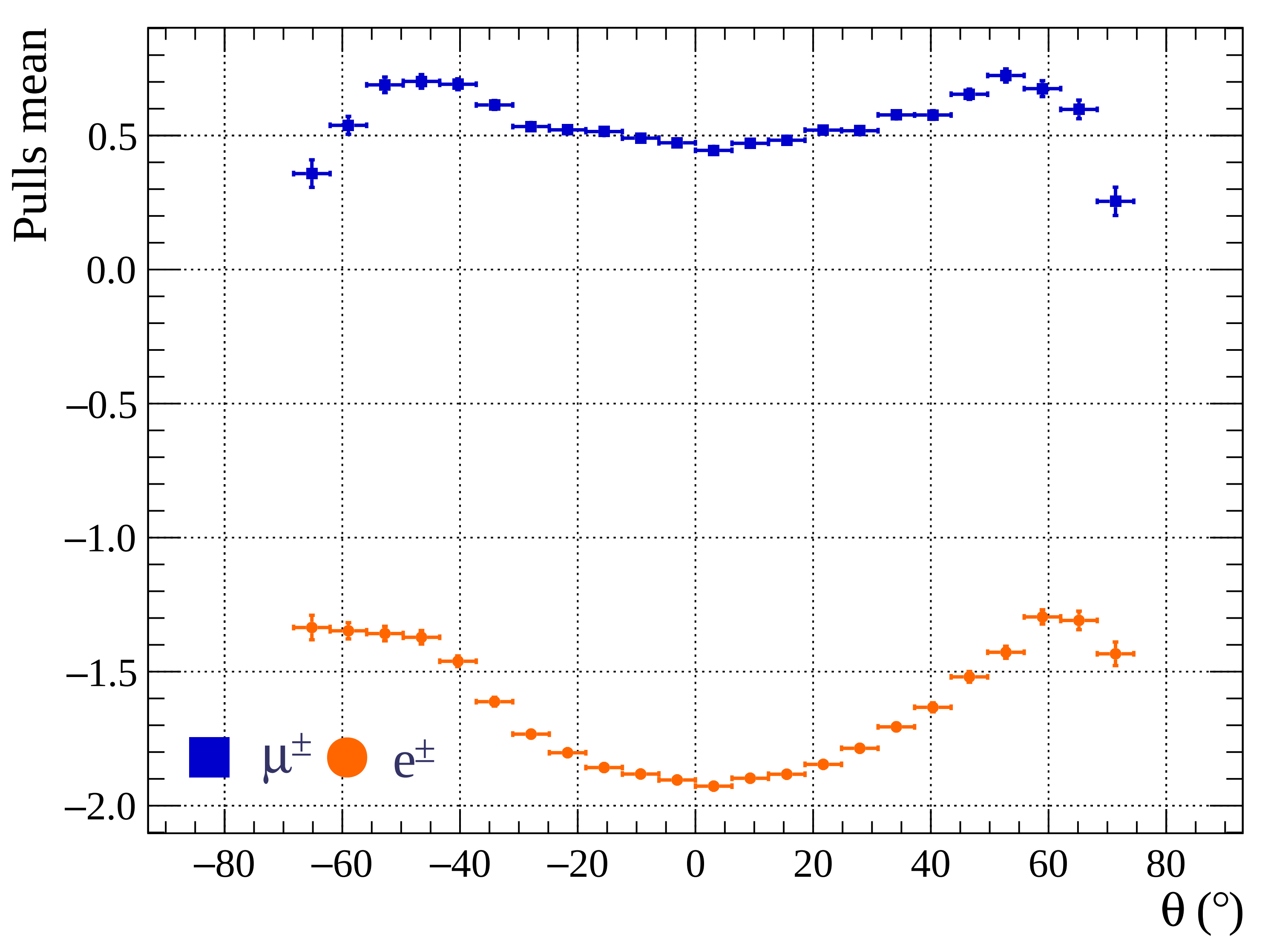

Count
momentum (MeV)
Both HATs
tHAT
bHAT
0
50
100
150
200
250
300
350
400
450
−2000
−1500
−1000
−500
0
500
1000
1500
2000



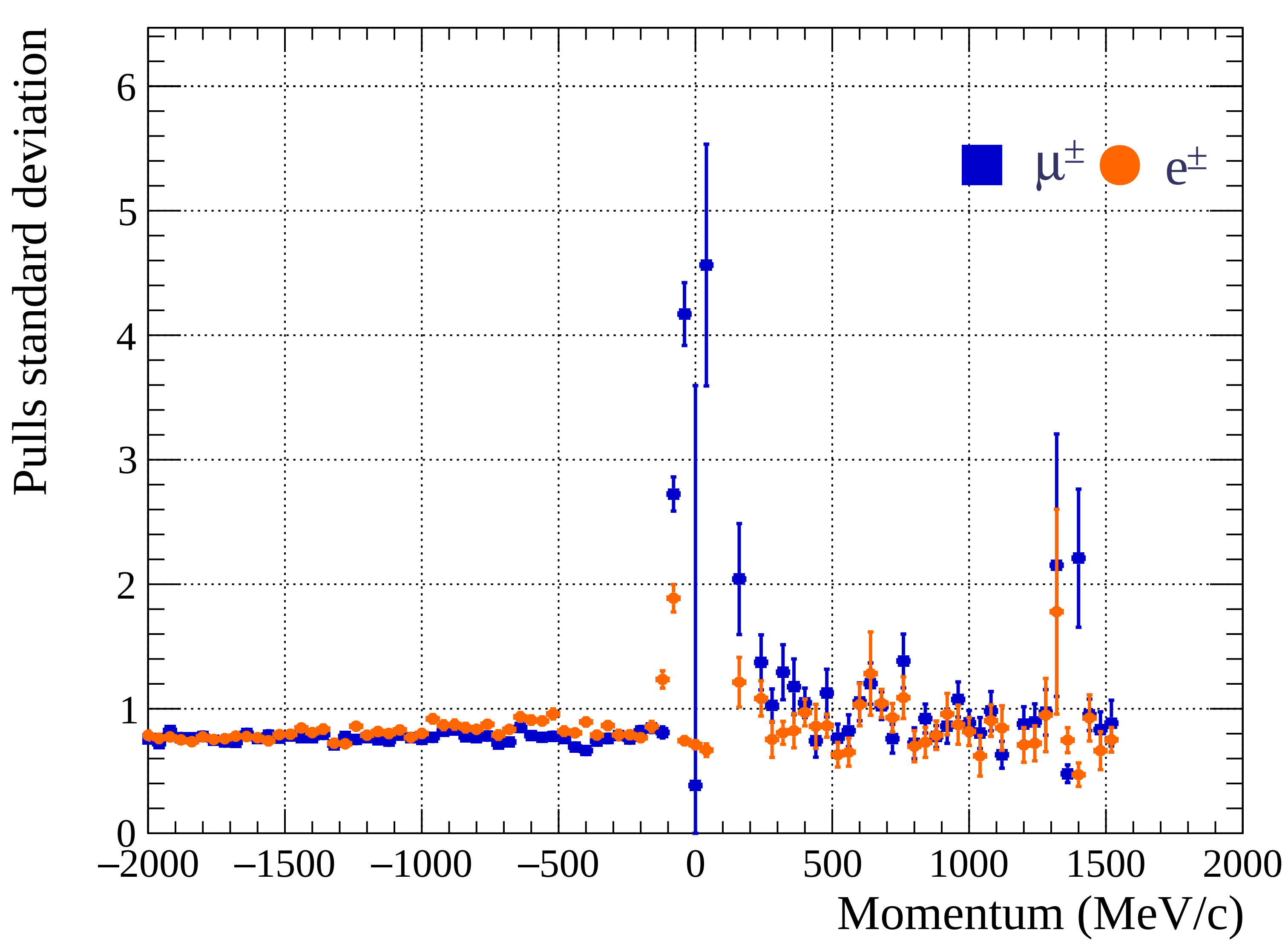
χ^2
0
5
10
15
20
25
30
35
40
45
50
−2000
−1500
−1000
−500
0
500
1000
1500
2000
momentum (MeV)
0
5
10
15
20
25
30
35



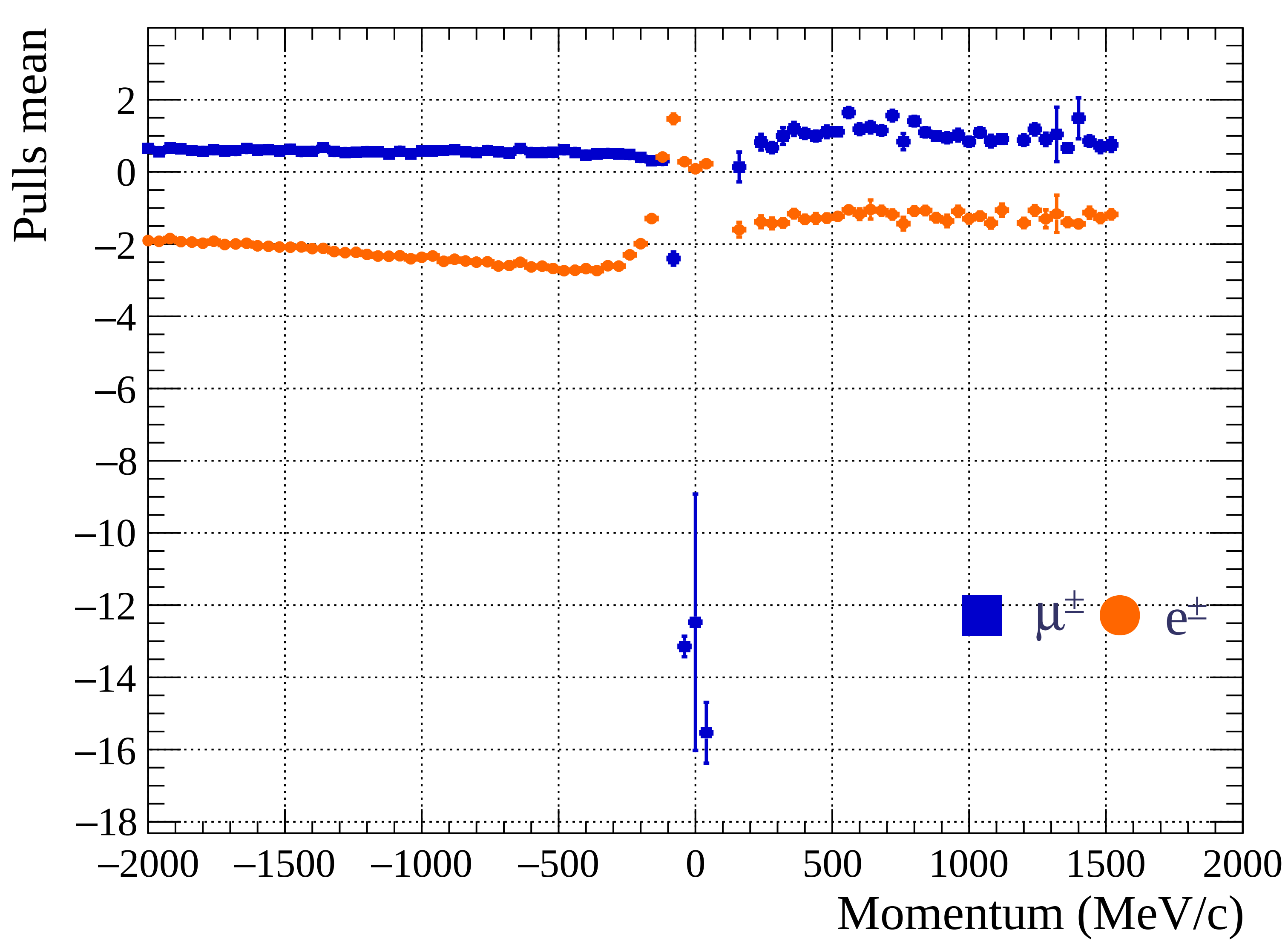
Pulls standard deviation
1.0
0.8
0.6
0.4
0.2
0.0
−80
−60
−40
−20
0
20
40
60
80
θ (°)
μ±
e±

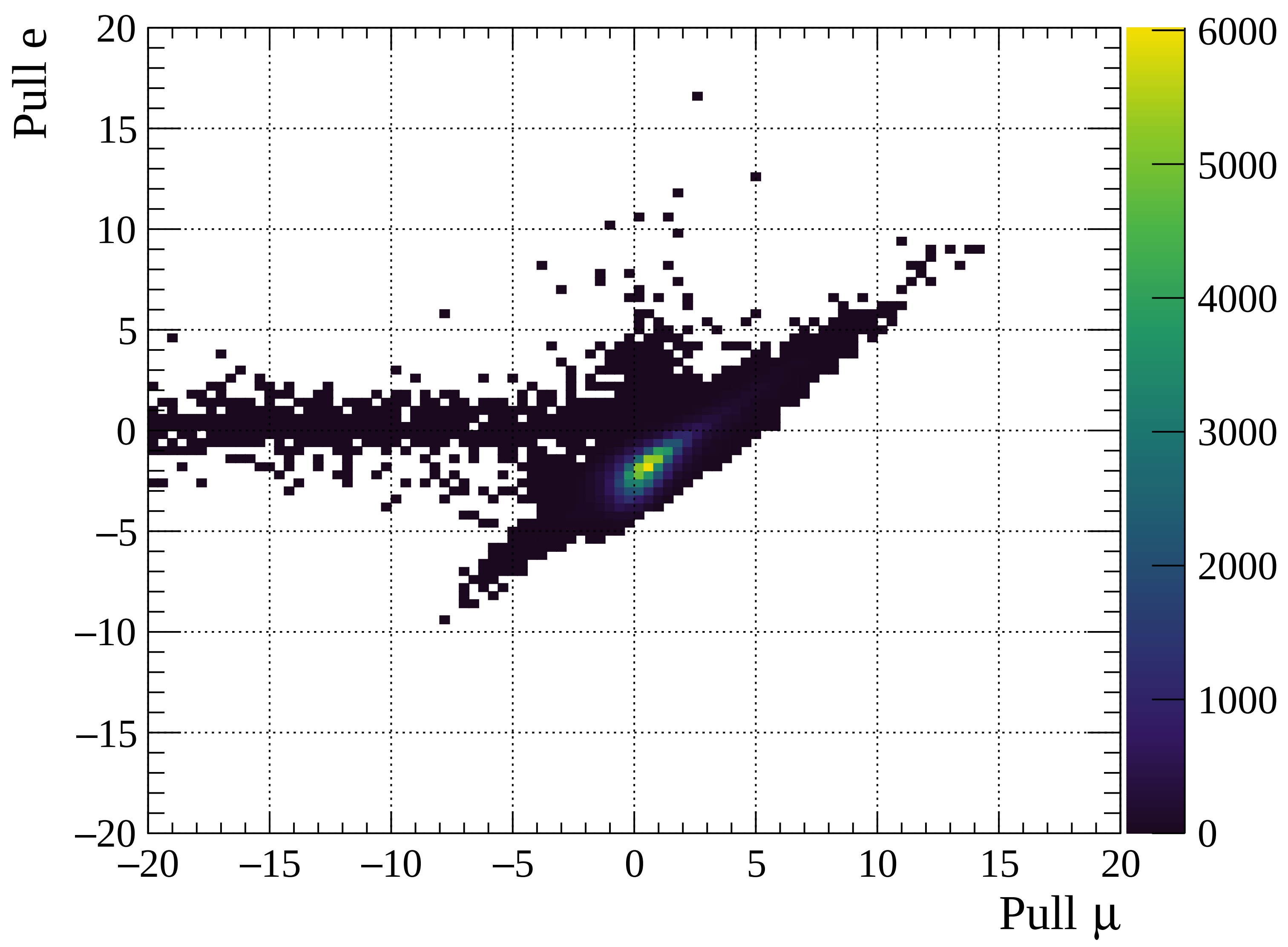


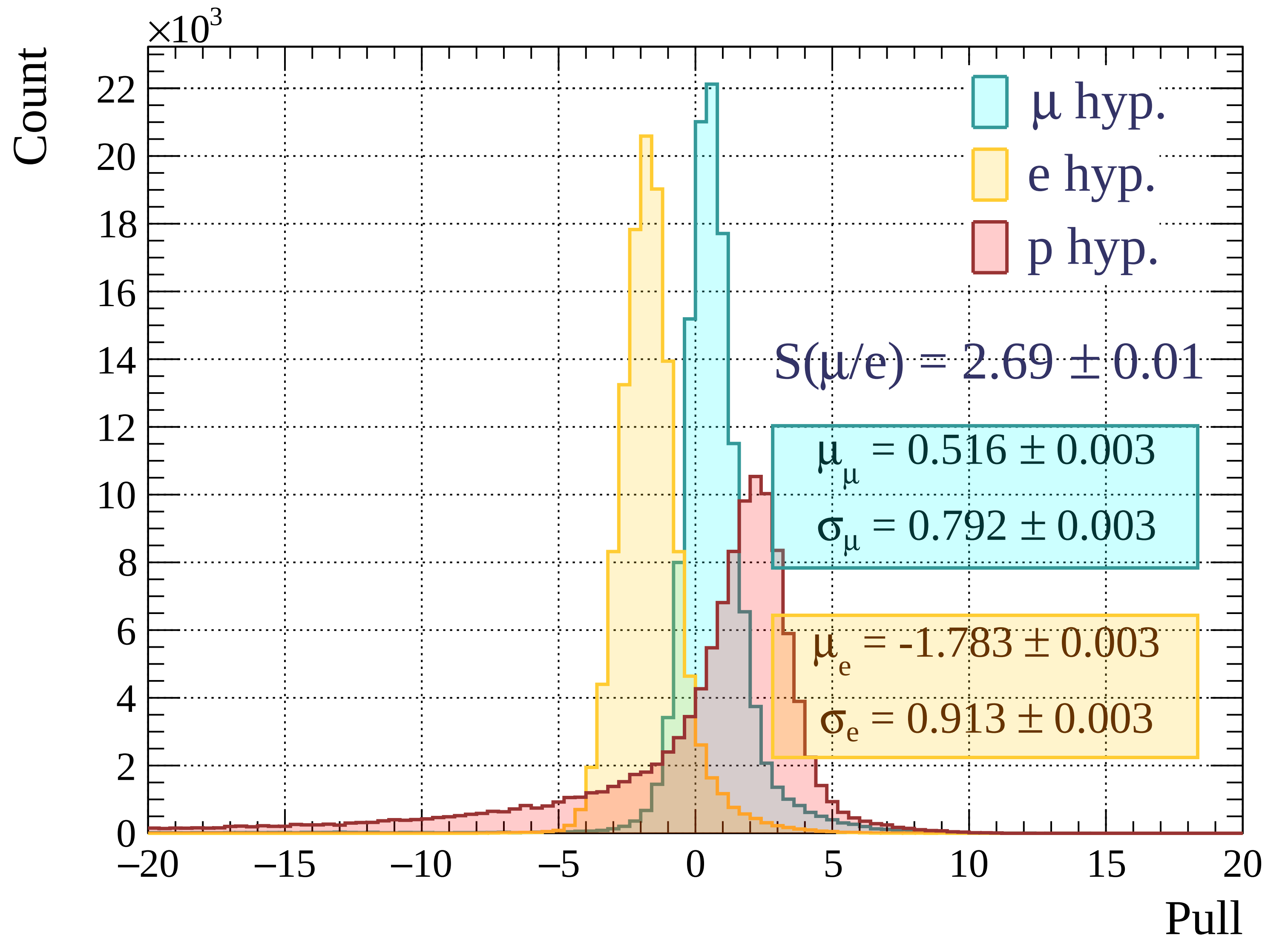
Pulls mean
0.5
0.0
−0.5
−1.0
−1.5
−2.0
−80
−60
−40
−20
0
20
40
60
80
θ (°)
μ±
e±

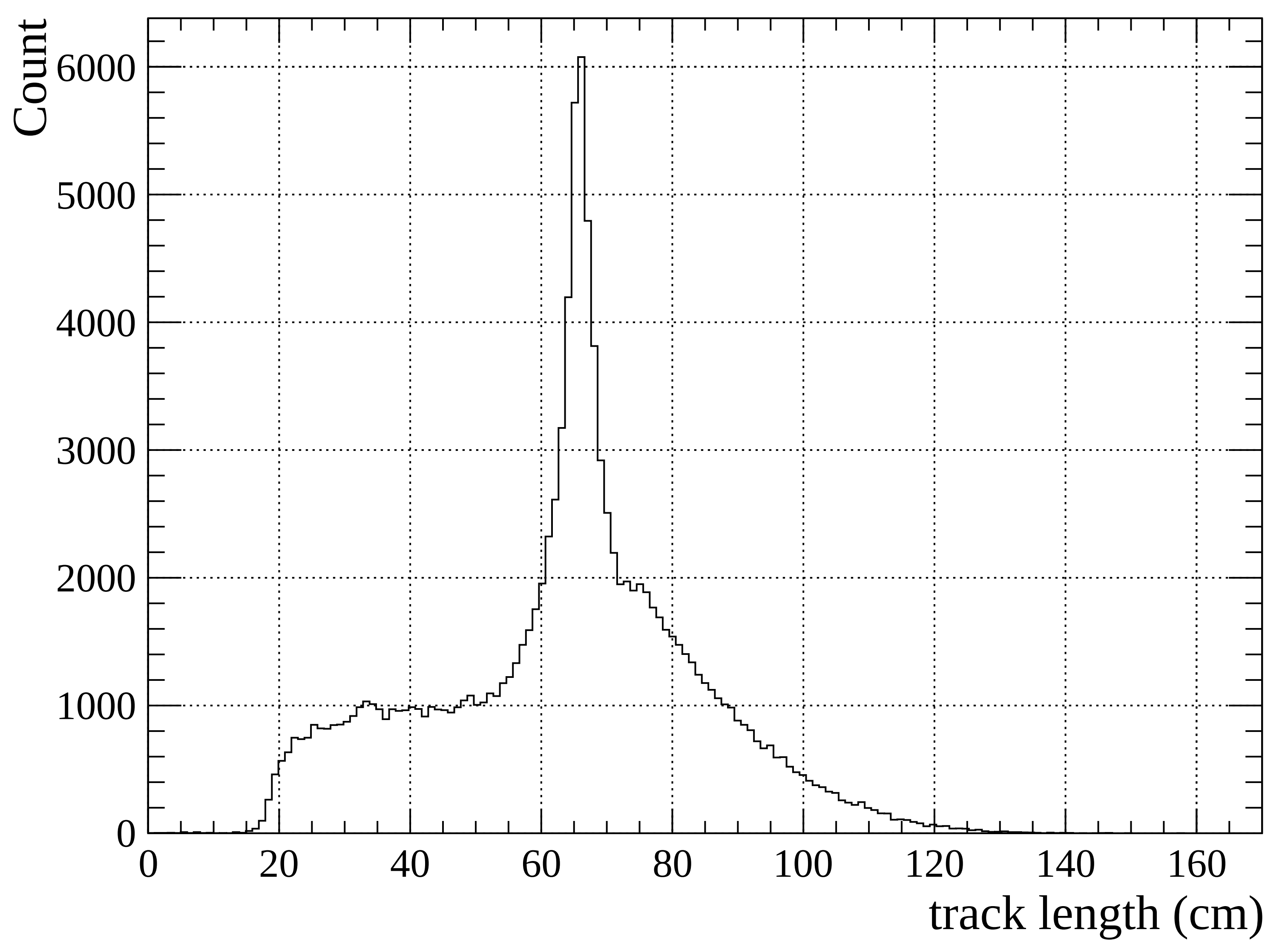


Pulls standard deviation
0
1
2
3
4
5
6
−2000
−1500
−1000
−500
0
500
1000
1500
2000
Momentum (MeV/c)
$\mu^{\pm}$
$e^{\pm}$

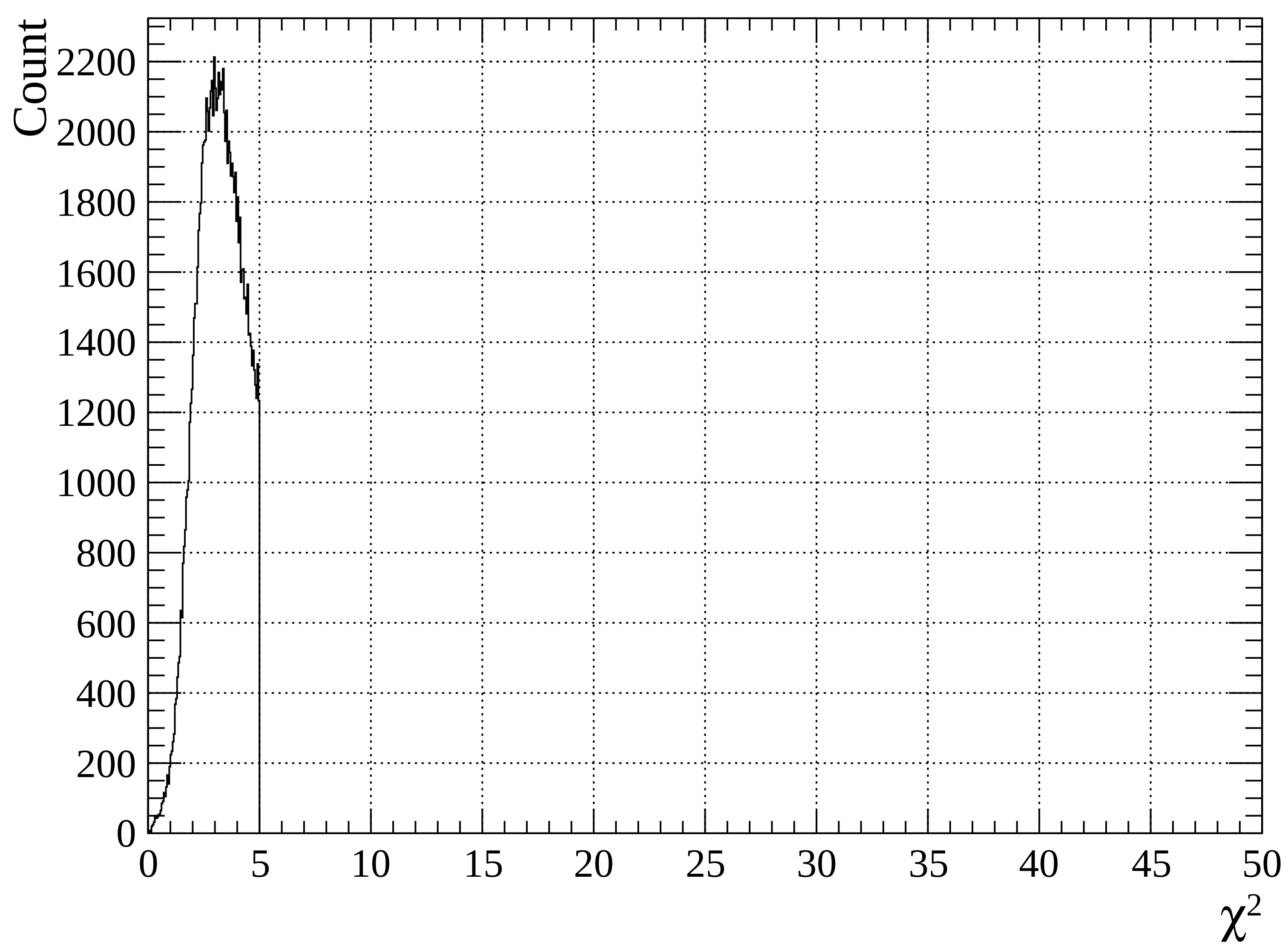

Pulls mean
2
0
−2
−4
−6
−8
−10
−12
−14
−16
−18
−2000
−1500
−1000
−500
0
500
1000
1500
2000
Momentum (MeV/c)
μ±
e±


Pull e
20
15
10
5
0
−5
−10
−15
−20
−20
−15
−10
−5
0
5
10
15
20
Pull μ
6000
5000
4000
3000
2000
1000
0

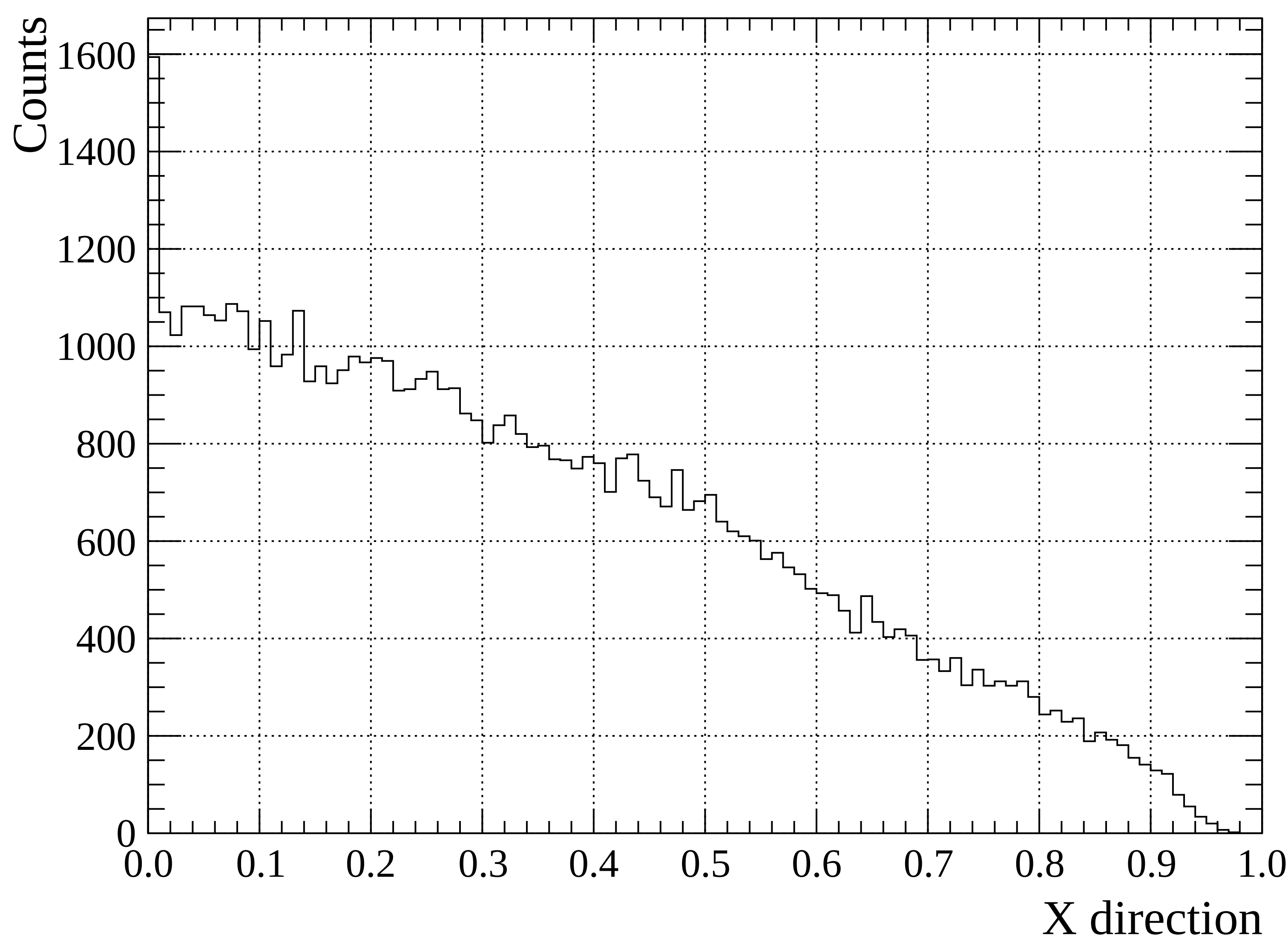

×10^3
Count
22
20
18
16
14
12
10
8
6
4
2
0
−20
−15
−10
−5
0
5
10
15
20
Pull
μ hyp.
e hyp.
p hyp.
S(μ/e) = 2.69 ± 0.01
μ_μ = 0.516 ± 0.003
σ_μ = 0.792 ± 0.003
μ_e = -1.783 ± 0.003
σ_e = 0.913 ± 0.003



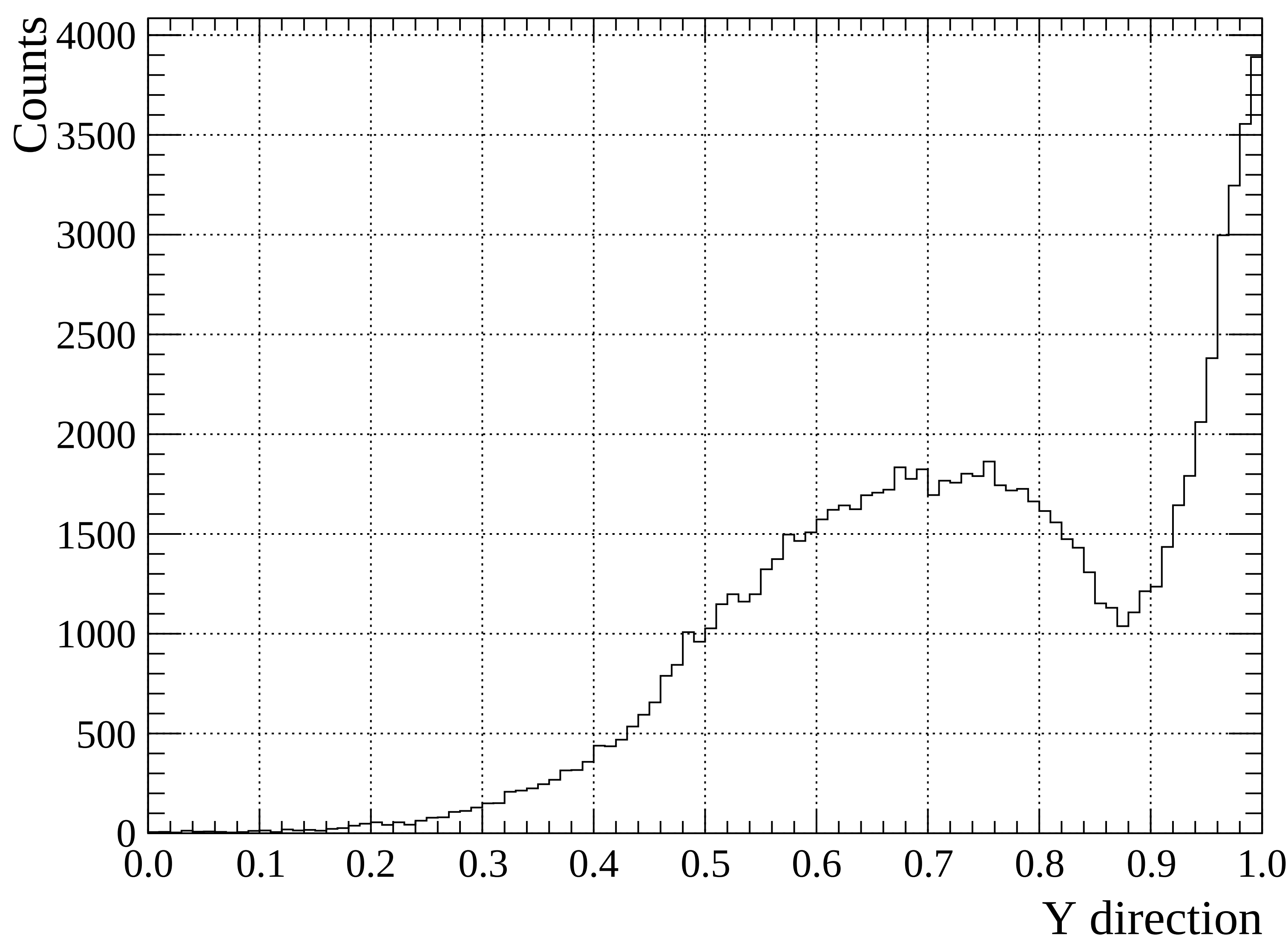
Count
6000
5000
4000
3000
2000
1000
0
0
20
40
60
80
100
120
140
160
track length (cm)

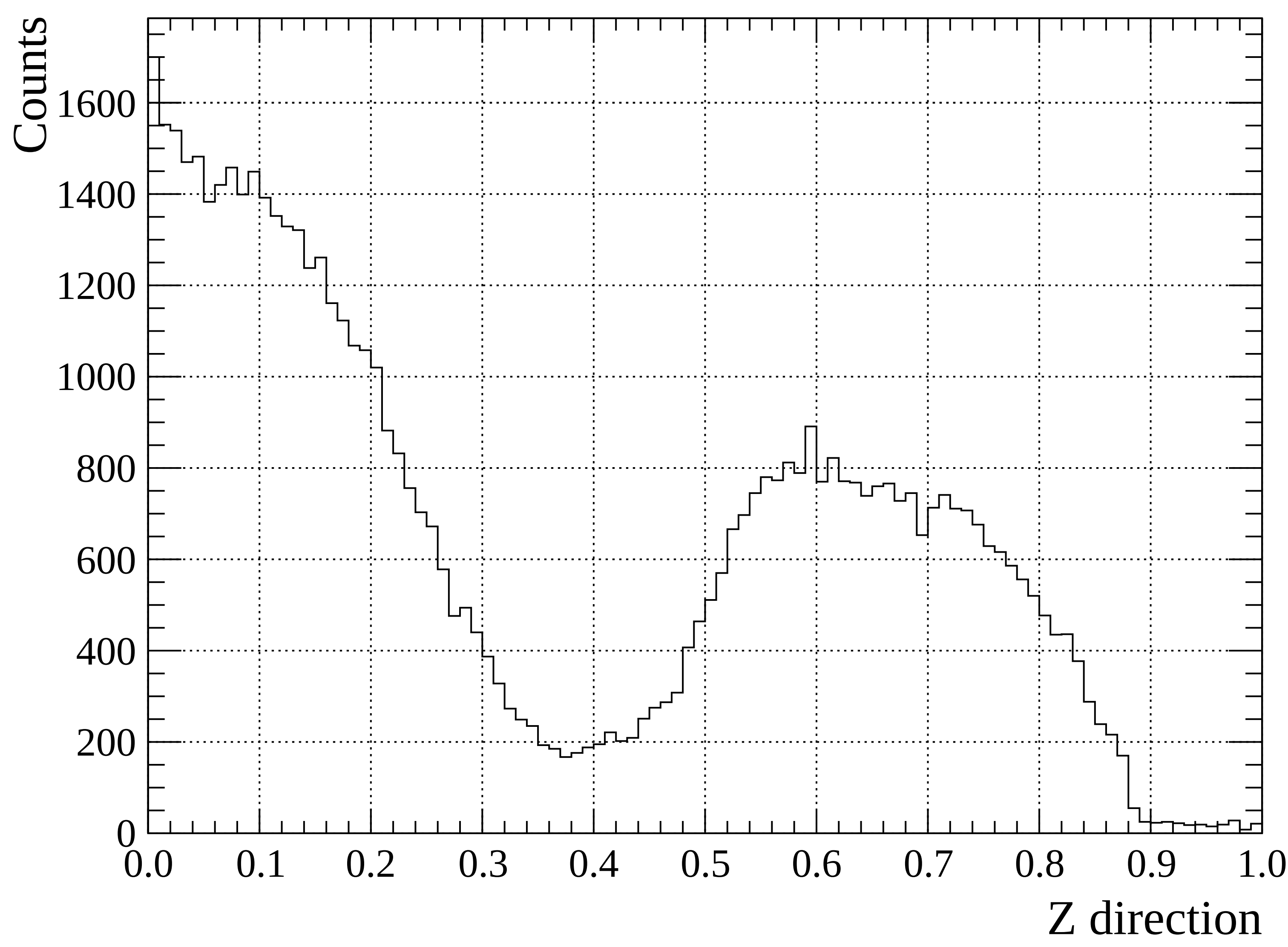


Count
2200
2000
1800
1600
1400
1200
1000
800
600
400
200
0
0
5
10
15
20
25
30
35
40
45
50
χ^2

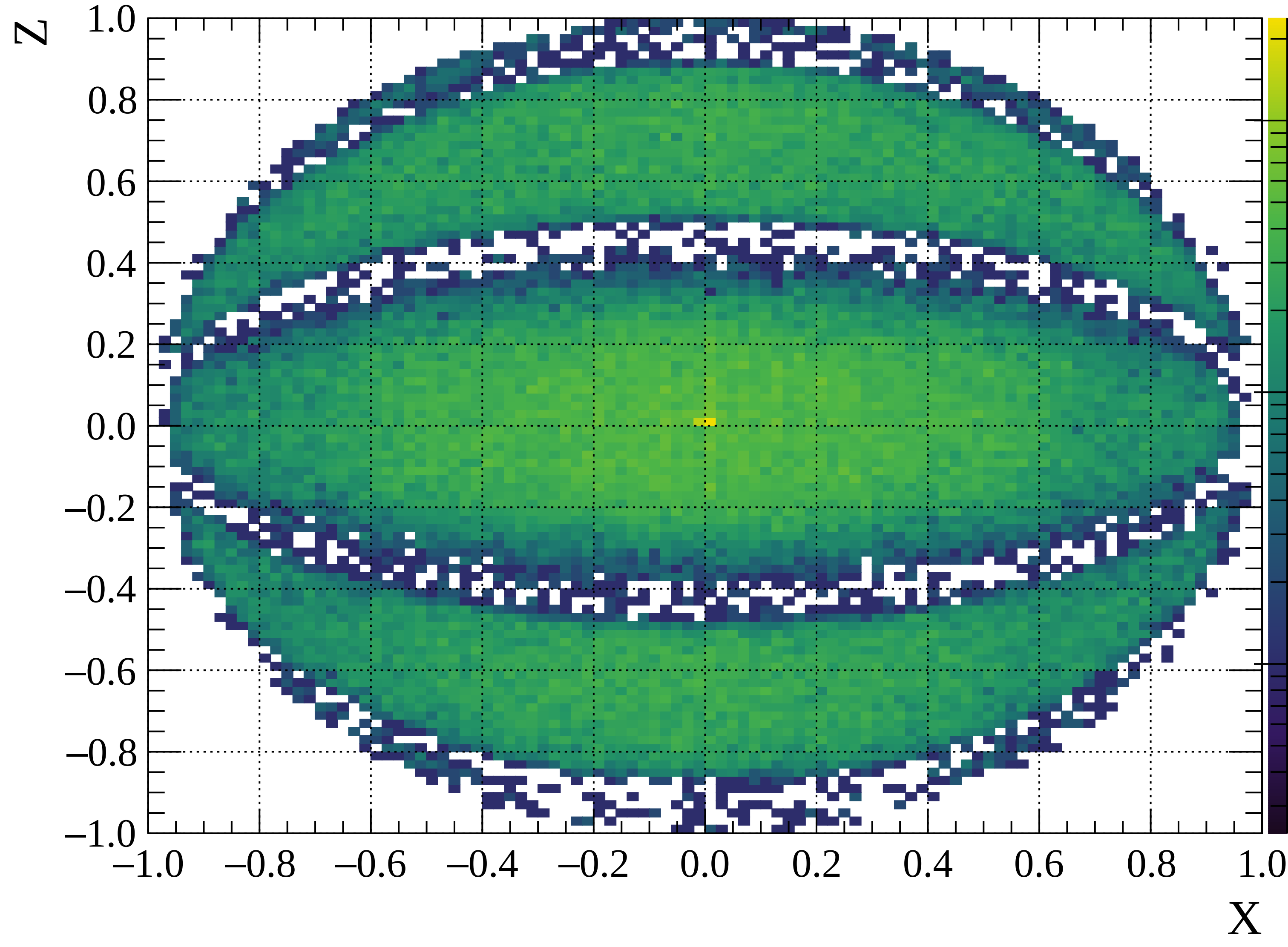


Counts
1600
1400
1200
1000
800
600
400
200
0
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
X direction

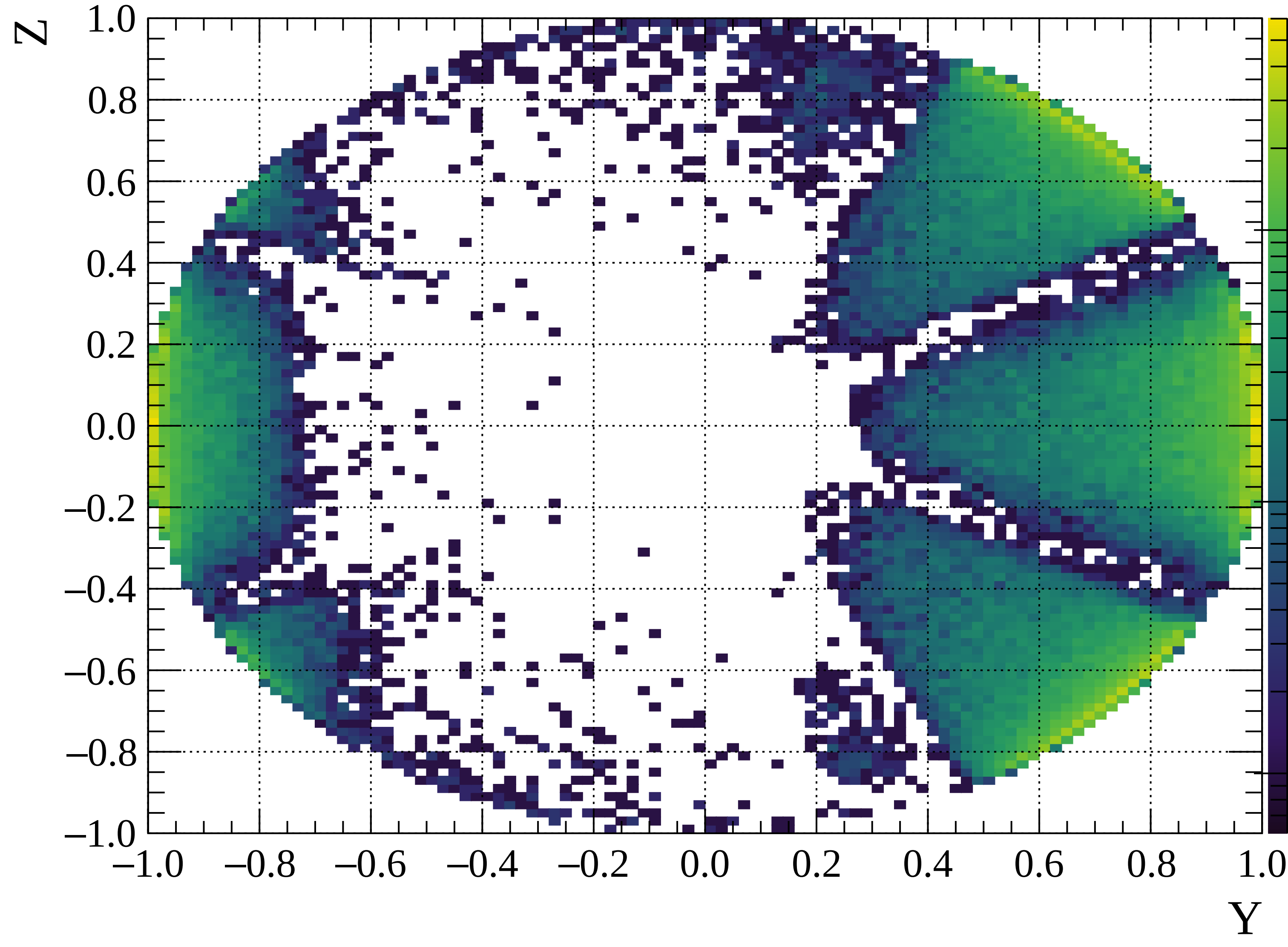

Counts
4000
3500
3000
2500
2000
1500
1000
500
0
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
Y direction



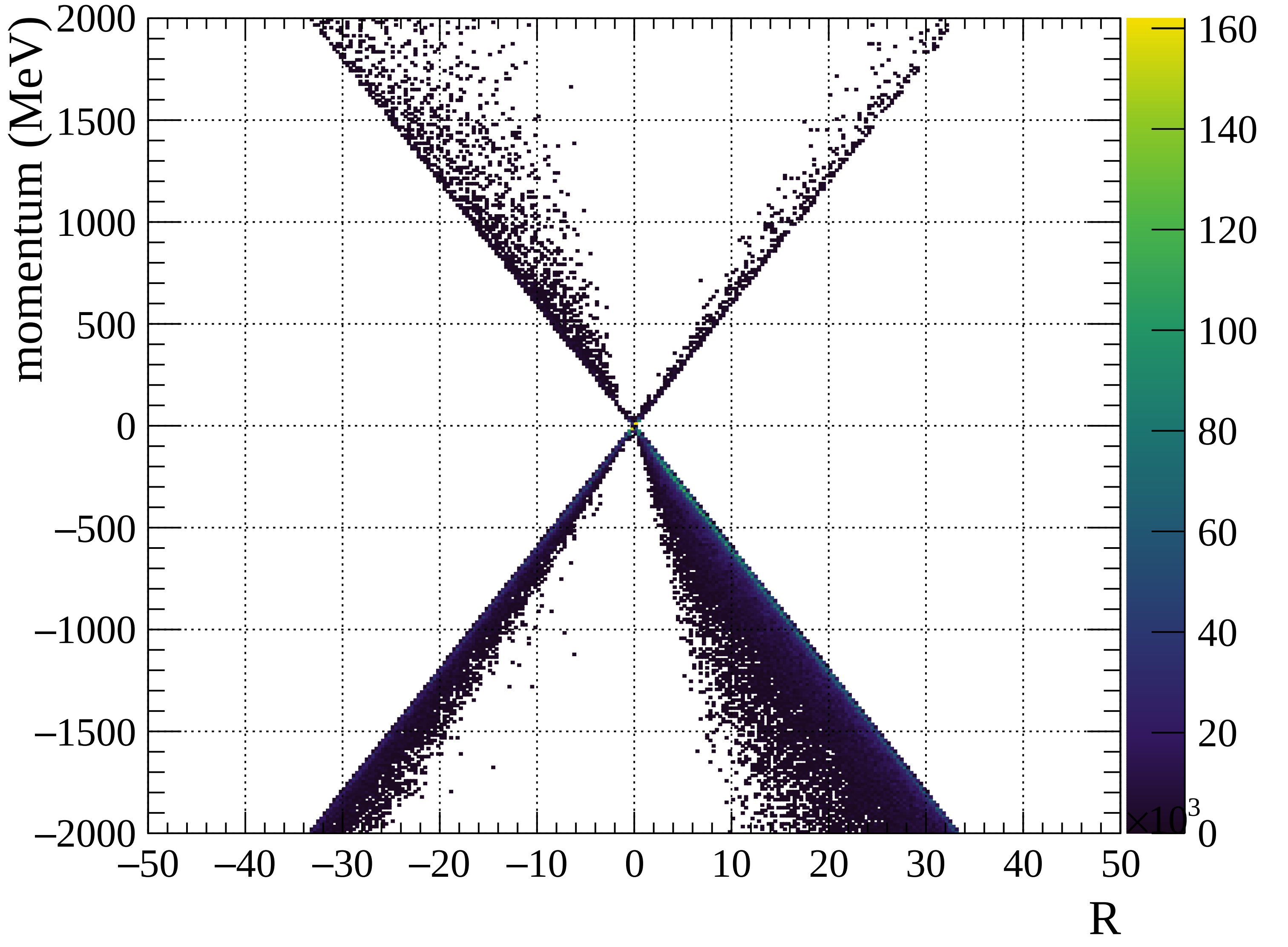
Counts
1600
1400
1200
1000
800
600
400
200
0
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
Z direction



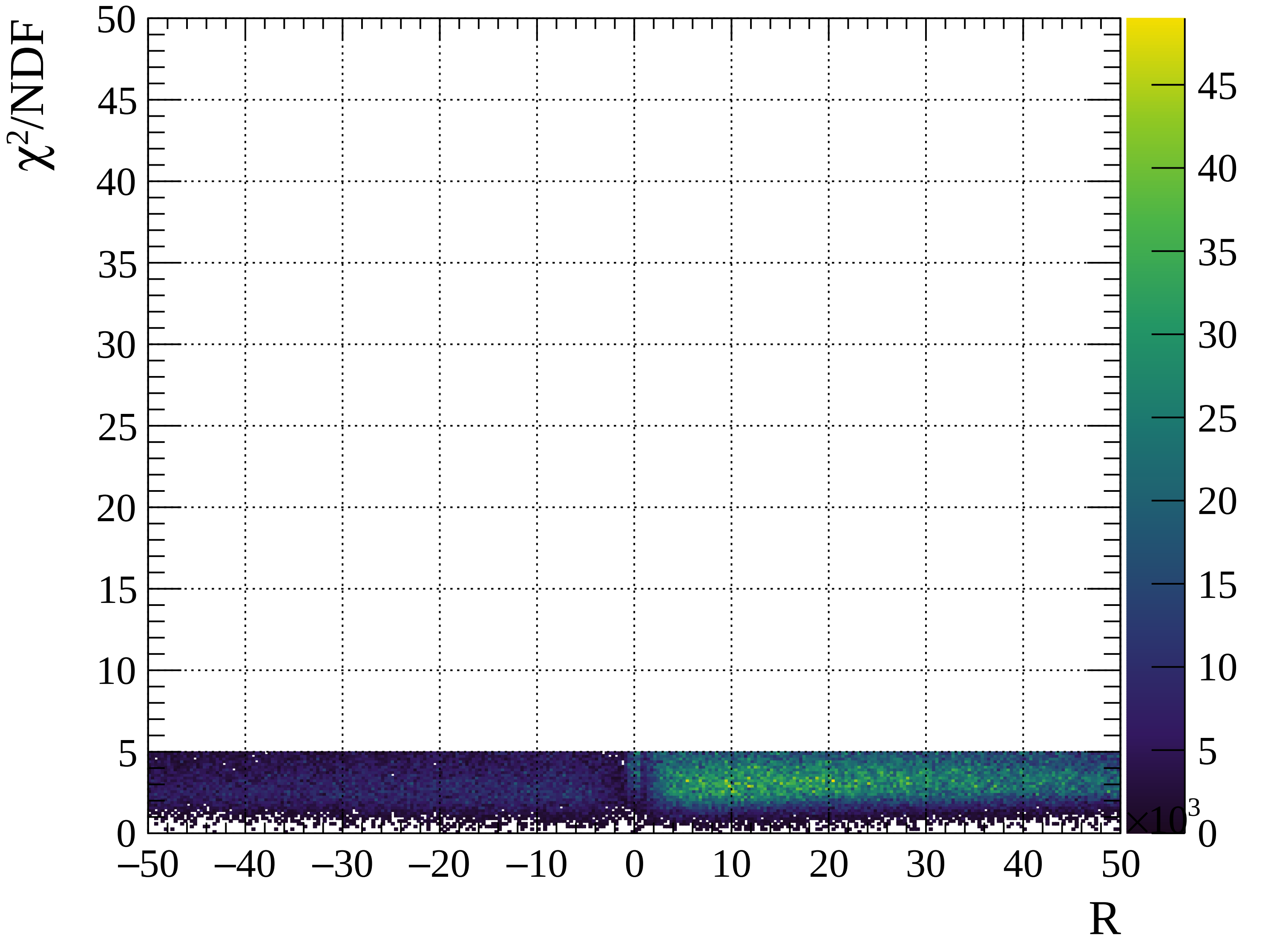
Z
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
X

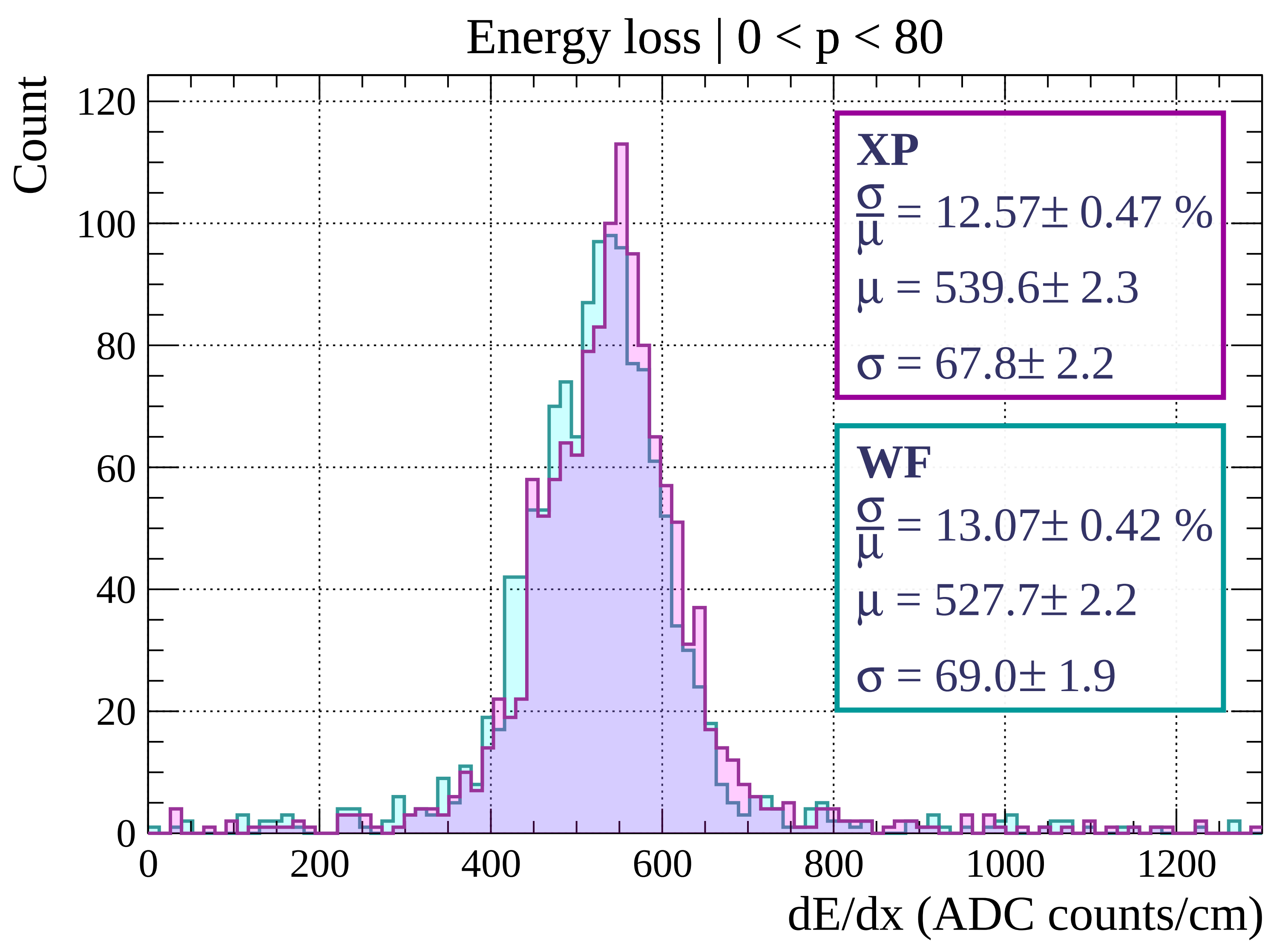


Z
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
Y

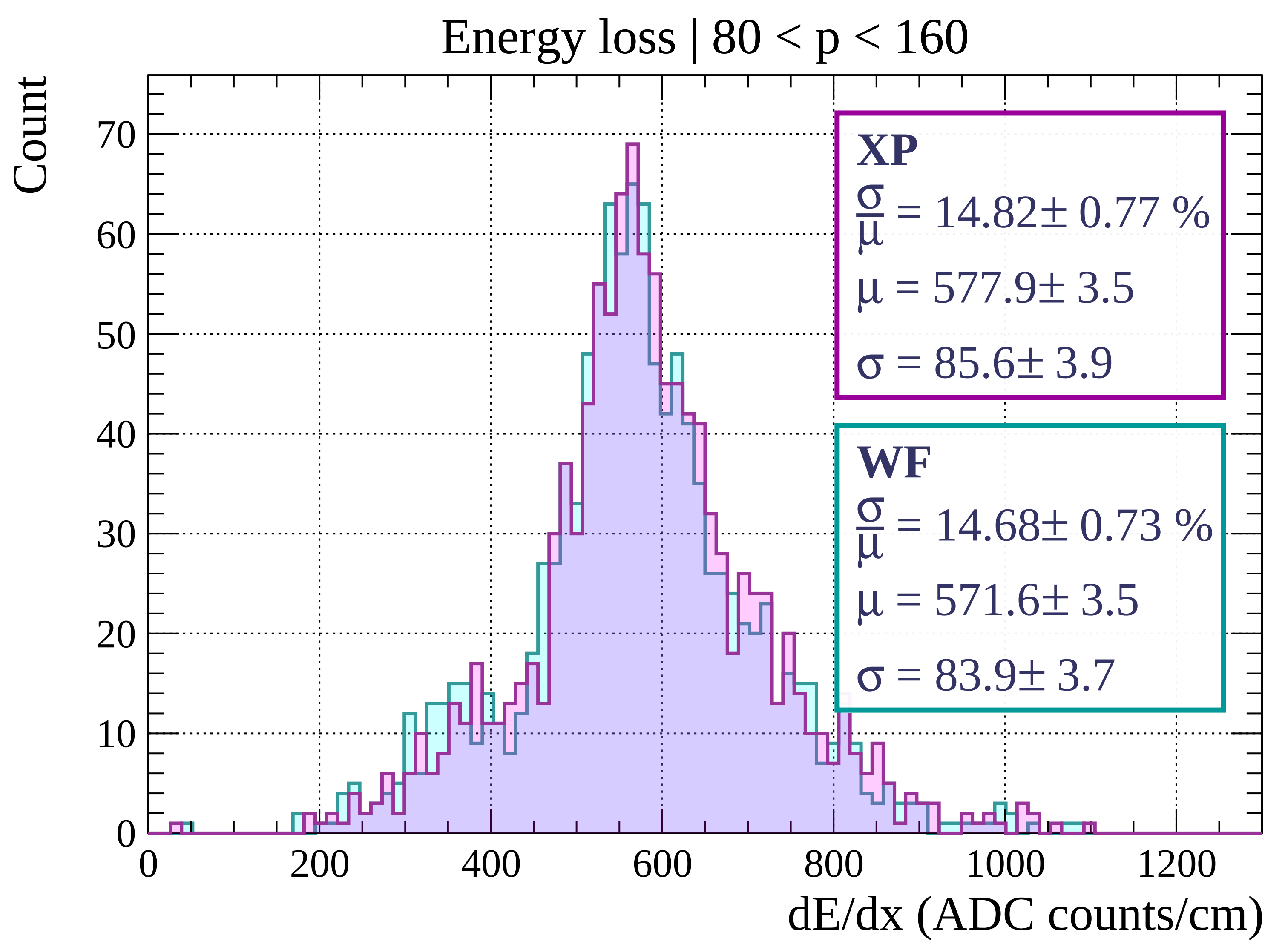


momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−50
−40
−30
−20
−10
0
10
20
30
40
50
R
160
140
120
100
80
60
40
20
0
×10^3

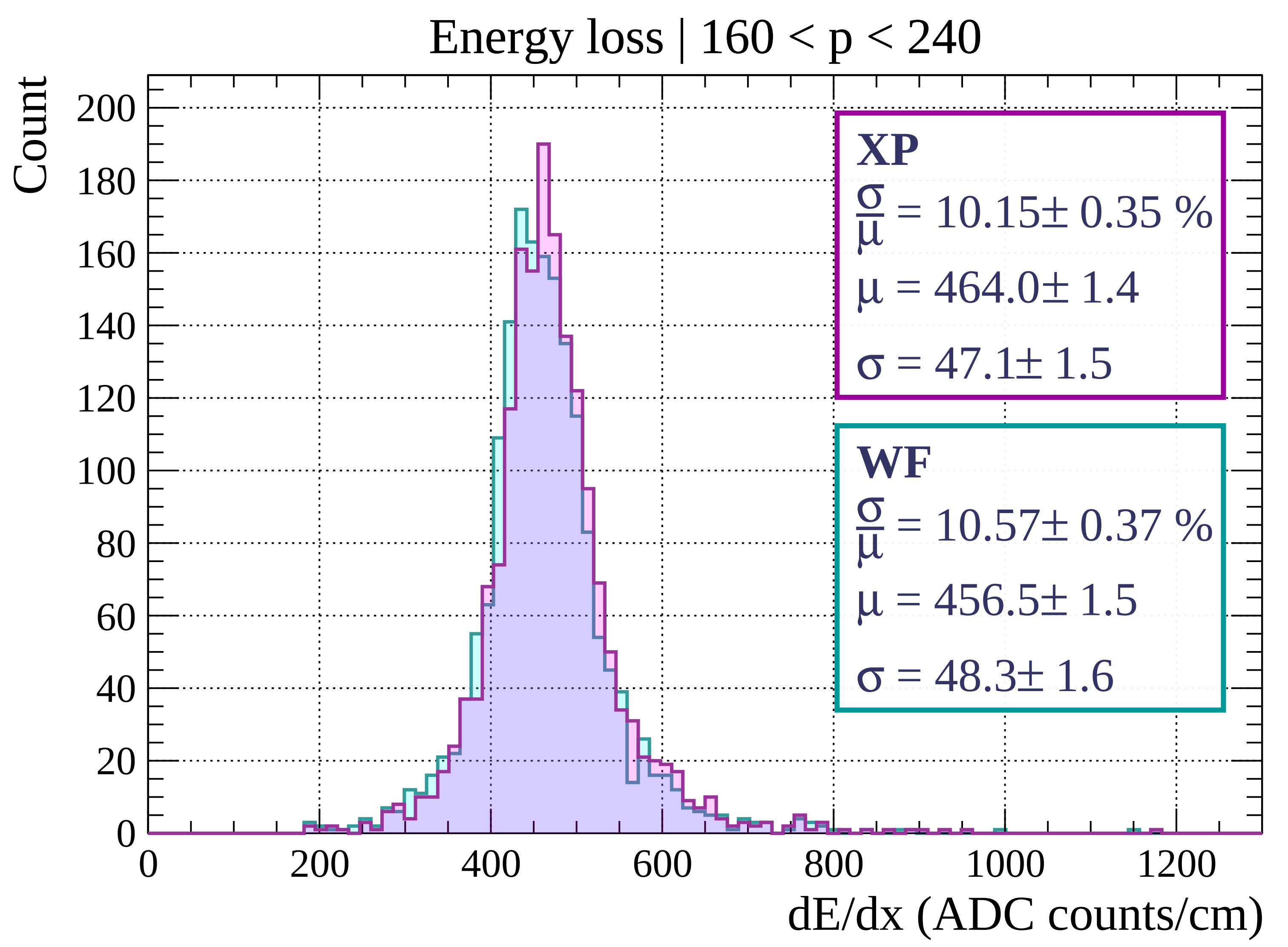

χ^2/NDF
R
$\times 10^3$



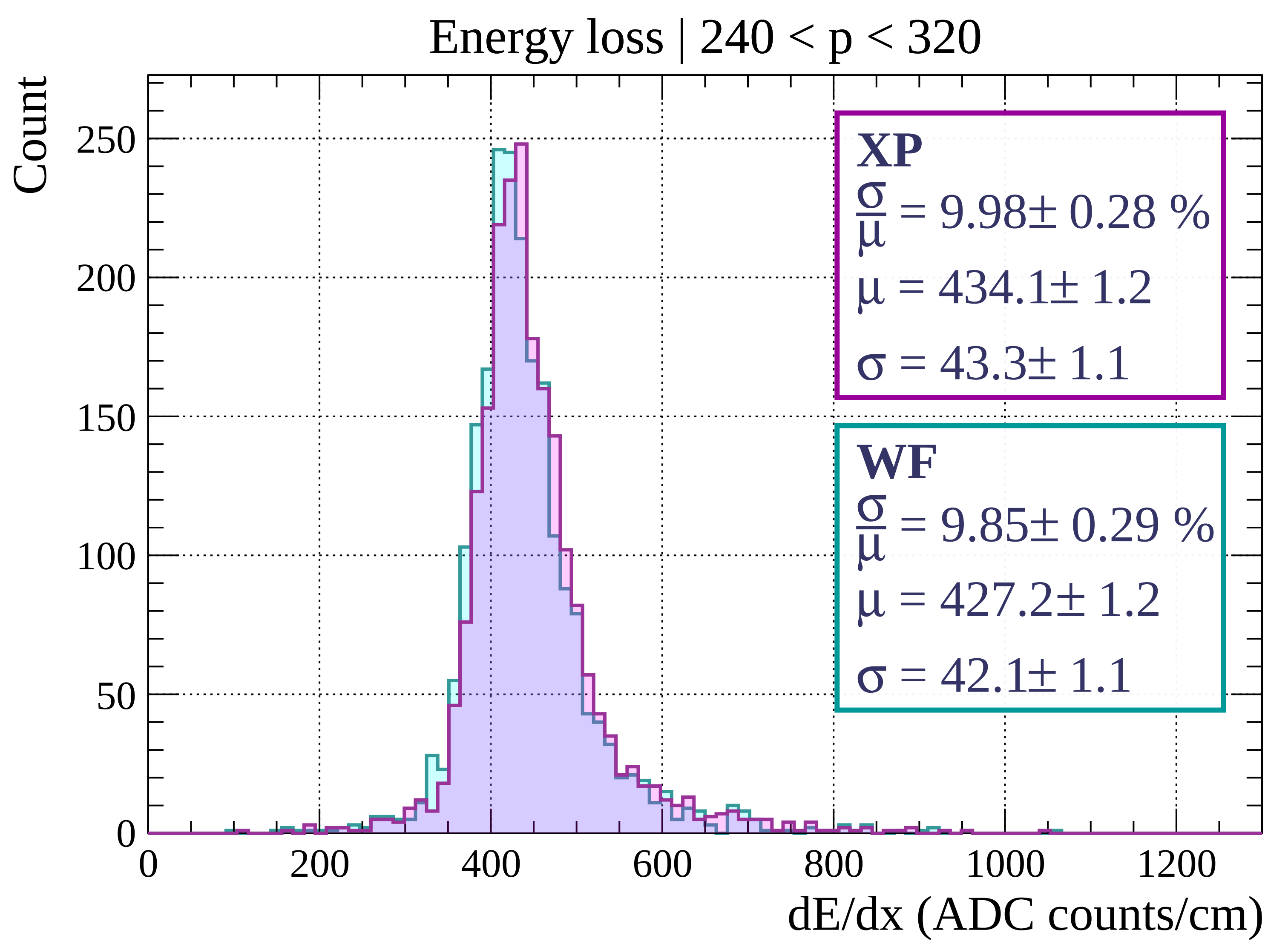
Energy loss | 0 < p < 80
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 12.57± 0.47 %
μ = 539.6± 2.3
σ = 67.8± 2.2
WF
$\frac{\sigma}{\mu}$ = 13.07± 0.42 %
μ = 527.7± 2.2
σ = 69.0± 1.9



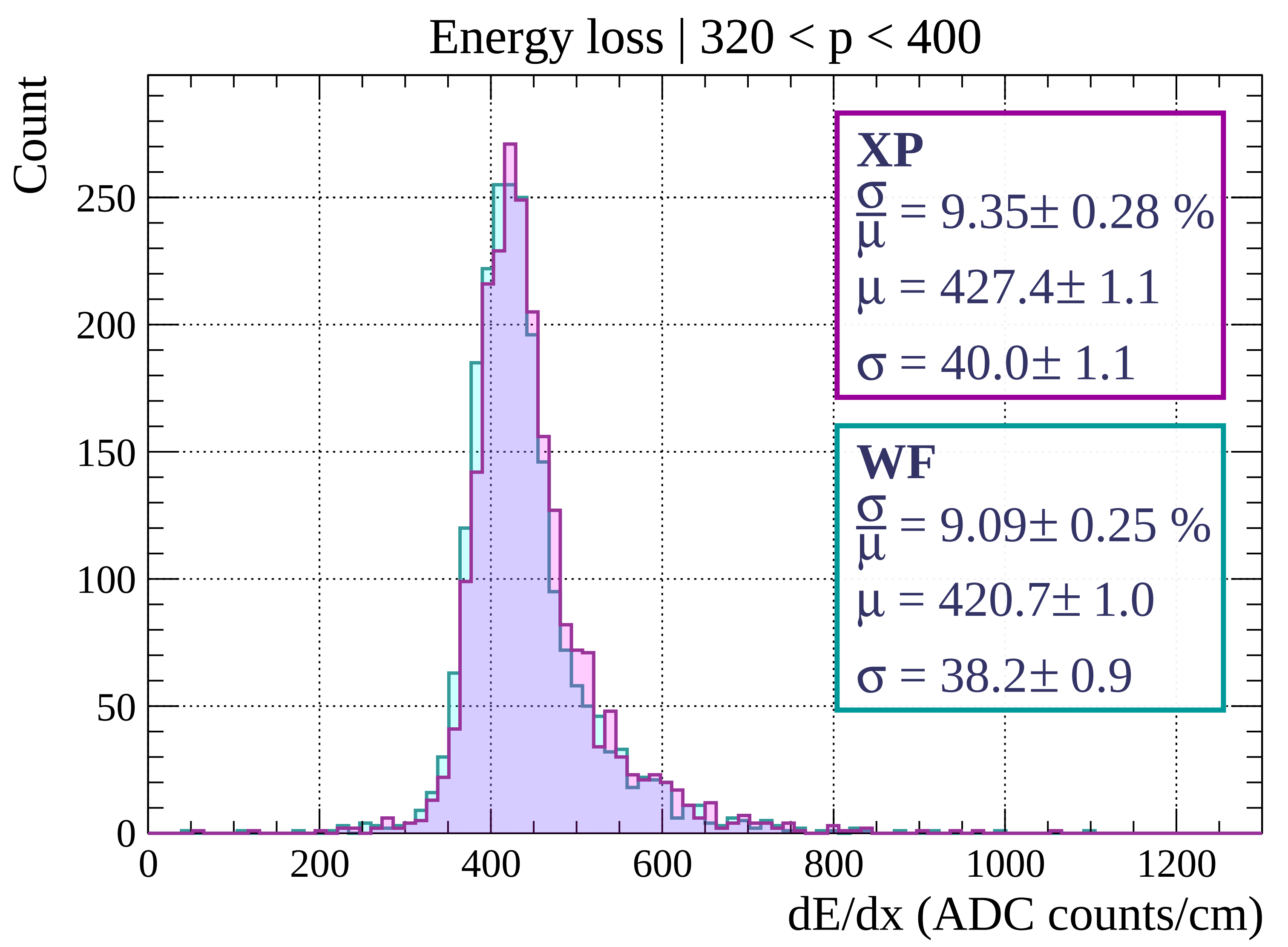
Energy loss | 80 < p < 160
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 14.82± 0.77 %
μ = 577.9± 3.5
σ = 85.6± 3.9
WF
σ/μ = 14.68± 0.73 %
μ = 571.6± 3.5
σ = 83.9± 3.7

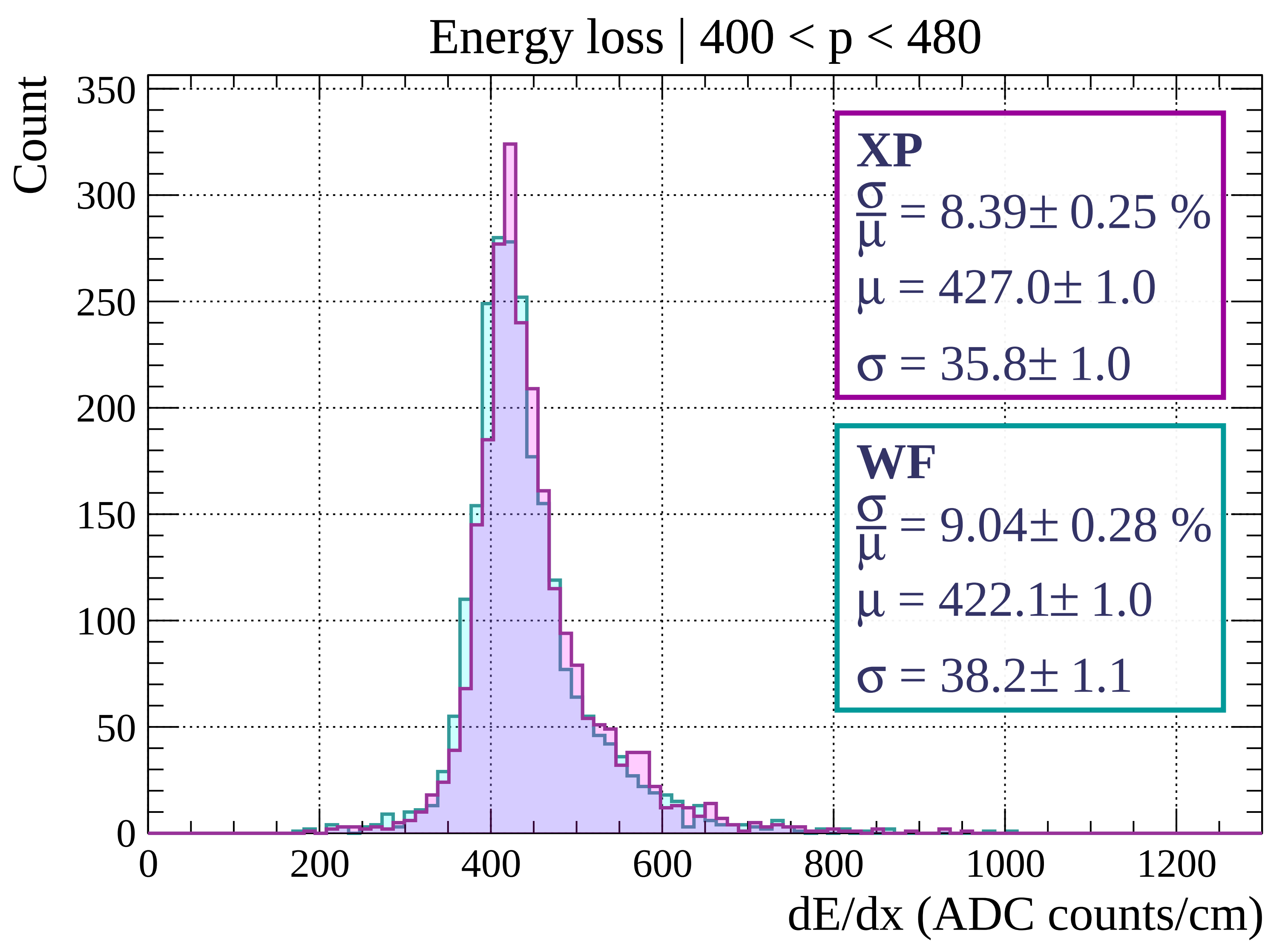


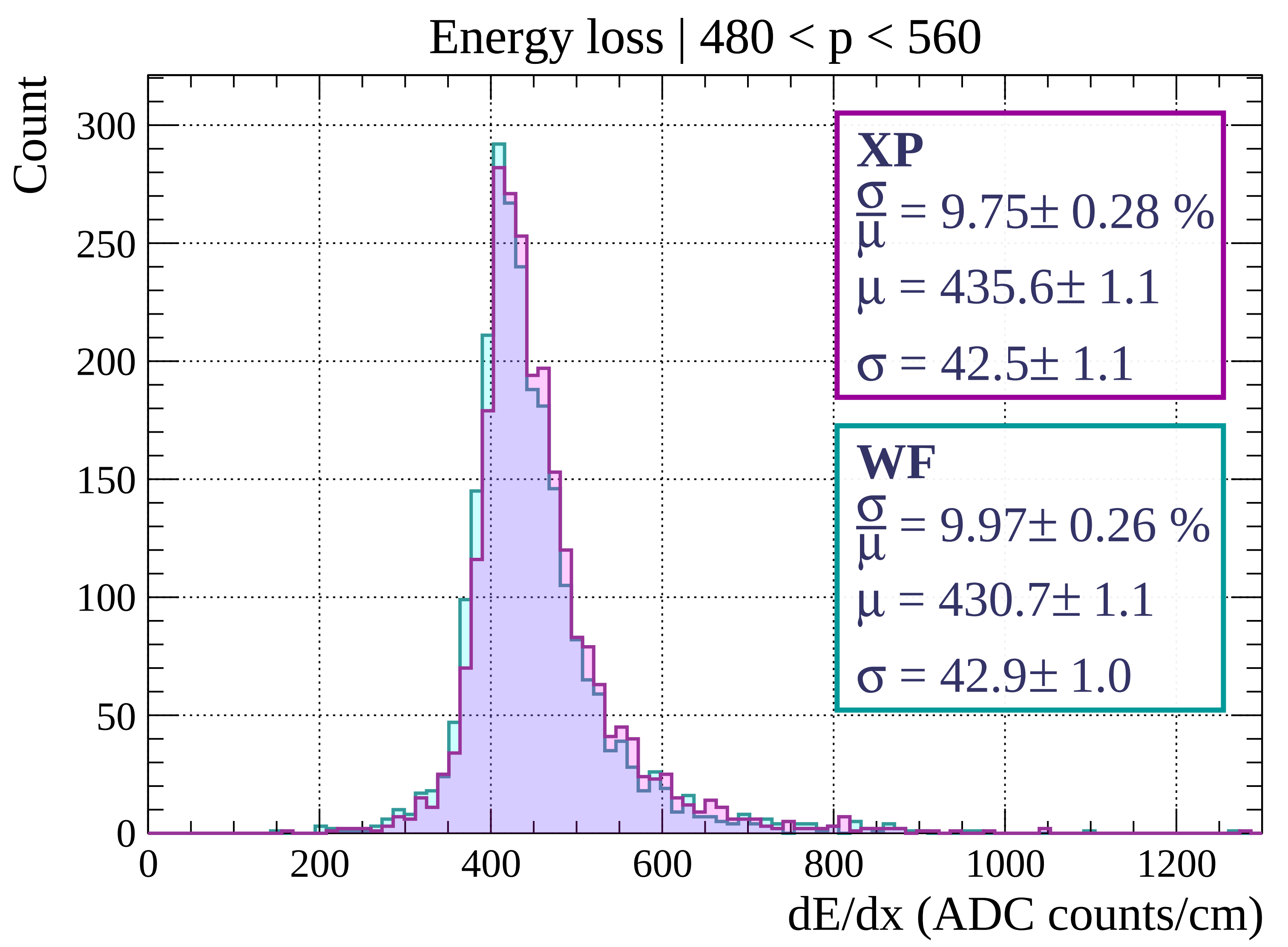
Energy loss | 160 < p < 240
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
XP
σ/μ = 10.15± 0.35 %
μ = 464.0± 1.4
σ = 47.1± 1.5
WF
σ/μ = 10.57± 0.37 %
μ = 456.5± 1.5
σ = 48.3± 1.6



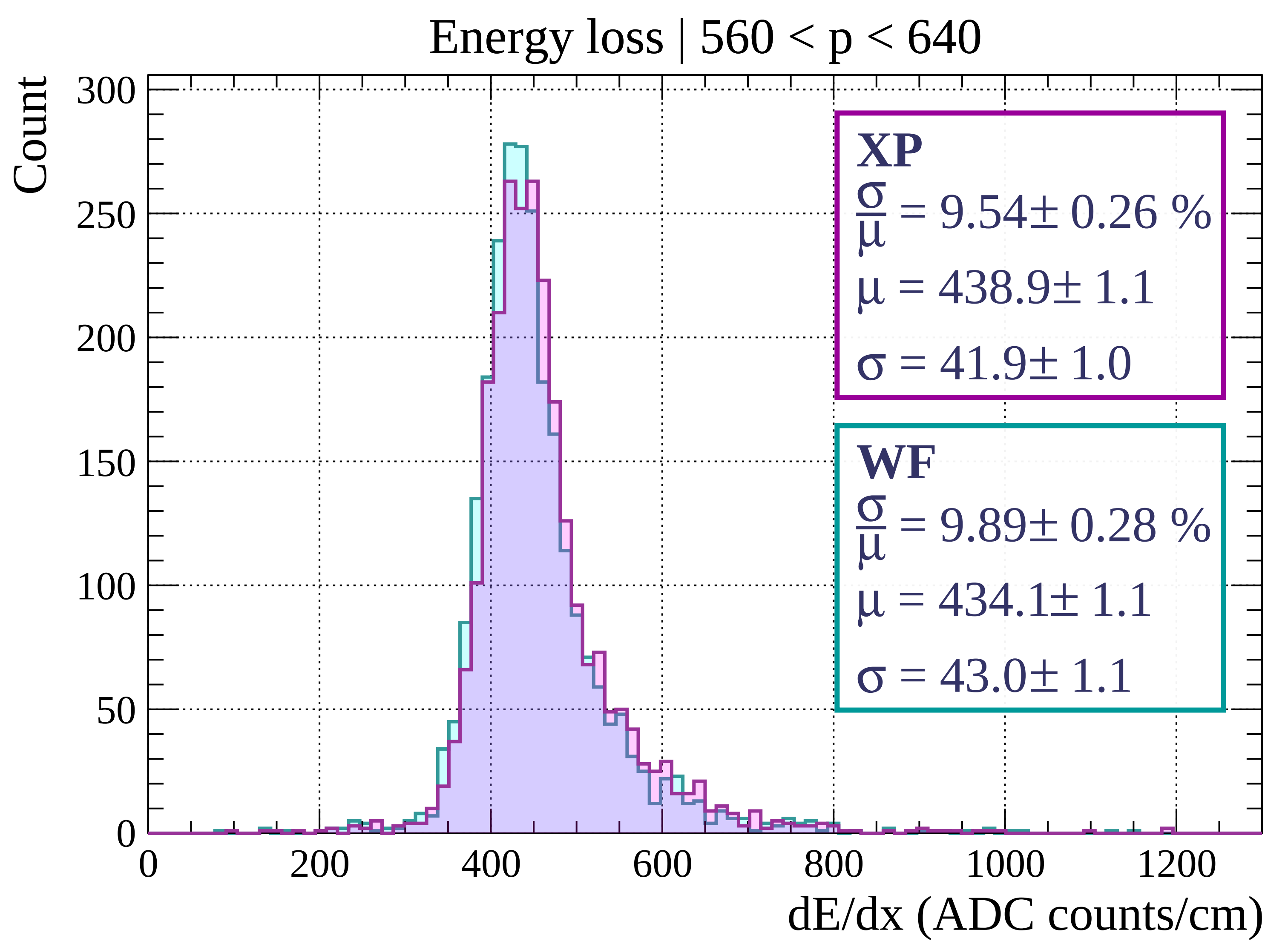
Energy loss | 240 < p < 320
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
σ/μ = 9.98± 0.28 %
μ = 434.1± 1.2
σ = 43.3± 1.1
WF
σ/μ = 9.85± 0.29 %
μ = 427.2± 1.2
σ = 42.1± 1.1


Energy loss | 320 < p < 400
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 9.35± 0.28 %
μ = 427.4± 1.1
σ = 40.0± 1.1
WF
σ/μ = 9.09± 0.25 %
μ = 420.7± 1.0
σ = 38.2± 0.9
0
50
100
150
200
250
0
200
400
600
800
1000
1200

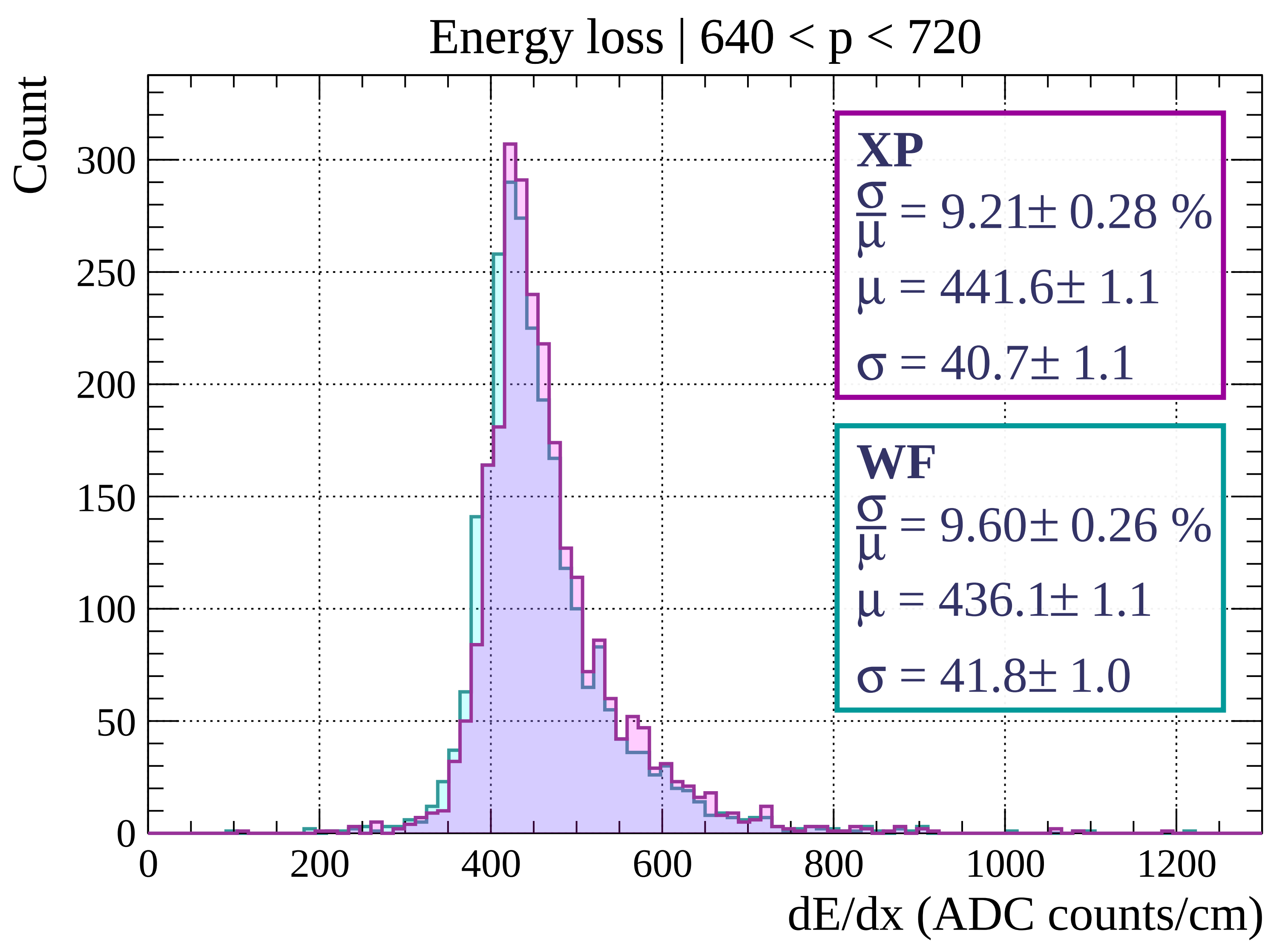

Energy loss | 400 < p < 480
Count
350
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 8.39± 0.25 %
μ = 427.0± 1.0
σ = 35.8± 1.0
WF
$\frac{\sigma}{\mu}$ = 9.04± 0.28 %
μ = 422.1± 1.0
σ = 38.2± 1.1



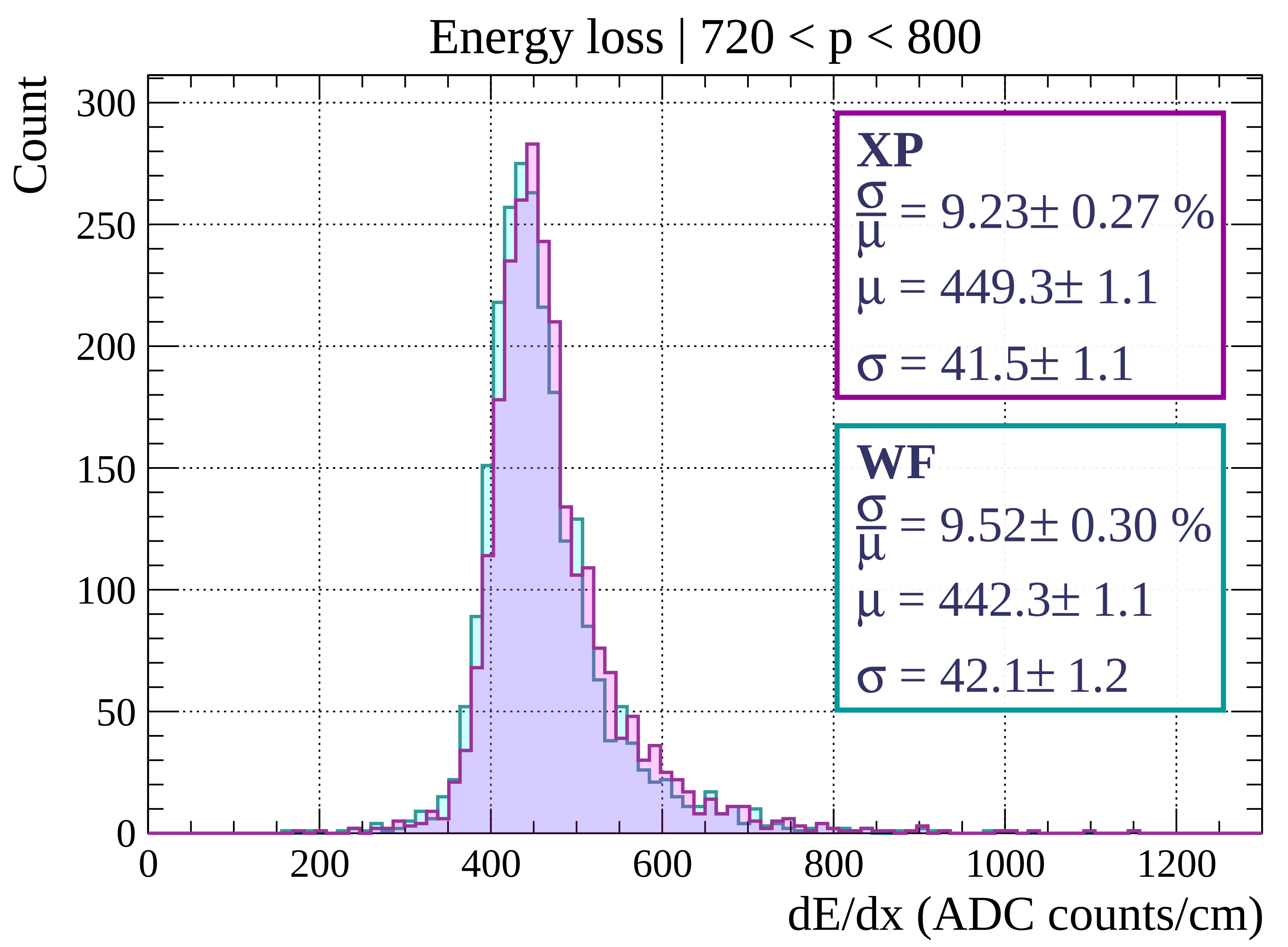
Energy loss | 480 < p < 560
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 9.75± 0.28 %
μ = 435.6± 1.1
σ = 42.5± 1.1
WF
$\frac{\sigma}{\mu}$ = 9.97± 0.26 %
μ = 430.7± 1.1
σ = 42.9± 1.0

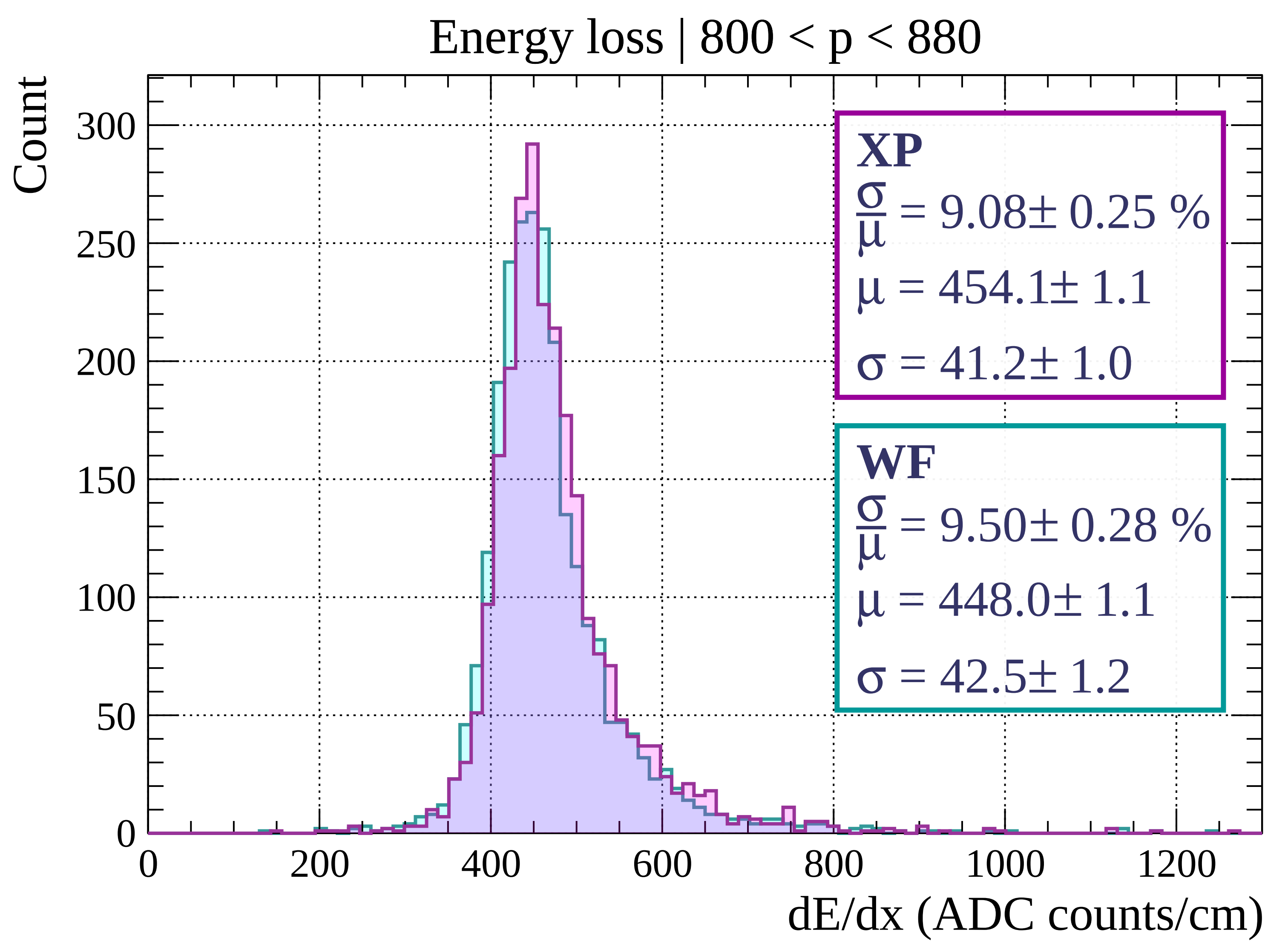


Energy loss | 560 < p < 640
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 9.54± 0.26 %
μ = 438.9± 1.1
σ = 41.9± 1.0
WF
$\frac{\sigma}{\mu}$ = 9.89± 0.28 %
μ = 434.1± 1.1
σ = 43.0± 1.1

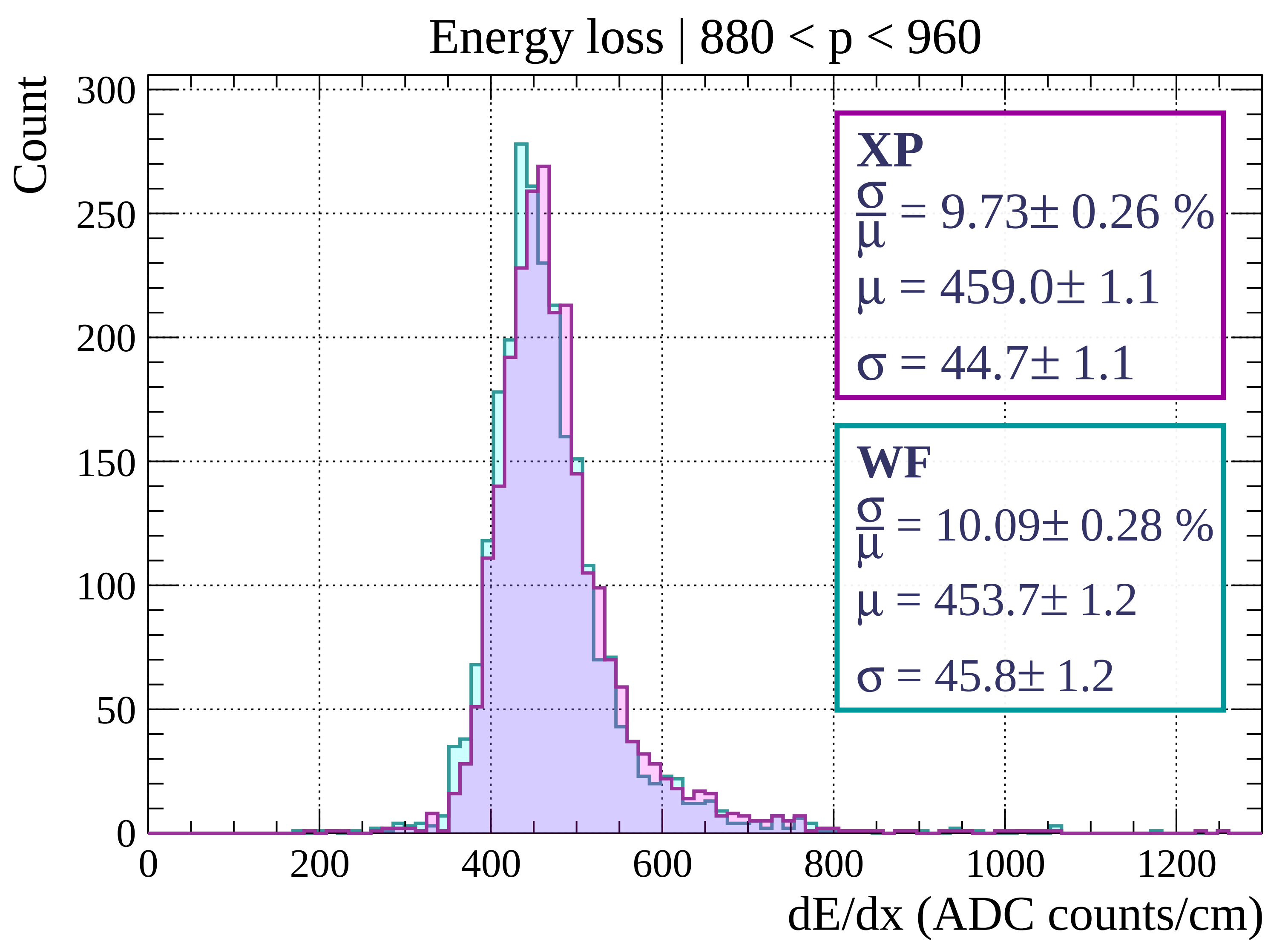


Energy loss | 640 < p < 720
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 9.21± 0.28 %
μ = 441.6± 1.1
σ = 40.7± 1.1
WF
$\frac{\sigma}{\mu}$ = 9.60± 0.26 %
μ = 436.1± 1.1
σ = 41.8± 1.0

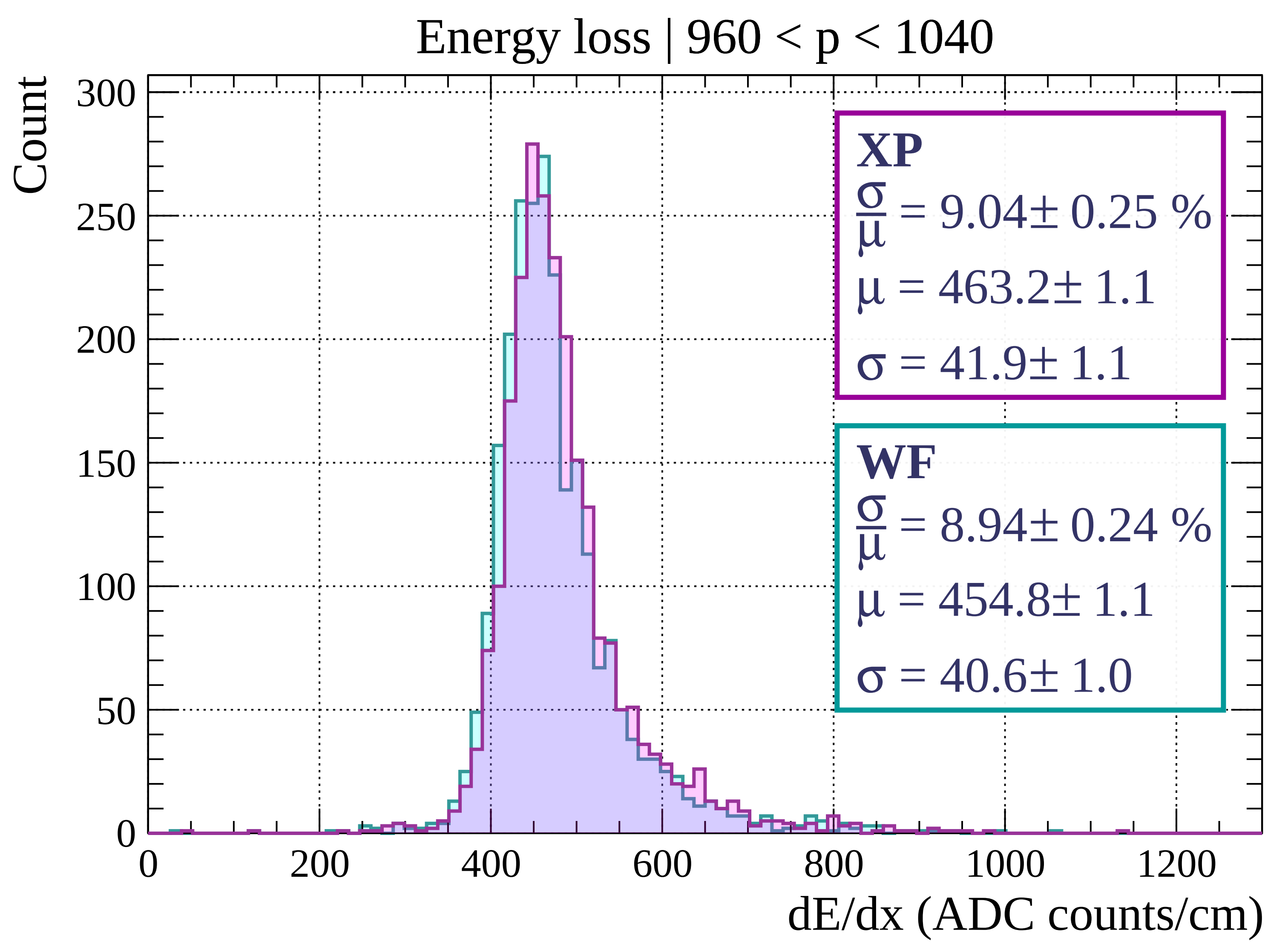

Energy loss | 720 < p < 800
Count
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.23± 0.27 %
μ = 449.3± 1.1
σ = 41.5± 1.1
WF
σ/μ = 9.52± 0.30 %
μ = 442.3± 1.1
σ = 42.1± 1.2



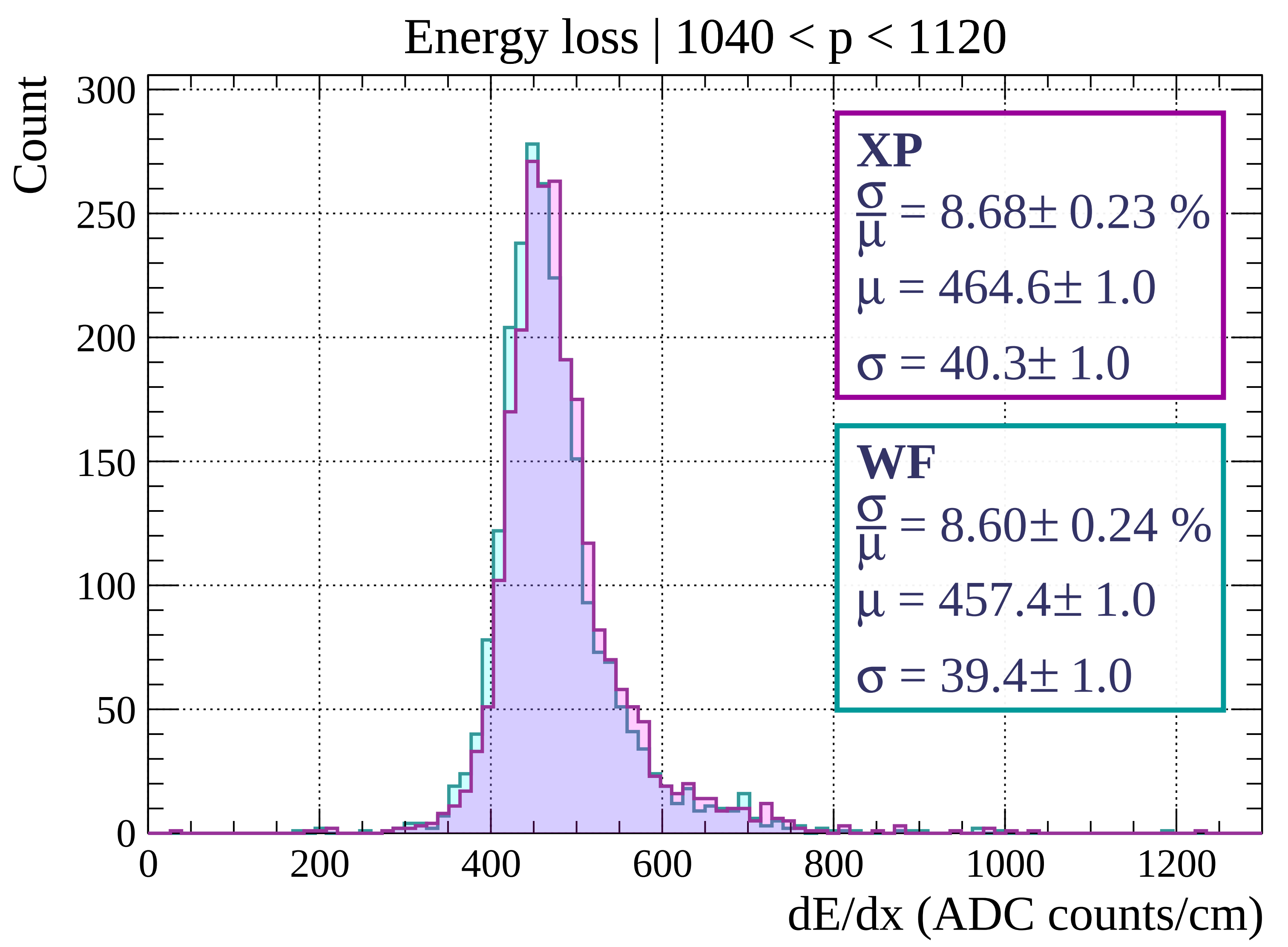
Energy loss | 800 < p < 880
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 9.08± 0.25 %
μ = 454.1± 1.1
σ = 41.2± 1.0
WF
$\frac{\sigma}{\mu}$ = 9.50± 0.28 %
μ = 448.0± 1.1
σ = 42.5± 1.2
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200



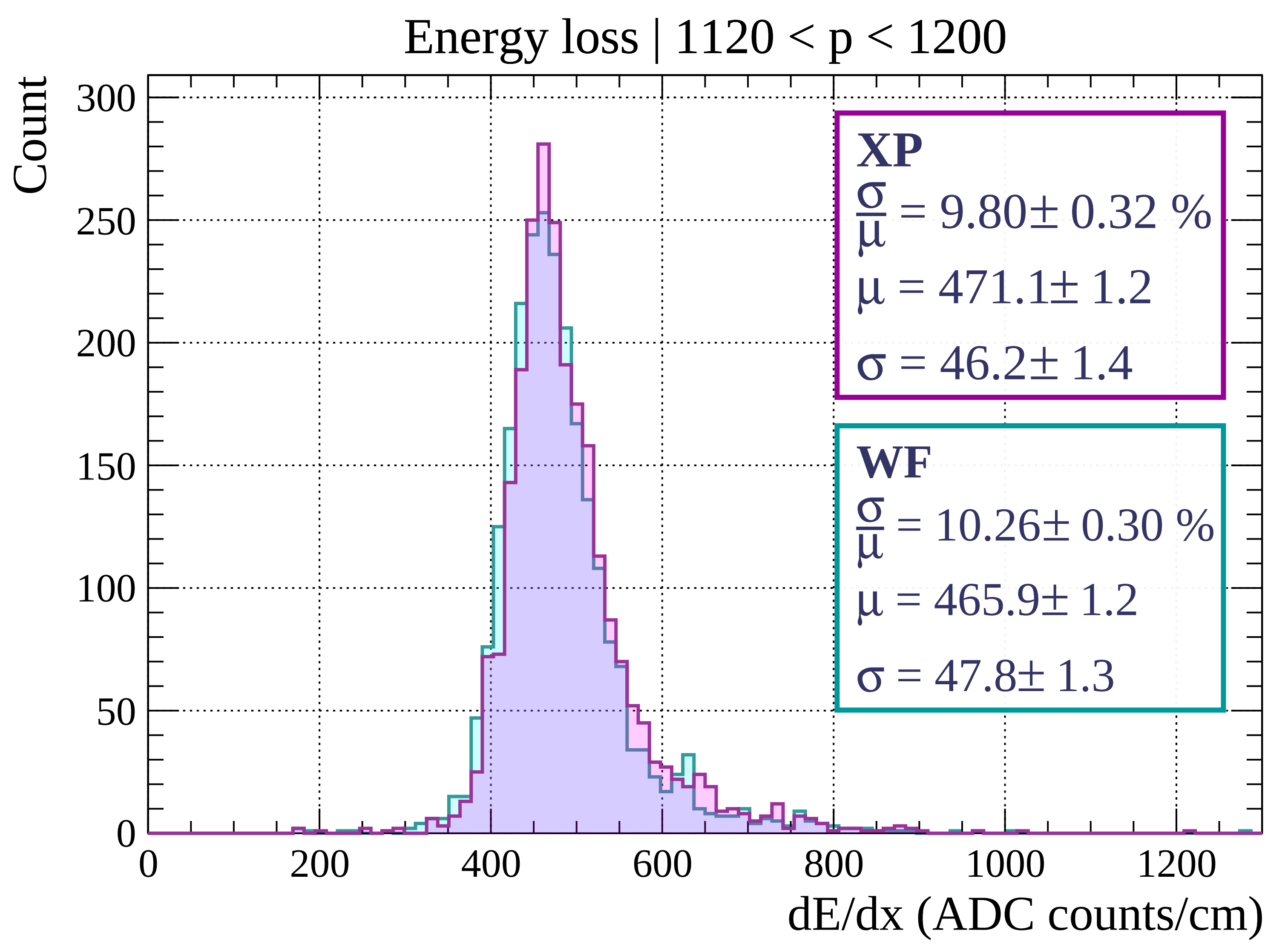
Energy loss | 880 < p < 960
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 9.73± 0.26 %
μ = 459.0± 1.1
σ = 44.7± 1.1
WF
$\frac{\sigma}{\mu}$ = 10.09± 0.28 %
μ = 453.7± 1.2
σ = 45.8± 1.2

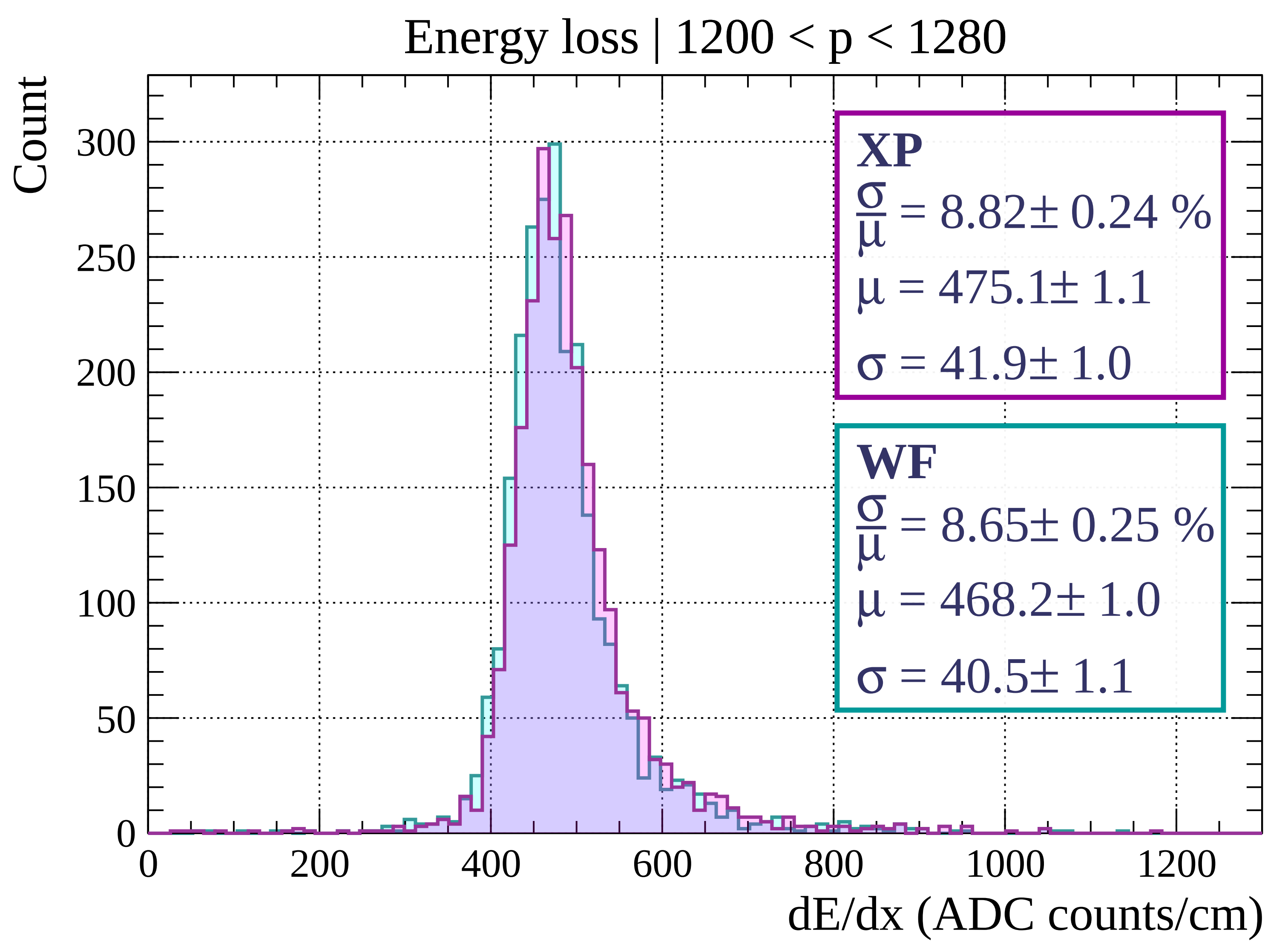


Energy loss | 960 < p < 1040
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 9.04± 0.25 %
μ = 463.2± 1.1
σ = 41.9± 1.1
WF
$\frac{\sigma}{\mu}$ = 8.94± 0.24 %
μ = 454.8± 1.1
σ = 40.6± 1.0

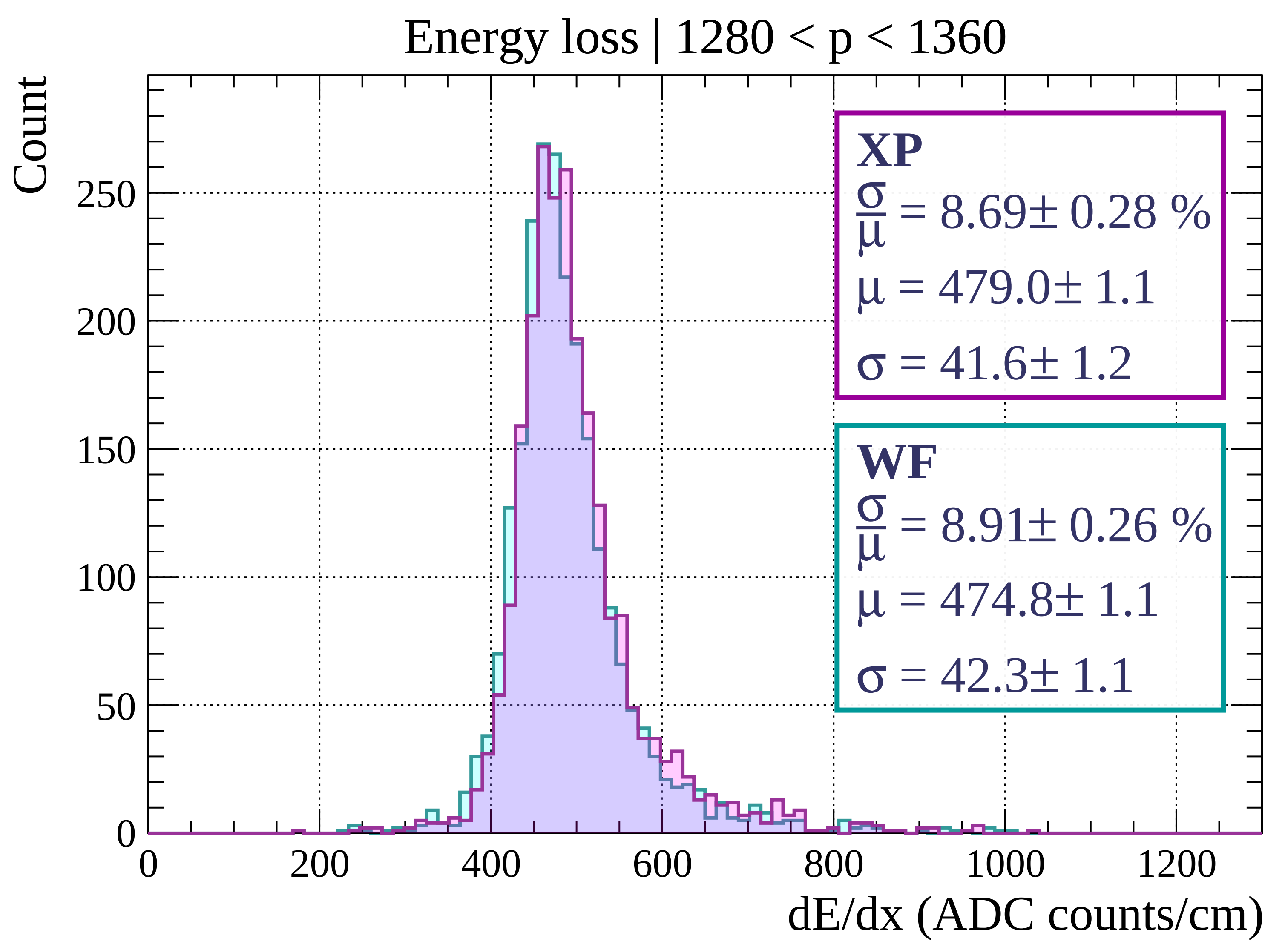


Energy loss | 1040 < p < 1120
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
50
100
150
200
250
300
XP
$\frac{\sigma}{\mu}$ = 8.68 ± 0.23 %
μ = 464.6 ± 1.0
σ = 40.3 ± 1.0
WF
$\frac{\sigma}{\mu}$ = 8.60 ± 0.24 %
μ = 457.4 ± 1.0
σ = 39.4 ± 1.0

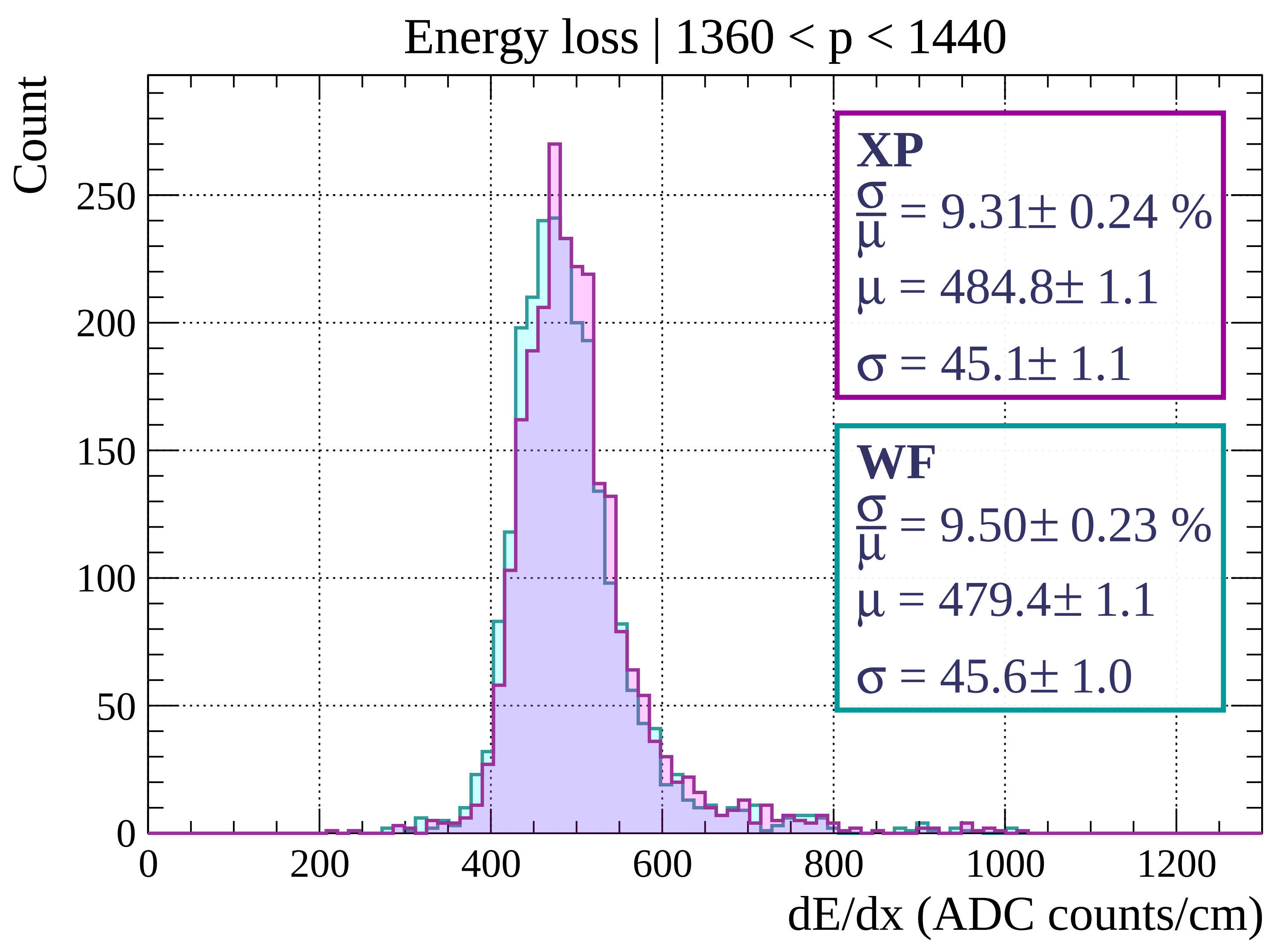

Energy loss | 1120 < p < 1200
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
50
100
150
200
250
300
XP
σ/μ = 9.80± 0.32 %
μ = 471.1± 1.2
σ = 46.2± 1.4
WF
σ/μ = 10.26± 0.30 %
μ = 465.9± 1.2
σ = 47.8± 1.3



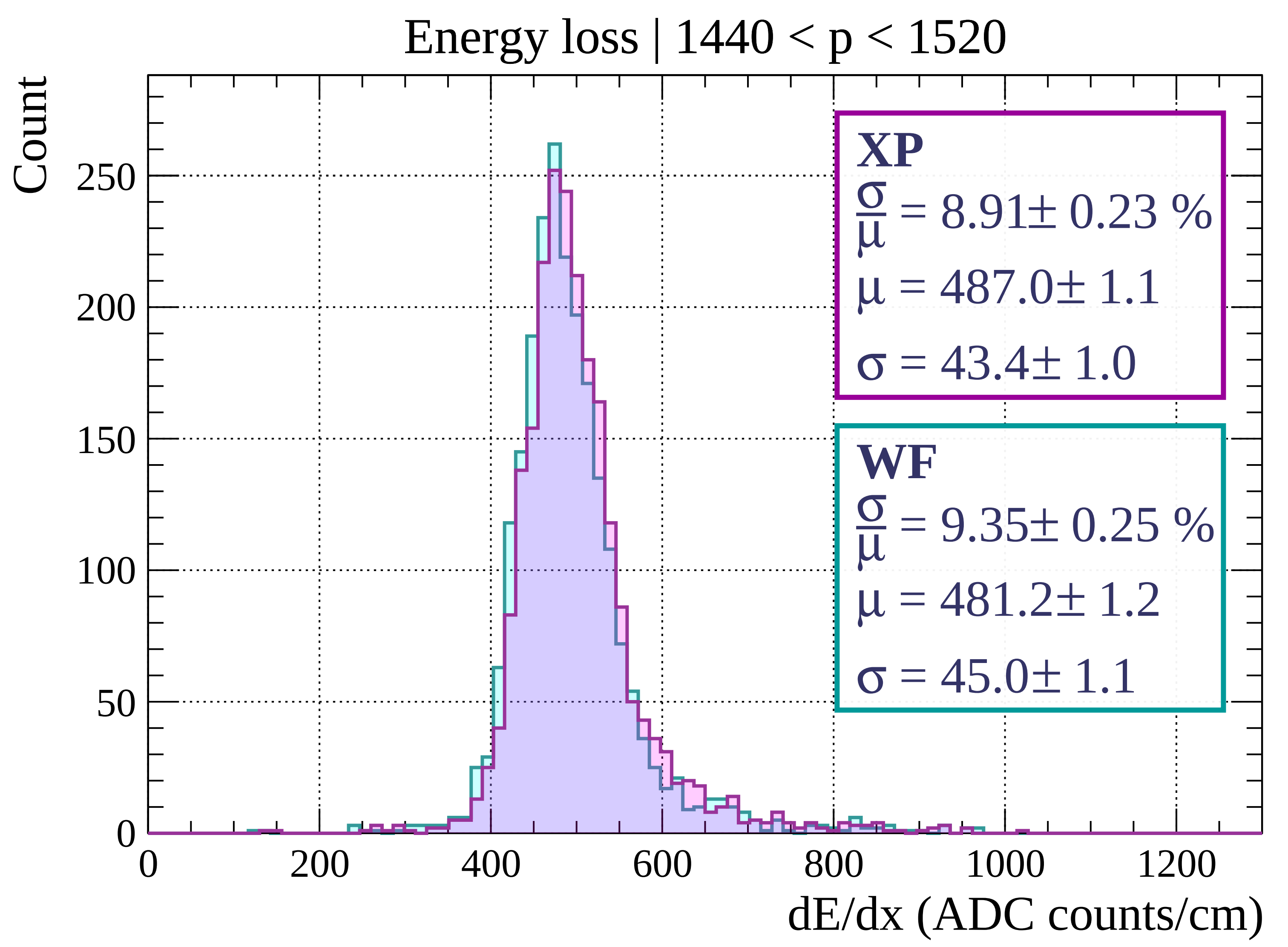
Energy loss | 1200 < p < 1280
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 8.82± 0.24 %
μ = 475.1± 1.1
σ = 41.9± 1.0
WF
σ/μ = 8.65± 0.25 %
μ = 468.2± 1.0
σ = 40.5± 1.1



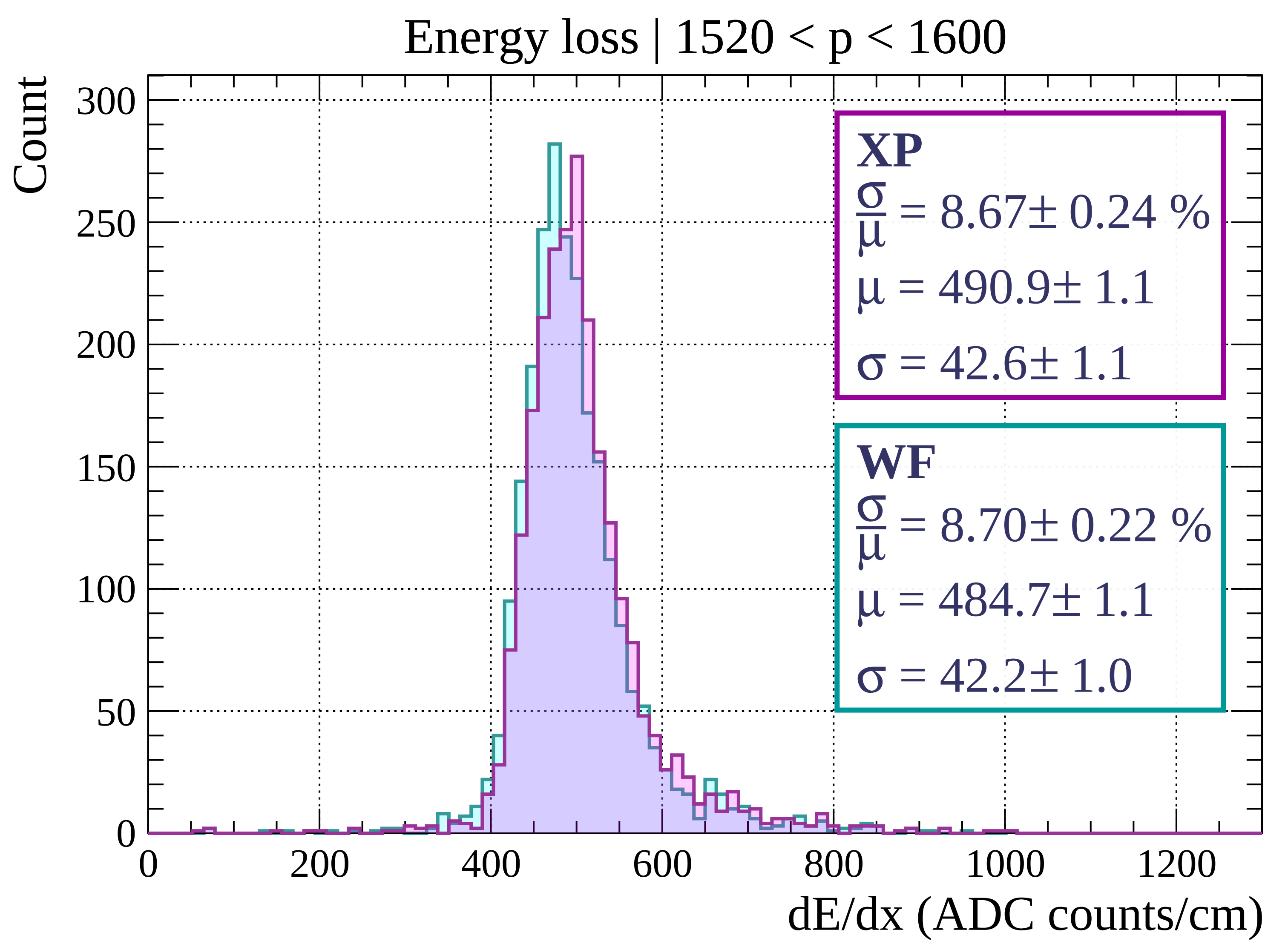
Energy loss | 1280 < p < 1360
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 8.69± 0.28 %
μ = 479.0± 1.1
σ = 41.6± 1.2
WF
$\frac{\sigma}{\mu}$ = 8.91± 0.26 %
μ = 474.8± 1.1
σ = 42.3± 1.1



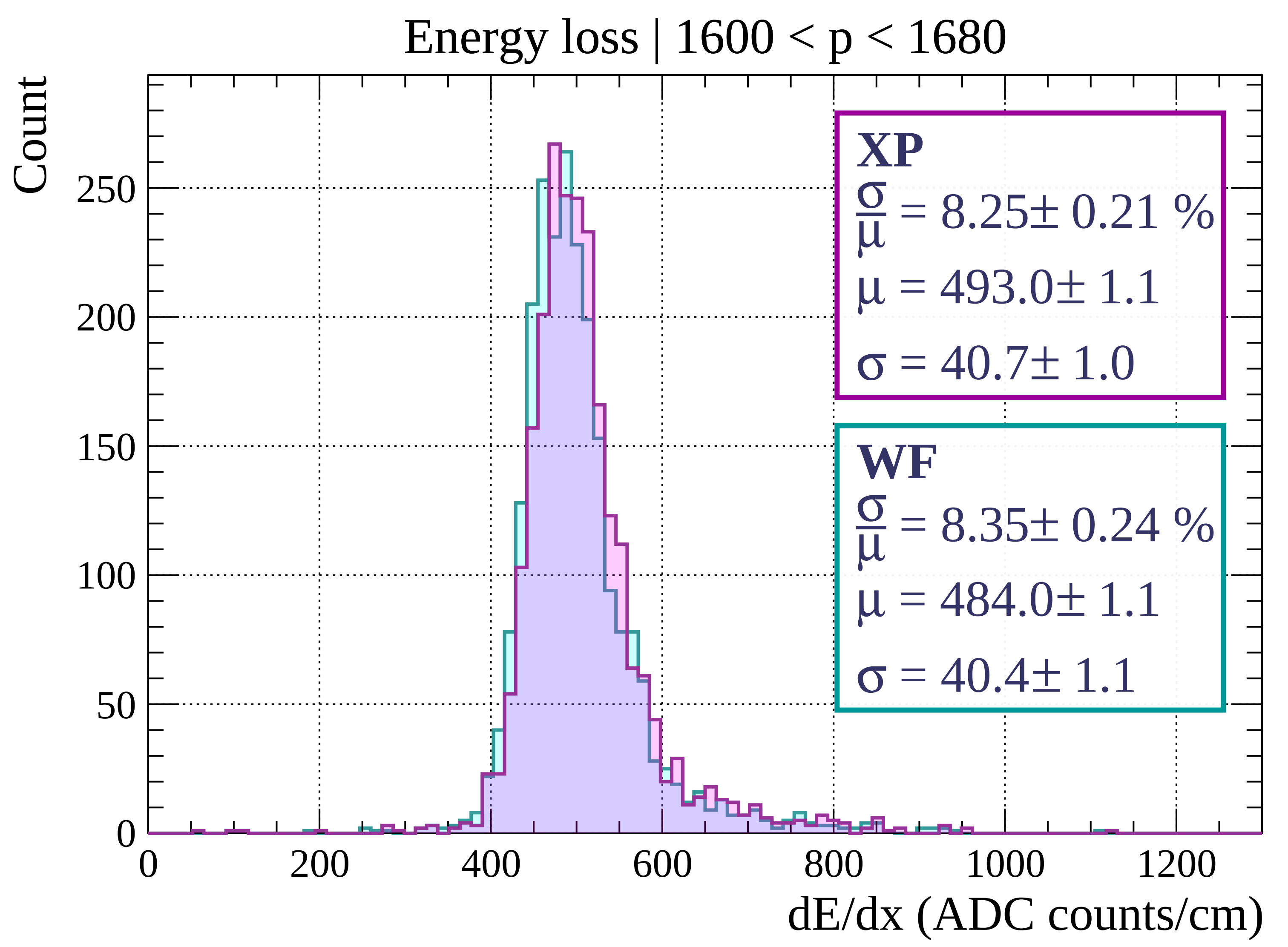
Energy loss | 1360 < p < 1440
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
σ/μ = 9.31± 0.24 %
μ = 484.8± 1.1
σ = 45.1± 1.1
WF
σ/μ = 9.50± 0.23 %
μ = 479.4± 1.1
σ = 45.6± 1.0

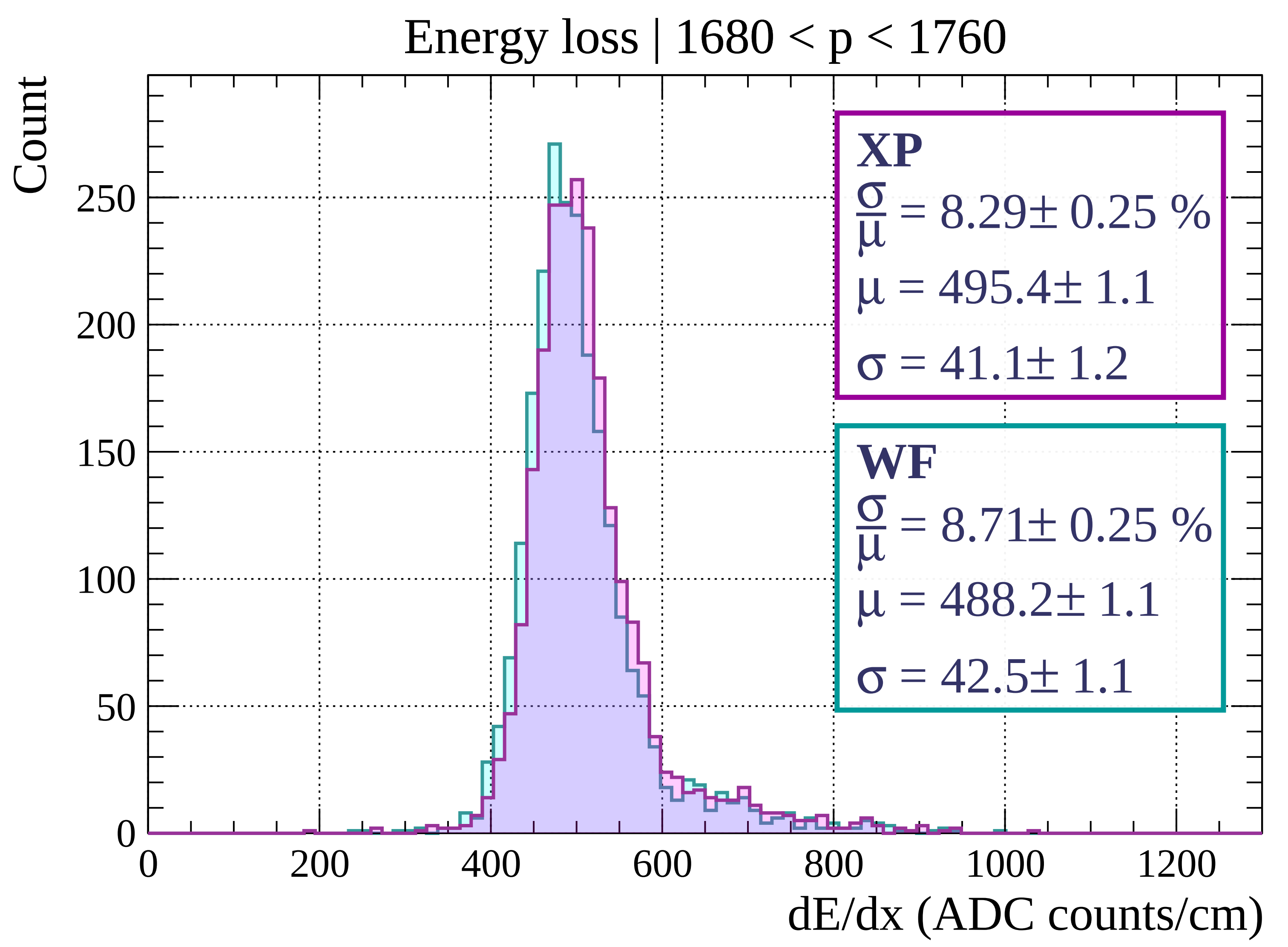


Energy loss | 1440 < p < 1520
Count
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 8.91± 0.23 %
μ = 487.0± 1.1
σ = 43.4± 1.0
WF
$\frac{\sigma}{\mu}$ = 9.35± 0.25 %
μ = 481.2± 1.2
σ = 45.0± 1.1

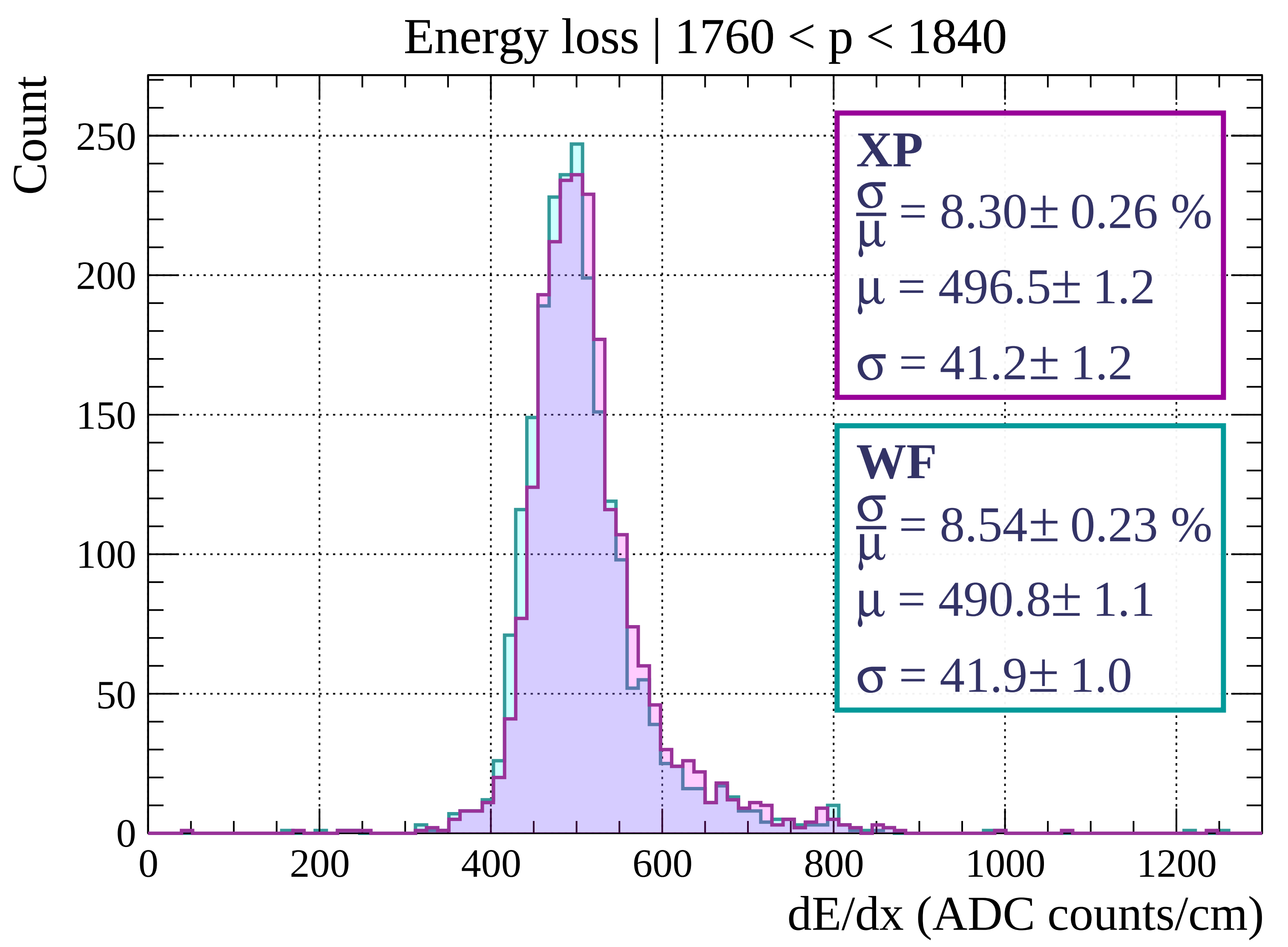


Energy loss | 1520 < p < 1600
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
50
100
150
200
250
300
XP
σ/μ = 8.67± 0.24 %
μ = 490.9± 1.1
σ = 42.6± 1.1
WF
σ/μ = 8.70± 0.22 %
μ = 484.7± 1.1
σ = 42.2± 1.0

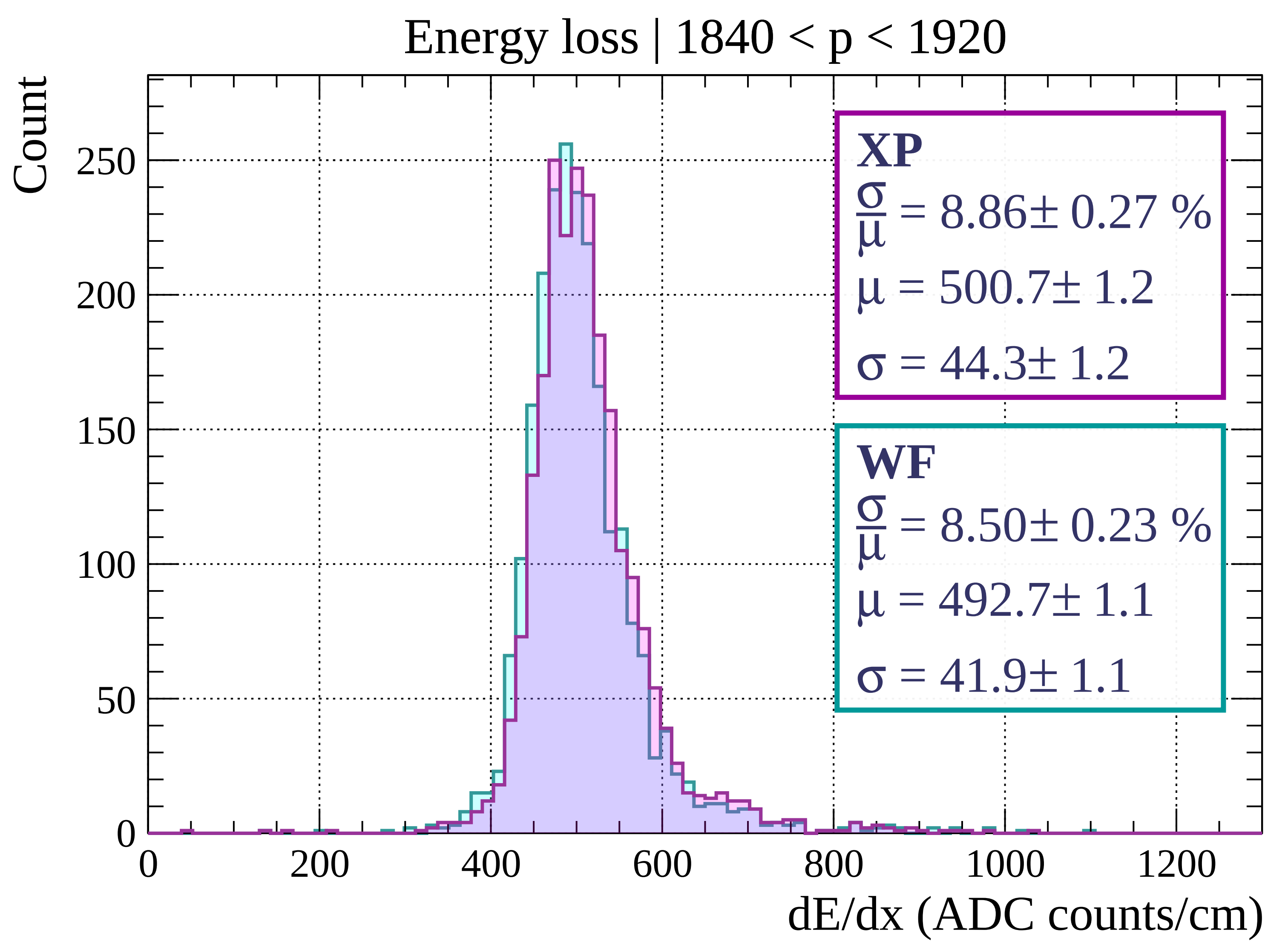

Energy loss | 1600 < p < 1680
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 8.25± 0.21 %
μ = 493.0± 1.1
σ = 40.7± 1.0
WF
$\frac{\sigma}{\mu}$ = 8.35± 0.24 %
μ = 484.0± 1.1
σ = 40.4± 1.1



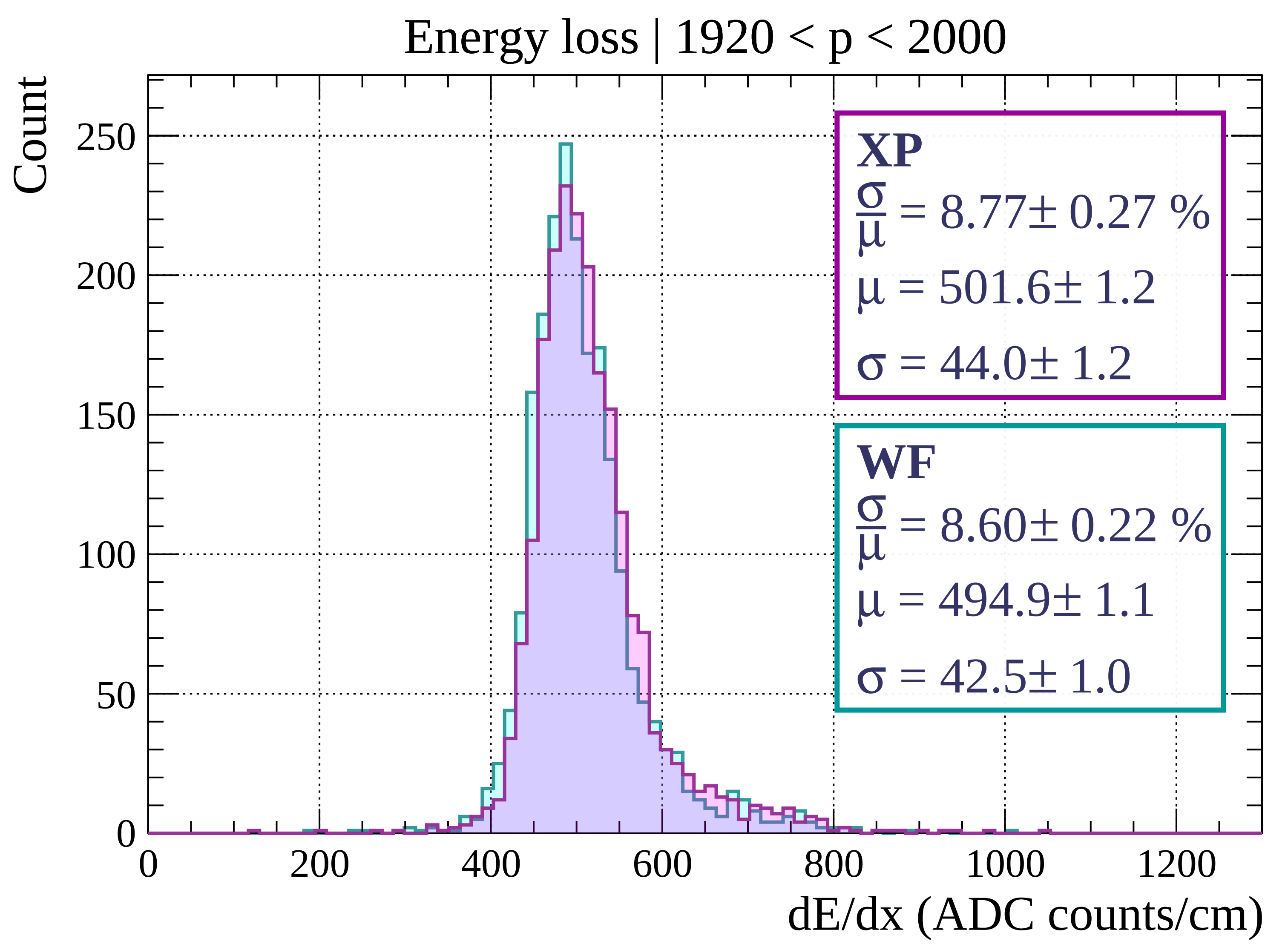
Energy loss | 1680 < p < 1760
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 8.29± 0.25 %
μ = 495.4± 1.1
σ = 41.1± 1.2
WF
$\frac{\sigma}{\mu}$ = 8.71± 0.25 %
μ = 488.2± 1.1
σ = 42.5± 1.1



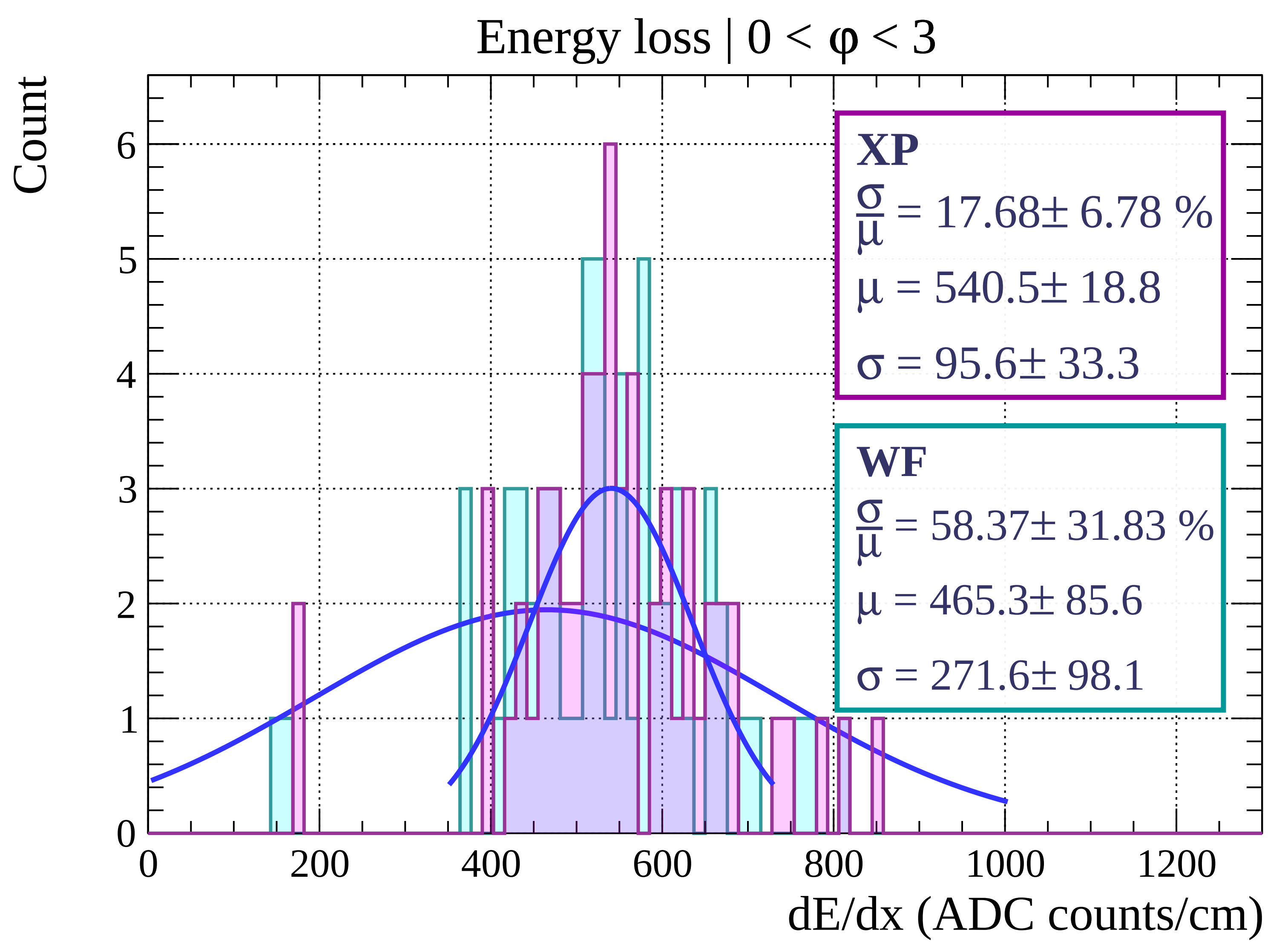
Energy loss | 1760 < p < 1840
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
50
100
150
200
250
XP
$\frac{\sigma}{\mu}$ = 8.30± 0.26 %
μ = 496.5± 1.2
σ = 41.2± 1.2
WF
$\frac{\sigma}{\mu}$ = 8.54± 0.23 %
μ = 490.8± 1.1
σ = 41.9± 1.0



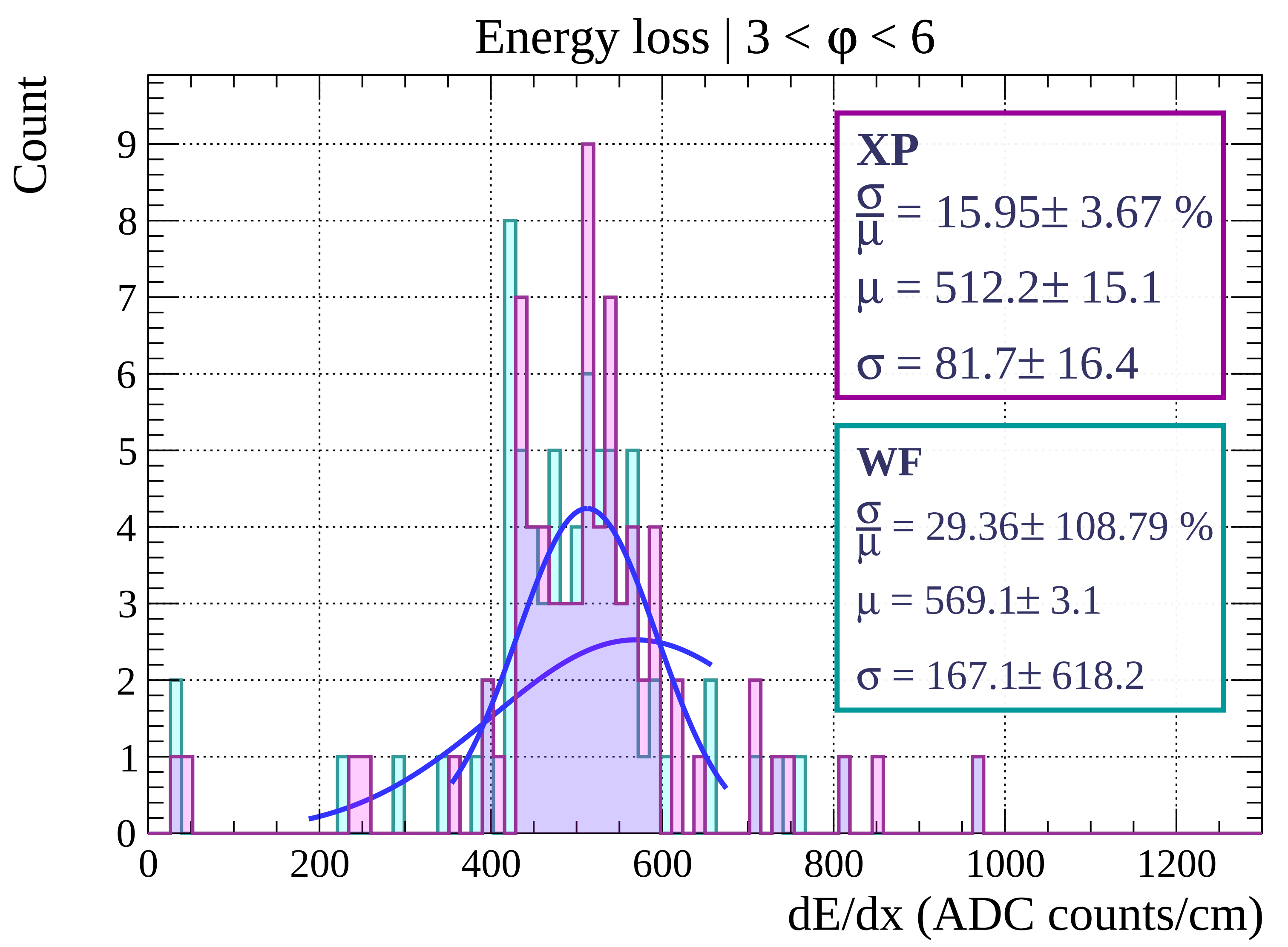
Energy loss | 1840 < p < 1920
Count
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 8.86± 0.27 %
μ = 500.7± 1.2
σ = 44.3± 1.2
WF
$\frac{\sigma}{\mu}$ = 8.50± 0.23 %
μ = 492.7± 1.1
σ = 41.9± 1.1



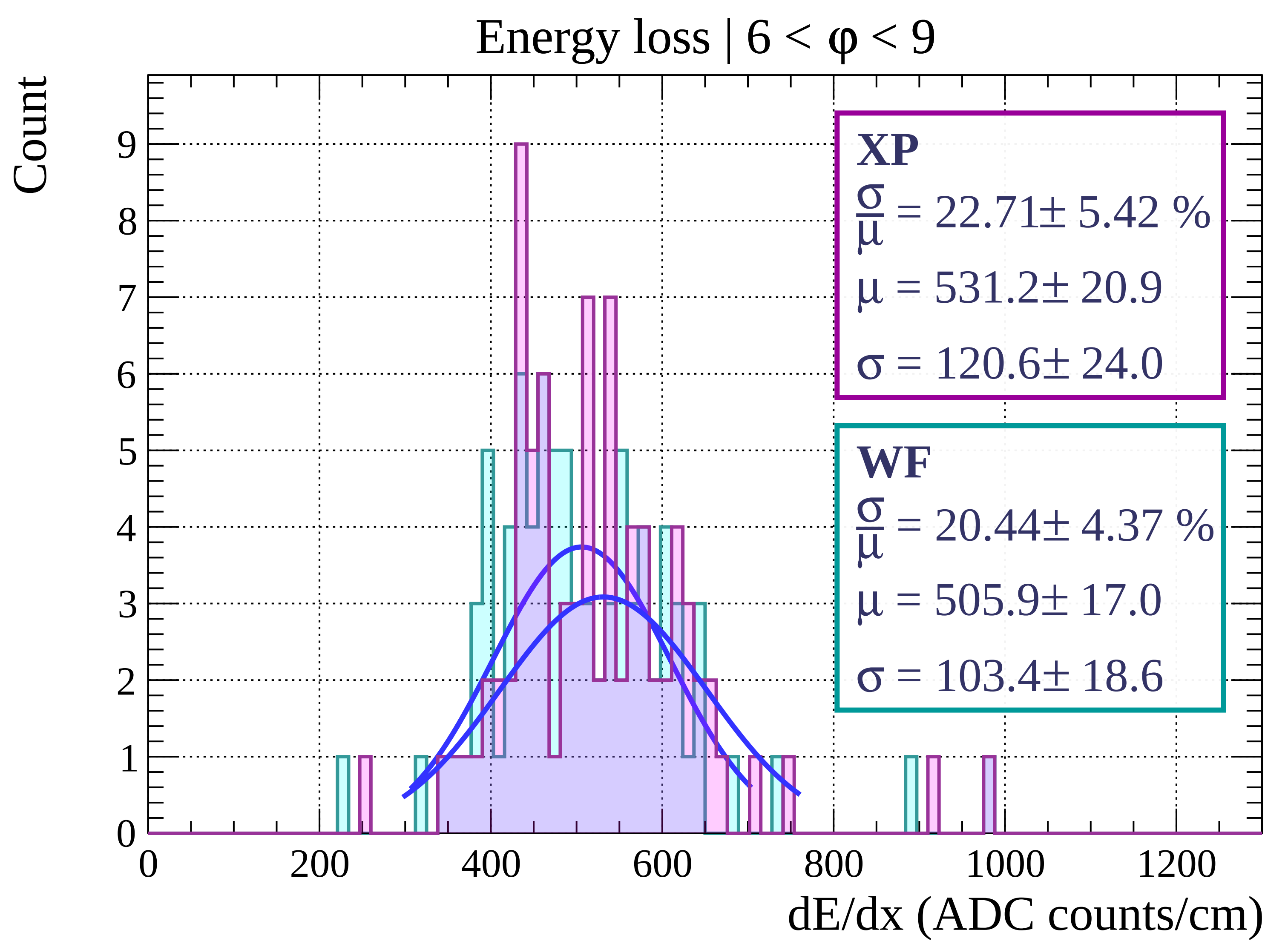
Energy loss | 1920 < p < 2000
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
50
100
150
200
250
XP
σ/μ = 8.77± 0.27 %
μ = 501.6± 1.2
σ = 44.0± 1.2
WF
σ/μ = 8.60± 0.22 %
μ = 494.9± 1.1
σ = 42.5± 1.0



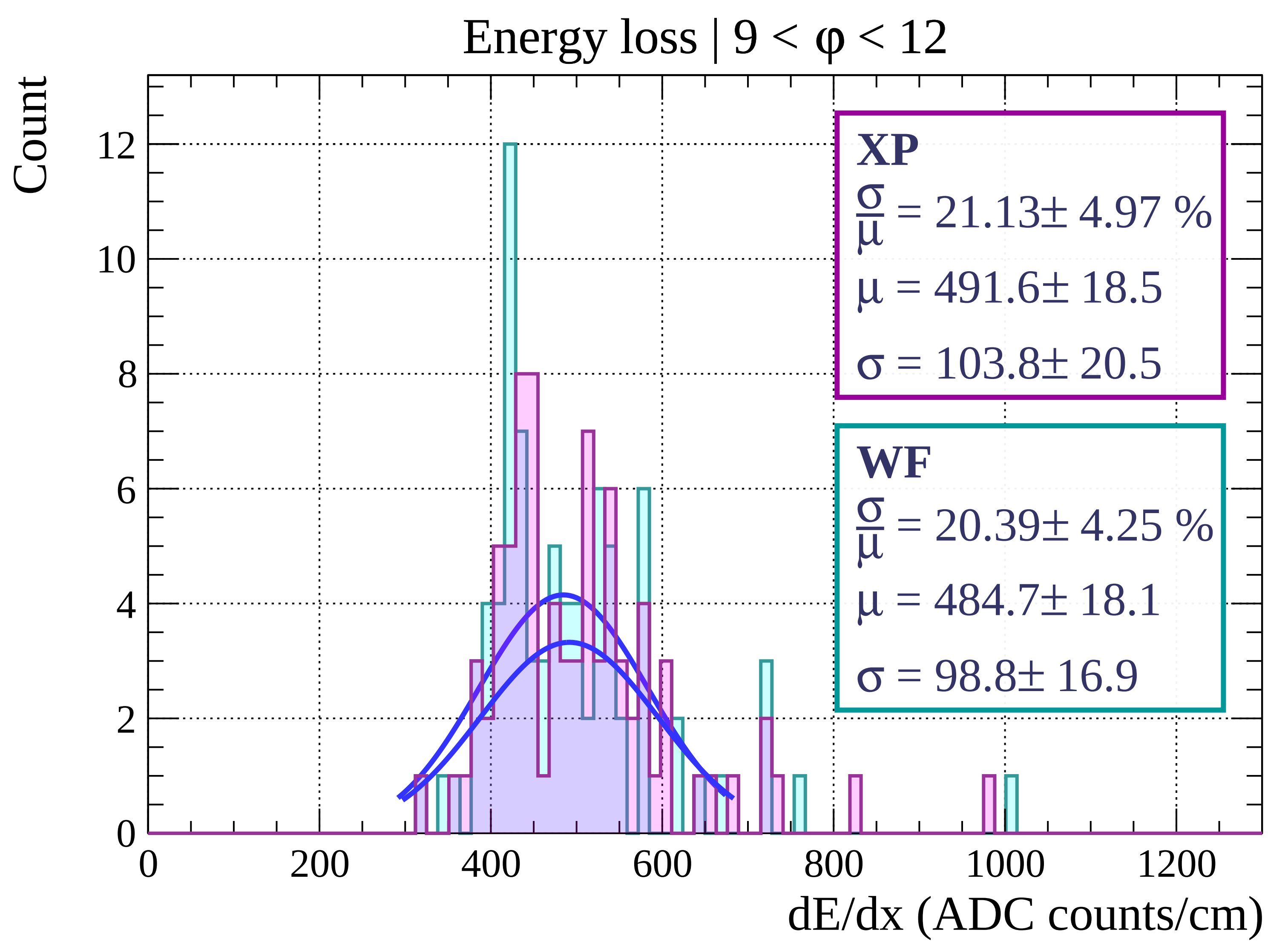
Energy loss | 0 < φ < 3
Count
dE/dx (ADC counts/cm)
0
1
2
3
4
5
6
0
200
400
600
800
1000
1200
XP
σ/μ = 17.68± 6.78 %
μ = 540.5± 18.8
σ = 95.6± 33.3
WF
σ/μ = 58.37± 31.83 %
μ = 465.3± 85.6
σ = 271.6± 98.1



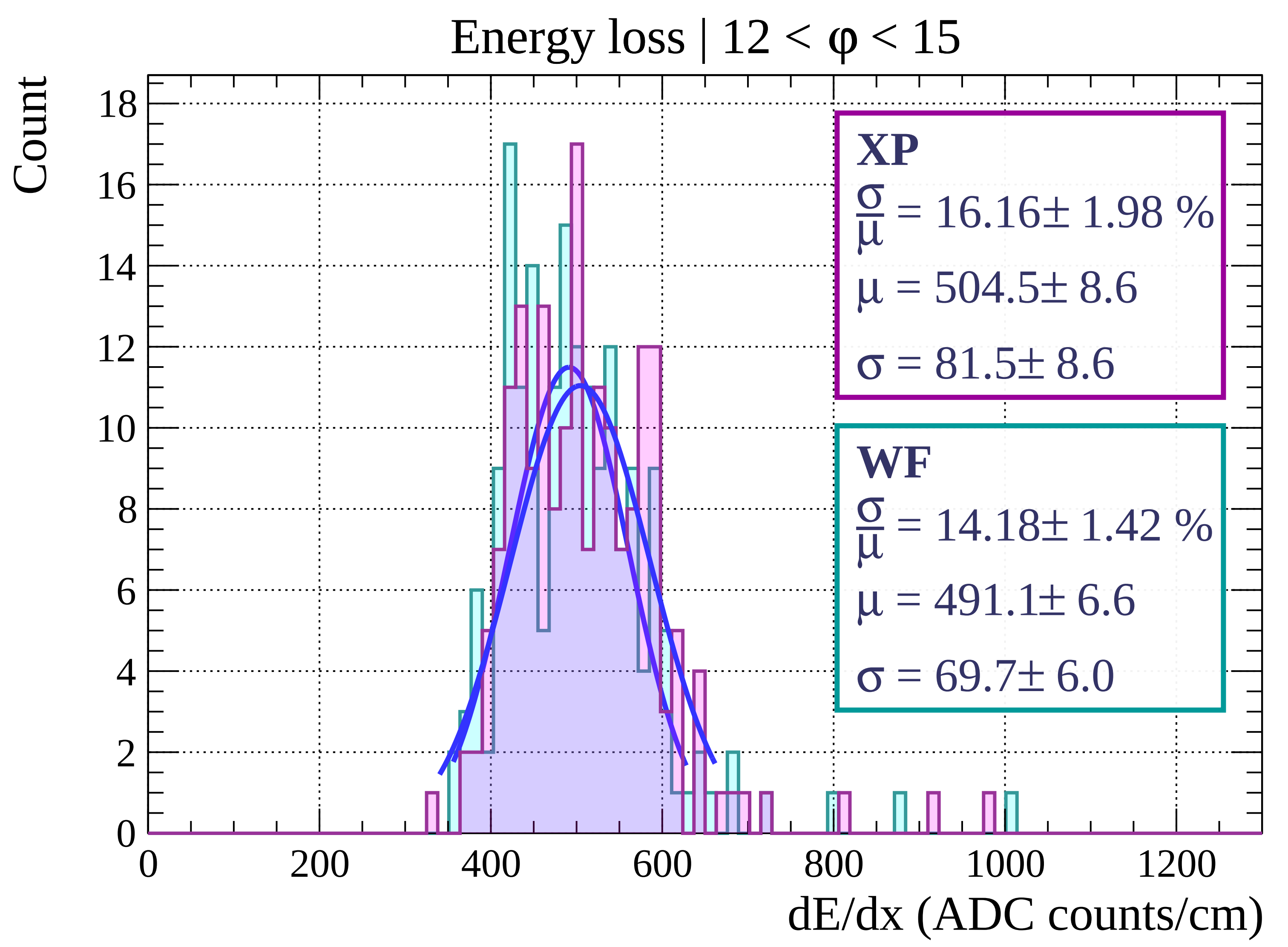
Energy loss | 3 < φ < 6
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
1
2
3
4
5
6
7
8
9
XP
σ/μ = 15.95± 3.67 %
μ = 512.2± 15.1
σ = 81.7± 16.4
WF
σ/μ = 29.36± 108.79 %
μ = 569.1± 3.1
σ = 167.1± 618.2



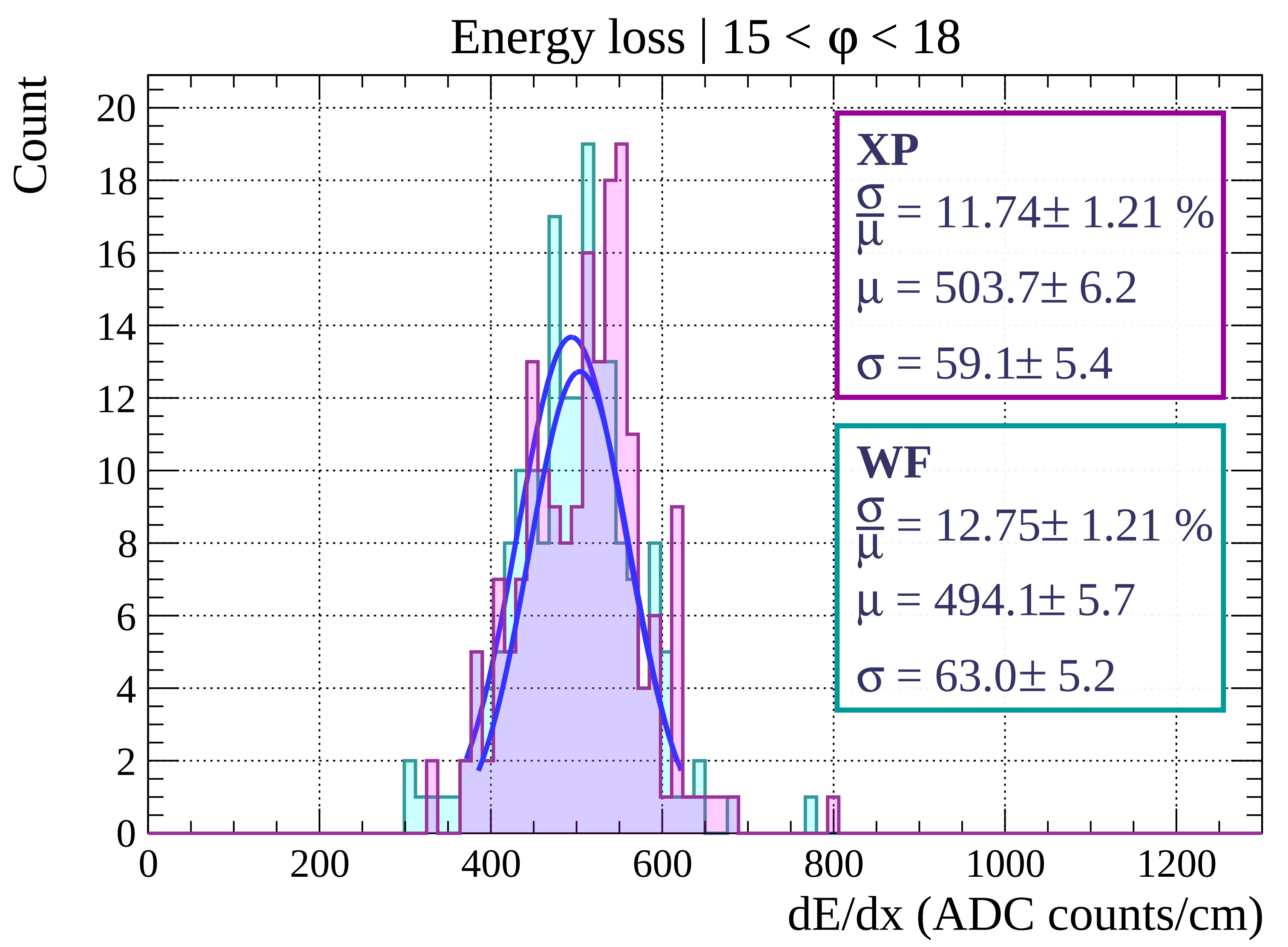
Energy loss | 6 < φ < 9
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
1
2
3
4
5
6
7
8
9
XP
σ/μ = 22.71± 5.42 %
μ = 531.2± 20.9
σ = 120.6± 24.0
WF
σ/μ = 20.44± 4.37 %
μ = 505.9± 17.0
σ = 103.4± 18.6



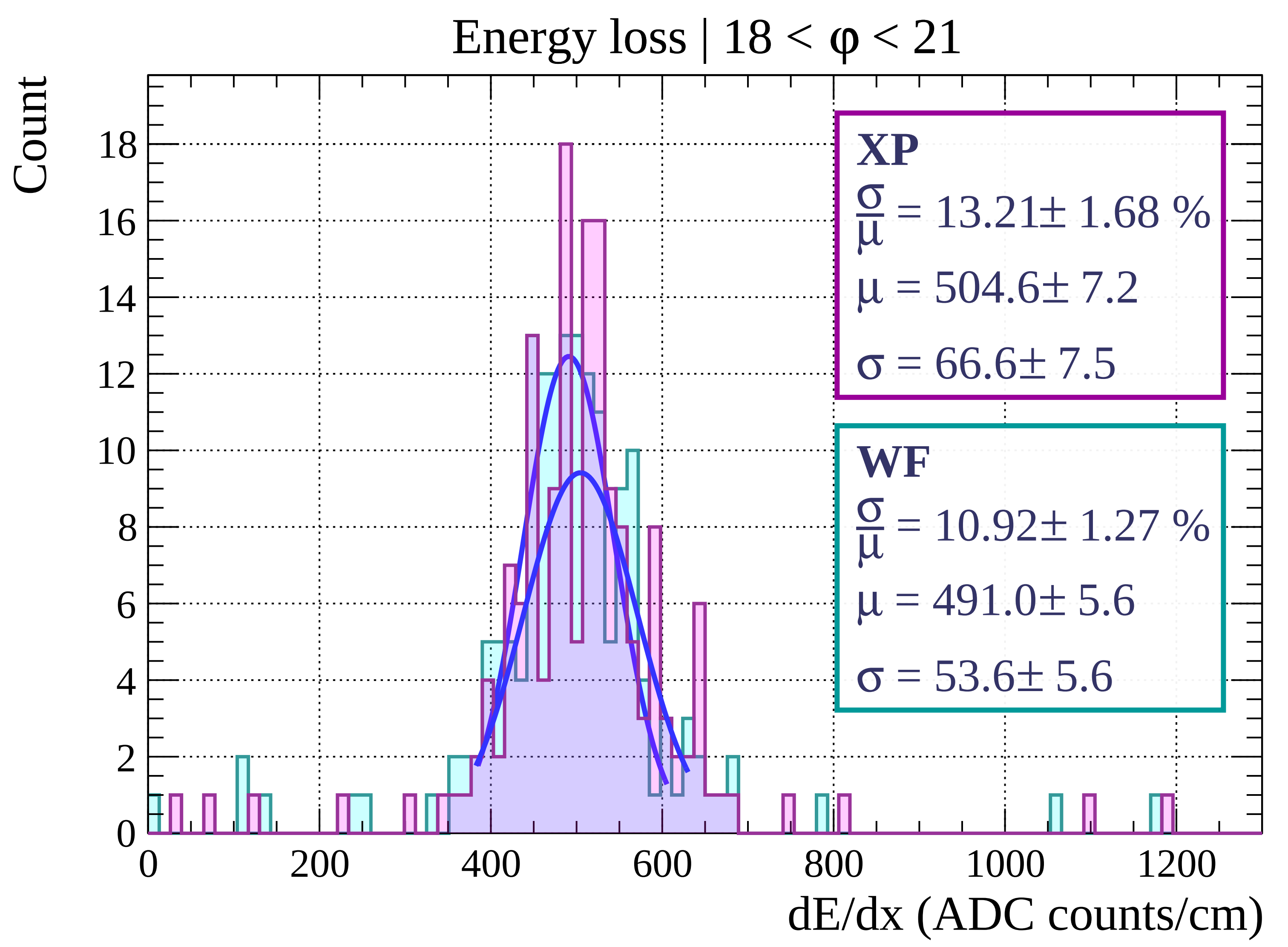
Energy loss | 9 < φ < 12
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
2
4
6
8
10
12
XP
σ/μ = 21.13± 4.97 %
μ = 491.6± 18.5
σ = 103.8± 20.5
WF
σ/μ = 20.39± 4.25 %
μ = 484.7± 18.1
σ = 98.8± 16.9

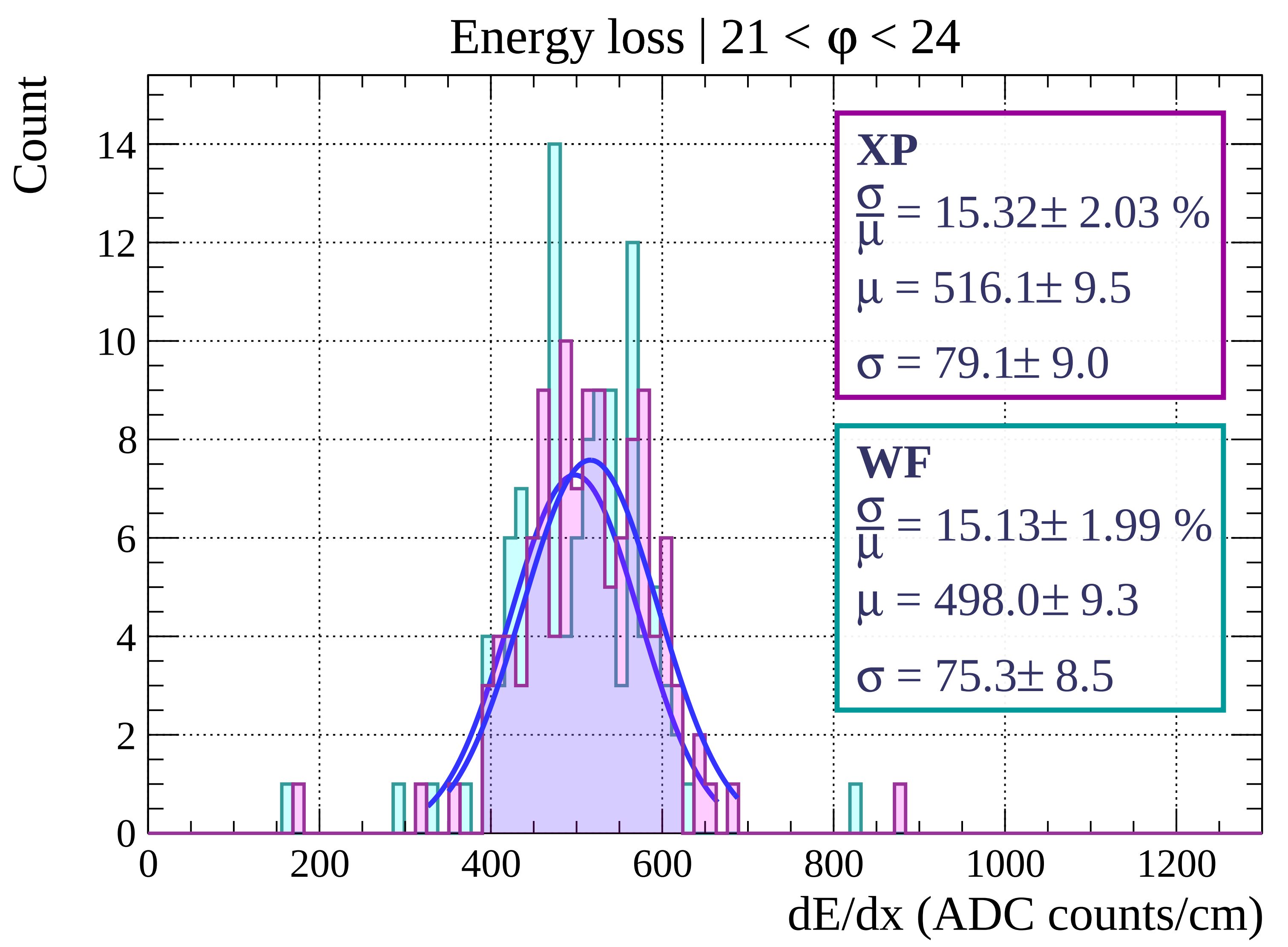


Energy loss | 12 < φ < 15
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 16.16± 1.98 %
μ = 504.5± 8.6
σ = 81.5± 8.6
WF
$\frac{\sigma}{\mu}$ = 14.18± 1.42 %
μ = 491.1± 6.6
σ = 69.7± 6.0

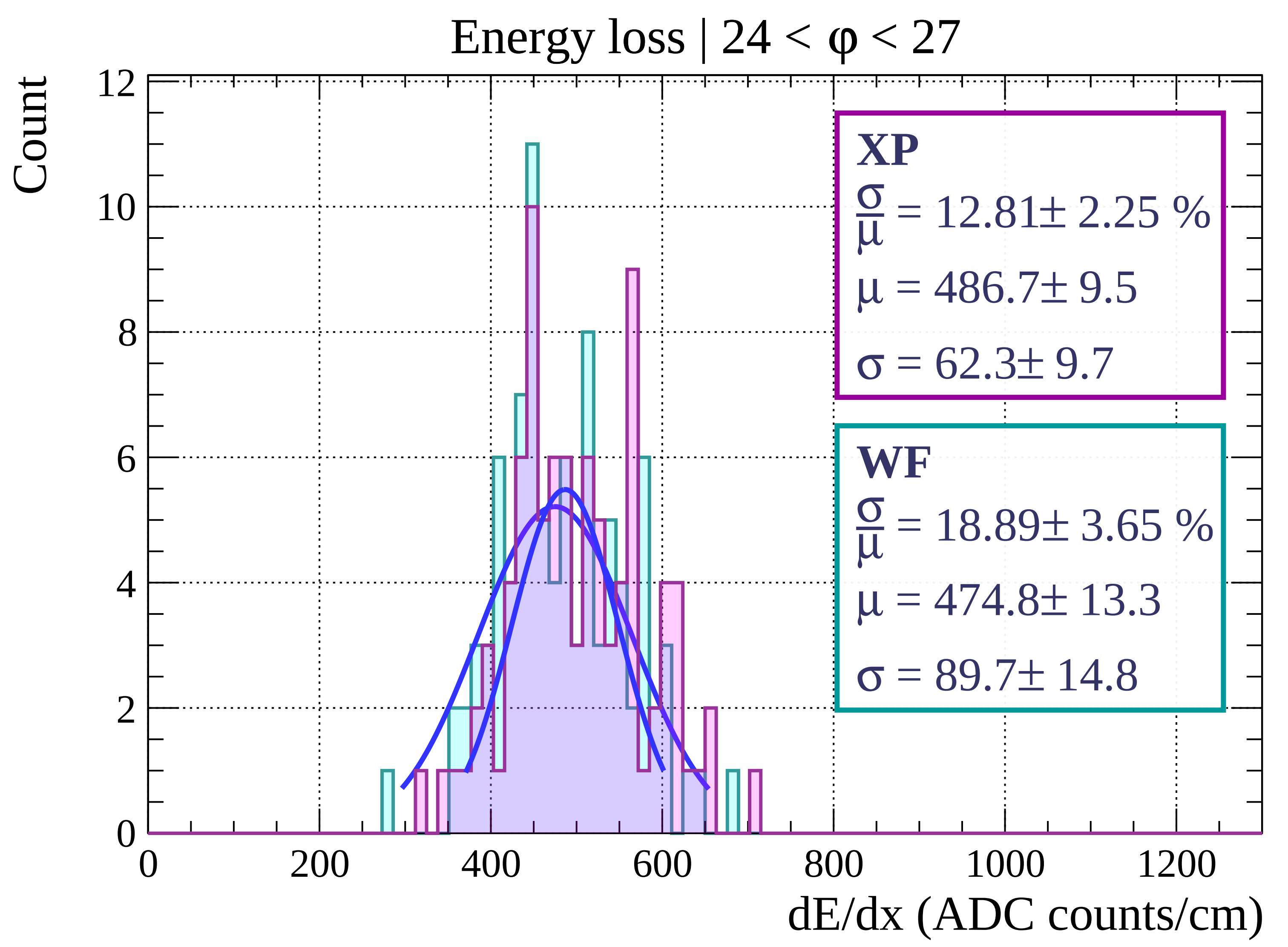


Energy loss | 15 < φ < 18
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
12
14
16
18
20
0
200
400
600
800
1000
1200
XP
σ/μ = 11.74± 1.21 %
μ = 503.7± 6.2
σ = 59.1± 5.4
WF
σ/μ = 12.75± 1.21 %
μ = 494.1± 5.7
σ = 63.0± 5.2

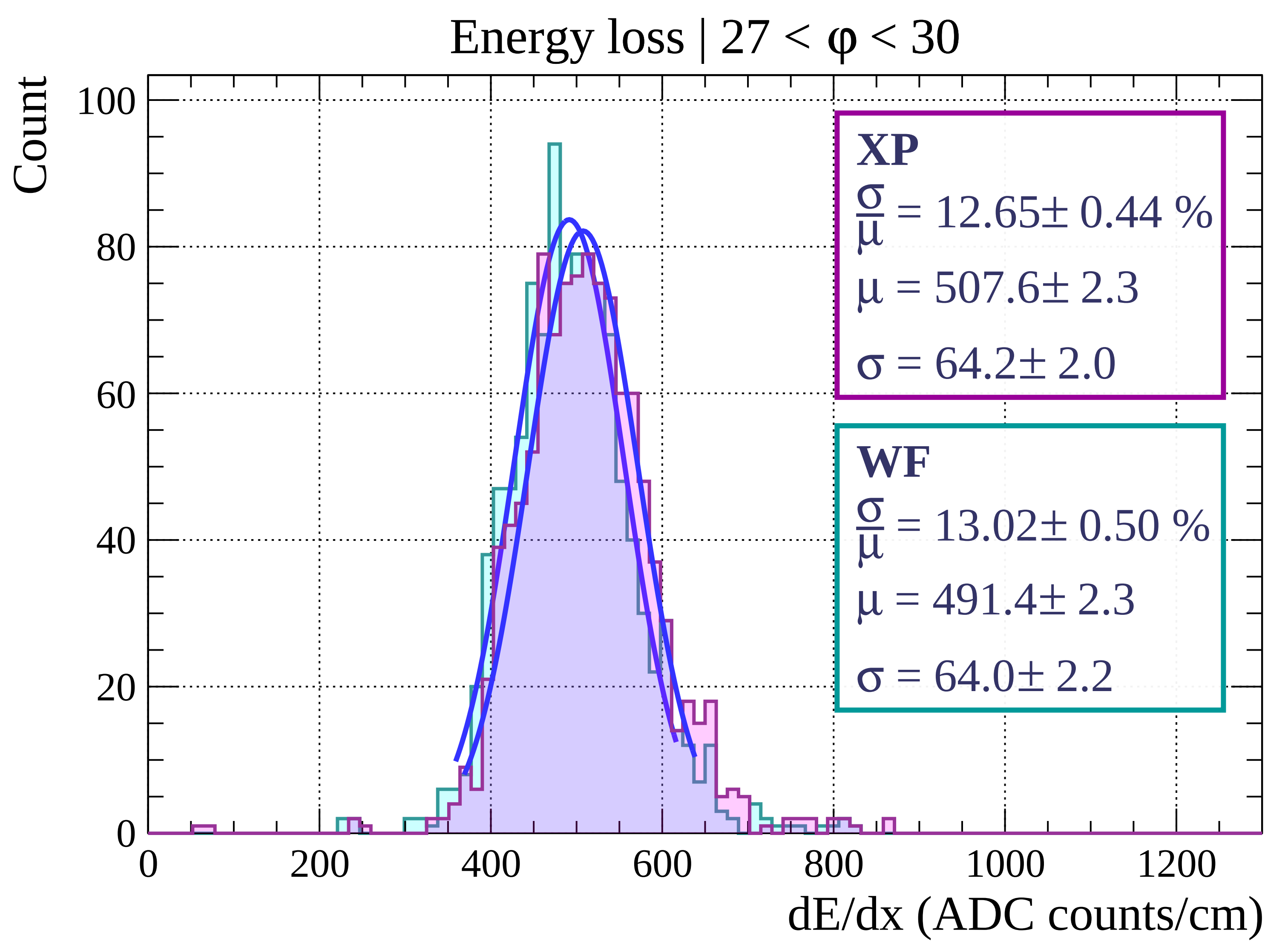

Energy loss | 18 < φ < 21
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 13.21± 1.68 %
μ = 504.6± 7.2
σ = 66.6± 7.5
WF
σ/μ = 10.92± 1.27 %
μ = 491.0± 5.6
σ = 53.6± 5.6
0
2
4
6
8
10
12
14
16
18
0
200
400
600
800
1000
1200



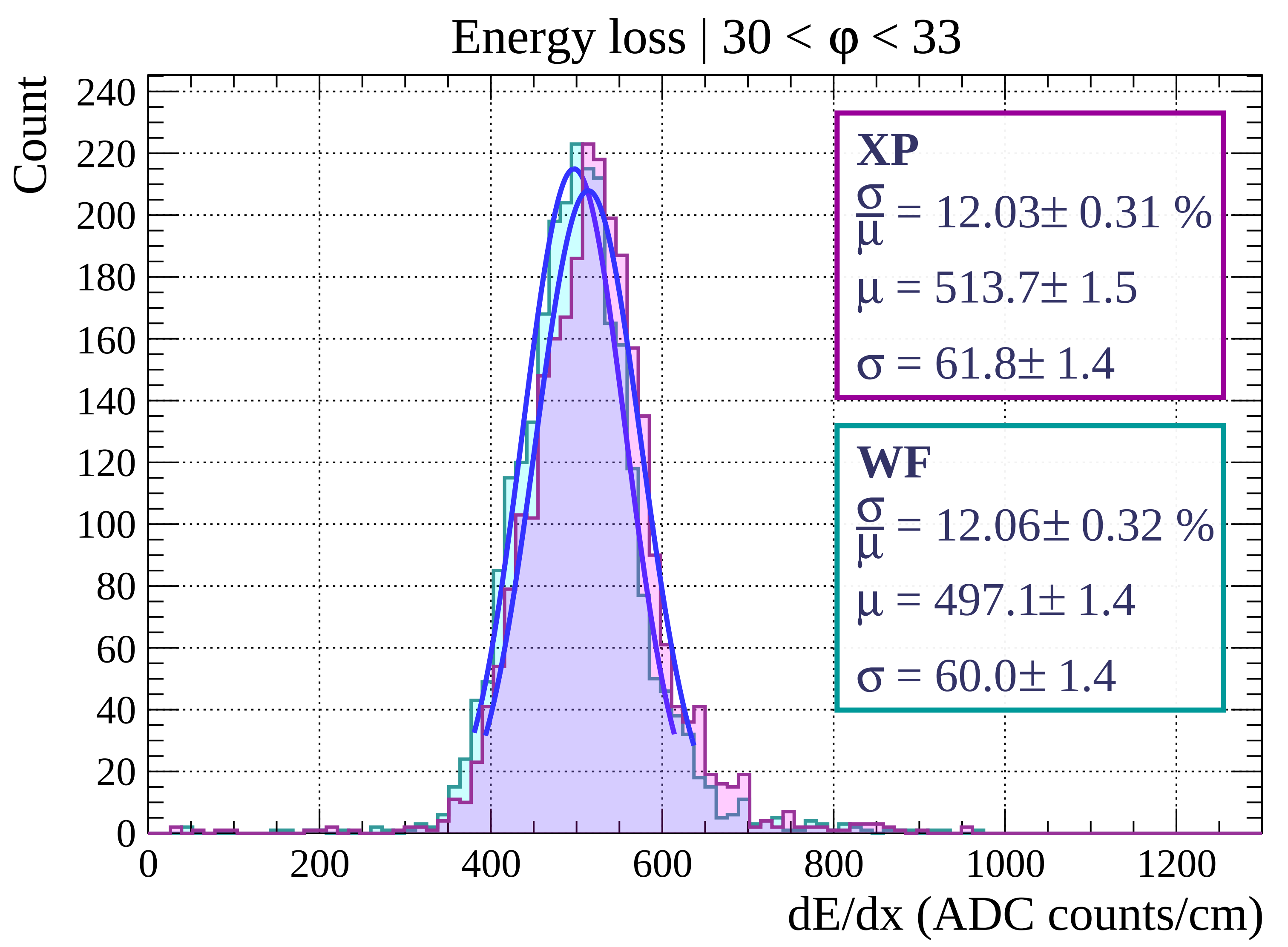
Energy loss | 21 < φ < 24
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
2
4
6
8
10
12
14
XP
σ/μ = 15.32± 2.03 %
μ = 516.1± 9.5
σ = 79.1± 9.0
WF
σ/μ = 15.13± 1.99 %
μ = 498.0± 9.3
σ = 75.3± 8.5

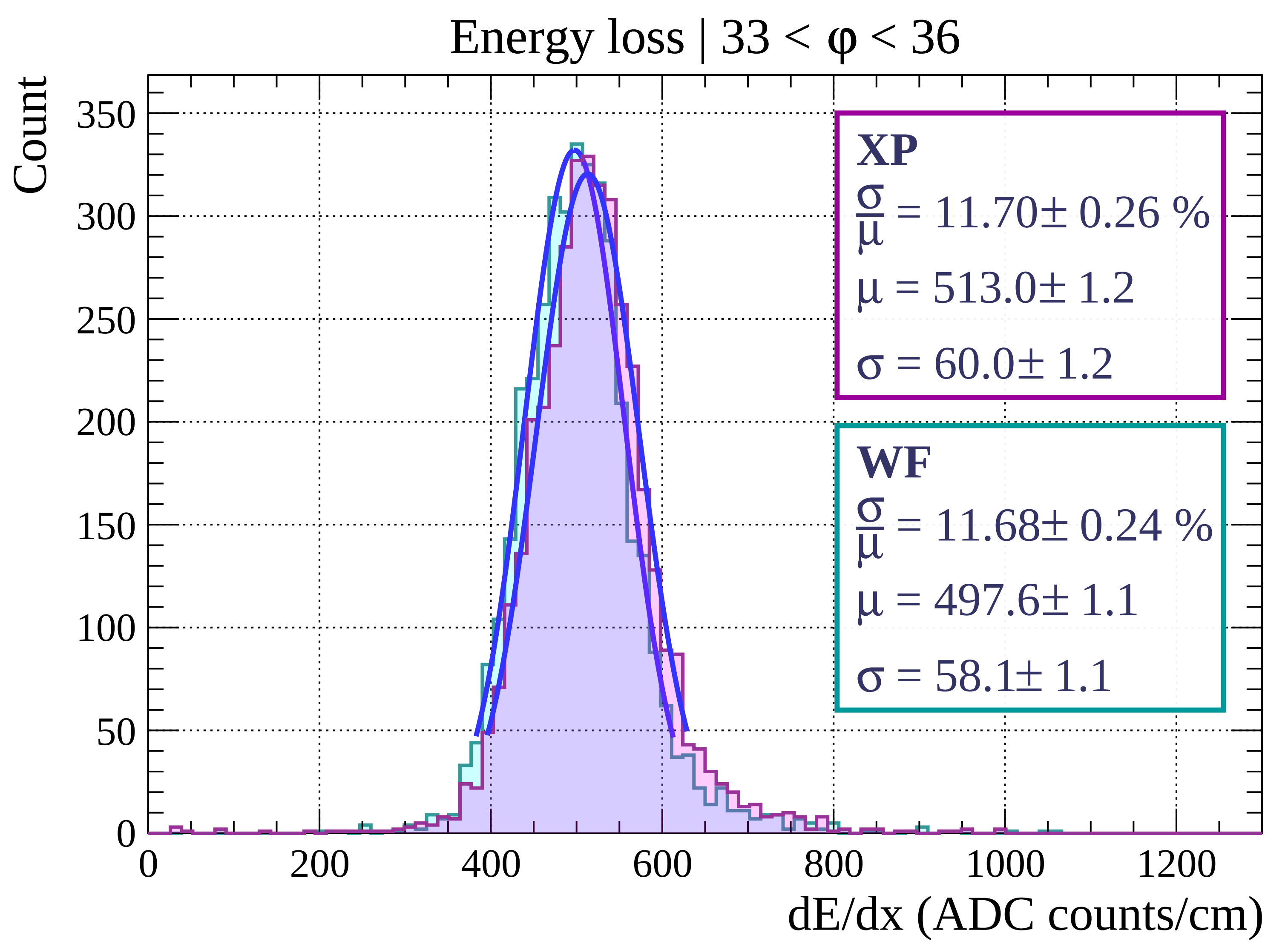


Energy loss | 24 < φ < 27
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
2
4
6
8
10
12
XP
σ/μ = 12.81± 2.25 %
μ = 486.7± 9.5
σ = 62.3± 9.7
WF
σ/μ = 18.89± 3.65 %
μ = 474.8± 13.3
σ = 89.7± 14.8

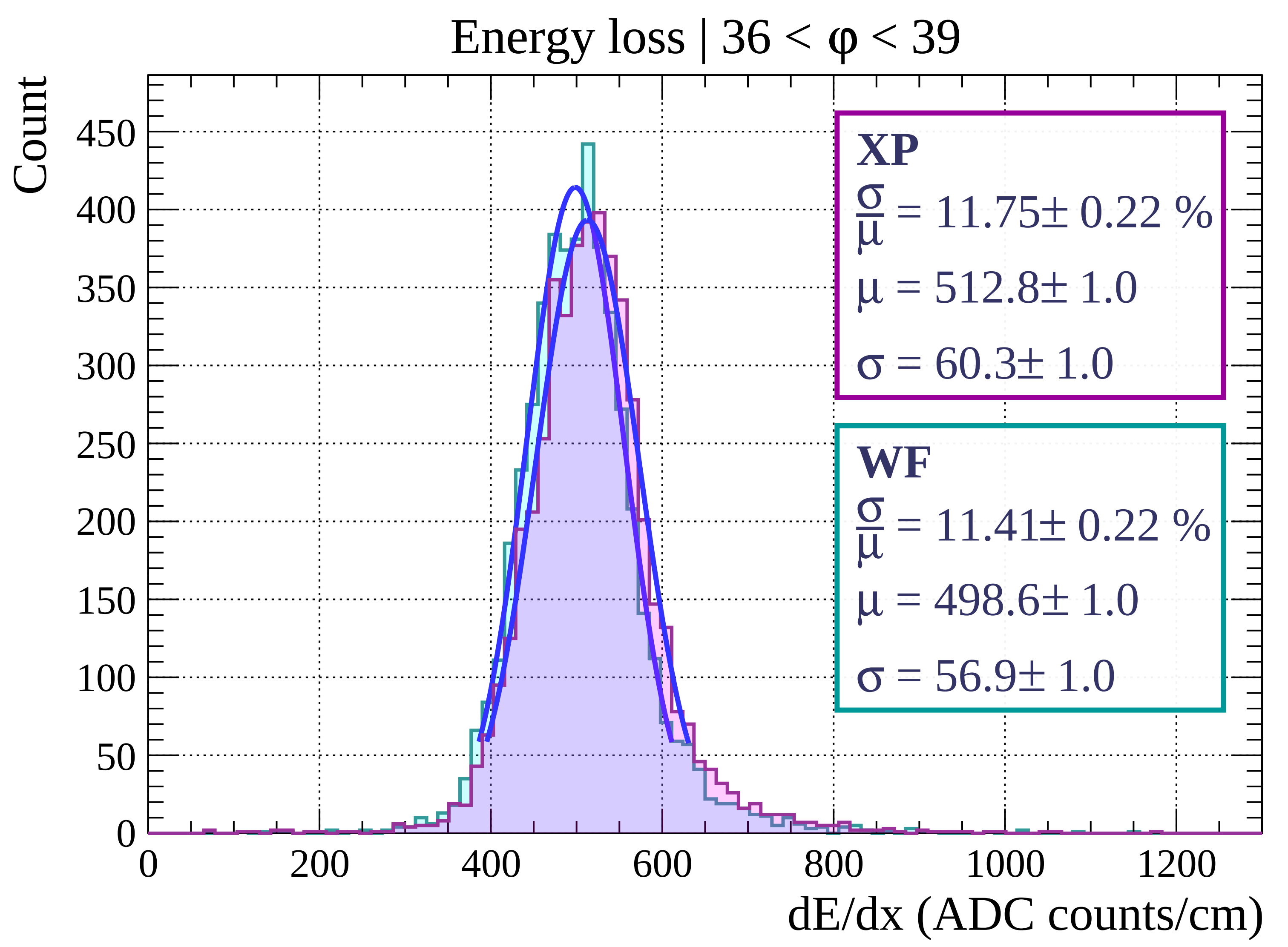


Energy loss | 27 < φ < 30
Count
0
20
40
60
80
100
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 12.65± 0.44 %
μ = 507.6± 2.3
σ = 64.2± 2.0
WF
σ/μ = 13.02± 0.50 %
μ = 491.4± 2.3
σ = 64.0± 2.2

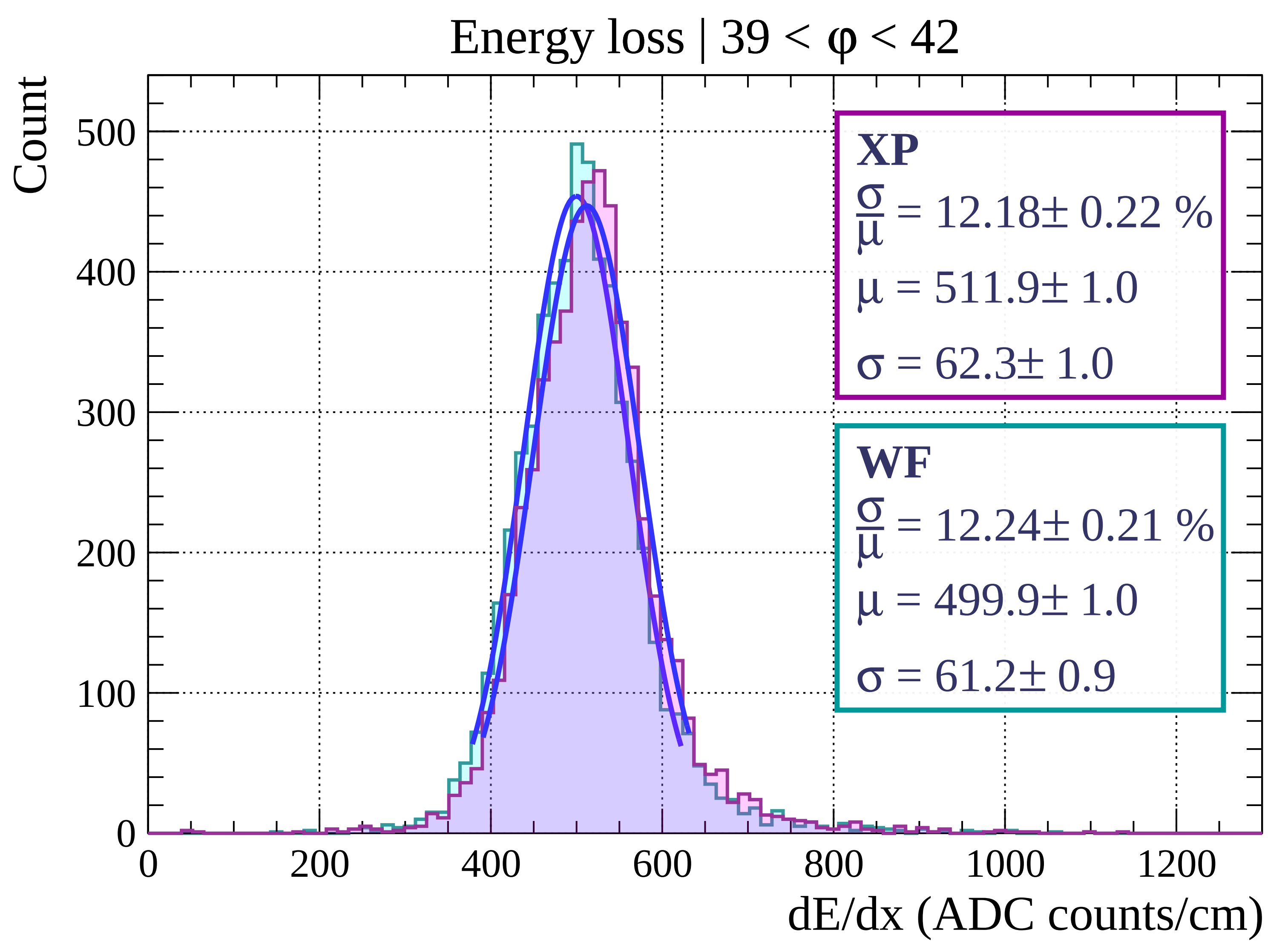

Energy loss | 30 < φ < 33
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
220
240
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 12.03± 0.31 %
μ = 513.7± 1.5
σ = 61.8± 1.4
WF
$\frac{\sigma}{\mu}$ = 12.06± 0.32 %
μ = 497.1± 1.4
σ = 60.0± 1.4



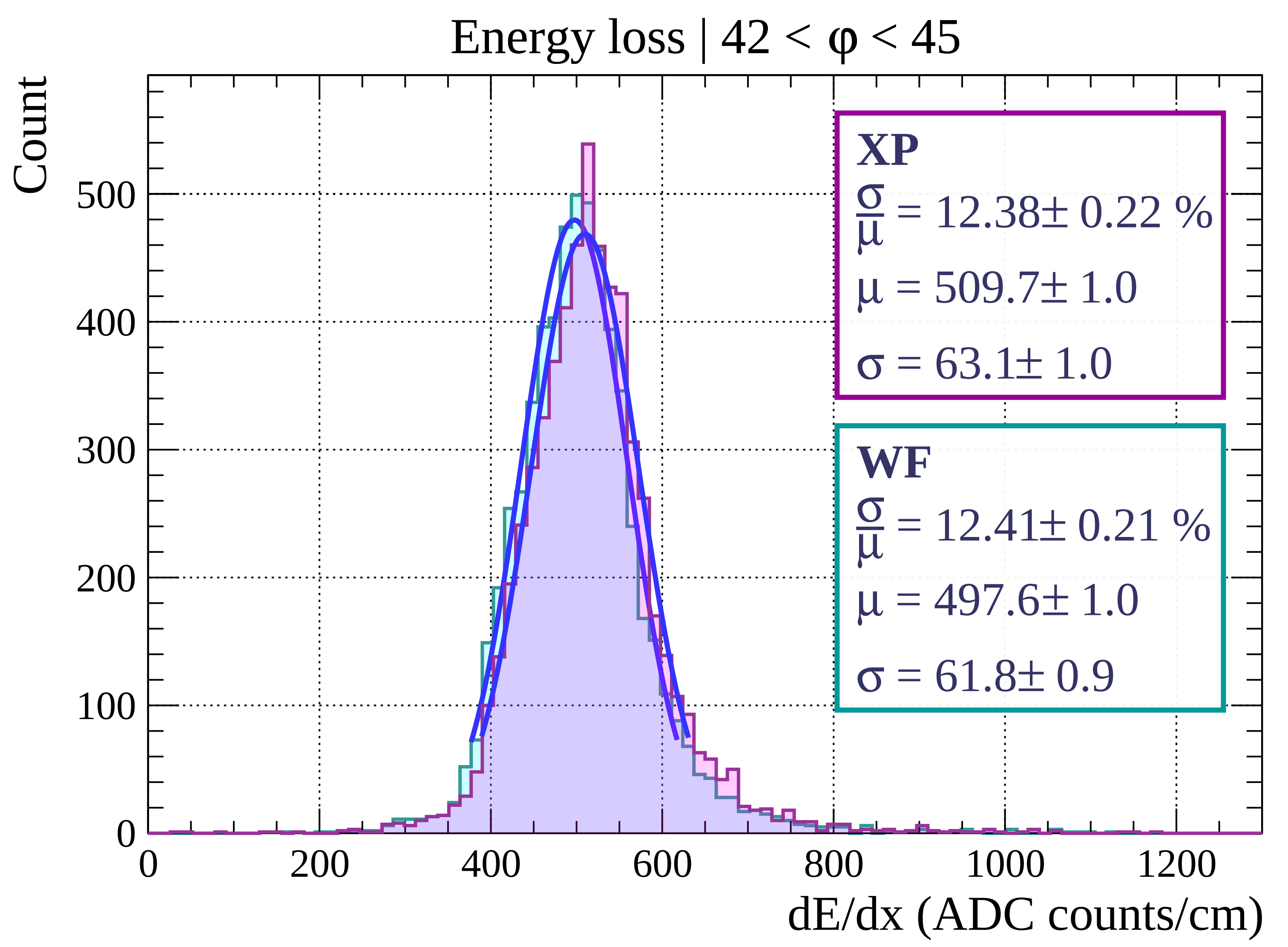
Energy loss | 33 < φ < 36
Count
350
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 11.70± 0.26 %
μ = 513.0± 1.2
σ = 60.0± 1.2
WF
σ/μ = 11.68± 0.24 %
μ = 497.6± 1.1
σ = 58.1± 1.1



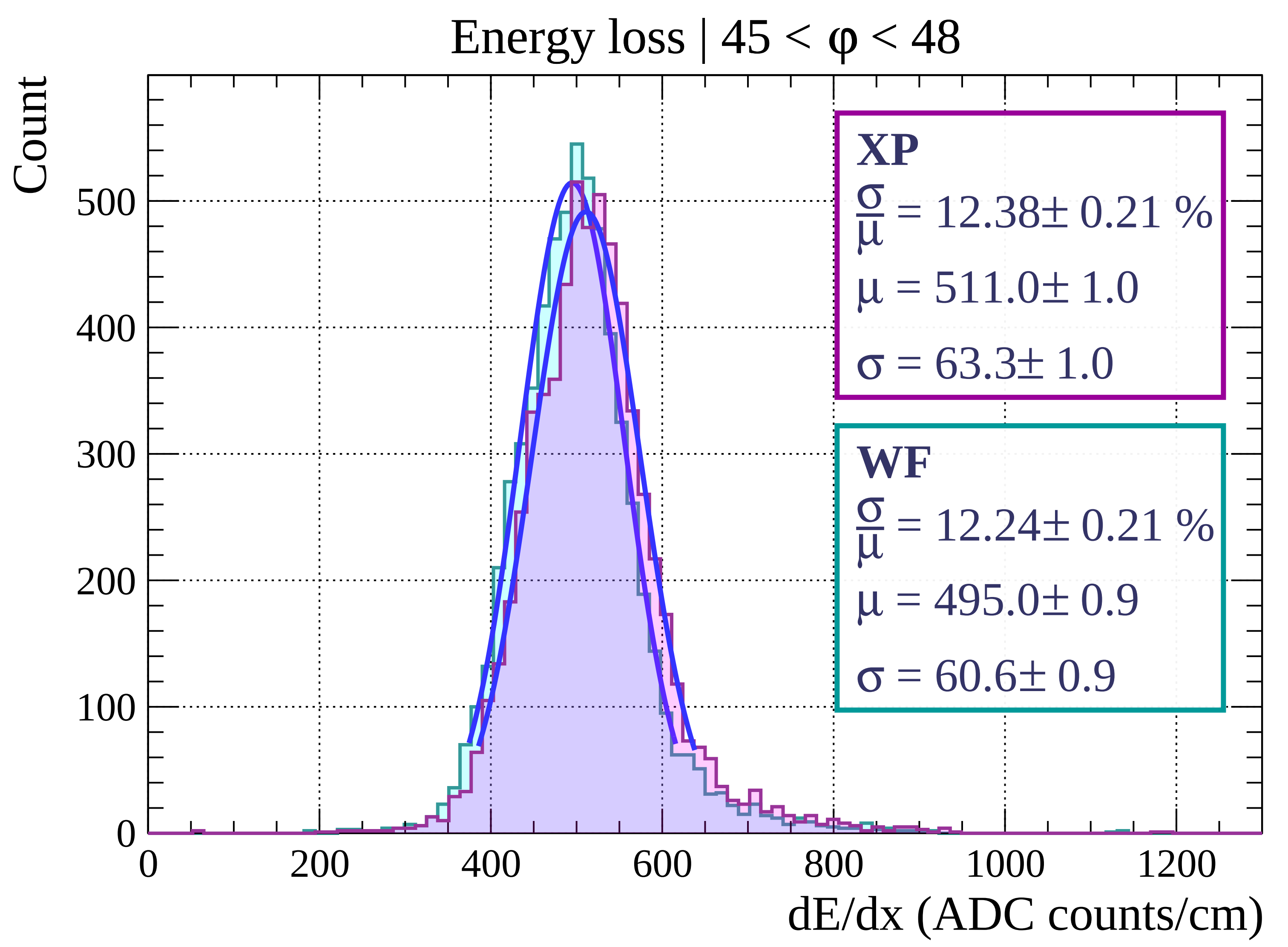
Energy loss | 36 < φ < 39
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 11.75± 0.22 %
μ = 512.8± 1.0
σ = 60.3± 1.0
WF
σ/μ = 11.41± 0.22 %
μ = 498.6± 1.0
σ = 56.9± 1.0
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200



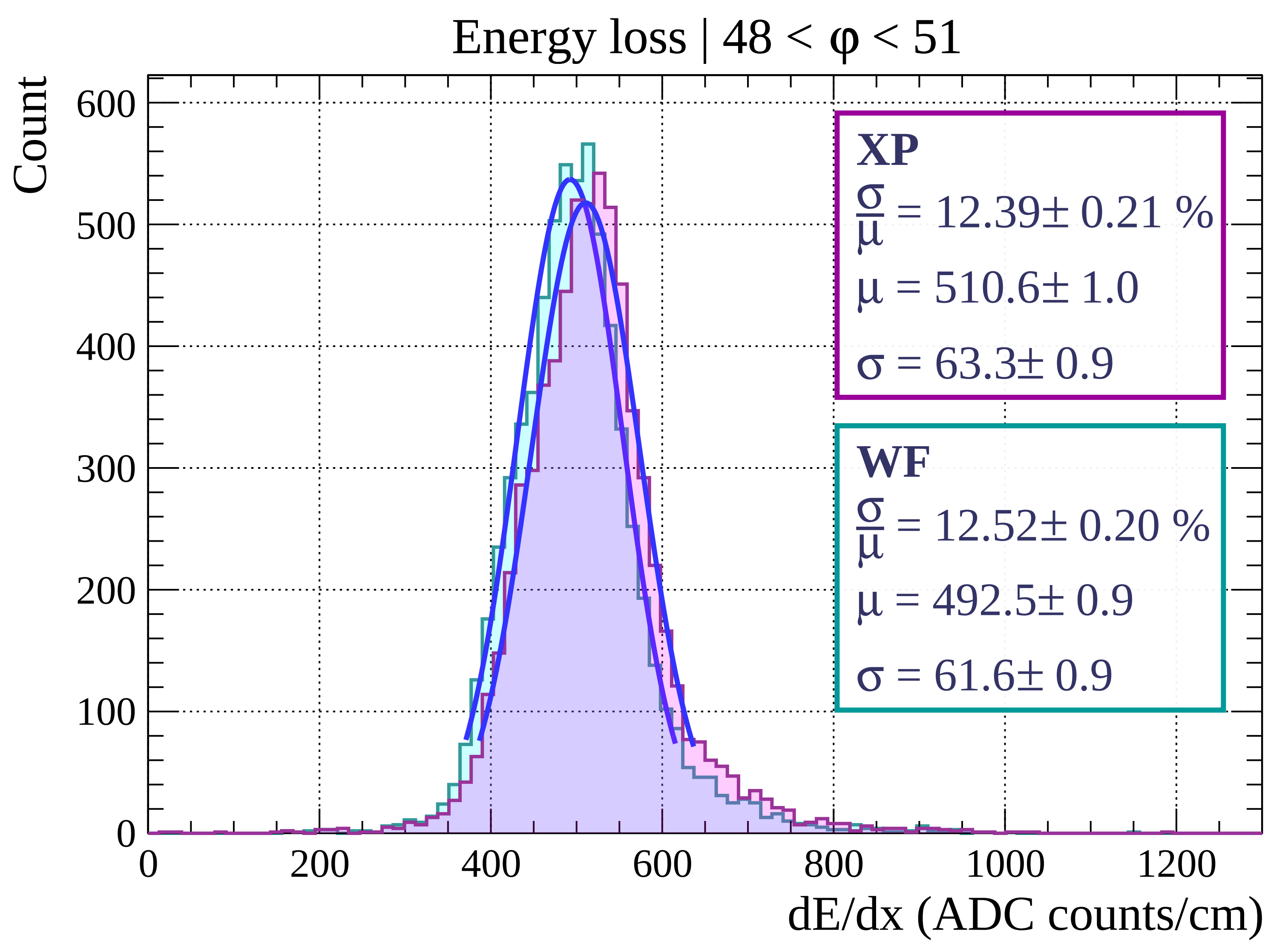
Energy loss | 39 < φ < 42
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 12.18± 0.22 %
μ = 511.9± 1.0
σ = 62.3± 1.0
WF
σ/μ = 12.24± 0.21 %
μ = 499.9± 1.0
σ = 61.2± 0.9



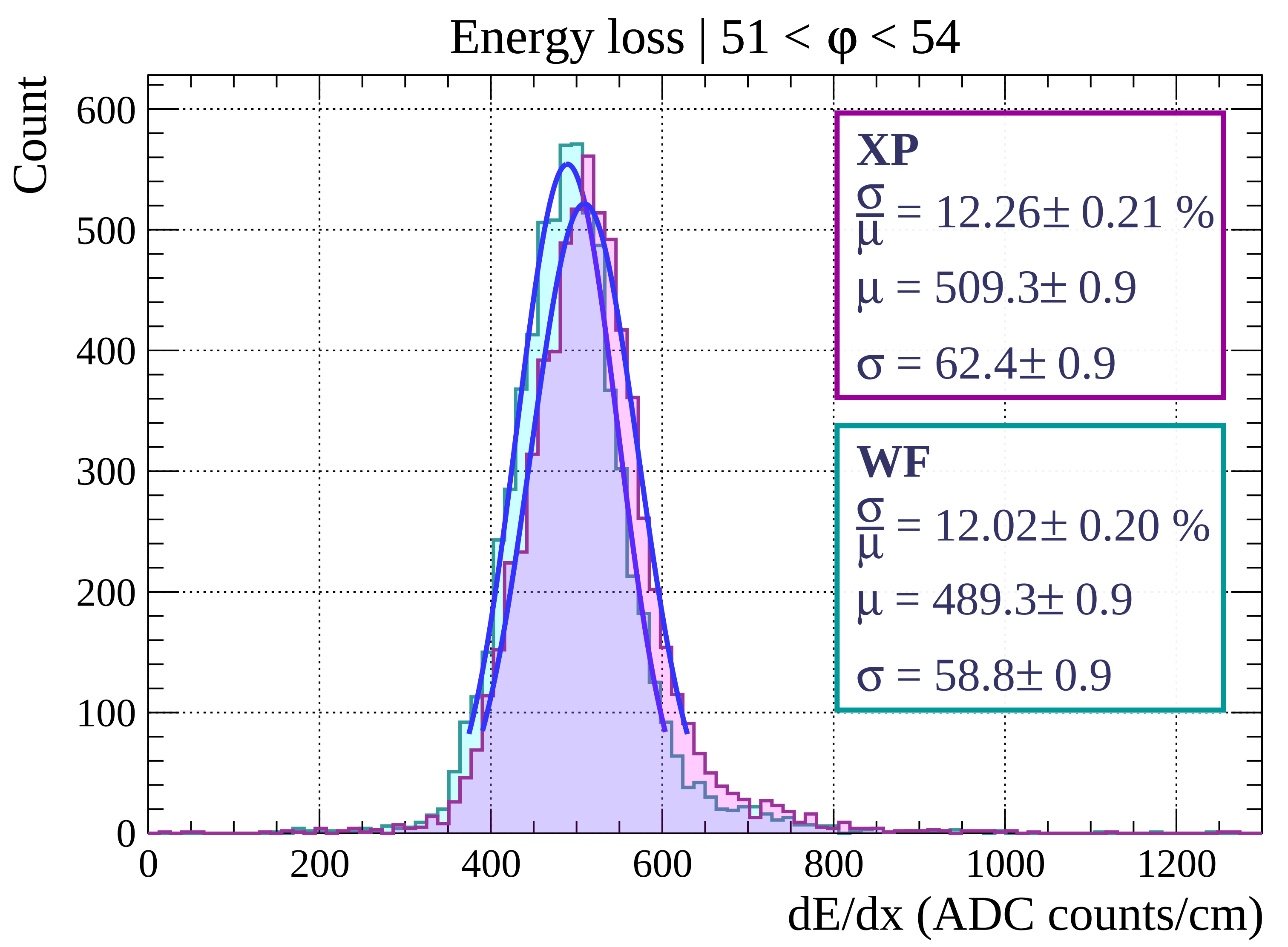
Energy loss | 42 < φ < 45
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
σ/μ = 12.38± 0.22 %
μ = 509.7± 1.0
σ = 63.1± 1.0
WF
σ/μ = 12.41± 0.21 %
μ = 497.6± 1.0
σ = 61.8± 0.9



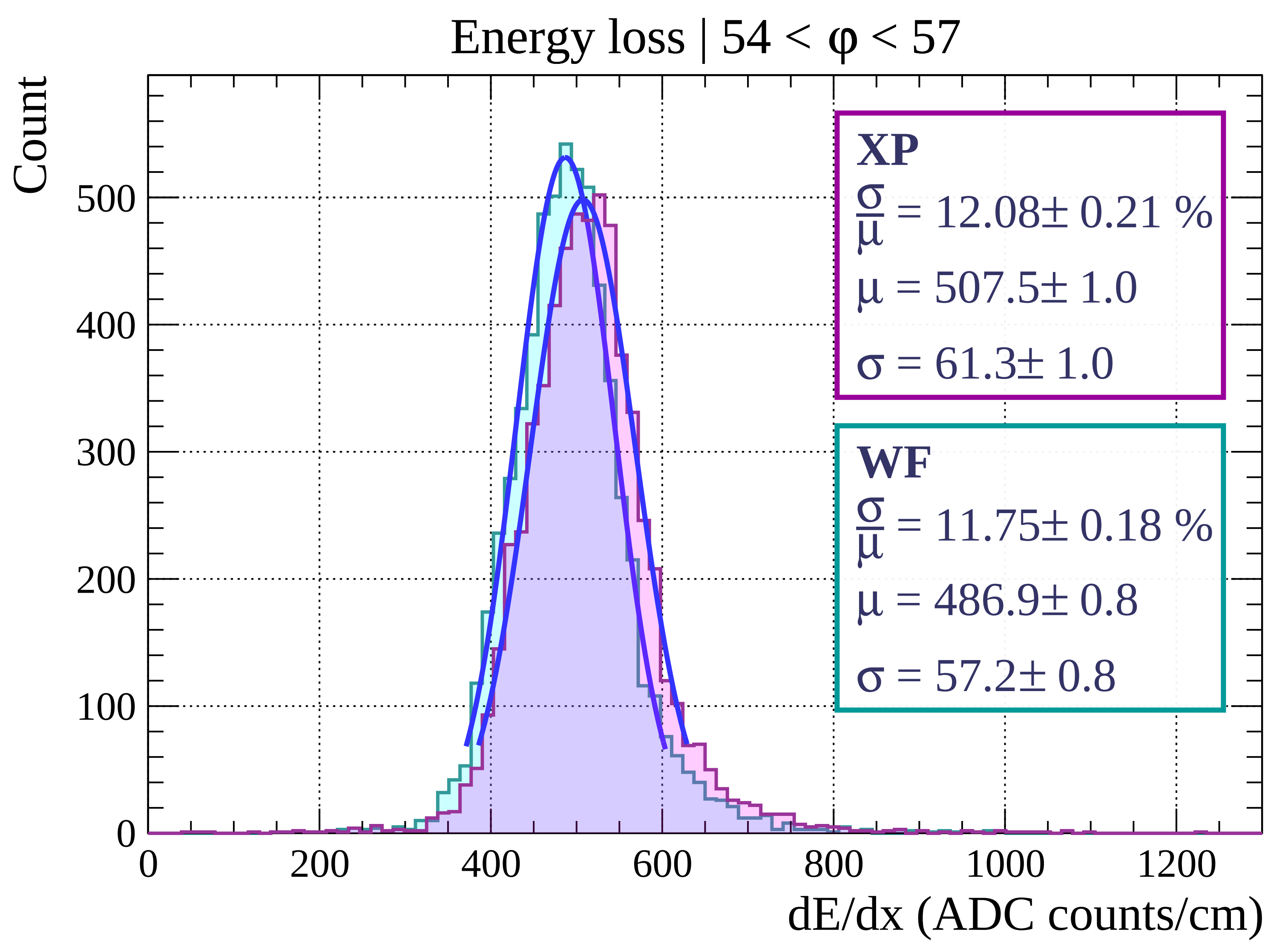
Energy loss | 45 < φ < 48
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
σ/μ = 12.38± 0.21 %
μ = 511.0± 1.0
σ = 63.3± 1.0
WF
σ/μ = 12.24± 0.21 %
μ = 495.0± 0.9
σ = 60.6± 0.9



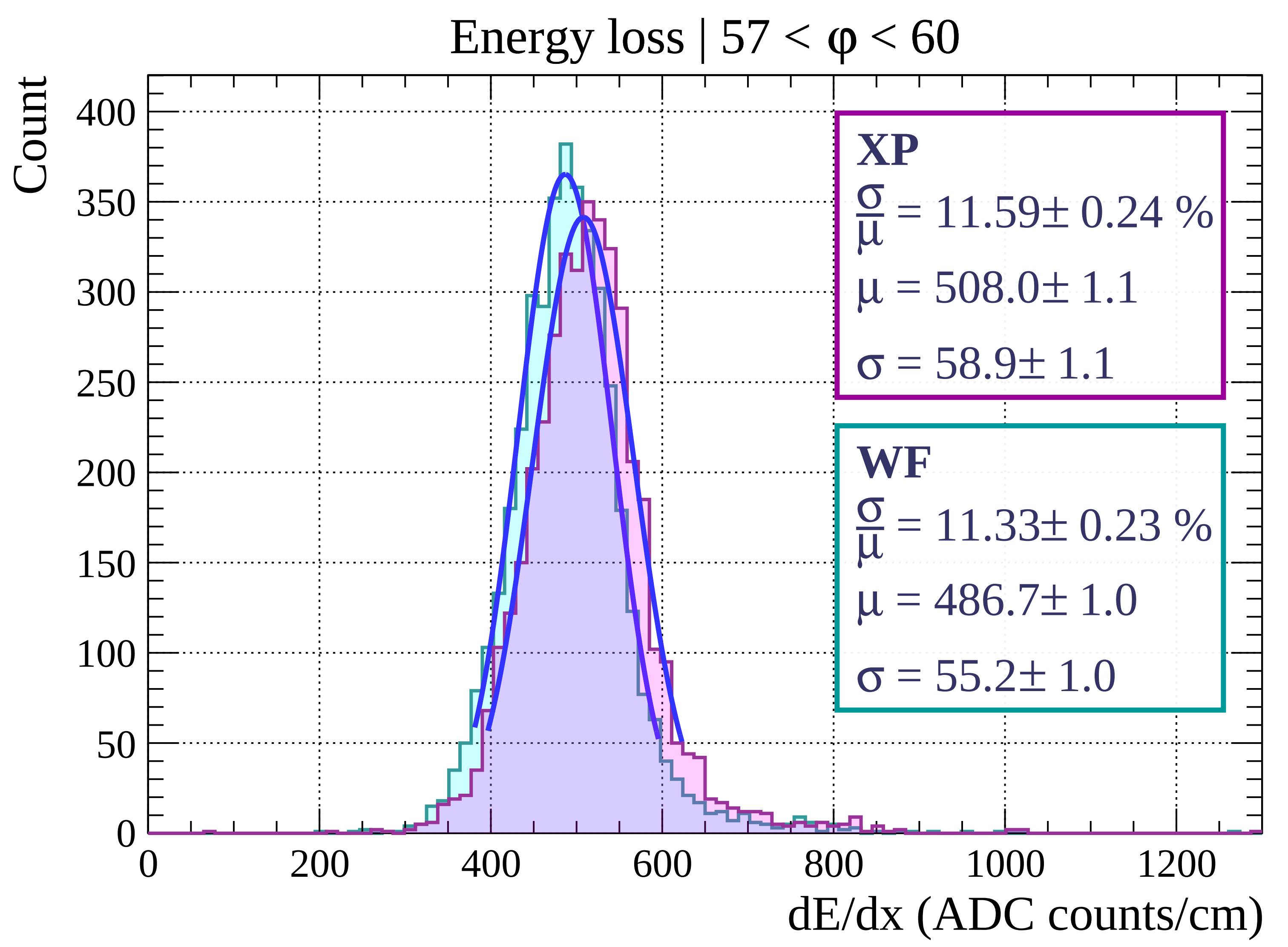
Energy loss | 48 < φ < 51
Count
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 12.39± 0.21 %
μ = 510.6± 1.0
σ = 63.3± 0.9
WF
σ/μ = 12.52± 0.20 %
μ = 492.5± 0.9
σ = 61.6± 0.9

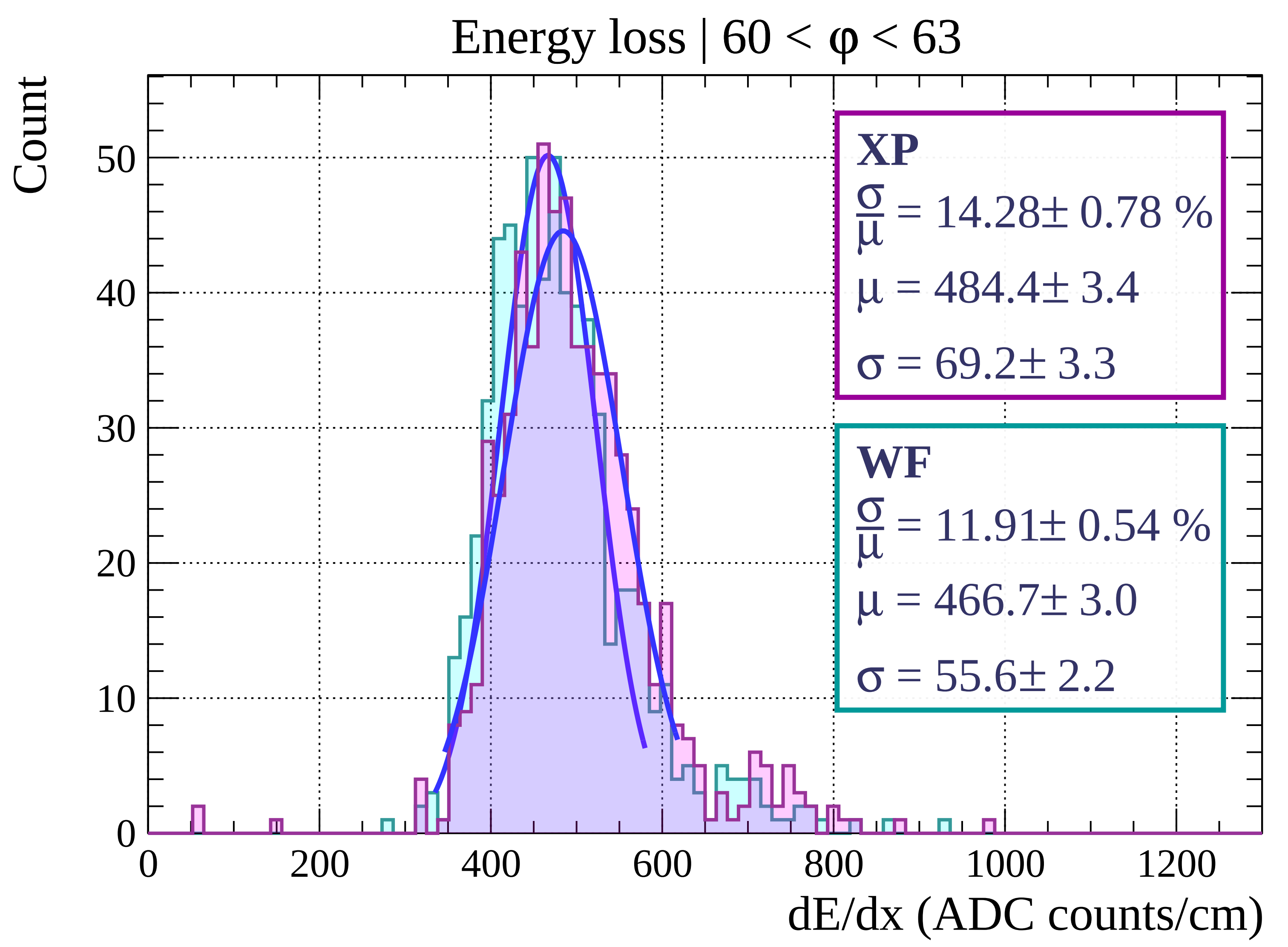


Energy loss | 51 < φ < 54
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP
σ/μ = 12.26± 0.21 %
μ = 509.3± 0.9
σ = 62.4± 0.9
WF
σ/μ = 12.02± 0.20 %
μ = 489.3± 0.9
σ = 58.8± 0.9

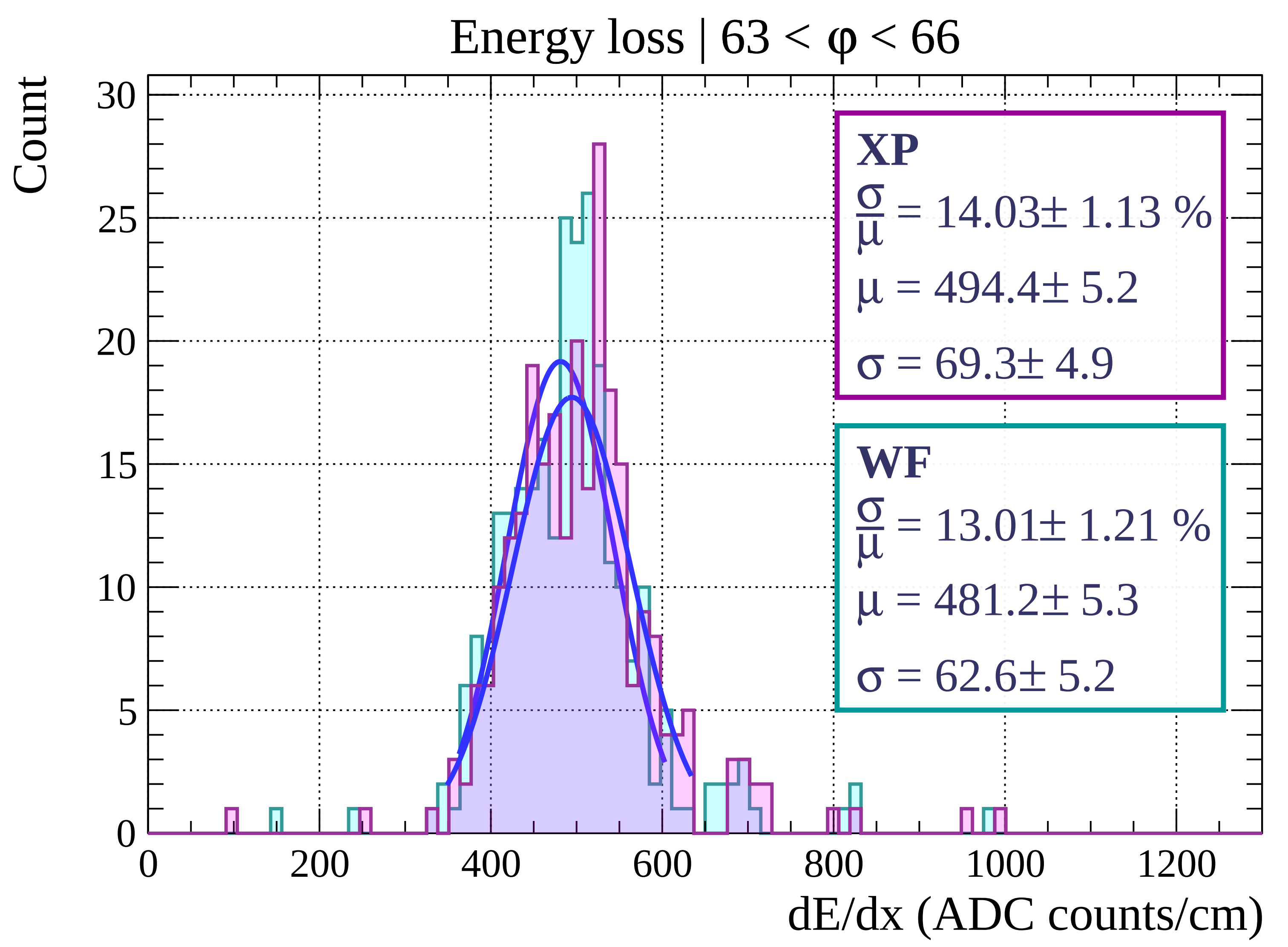


Energy loss | 54 < φ < 57
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 12.08± 0.21 %
μ = 507.5± 1.0
σ = 61.3± 1.0
WF
$\frac{\sigma}{\mu}$ = 11.75± 0.18 %
μ = 486.9± 0.8
σ = 57.2± 0.8

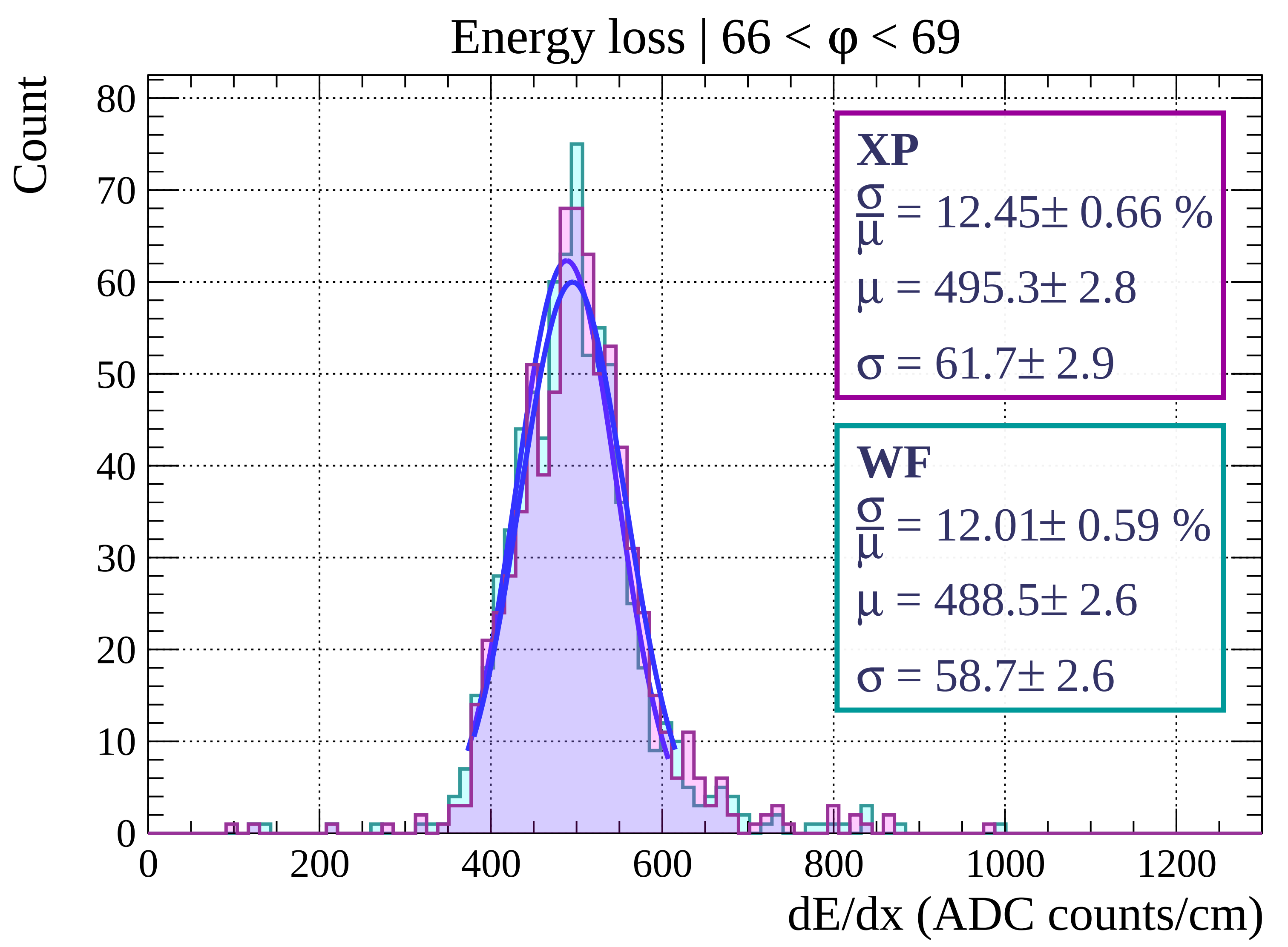

Energy loss | 57 < φ < 60
Count
XP
σ/μ = 11.59± 0.24 %
μ = 508.0± 1.1
σ = 58.9± 1.1
WF
σ/μ = 11.33± 0.23 %
μ = 486.7± 1.0
σ = 55.2± 1.0
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200

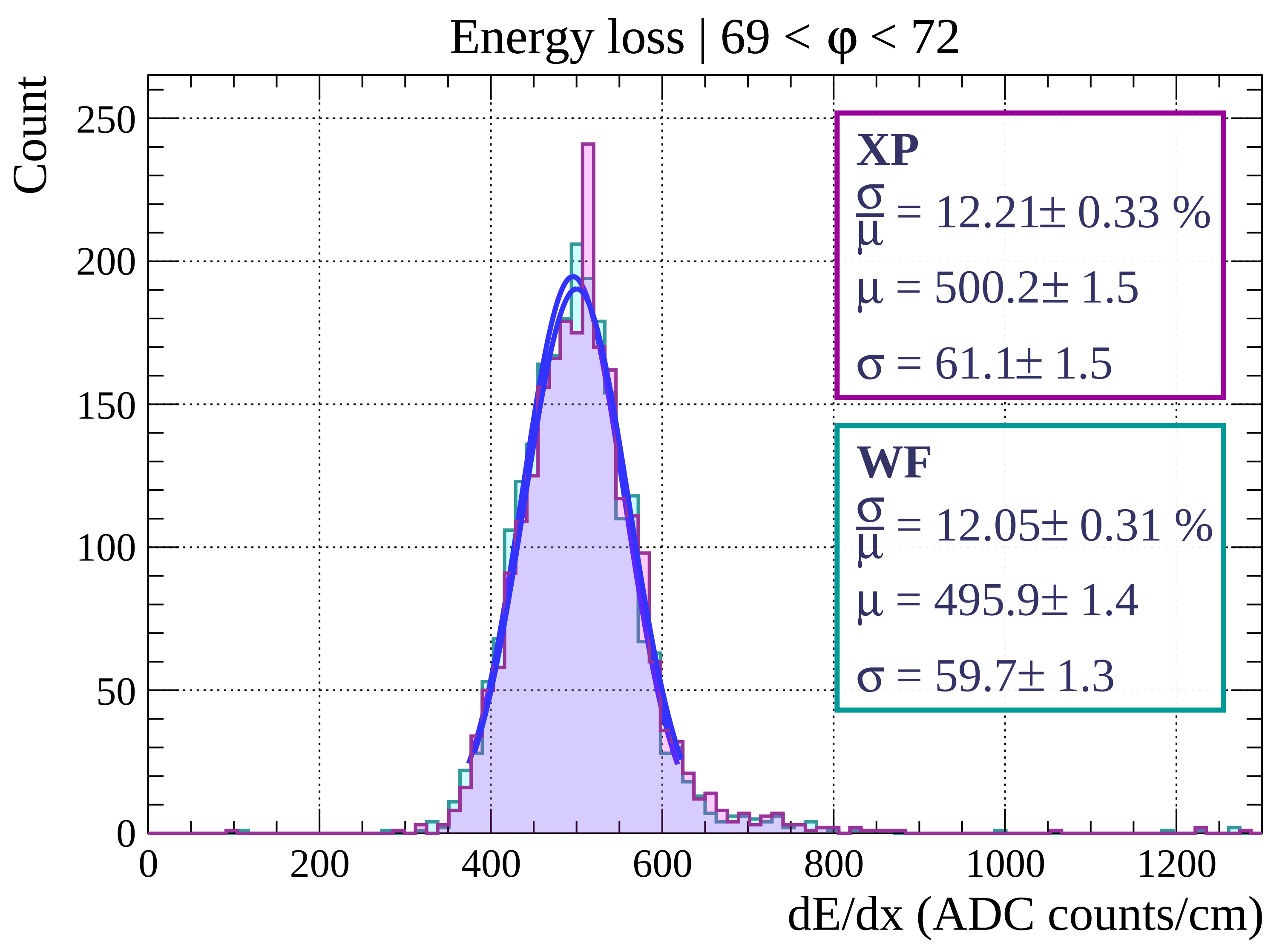


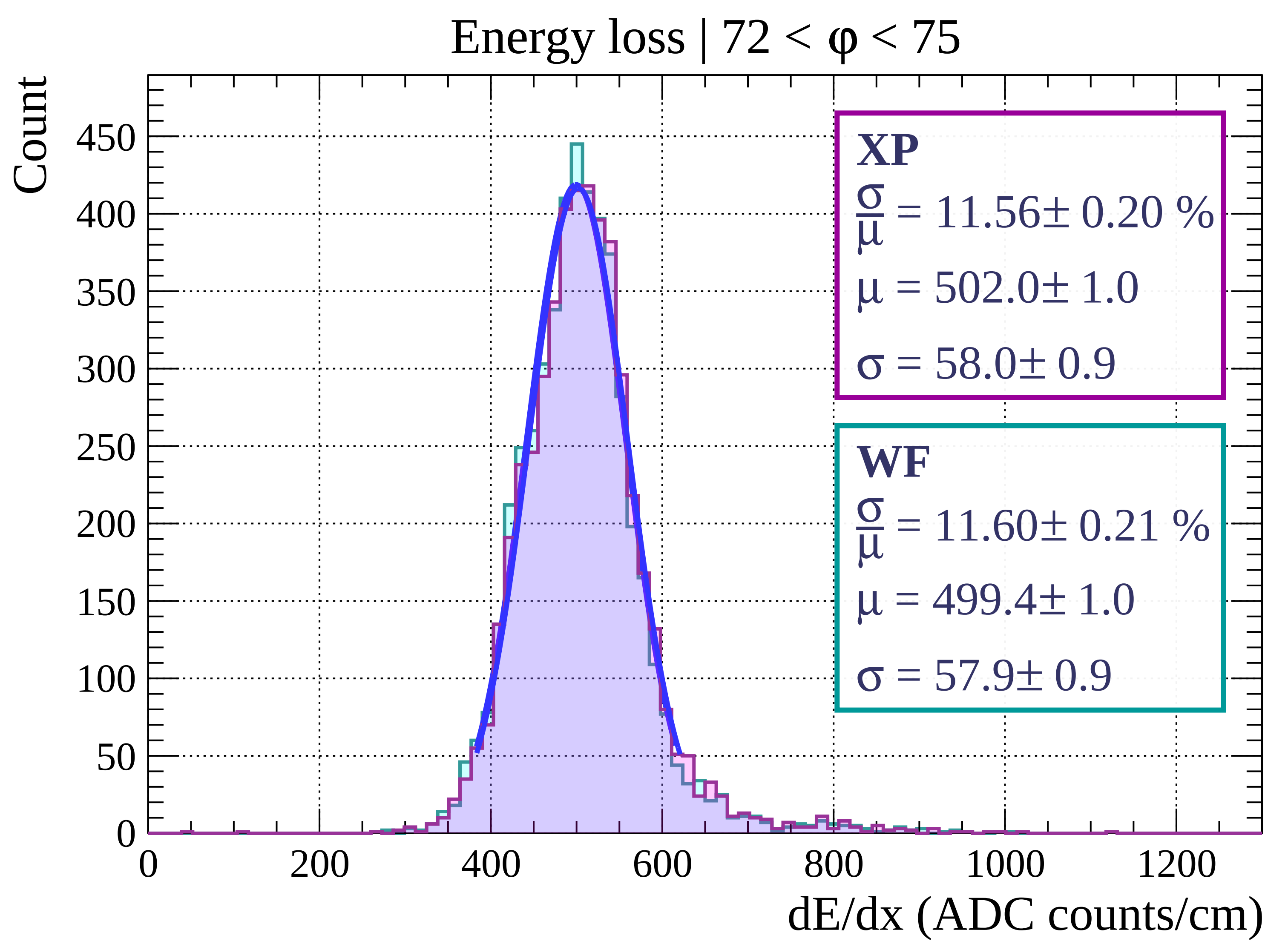
Energy loss | 60 < φ < 63
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
0
200
400
600
800
1000
1200
XP
σ/μ = 14.28± 0.78 %
μ = 484.4± 3.4
σ = 69.2± 3.3
WF
σ/μ = 11.91± 0.54 %
μ = 466.7± 3.0
σ = 55.6± 2.2



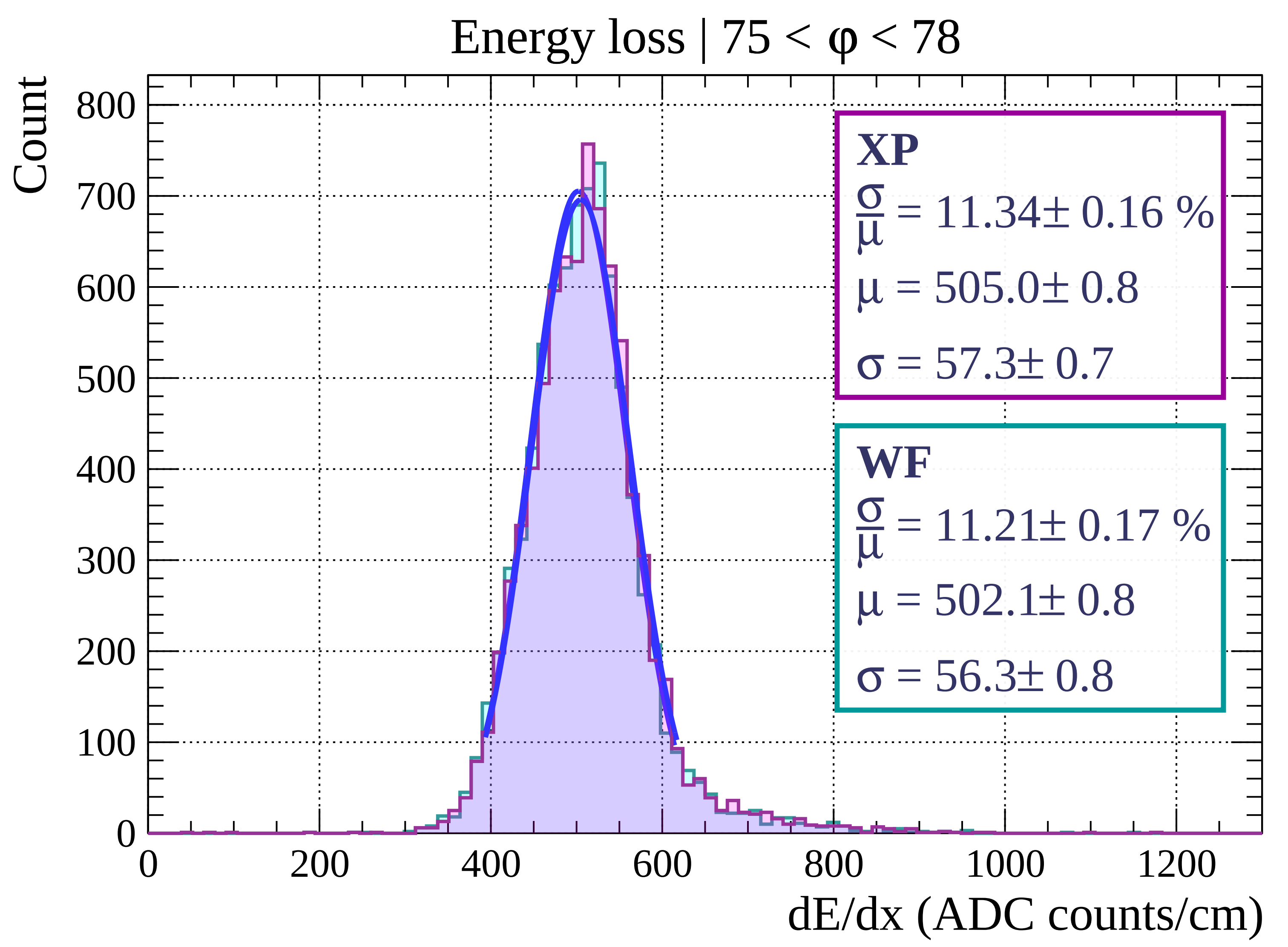
Energy loss | 63 < φ < 66
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
0
200
400
600
800
1000
1200
XP
σ/μ = 14.03± 1.13 %
μ = 494.4± 5.2
σ = 69.3± 4.9
WF
σ/μ = 13.01± 1.21 %
μ = 481.2± 5.3
σ = 62.6± 5.2


Energy loss | 66 < φ < 69
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 12.45± 0.66 %
μ = 495.3± 2.8
σ = 61.7± 2.9
WF
σ/μ = 12.01± 0.59 %
μ = 488.5± 2.6
σ = 58.7± 2.6
0
10
20
30
40
50
60
70
80
0
200
400
600
800
1000
1200

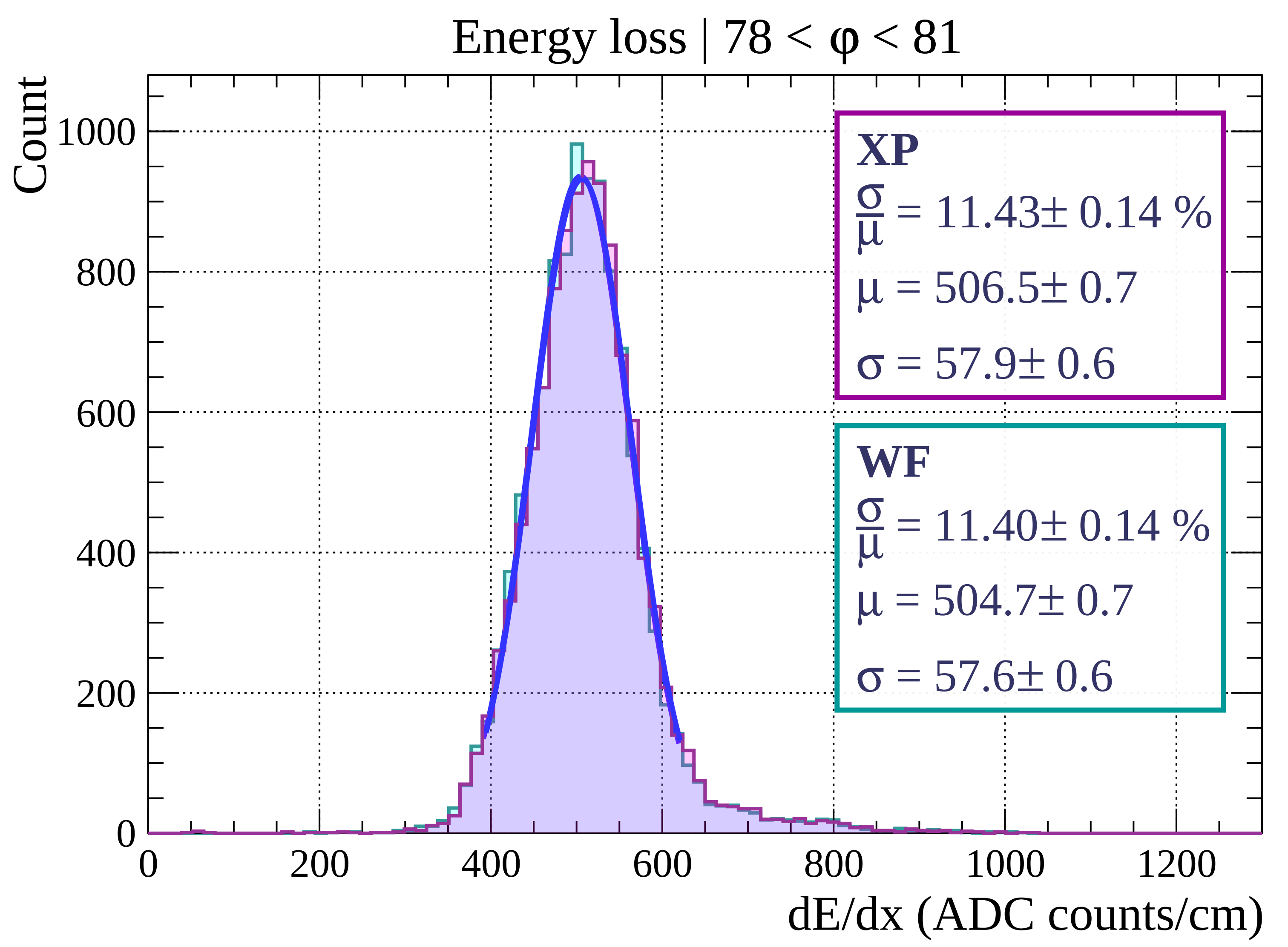

Energy loss | 69 < φ < 72
Count
0
50
100
150
200
250
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 12.21± 0.33 %
μ = 500.2± 1.5
σ = 61.1± 1.5
WF
σ/μ = 12.05± 0.31 %
μ = 495.9± 1.4
σ = 59.7± 1.3

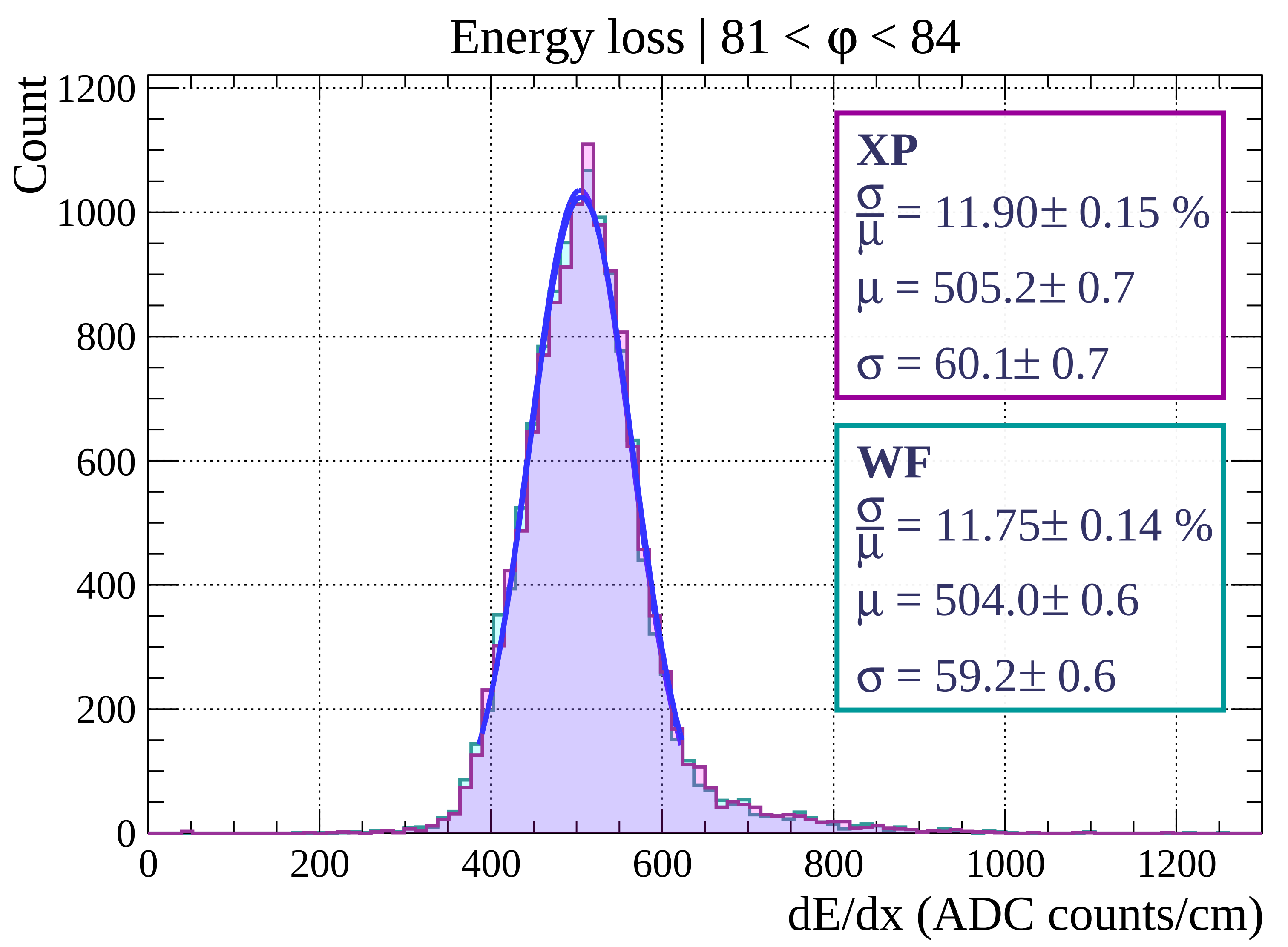


Energy loss | 72 < φ < 75
Count
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 11.56± 0.20 %
μ = 502.0± 1.0
σ = 58.0± 0.9
WF
σ/μ = 11.60± 0.21 %
μ = 499.4± 1.0
σ = 57.9± 0.9

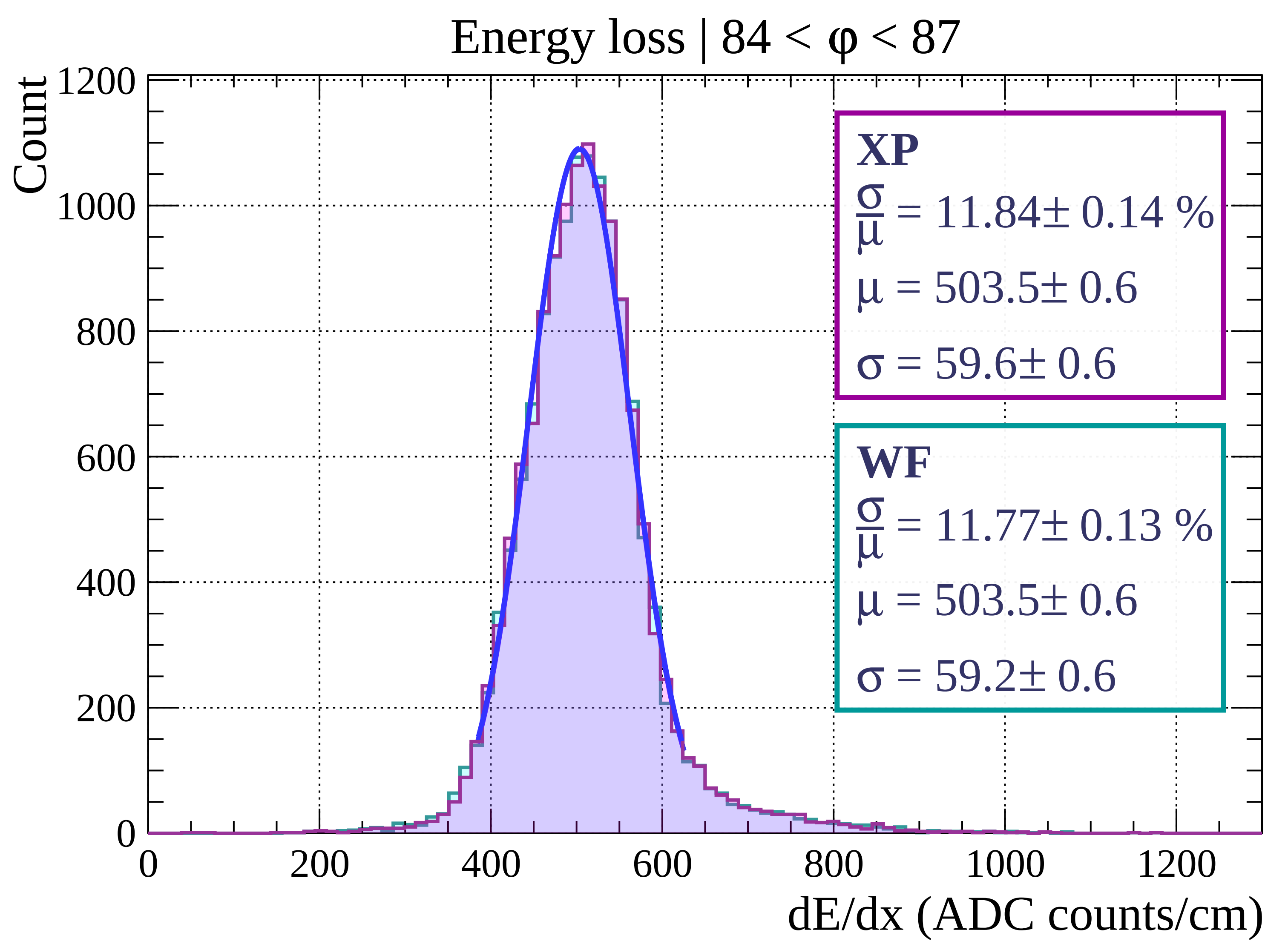


Energy loss | 75 < φ < 78
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200
XP
σ/μ = 11.34± 0.16 %
μ = 505.0± 0.8
σ = 57.3± 0.7
WF
σ/μ = 11.21± 0.17 %
μ = 502.1± 0.8
σ = 56.3± 0.8

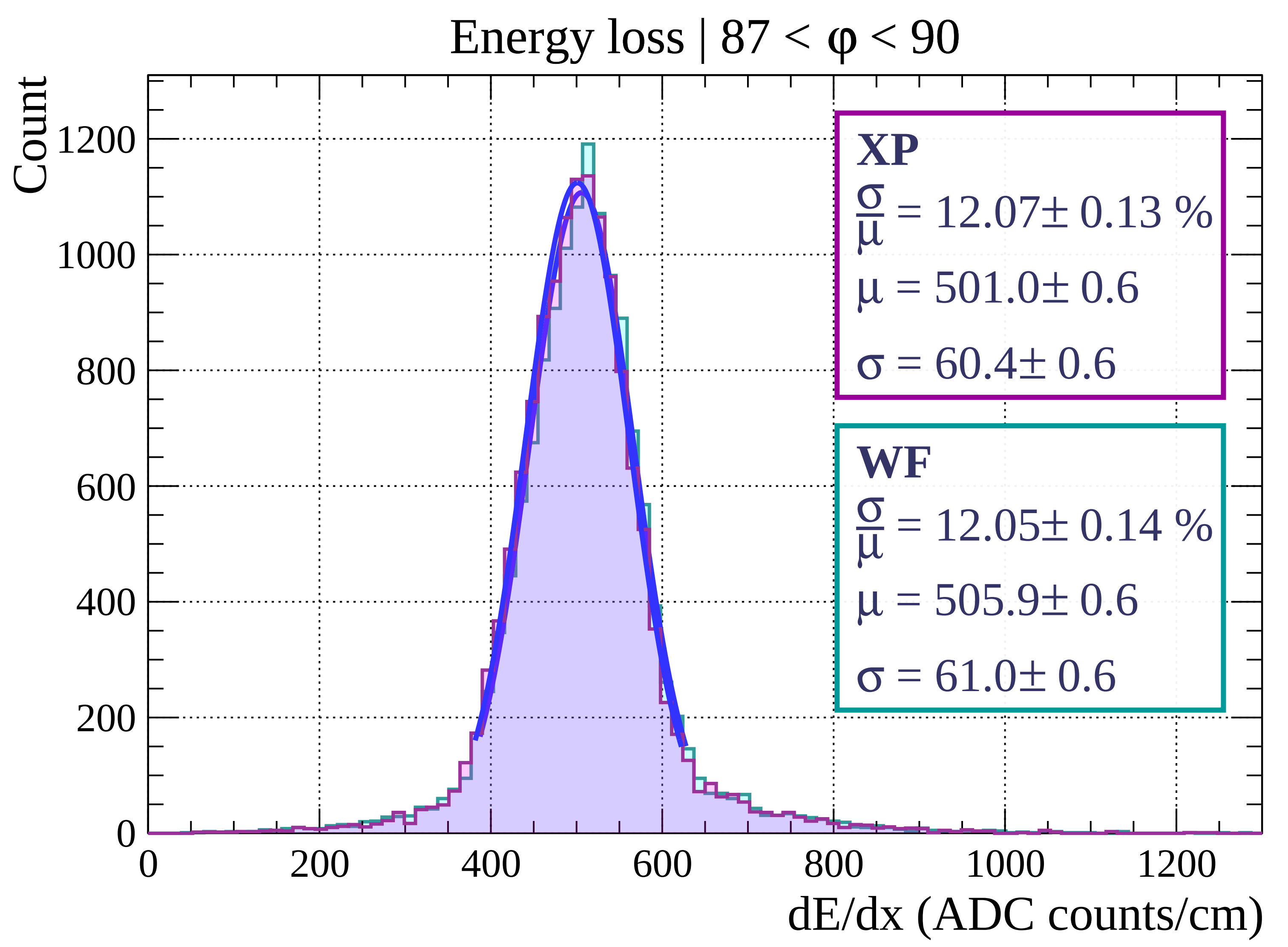

Energy loss | 78 < φ < 81
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP
σ/μ = 11.43± 0.14 %
μ = 506.5± 0.7
σ = 57.9± 0.6
WF
σ/μ = 11.40± 0.14 %
μ = 504.7± 0.7
σ = 57.6± 0.6



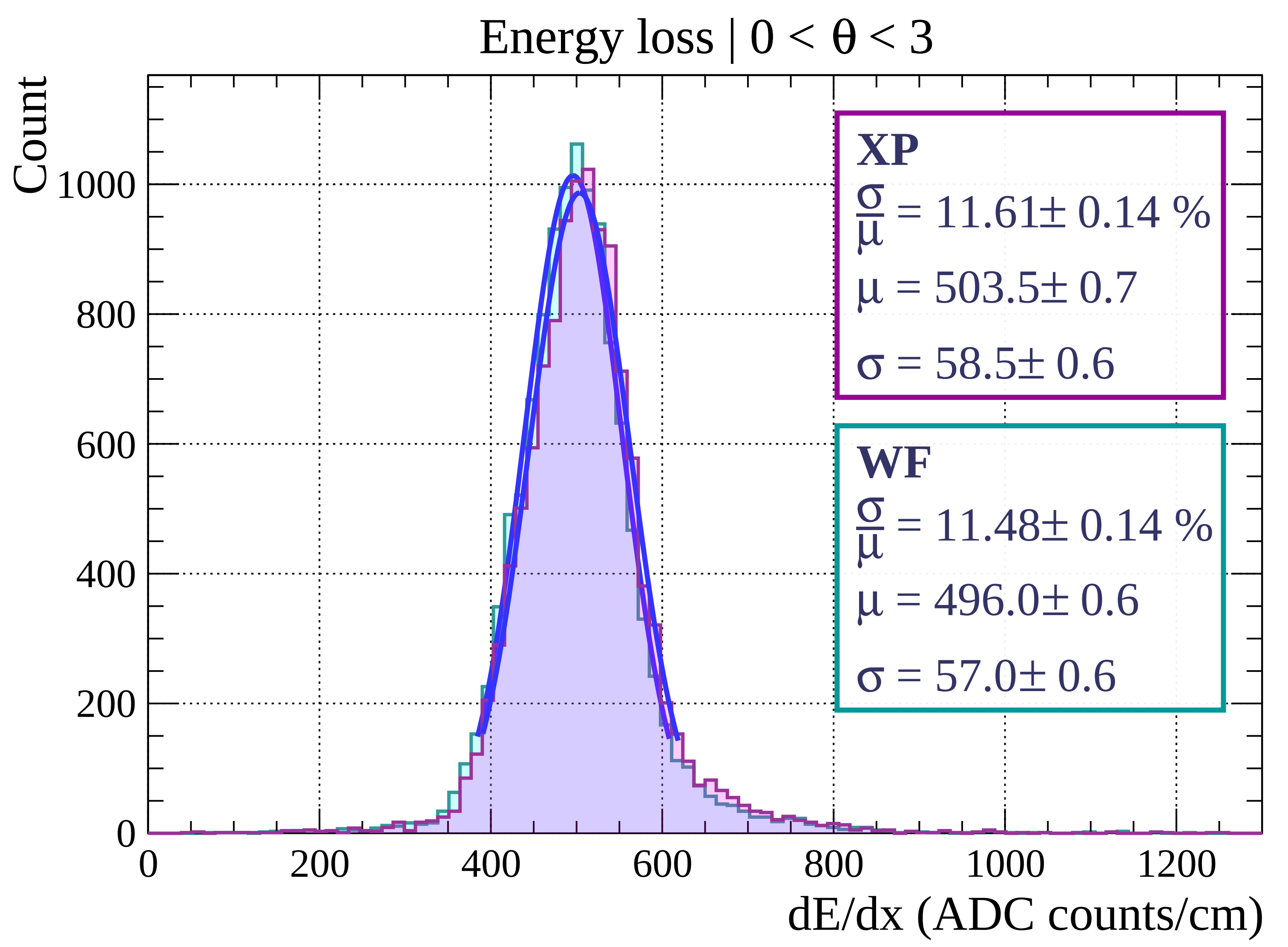
Energy loss | 81 < φ < 84
Count
XP
σ/μ = 11.90± 0.15 %
μ = 505.2± 0.7
σ = 60.1± 0.7
WF
σ/μ = 11.75± 0.14 %
μ = 504.0± 0.6
σ = 59.2± 0.6
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200



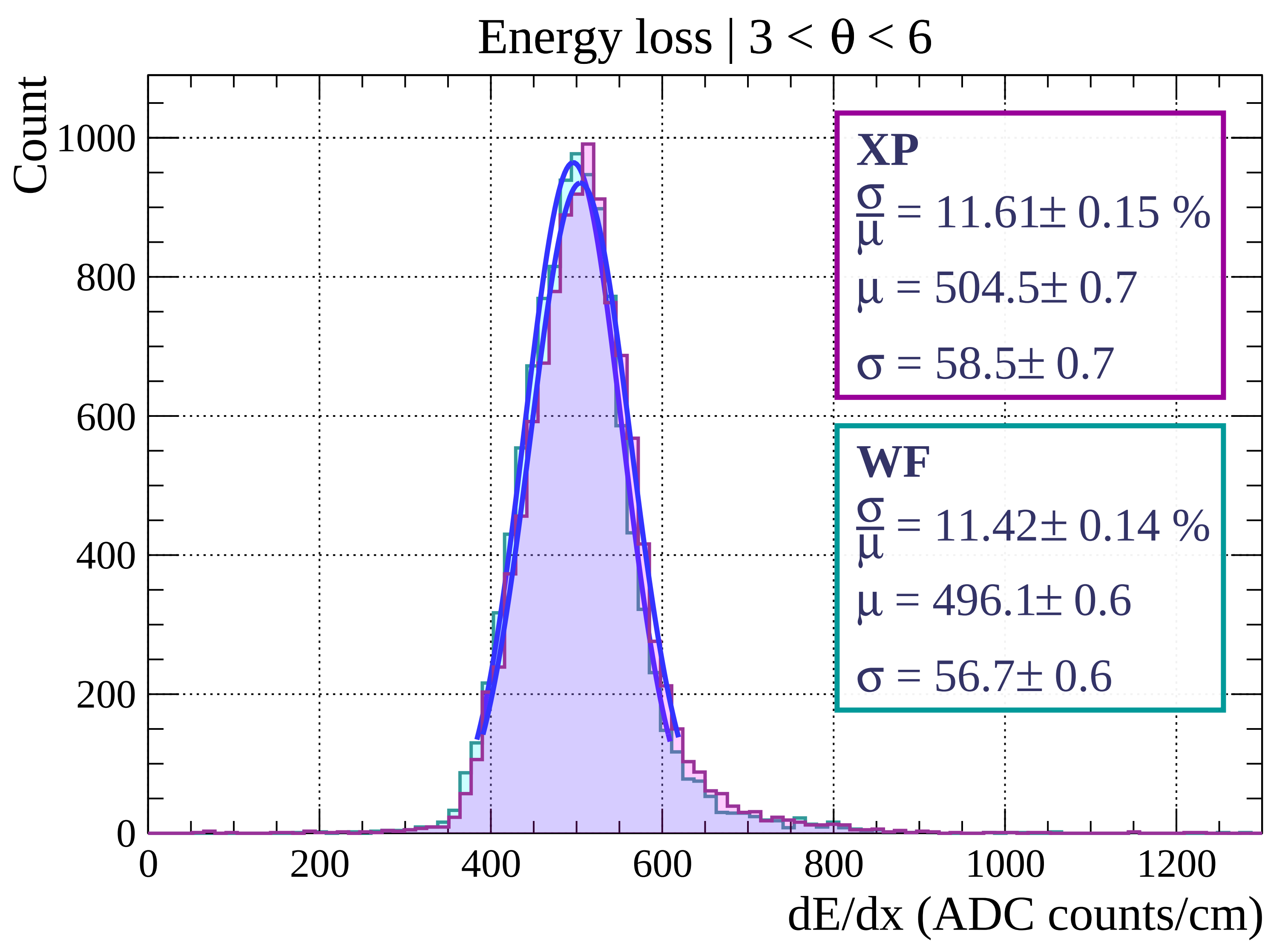
Energy loss | 84 < φ < 87
Count
1200
1000
800
600
400
200
0
XP
σ/μ = 11.84± 0.14 %
μ = 503.5± 0.6
σ = 59.6± 0.6
WF
σ/μ = 11.77± 0.13 %
μ = 503.5± 0.6
σ = 59.2± 0.6
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)



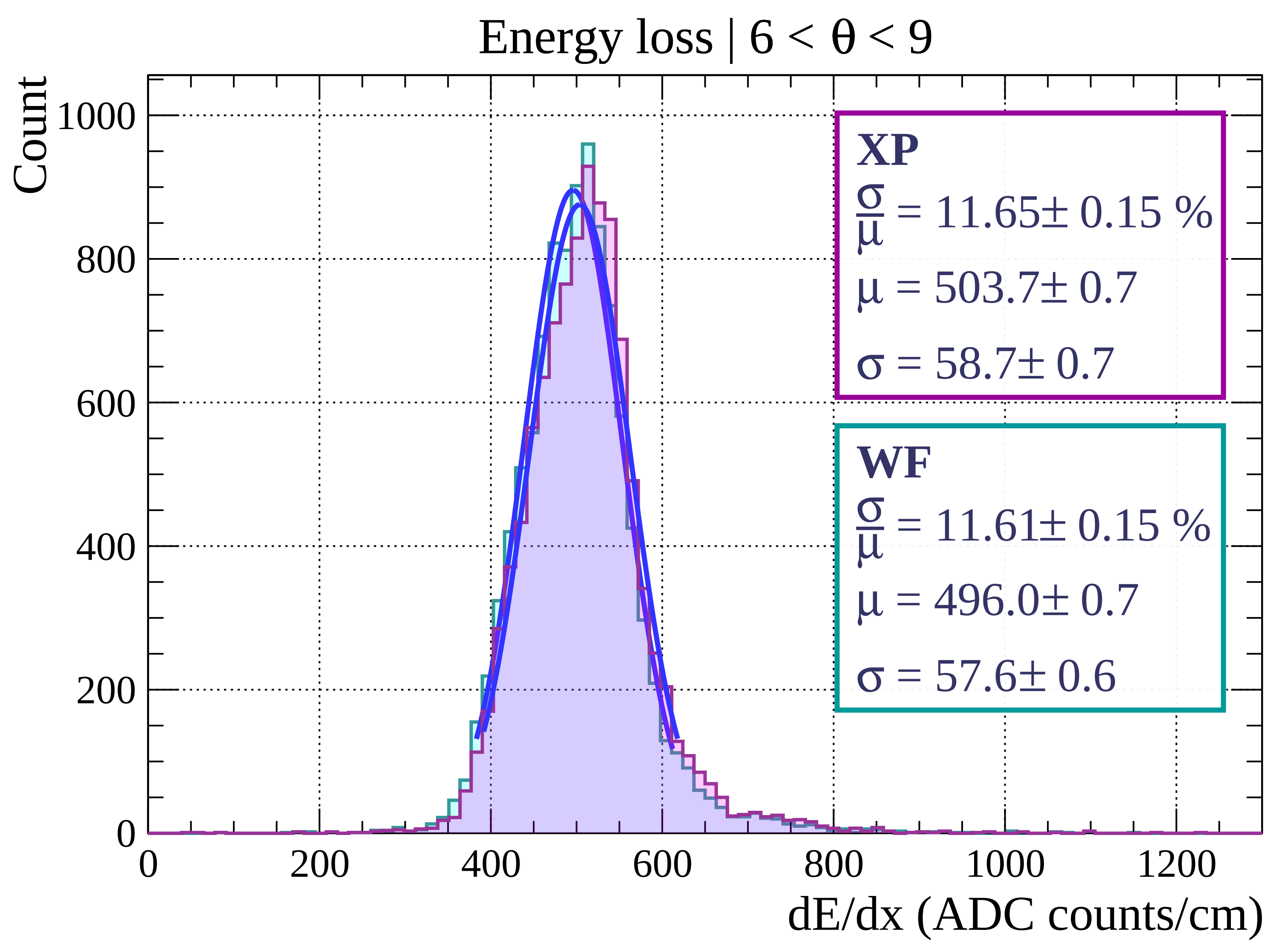
Energy loss | 87 < φ < 90
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP
σ/μ = 12.07± 0.13 %
μ = 501.0± 0.6
σ = 60.4± 0.6
WF
σ/μ = 12.05± 0.14 %
μ = 505.9± 0.6
σ = 61.0± 0.6

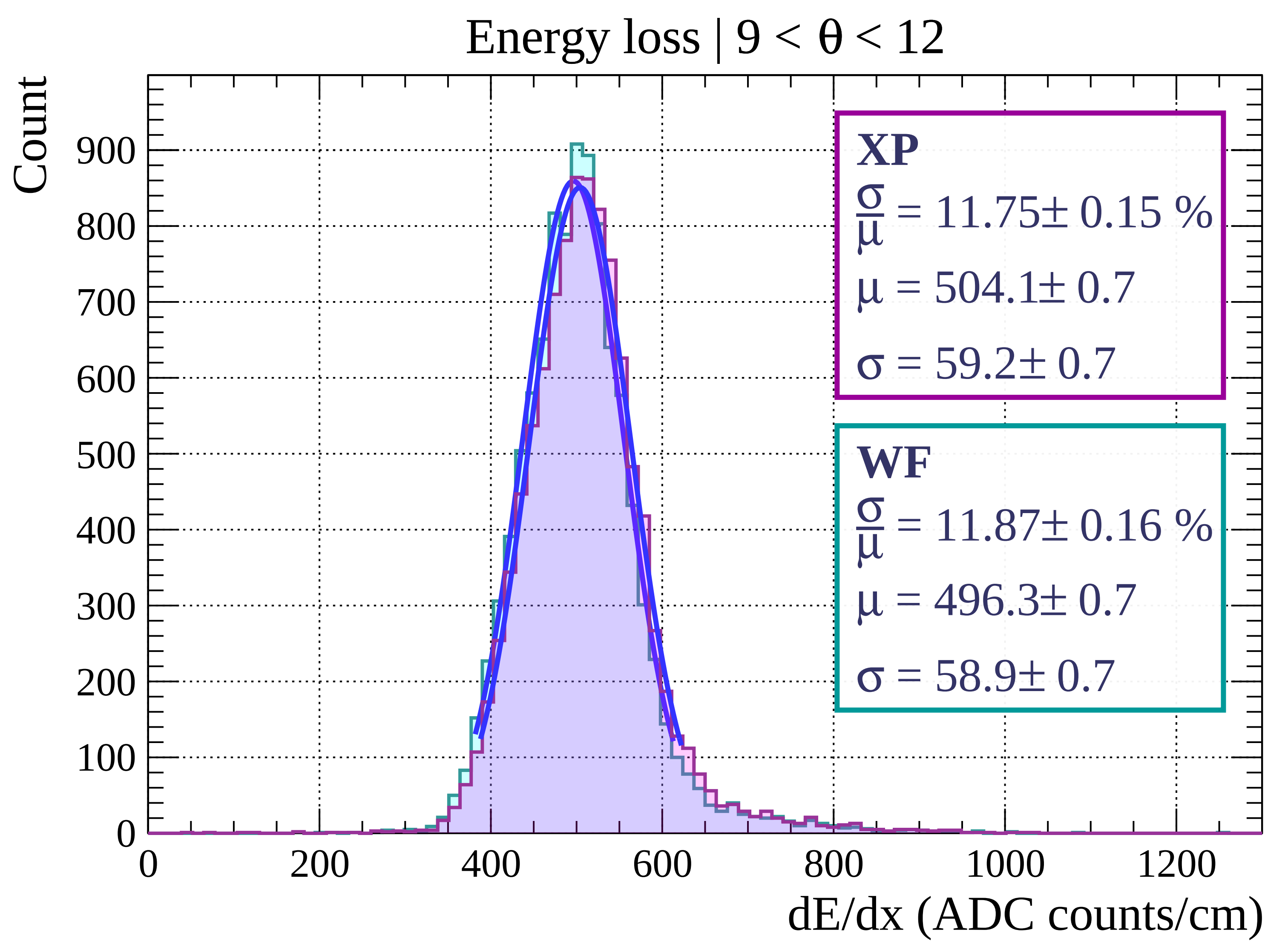


Energy loss | 0 < θ < 3
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 11.61± 0.14 %
μ = 503.5± 0.7
σ = 58.5± 0.6
WF
$\frac{\sigma}{\mu}$ = 11.48± 0.14 %
μ = 496.0± 0.6
σ = 57.0± 0.6

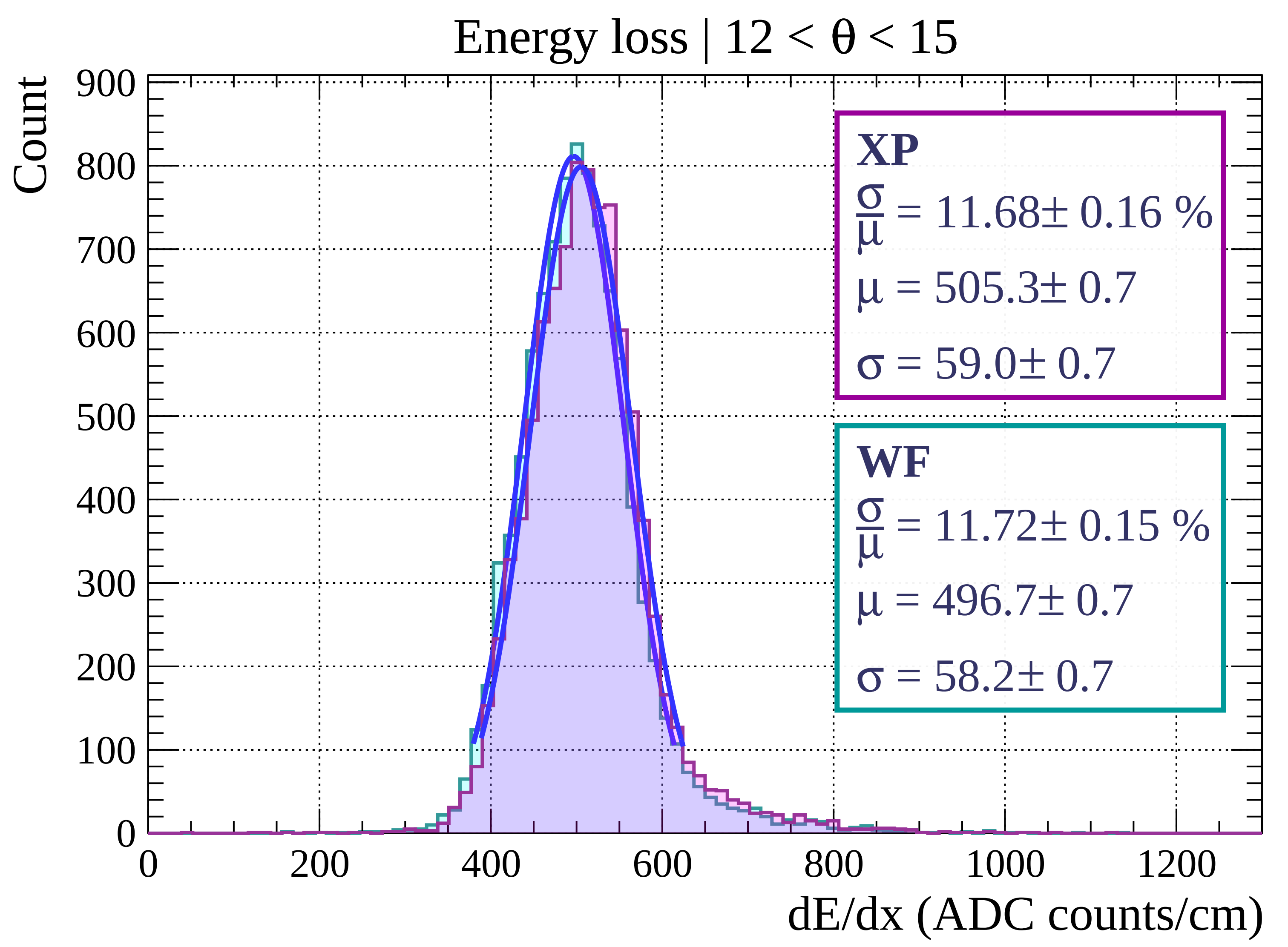


Energy loss | 3 < θ < 6
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 11.61± 0.15 %
μ = 504.5± 0.7
σ = 58.5± 0.7
WF
σ/μ = 11.42± 0.14 %
μ = 496.1± 0.6
σ = 56.7± 0.6
0
200
400
600
800
1000
1200

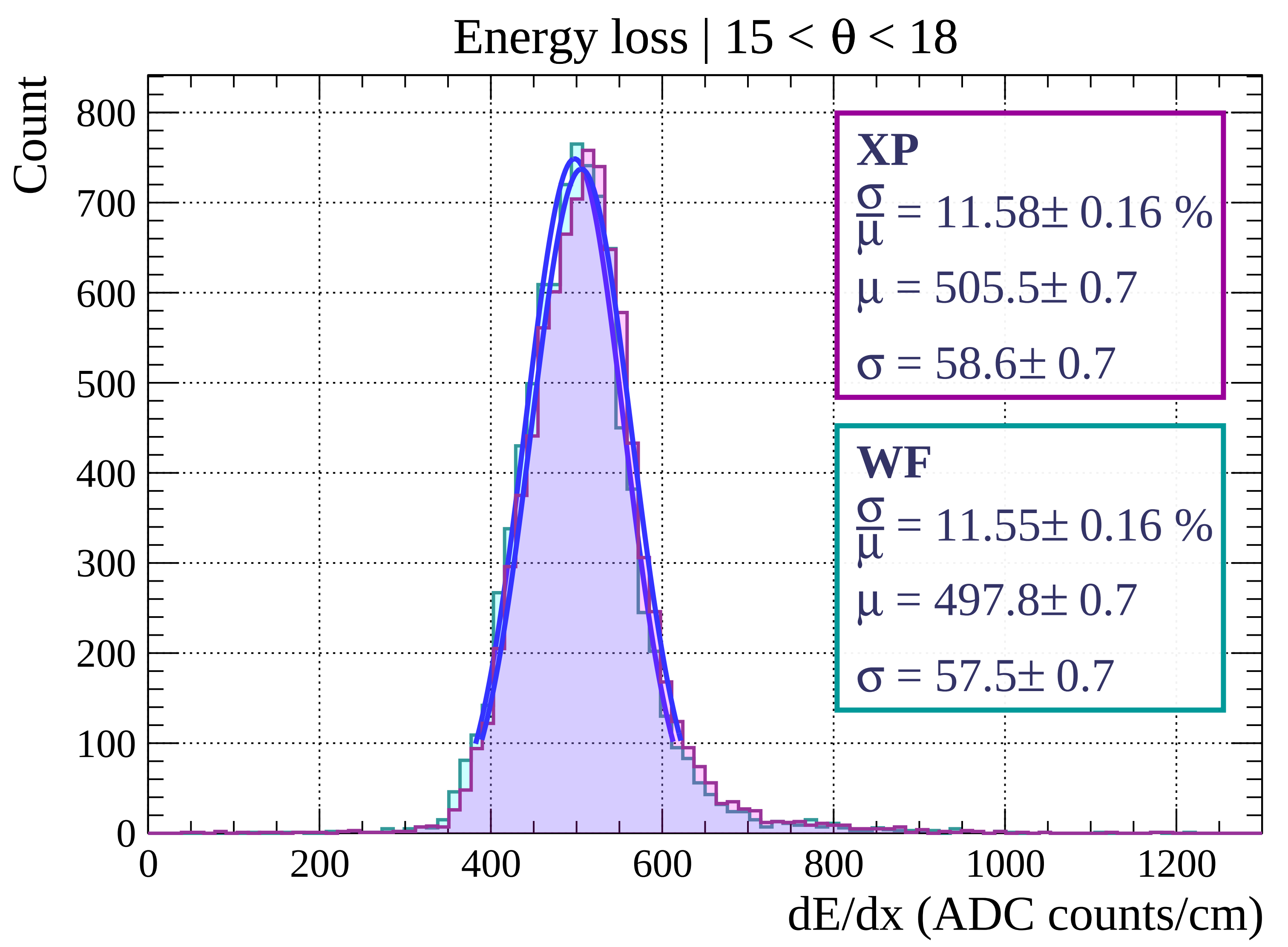

Energy loss | 6 < θ < 9
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP
σ/μ = 11.65± 0.15 %
μ = 503.7± 0.7
σ = 58.7± 0.7
WF
σ/μ = 11.61± 0.15 %
μ = 496.0± 0.7
σ = 57.6± 0.6



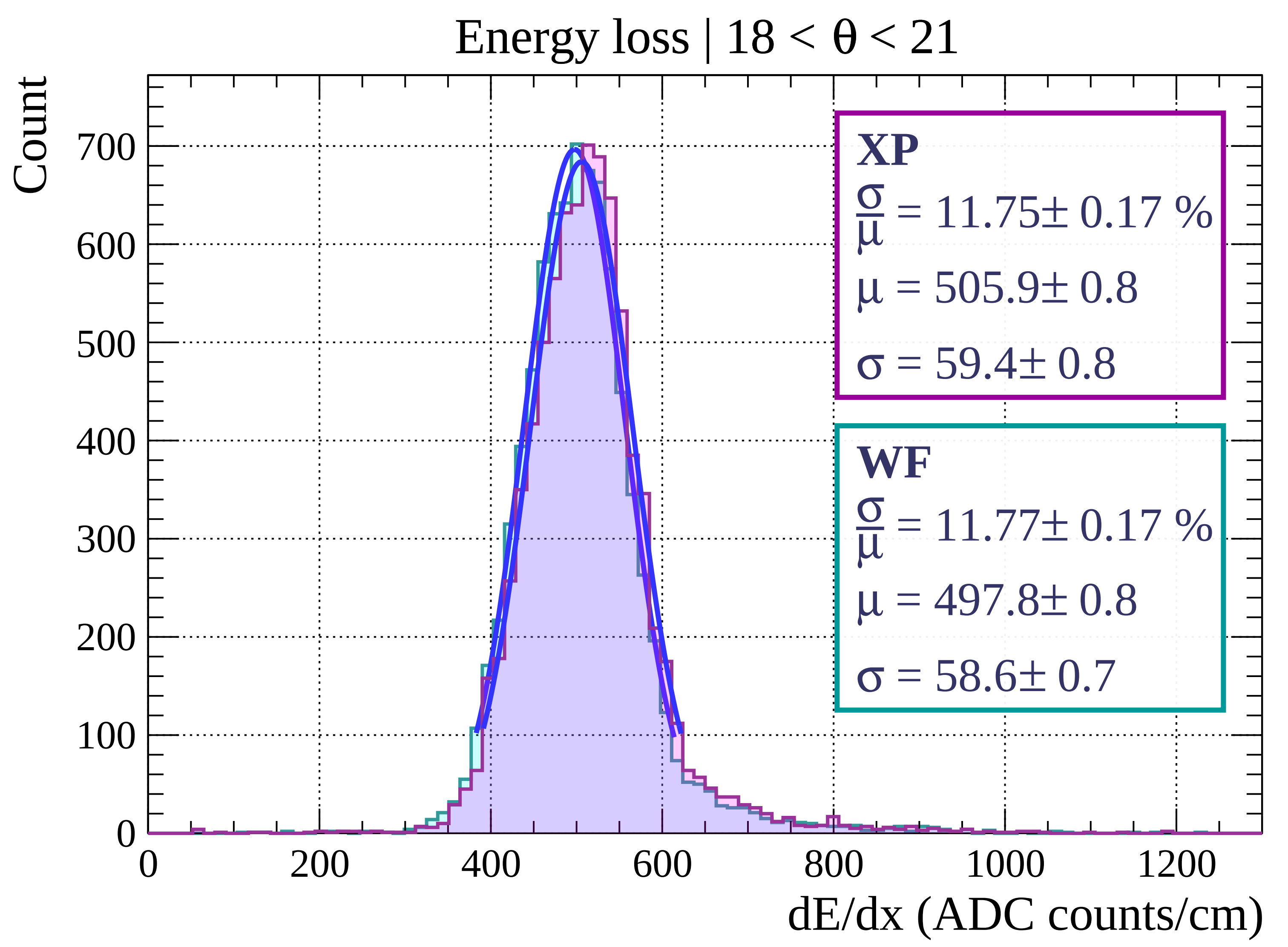
Energy loss | 9 < θ < 12
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200
XP
σ/μ = 11.75± 0.15 %
μ = 504.1± 0.7
σ = 59.2± 0.7
WF
σ/μ = 11.87± 0.16 %
μ = 496.3± 0.7
σ = 58.9± 0.7



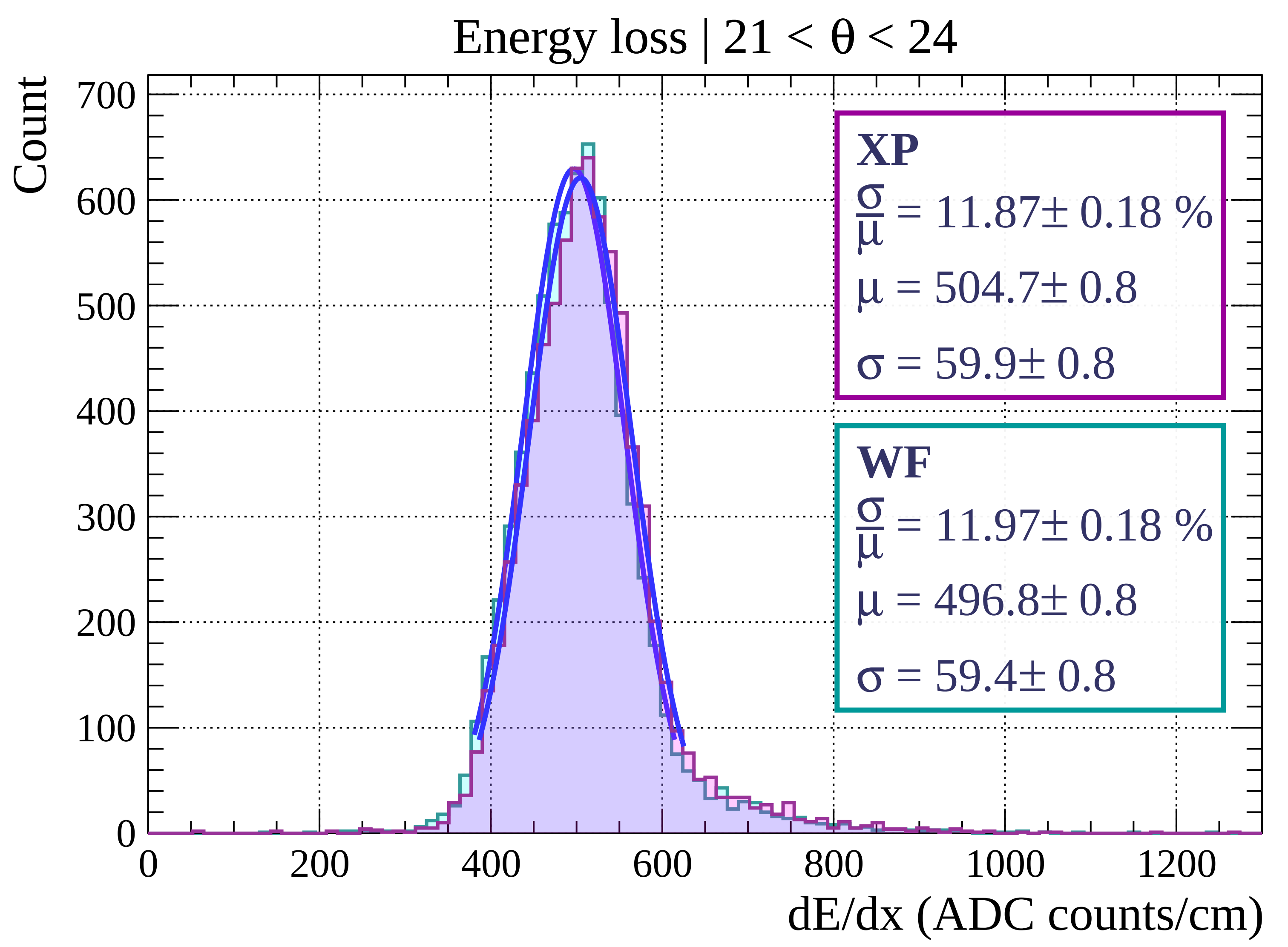
Energy loss | 12 < θ < 15
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 11.68± 0.16 %
μ = 505.3± 0.7
σ = 59.0± 0.7
WF
σ/μ = 11.72± 0.15 %
μ = 496.7± 0.7
σ = 58.2± 0.7

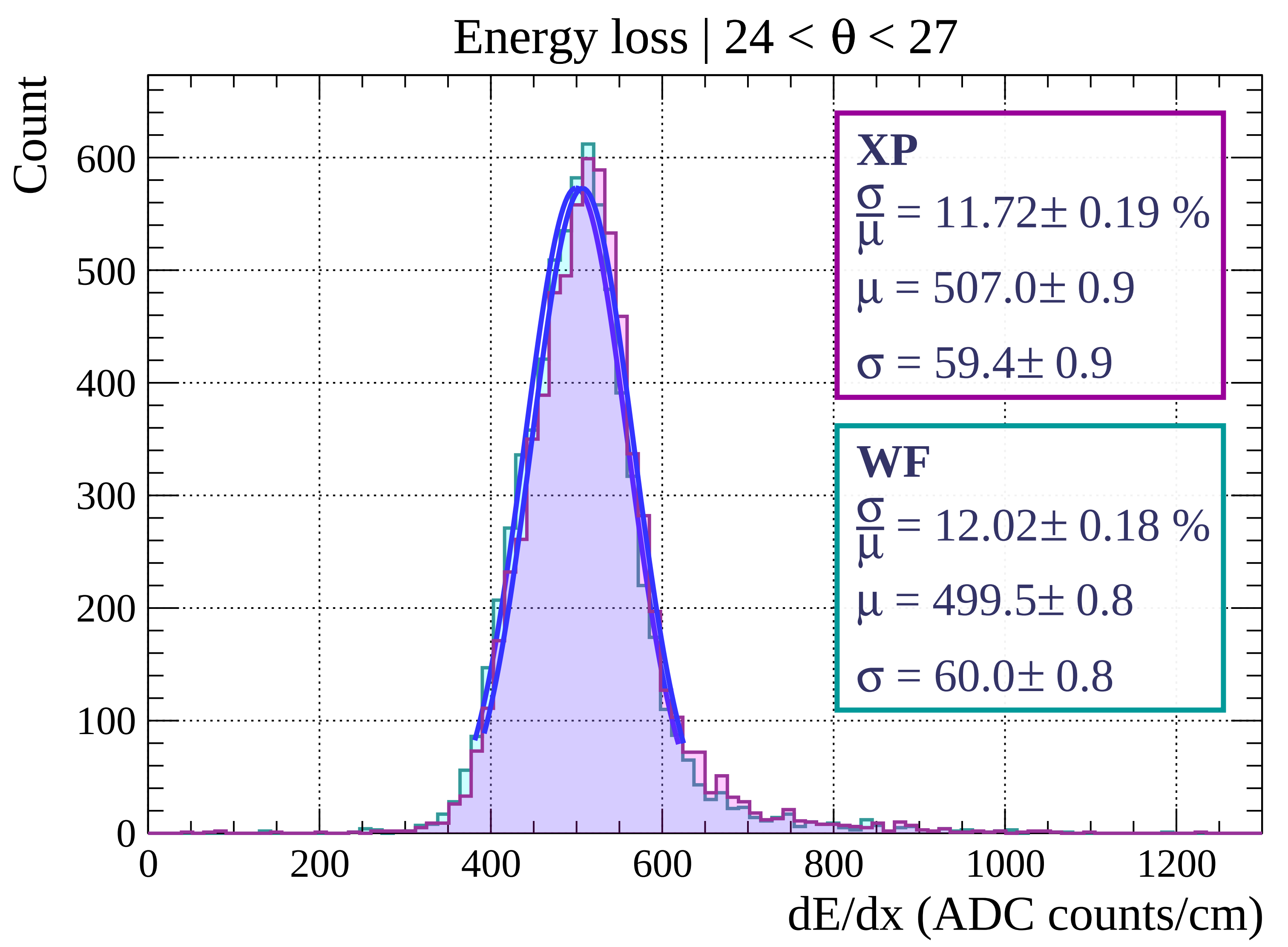


Energy loss | 15 < θ < 18
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200
XP
σ/μ = 11.58± 0.16 %
μ = 505.5± 0.7
σ = 58.6± 0.7
WF
σ/μ = 11.55± 0.16 %
μ = 497.8± 0.7
σ = 57.5± 0.7

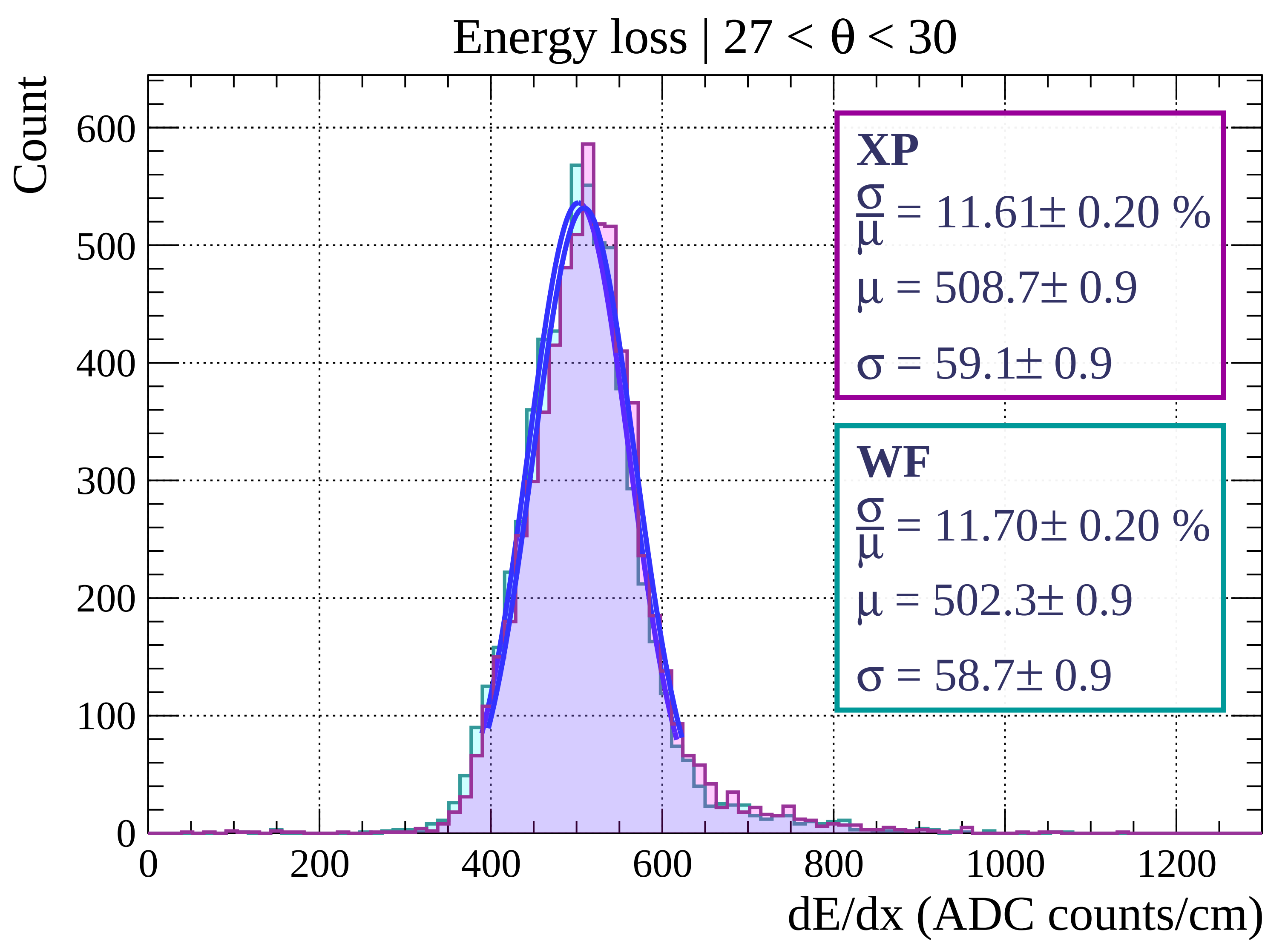


Energy loss | 18 < θ < 21
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP
σ/μ = 11.75± 0.17 %
μ = 505.9± 0.8
σ = 59.4± 0.8
WF
σ/μ = 11.77± 0.17 %
μ = 497.8± 0.8
σ = 58.6± 0.7

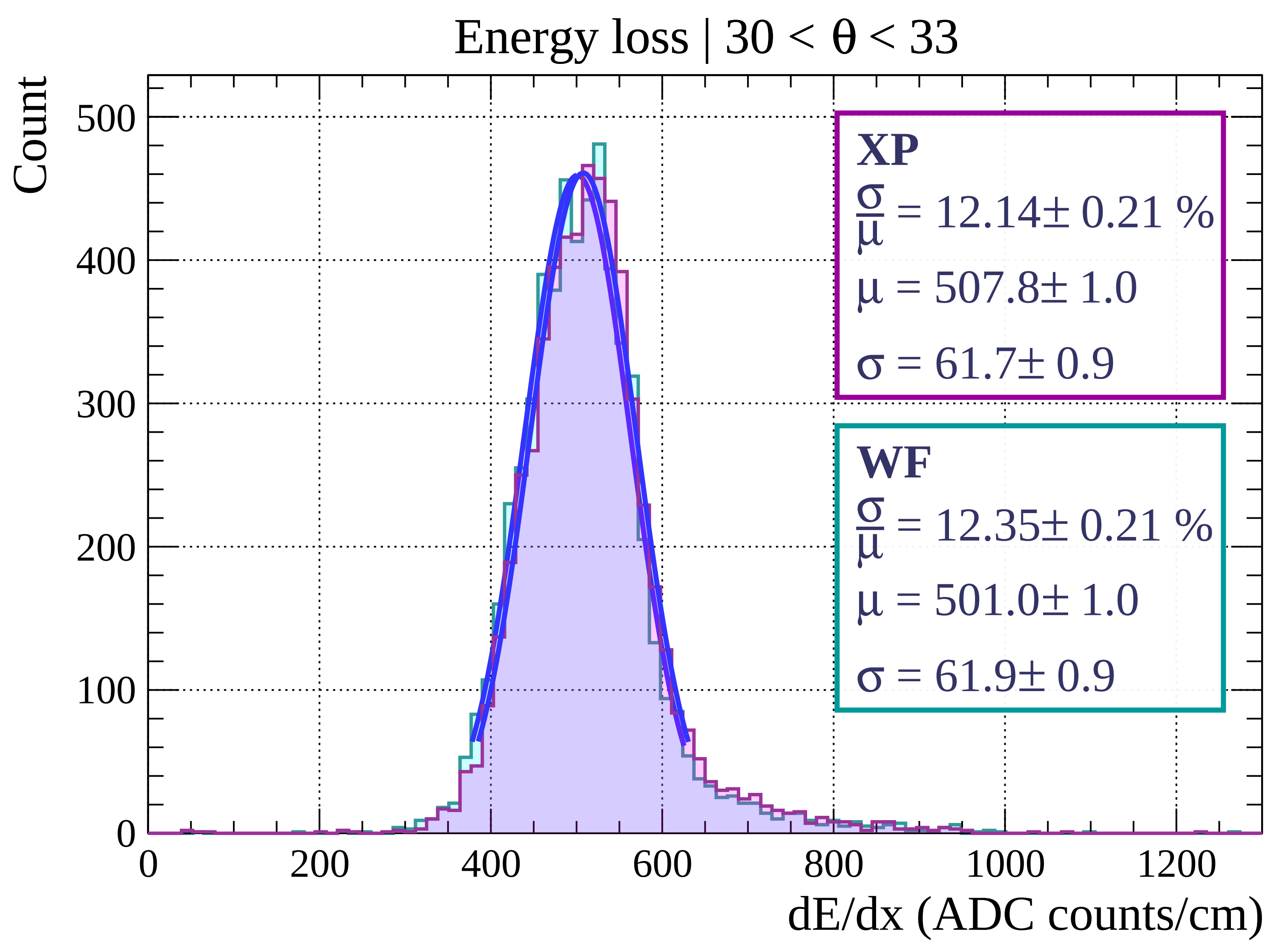

Energy loss | 21 < θ < 24
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP
σ/μ = 11.87± 0.18 %
μ = 504.7± 0.8
σ = 59.9± 0.8
WF
σ/μ = 11.97± 0.18 %
μ = 496.8± 0.8
σ = 59.4± 0.8



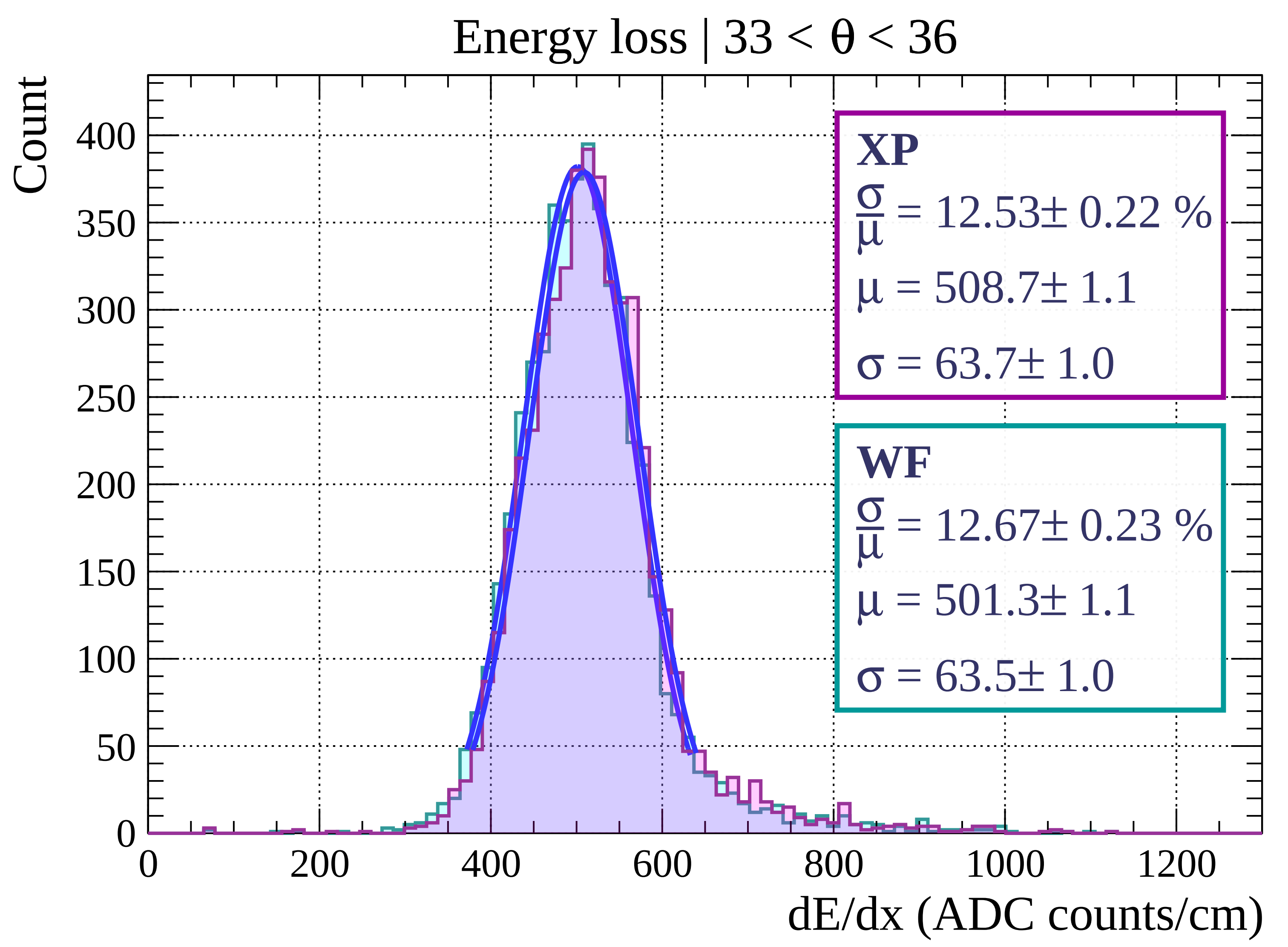
Energy loss | 24 < θ < 27
Count
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 11.72± 0.19 %
μ = 507.0± 0.9
σ = 59.4± 0.9
WF
σ/μ = 12.02± 0.18 %
μ = 499.5± 0.8
σ = 60.0± 0.8



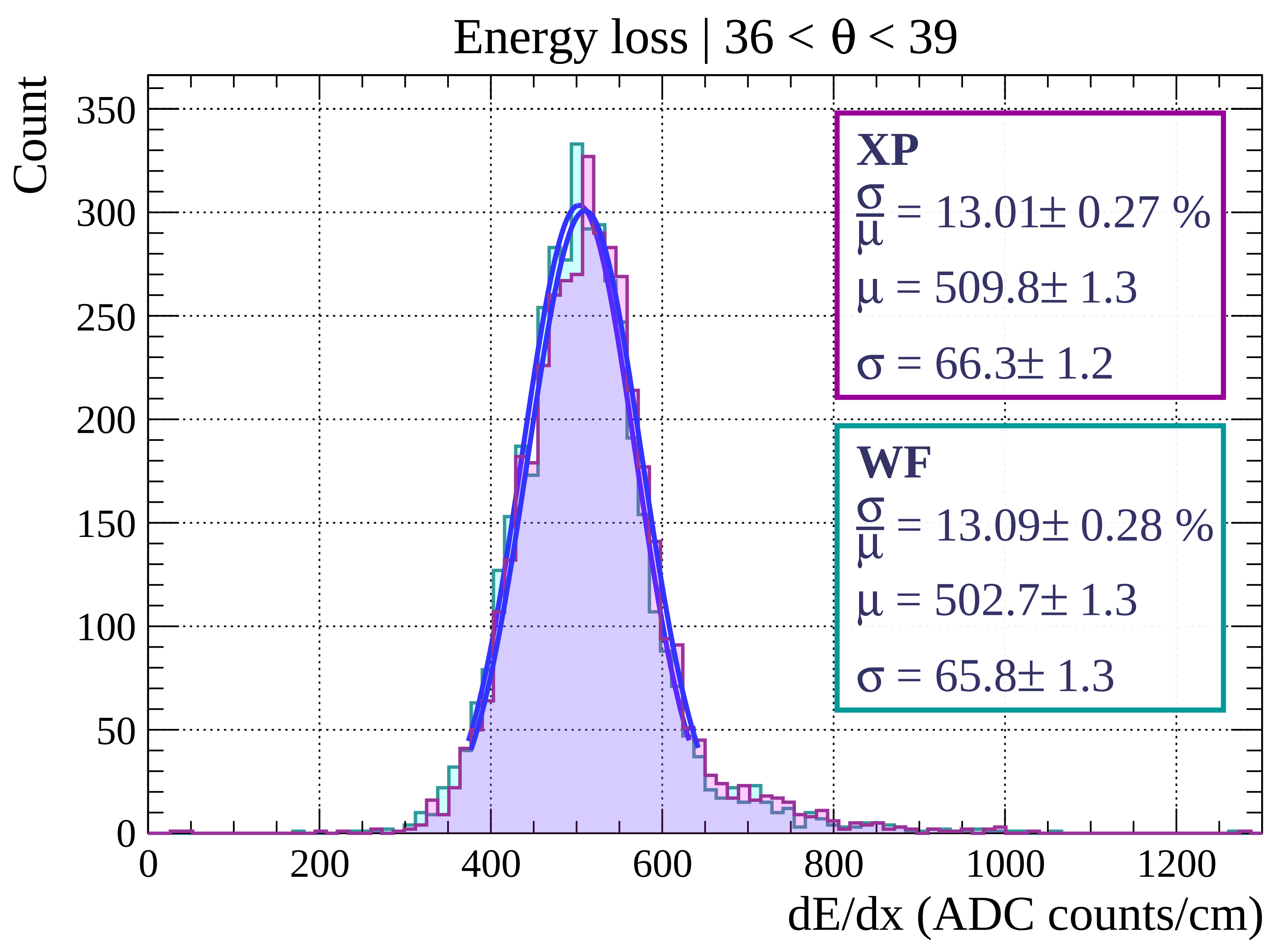
Energy loss | 27 < θ < 30
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP
σ/μ = 11.61± 0.20 %
μ = 508.7± 0.9
σ = 59.1± 0.9
WF
σ/μ = 11.70± 0.20 %
μ = 502.3± 0.9
σ = 58.7± 0.9



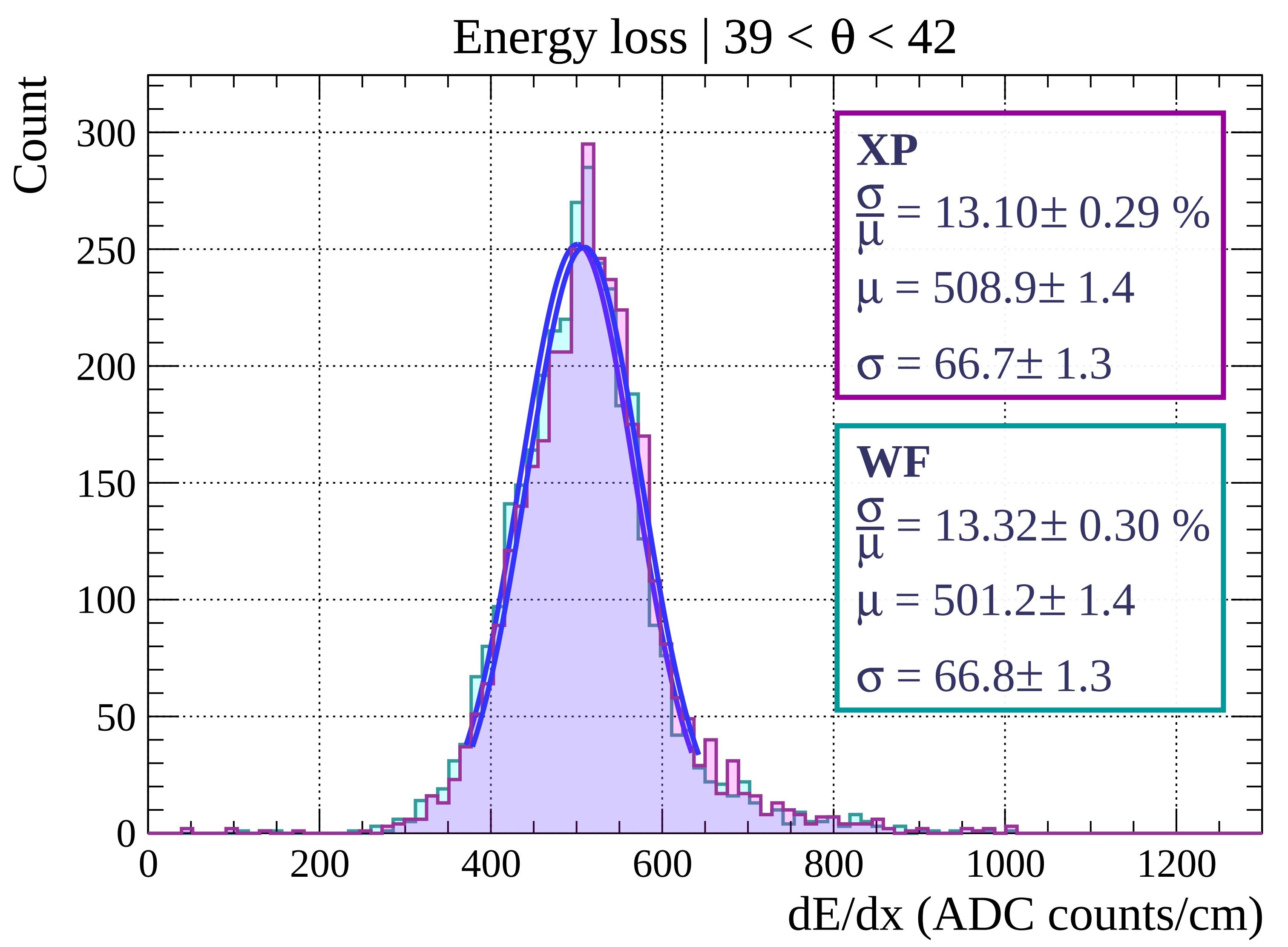
Energy loss | 30 < θ < 33
Count
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 12.14± 0.21 %
μ = 507.8± 1.0
σ = 61.7± 0.9
WF
σ/μ = 12.35± 0.21 %
μ = 501.0± 1.0
σ = 61.9± 0.9



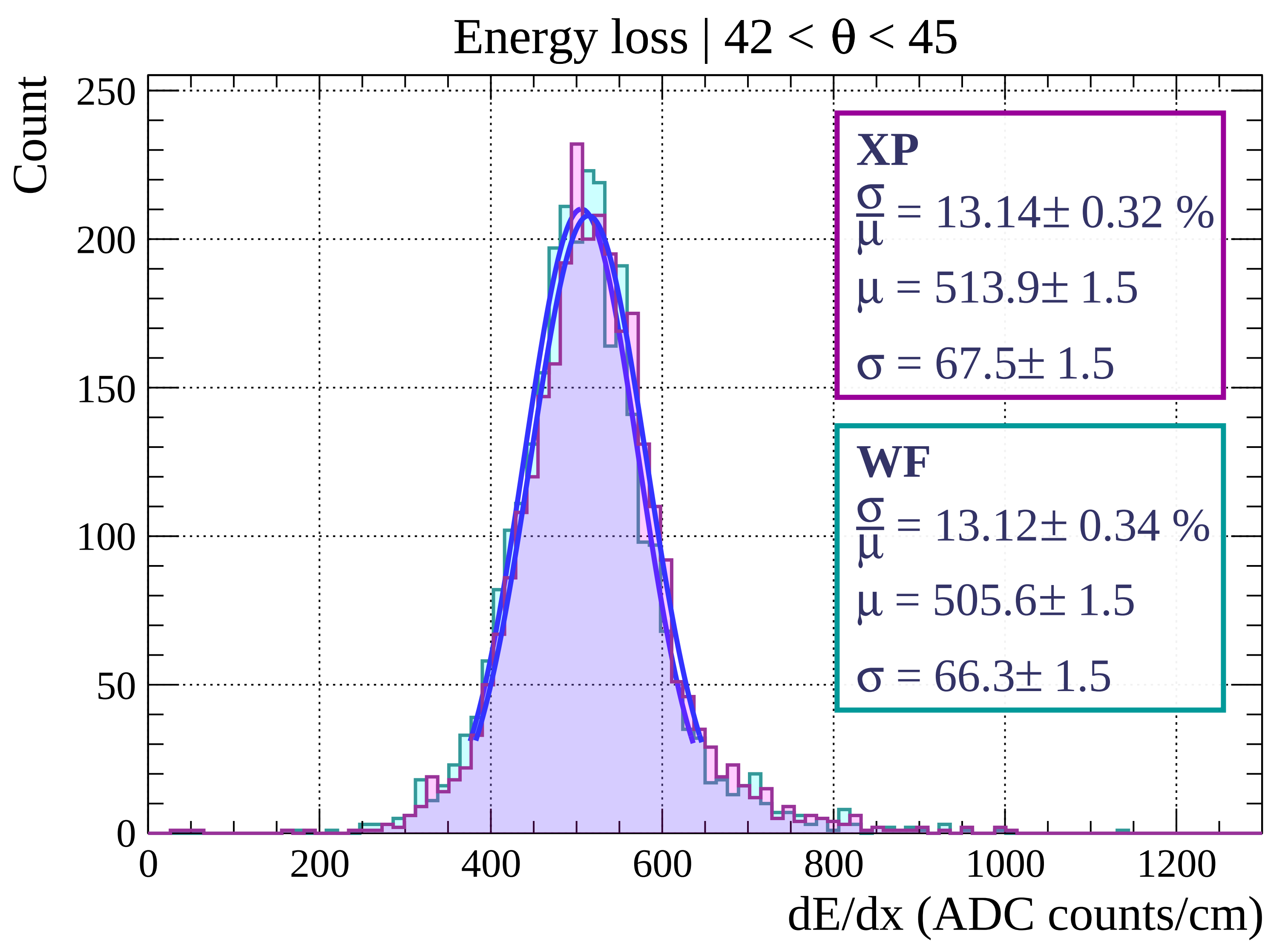
Energy loss | 33 < θ < 36
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
50
100
150
200
250
300
350
400
XP
σ/μ = 12.53± 0.22 %
μ = 508.7± 1.1
σ = 63.7± 1.0
WF
σ/μ = 12.67± 0.23 %
μ = 501.3± 1.1
σ = 63.5± 1.0



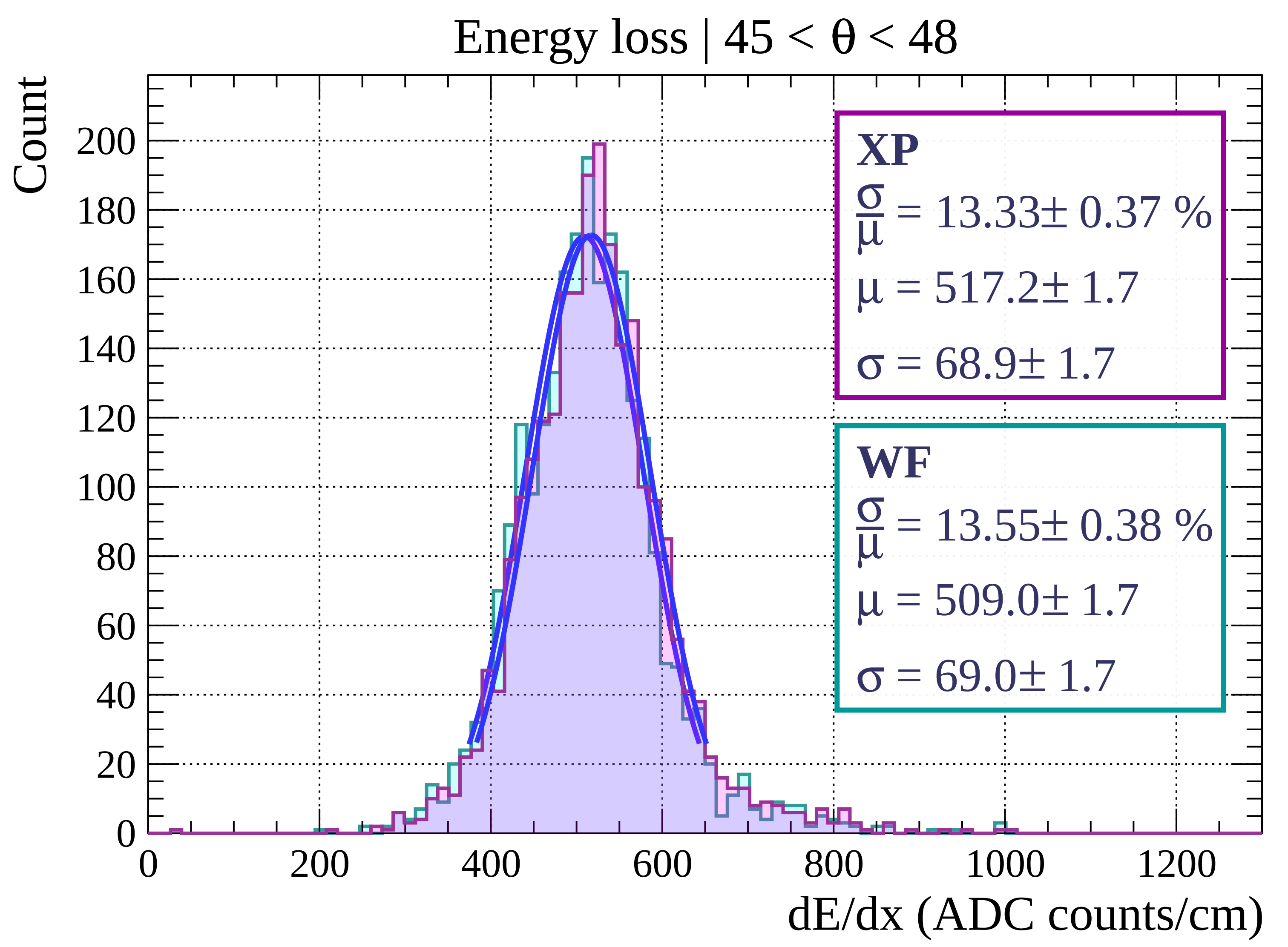
Energy loss | 36 < θ < 39
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP
σ/μ = 13.01± 0.27 %
μ = 509.8± 1.3
σ = 66.3± 1.2
WF
σ/μ = 13.09± 0.28 %
μ = 502.7± 1.3
σ = 65.8± 1.3



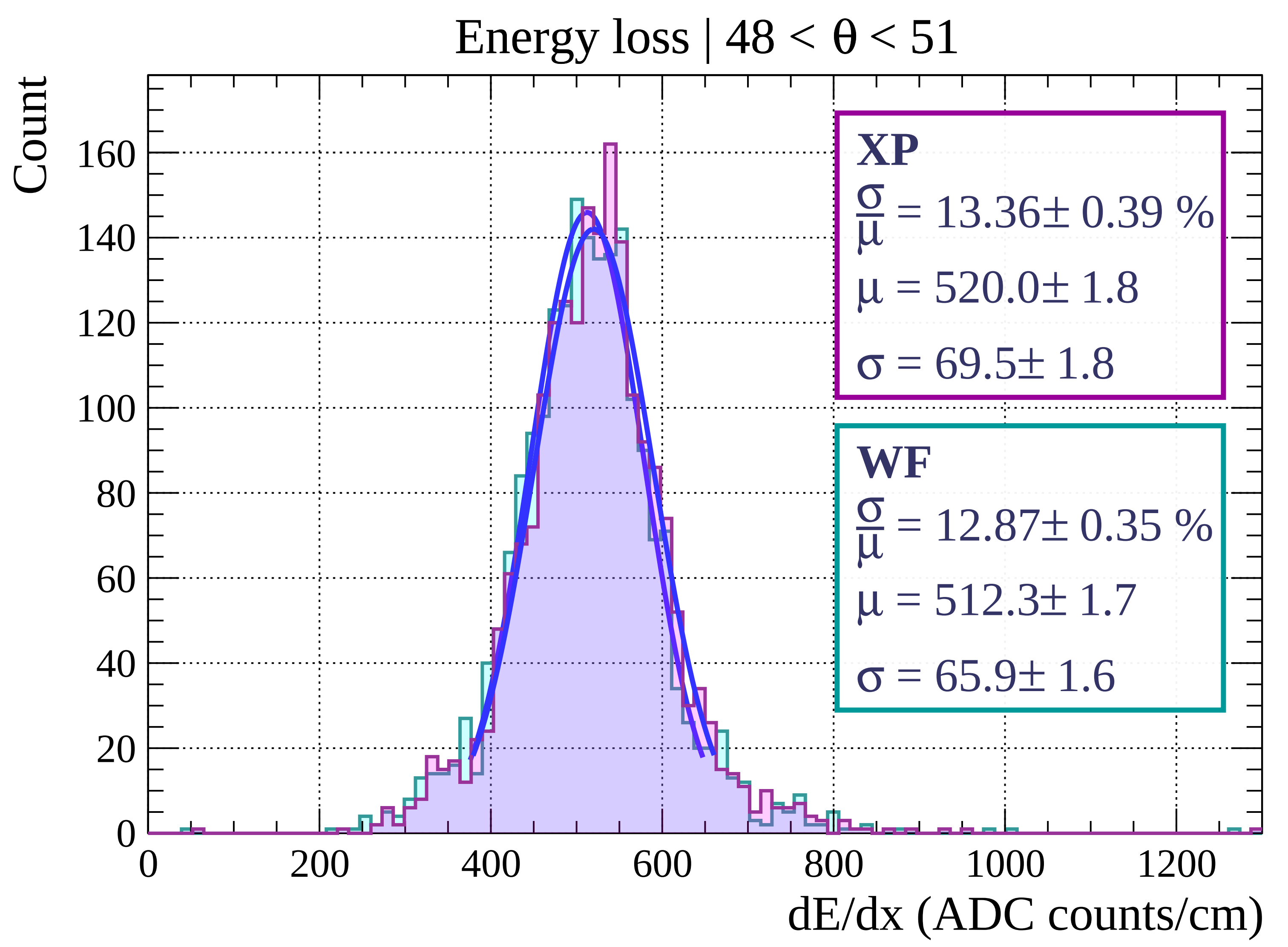
Energy loss | 39 < θ < 42
Count
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 13.10± 0.29 %
μ = 508.9± 1.4
σ = 66.7± 1.3
WF
σ/μ = 13.32± 0.30 %
μ = 501.2± 1.4
σ = 66.8± 1.3



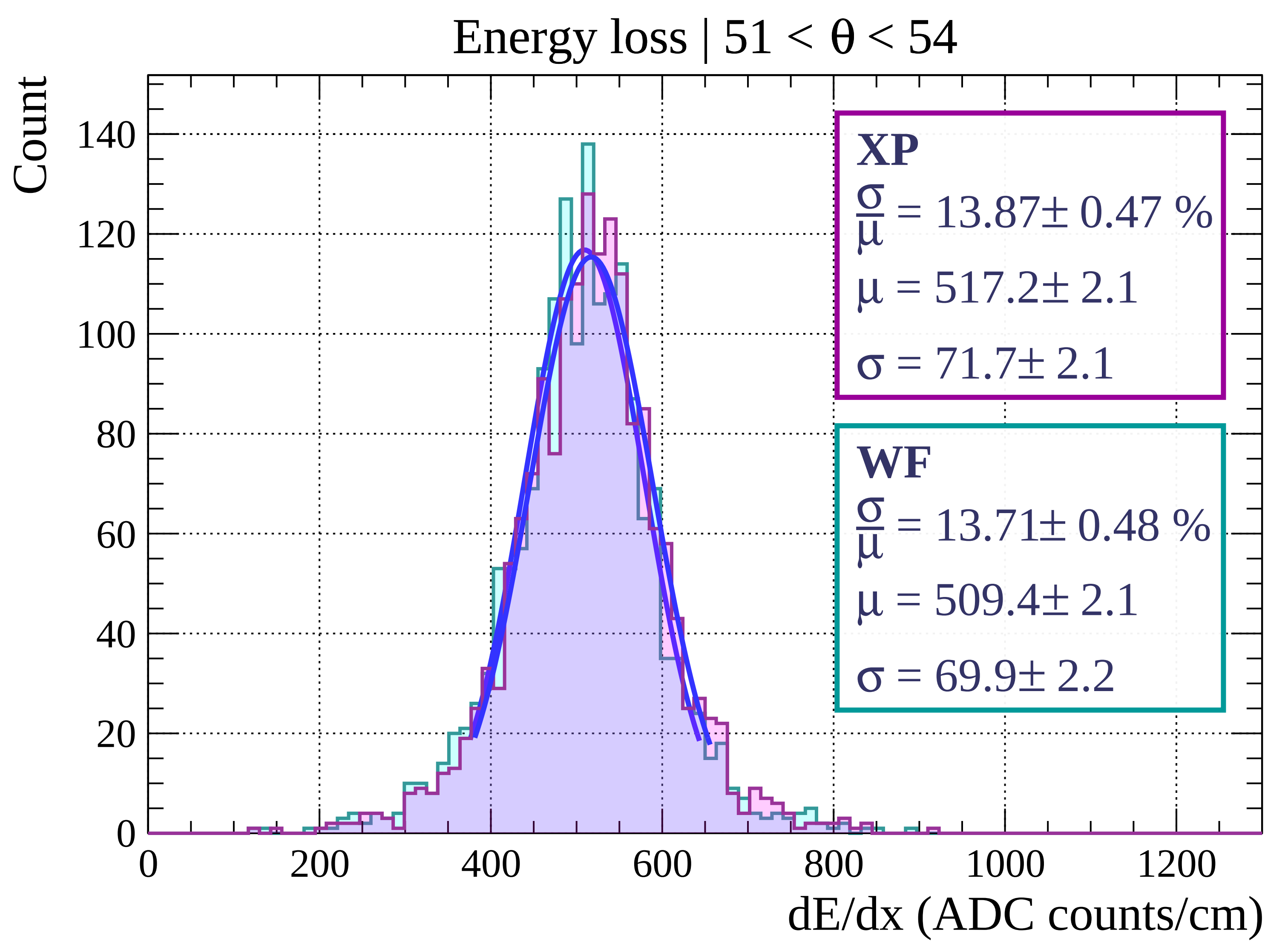
Energy loss | 42 < θ < 45
Count
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 13.14± 0.32 %
μ = 513.9± 1.5
σ = 67.5± 1.5
WF
σ/μ = 13.12± 0.34 %
μ = 505.6± 1.5
σ = 66.3± 1.5



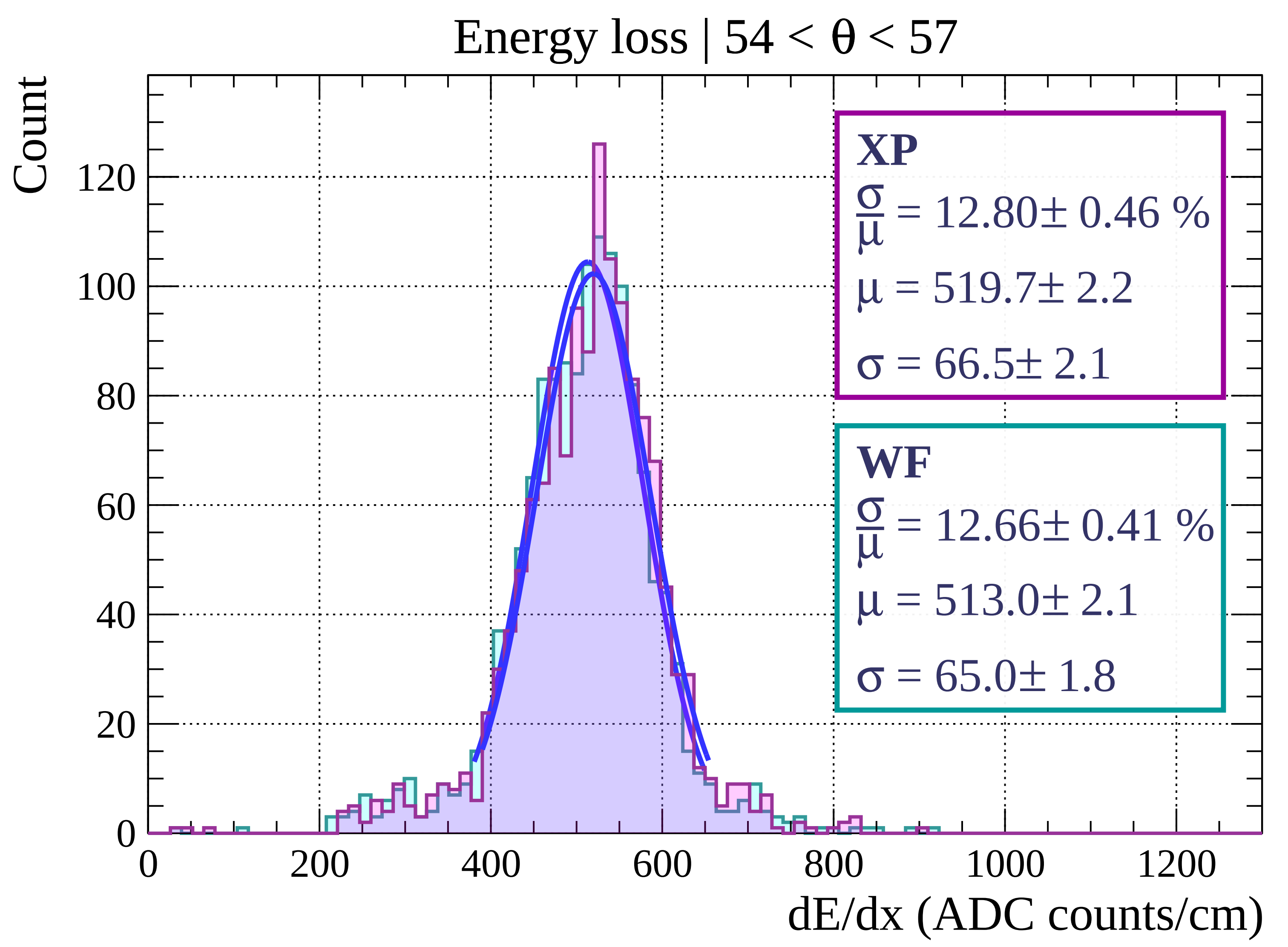
Energy loss | 45 < θ < 48
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
200
XP
σ/μ = 13.33± 0.37 %
μ = 517.2± 1.7
σ = 68.9± 1.7
WF
σ/μ = 13.55± 0.38 %
μ = 509.0± 1.7
σ = 69.0± 1.7



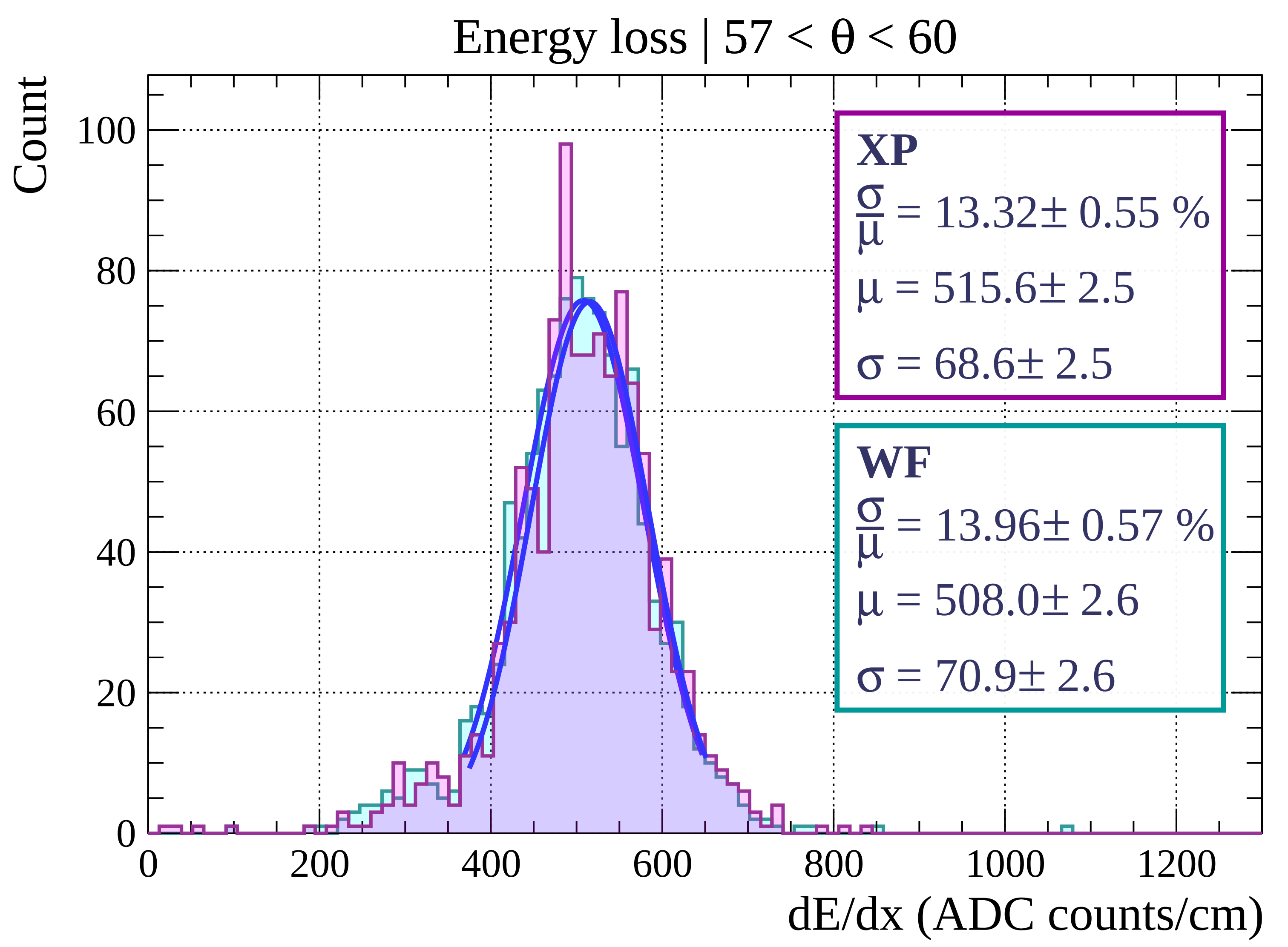
Energy loss | 48 < θ < 51
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200
XP
σ/μ = 13.36± 0.39 %
μ = 520.0± 1.8
σ = 69.5± 1.8
WF
σ/μ = 12.87± 0.35 %
μ = 512.3± 1.7
σ = 65.9± 1.6



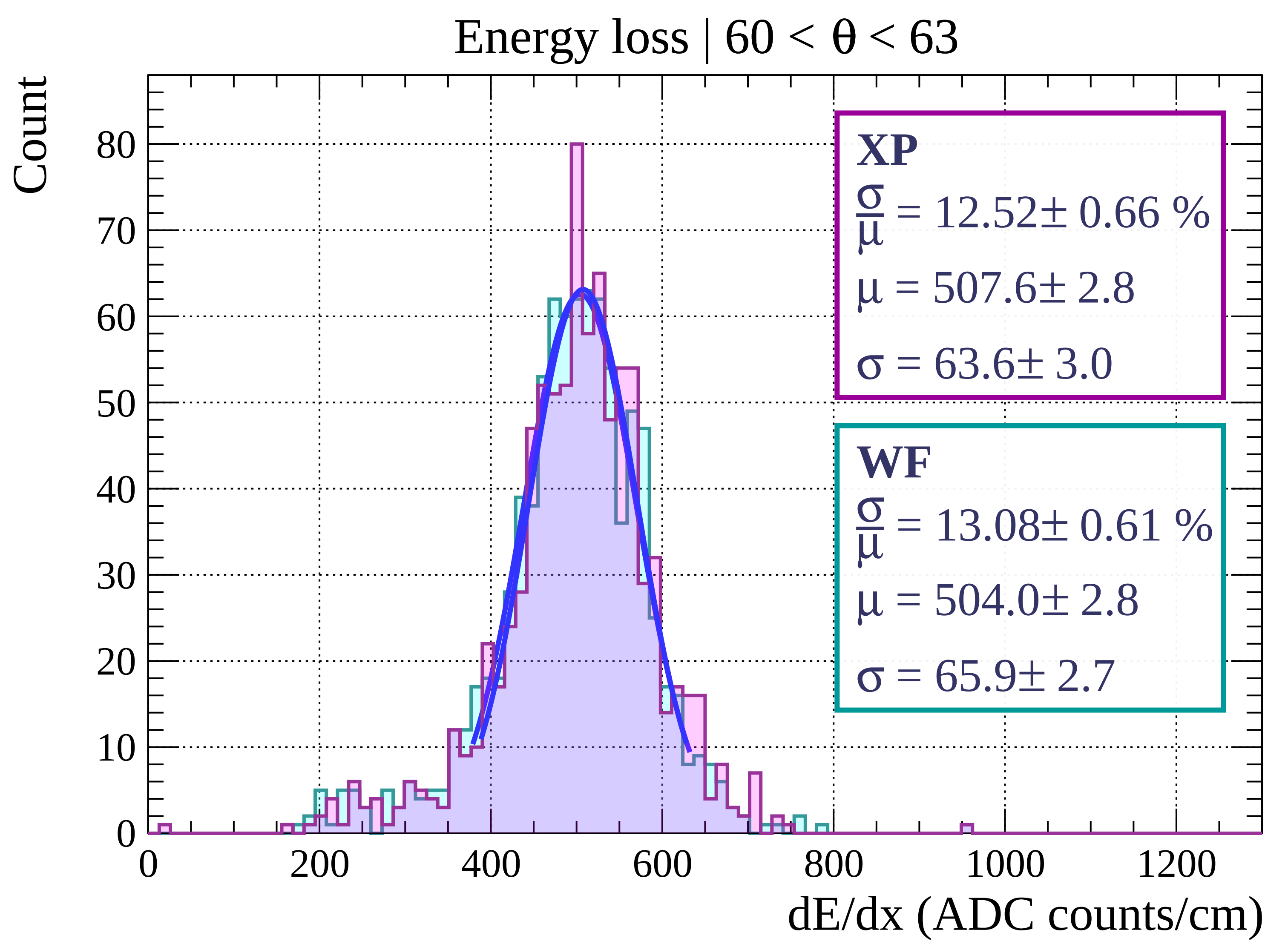
Energy loss | 51 < θ < 54
Count
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 13.87± 0.47 %
μ = 517.2± 2.1
σ = 71.7± 2.1
WF
σ/μ = 13.71± 0.48 %
μ = 509.4± 2.1
σ = 69.9± 2.2



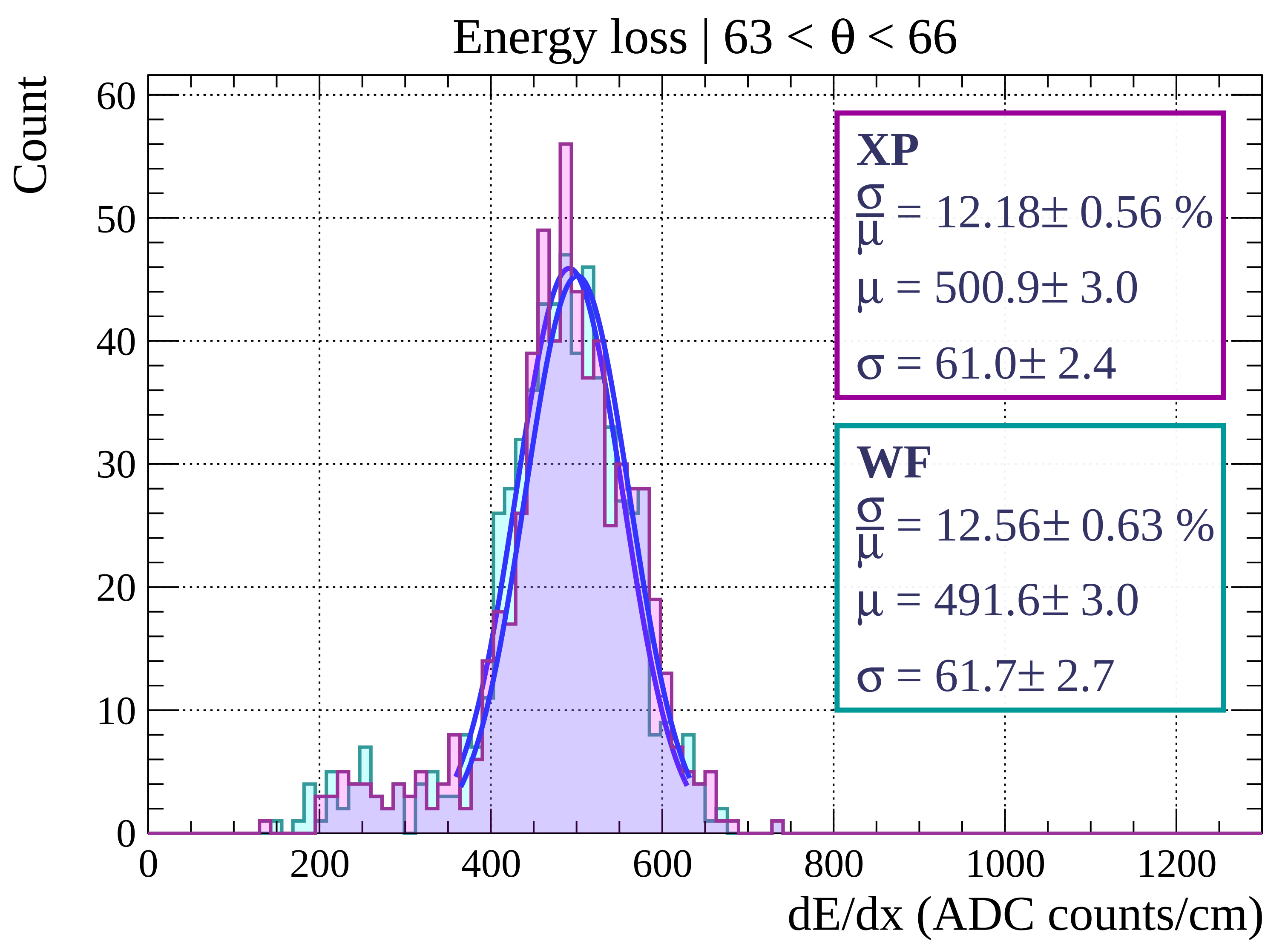
Energy loss | 54 < θ < 57
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP
σ/μ = 12.80± 0.46 %
μ = 519.7± 2.2
σ = 66.5± 2.1
WF
σ/μ = 12.66± 0.41 %
μ = 513.0± 2.1
σ = 65.0± 1.8



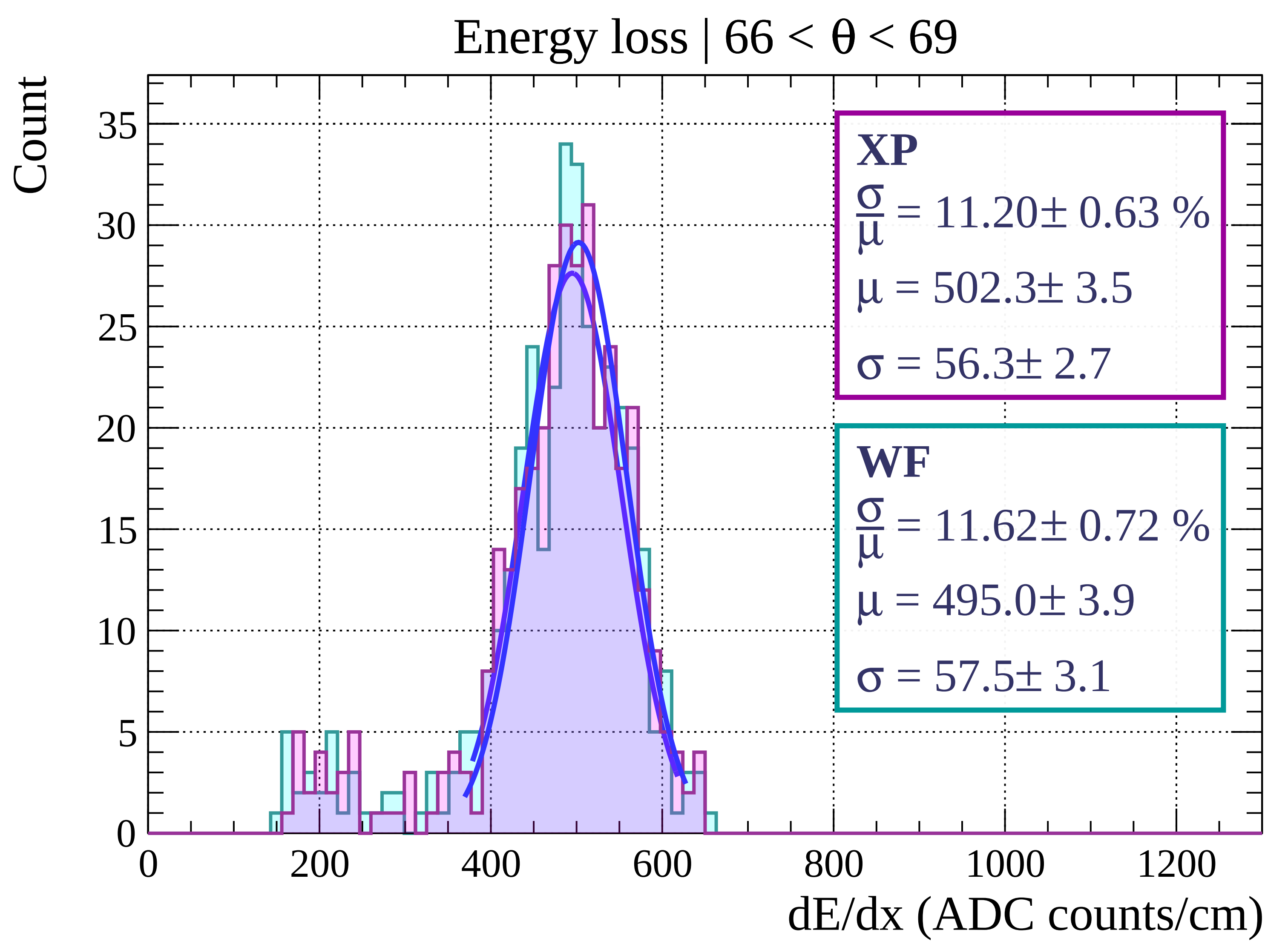
Energy loss | 57 < θ < 60
Count
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 13.32± 0.55 %
μ = 515.6± 2.5
σ = 68.6± 2.5
WF
σ/μ = 13.96± 0.57 %
μ = 508.0± 2.6
σ = 70.9± 2.6



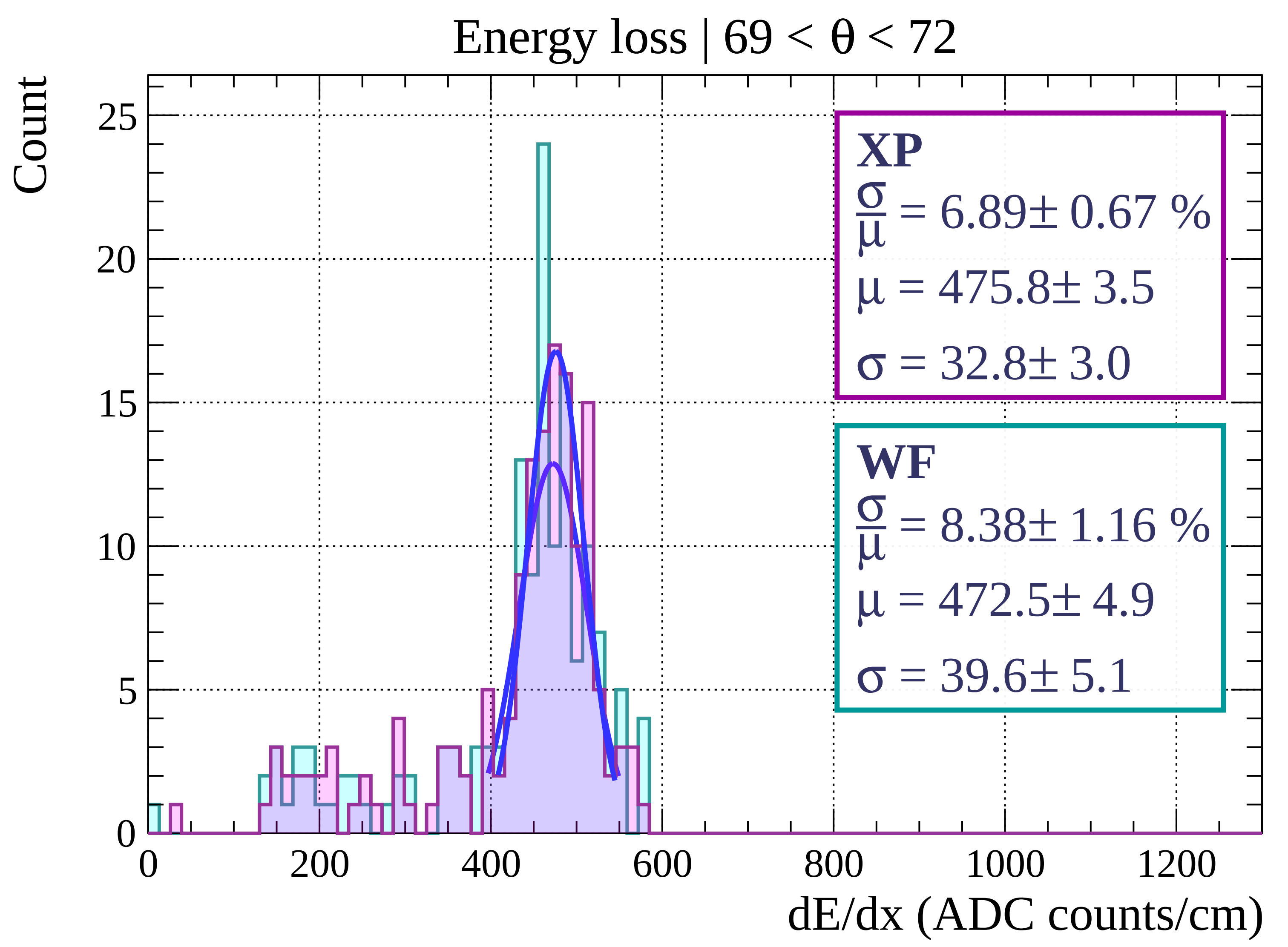
Energy loss | 60 < θ < 63
Count
0
10
20
30
40
50
60
70
80
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 12.52± 0.66 %
μ = 507.6± 2.8
σ = 63.6± 3.0
WF
σ/μ = 13.08± 0.61 %
μ = 504.0± 2.8
σ = 65.9± 2.7

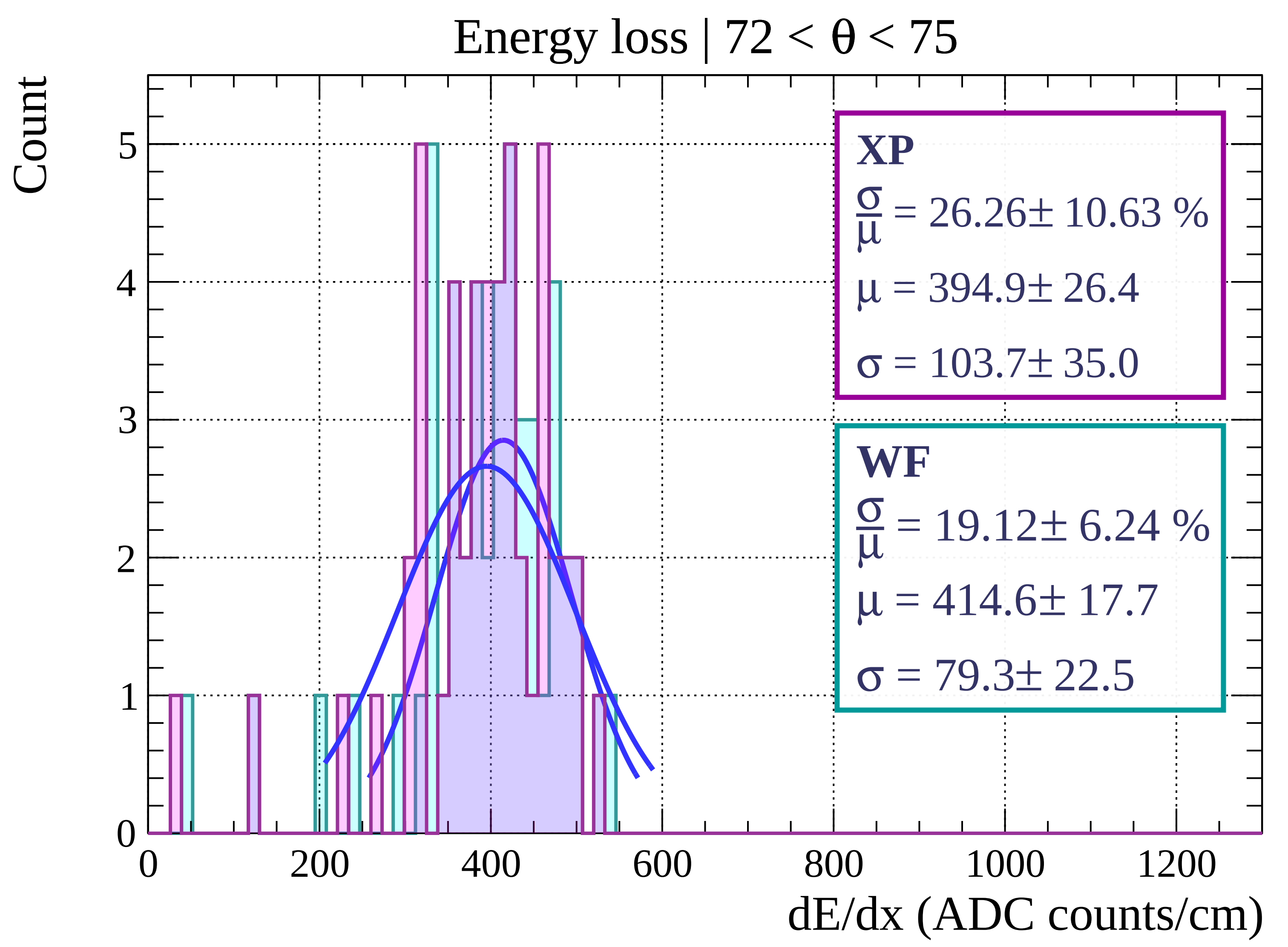


Energy loss | 63 < θ < 66
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200
XP
σ/μ = 12.18± 0.56 %
μ = 500.9± 3.0
σ = 61.0± 2.4
WF
σ/μ = 12.56± 0.63 %
μ = 491.6± 3.0
σ = 61.7± 2.7

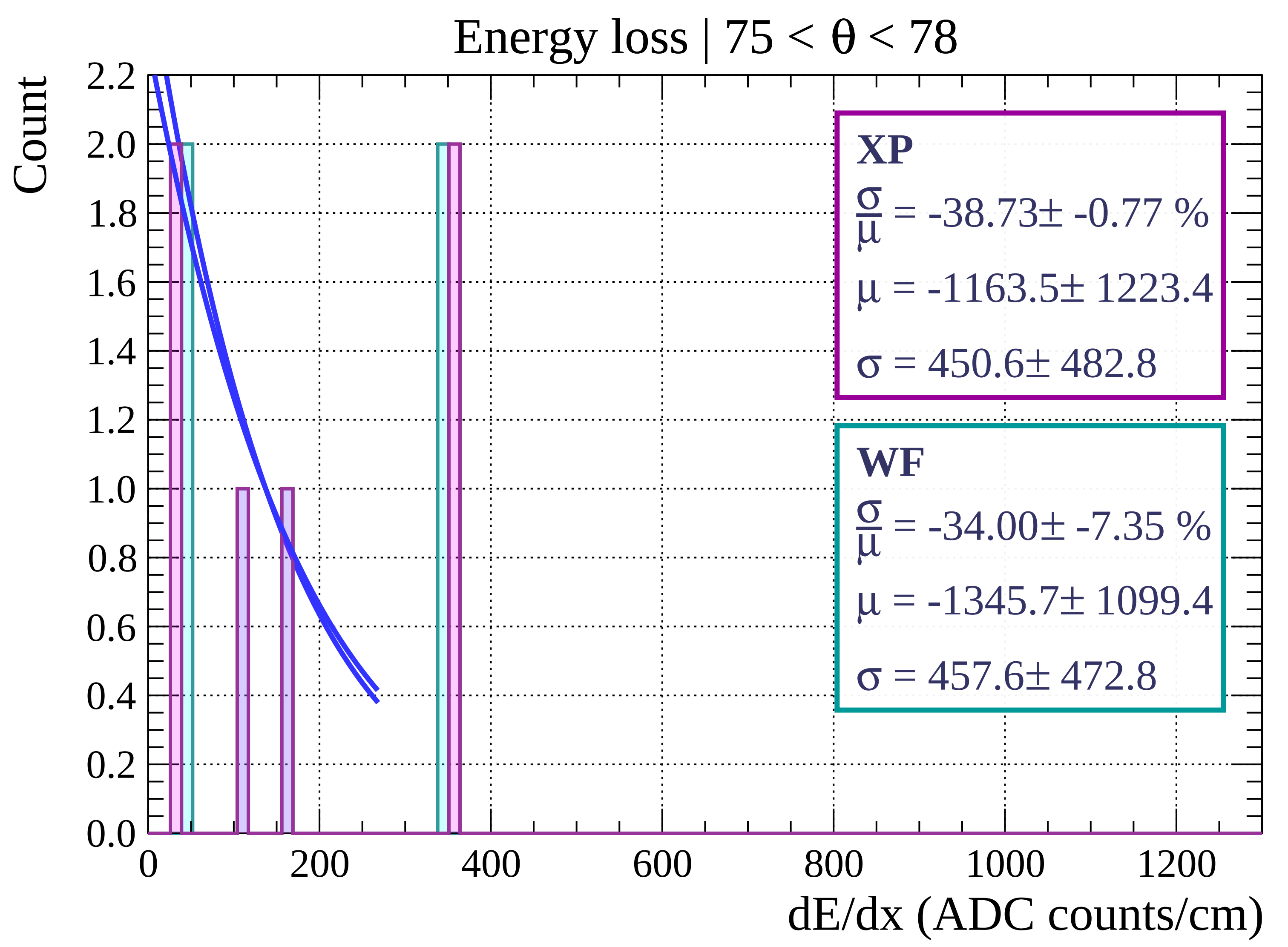


Energy loss | 66 < θ < 69
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
5
10
15
20
25
30
35
XP
σ/μ = 11.20± 0.63 %
μ = 502.3± 3.5
σ = 56.3± 2.7
WF
σ/μ = 11.62± 0.72 %
μ = 495.0± 3.9
σ = 57.5± 3.1

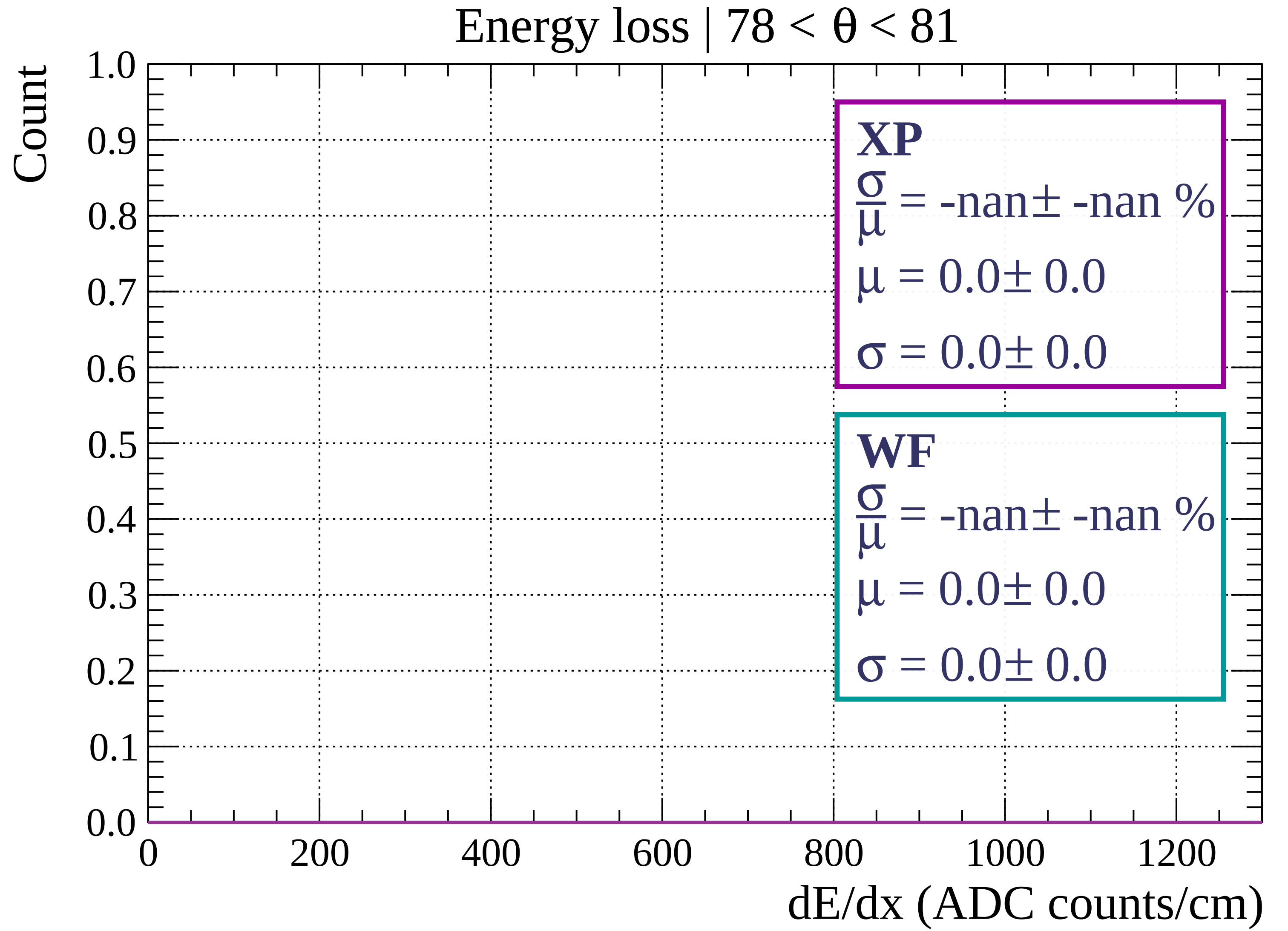

Energy loss | 69 < θ < 72
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
0
200
400
600
800
1000
1200
XP
σ/μ = 6.89± 0.67 %
μ = 475.8± 3.5
σ = 32.8± 3.0
WF
σ/μ = 8.38± 1.16 %
μ = 472.5± 4.9
σ = 39.6± 5.1



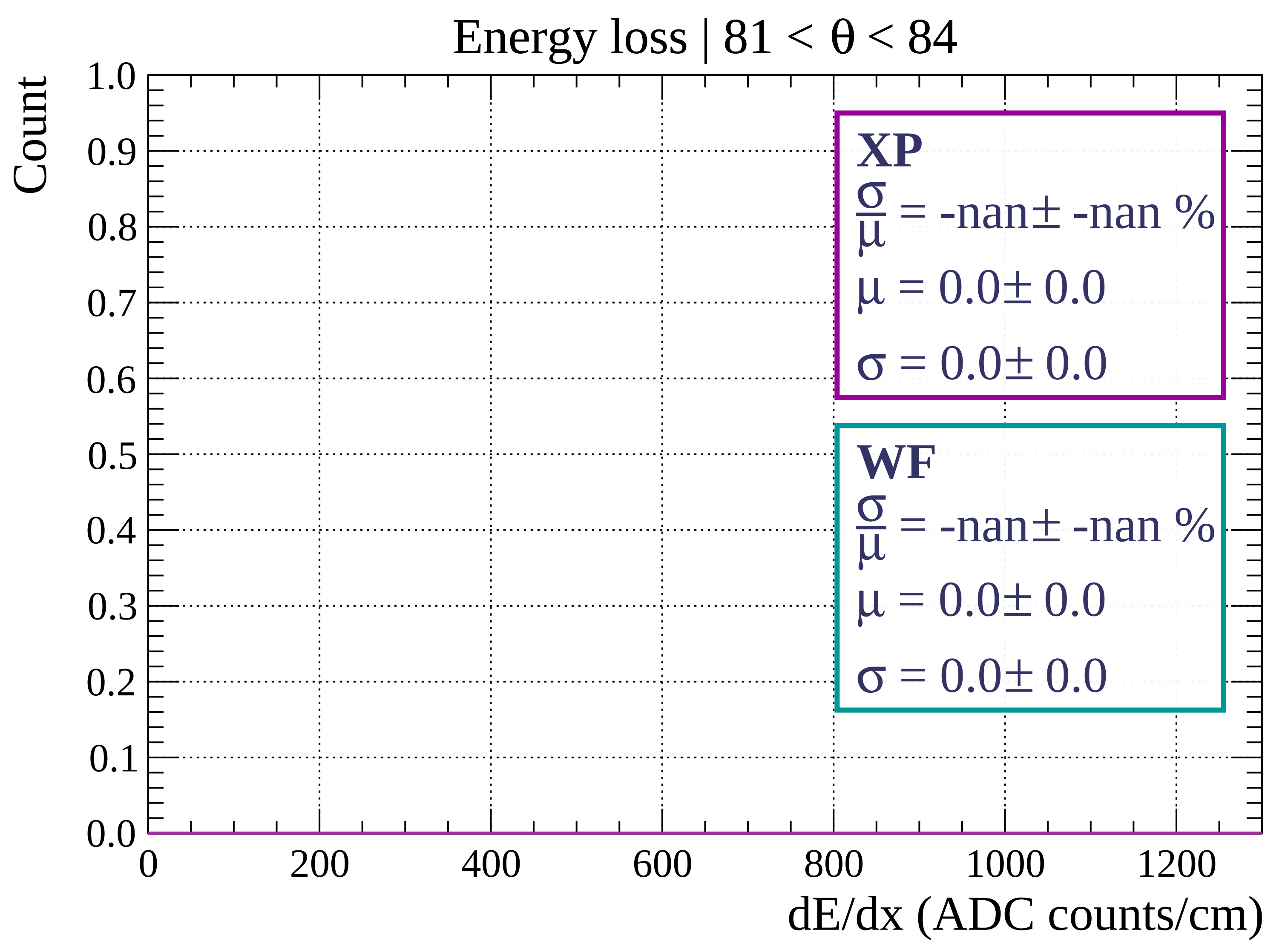
Energy loss | 72 < θ < 75
Count
dE/dx (ADC counts/cm)
0
1
2
3
4
5
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 26.26± 10.63 %
μ = 394.9± 26.4
σ = 103.7± 35.0
WF
$\frac{\sigma}{\mu}$ = 19.12± 6.24 %
μ = 414.6± 17.7
σ = 79.3± 22.5



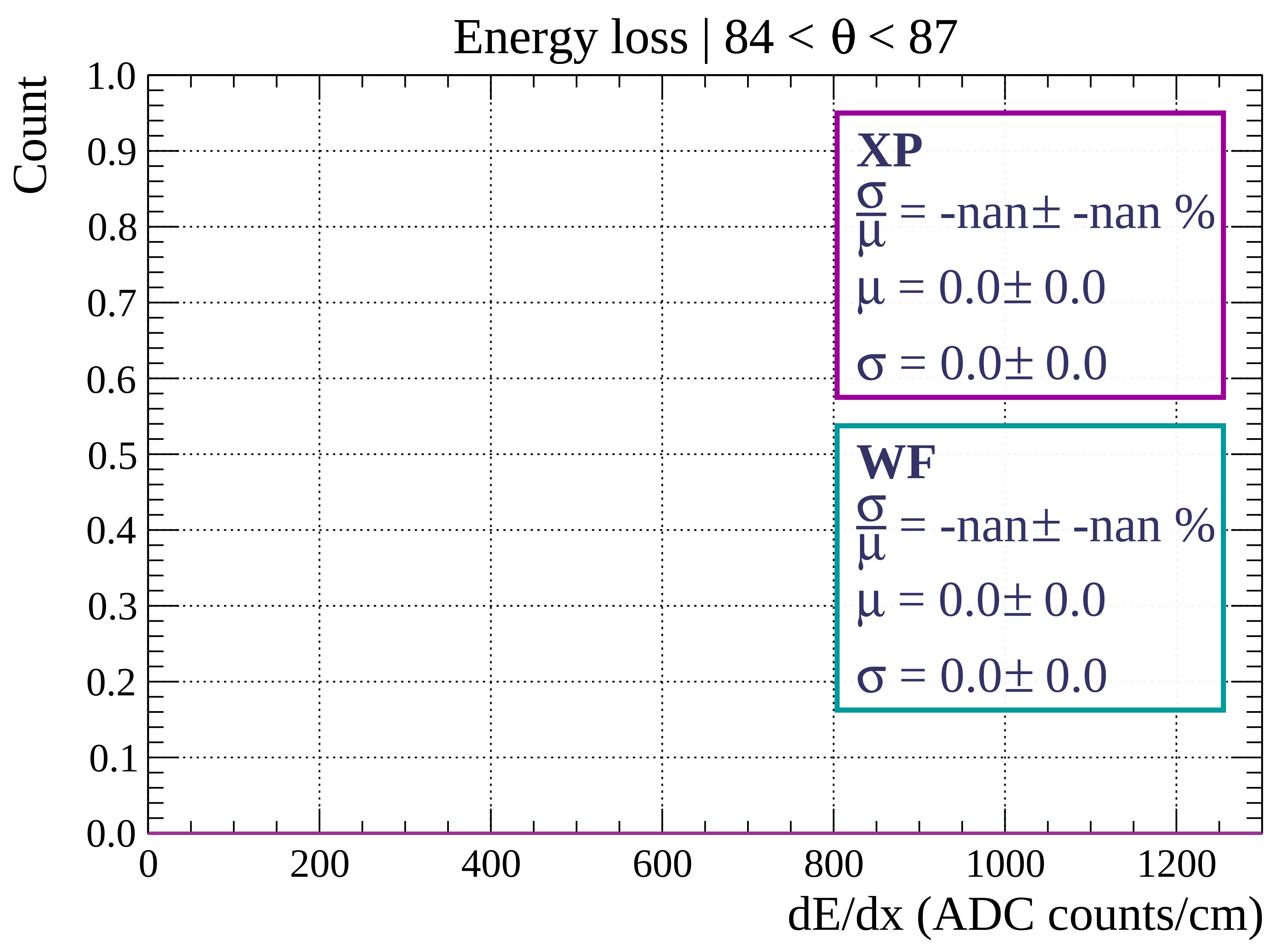
Energy loss | 75 < θ < 78
Count
dE/dx (ADC counts/cm)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
2.2
0
200
400
600
800
1000
1200
XP
σ/μ = -38.73± -0.77 %
μ = -1163.5± 1223.4
σ = 450.6± 482.8
WF
σ/μ = -34.00± -7.35 %
μ = -1345.7± 1099.4
σ = 457.6± 472.8

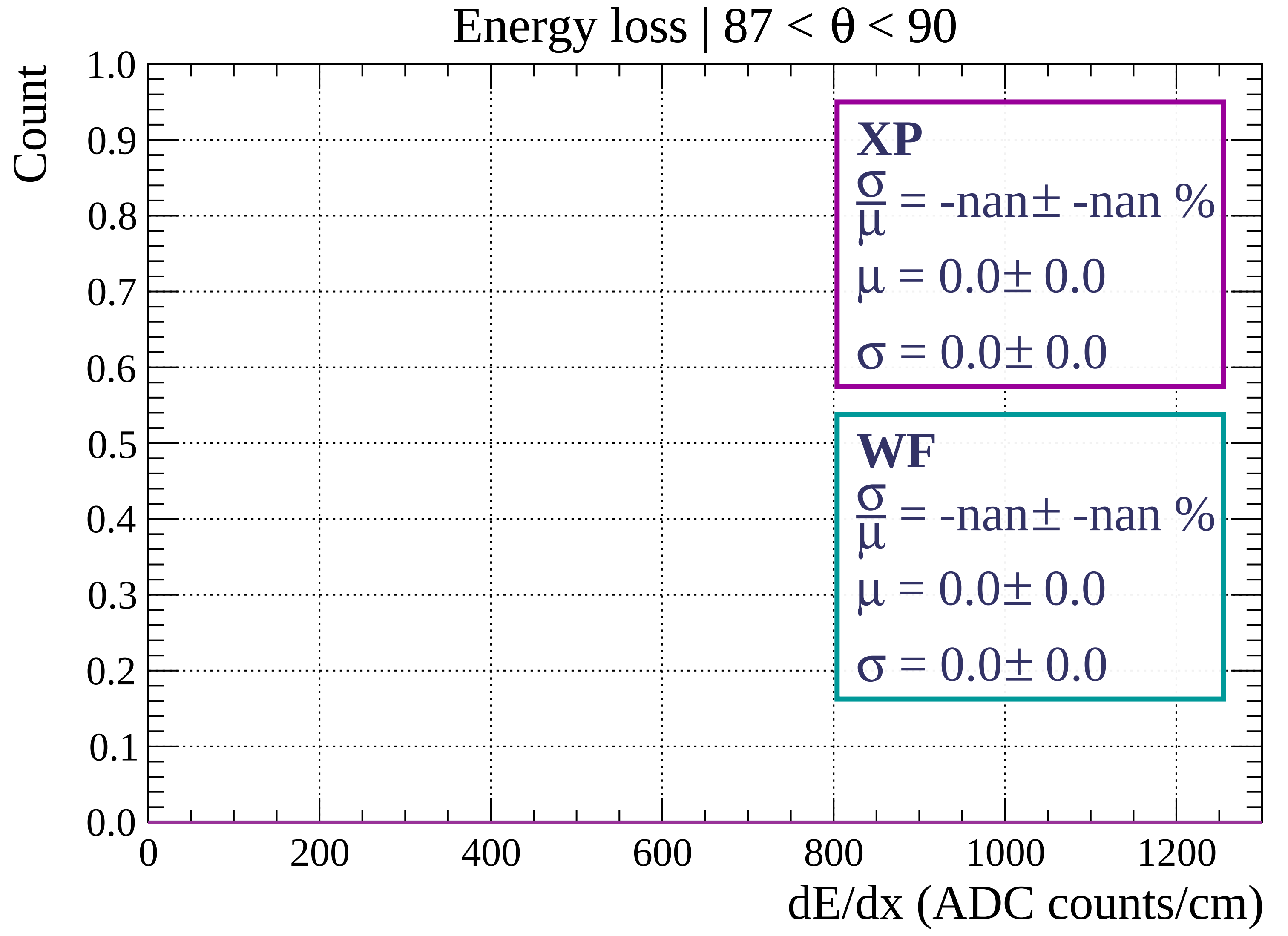


Energy loss | 78 < θ < 81
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

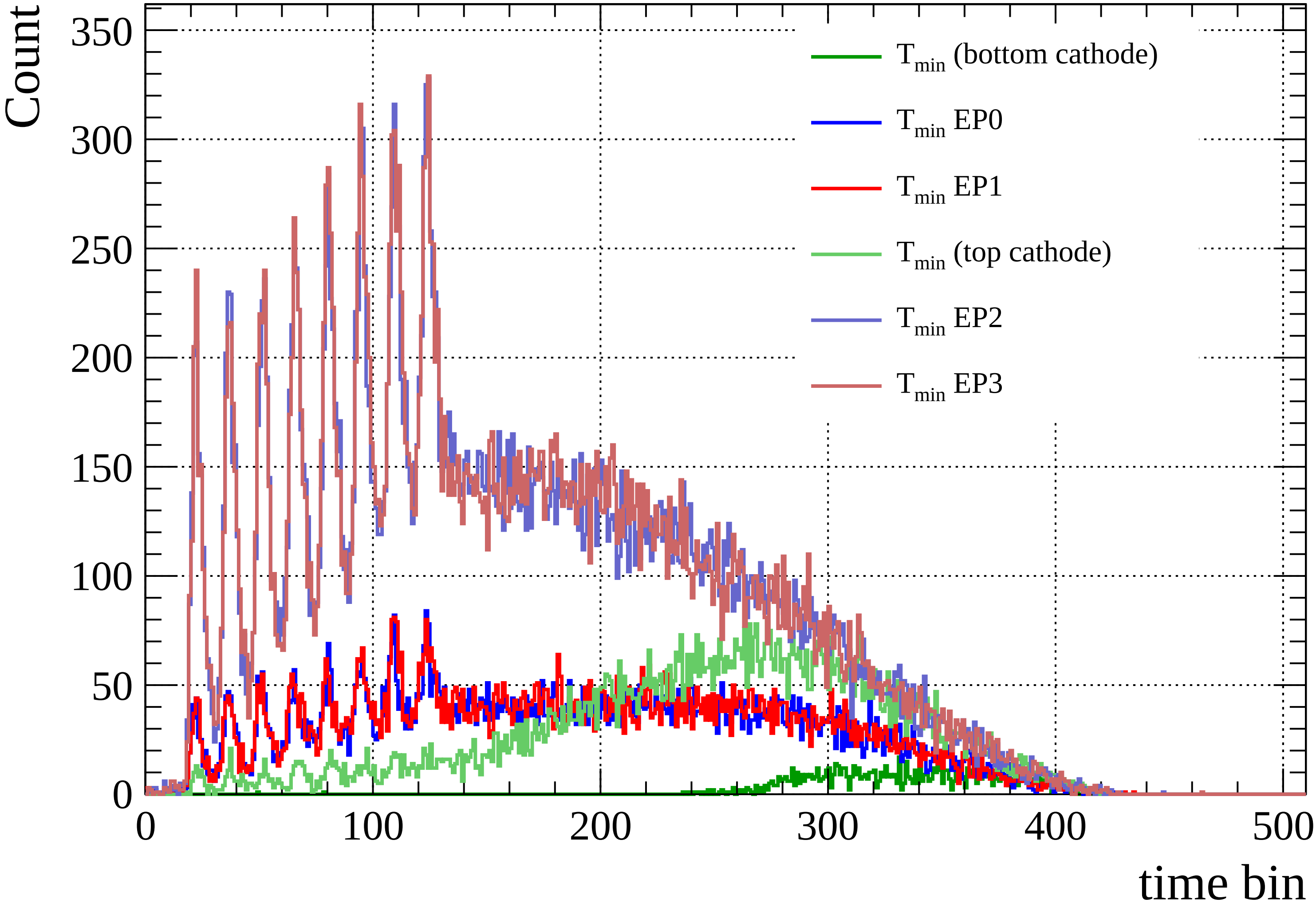


Energy loss | 81 < θ < 84
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

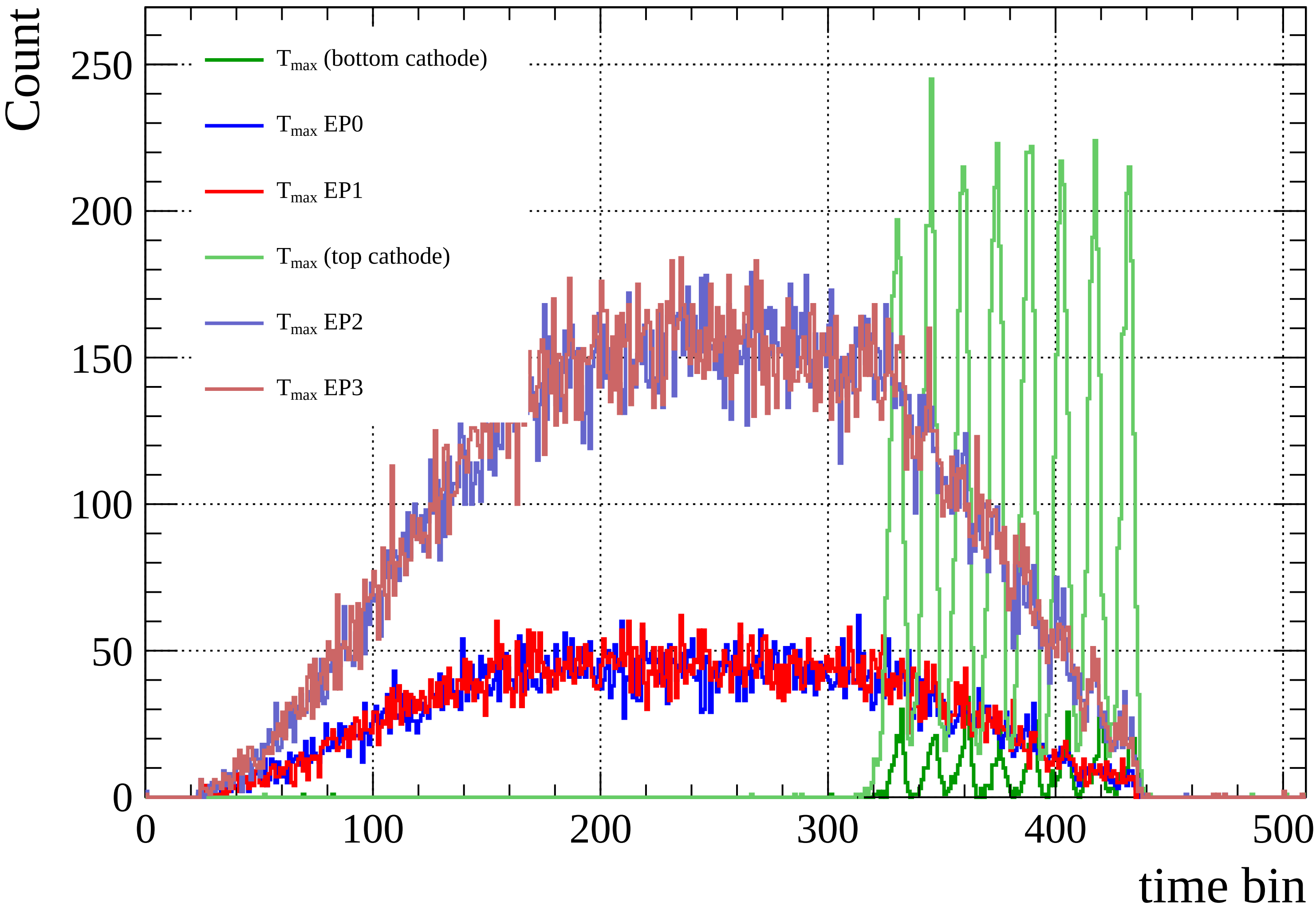

Energy loss | 84 < θ < 87
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



Energy loss | 87 < θ < 90
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



Count
time bin
0
100
200
300
400
500
50
150
250
350
T_{min} (bottom cathode)
T_{min} EP0
T_{min} EP1
T_{min} (top cathode)
T_{min} EP2
T_{min} EP3



Count
T_{max} (bottom cathode)
T_{max} EP0
T_{max} EP1
T_{max} (top cathode)
T_{max} EP2
T_{max} EP3
250
200
150
100
50
0
0
100
200
300
400
500
time bin